

**THE CSI FRAMEWORK FOR THE EVALUATION OF
BINARY CLASSIFICATIONS IN RARE EVENTS:
APPLICATIONS TO LAND USE CHANGE MODELS AND
ENVIRONMENTAL VALUATION**

Author:

Julen Perales Barriendo

Advisors:

Dr. María Jesús Campión Arrastia

Dr. Emilio Domínguez Irastorza

Dr. Nuria Osés Eraso



Faculty of Economics and Business Administration

Department of Economics

Public University of Navarra

May 2026

Agradecimientos

Mi parte favorita de cualquier tesis siempre han sido los agradecimientos. Me parece precioso ver, e incluso vivir en segunda persona, los procesos, inquietudes, motivaciones y reflexiones detrás de esos documentos tan asépticos y a la vez personales que son las tesis. Si algo he aprendido en estos años, es que este es uno de los trabajos más sociales que existen. Son las personas las que hacen que la ciencia avance. No la complejidad de las ideas, ni la magnitud de las publicaciones, ni el prestigio de las instituciones, sino las personas. En los siguientes párrafos quiero agradecer a las personas que han sido coautores de mi trabajo personal y profesional.

Primero de todo, me gustaría dar las gracias a mis directores de tesis. Me habéis cuidado, guiado, escuchado, protegido, enseñado y motivado sin esperar absolutamente nada a cambio. La explicación a esa generosidad, la de regalarme su tiempo y esfuerzo, es algo que no he podido resolver en esta tesis. Gracias Emilio, por tu brillantez (esférica), por tu talento, por tu contagiosa curiosidad, y por siempre creer que las personas son lo más importante. Gracias Nuria por tu pasión, por tu excelencia, por tu constancia, y por siempre creer que las personas son lo más importante. Gracias María Jesús, por tu humanidad, tu generosidad, tu vehemencia, y por siempre creer que las personas son lo más importante. María Jesús, contigo empezó todo, gracias a ti ha sido posible, y por ti seguirá siendo.

Gracias a Gil por acogerme y enseñarme tantas cosas. Cambiaste completamente el rumbo de mi tesis, pero también el de mi vida. En cierto modo, entiendo esta tesis como un homenaje a tu trayectoria profesional, sin la cual las contribuciones de este trabajo no se explican. Esta tesis tampoco se explica sin tus enseñanzas personales,

principalmente que la diversión es una parte primordial de este trabajo, y que es en las diferencias donde está el verdadero potencial para el aprendizaje.

Me gustaría hacer una mención especial a mis compañeros de sala, los doctorandos. Habéis sido una parte muy intensa de este proceso. Quiero agradecer, especialmente, a Sara y Álvaro, que habéis sido un soplo de aire fresco, y os habéis convertido en mis referentes para tantas cosas. A Alaitz, por darme el jamón de tus pintxos. Pero, sobre todo, gracias Izaskun. Gracias por poder y querer verme. Somos del tamaño de las cosas que vemos, y no te imaginas cuánto de lo que soy es por poder mirarte.

Para la gente del mundo real, concretamente amigos y familia, una breve reflexión sobre algo que nos toca vivir a muchos doctorandos y a sus entornos. Utilizando la jerga de esta tesis, me gustaría reflexionar sobre mis presencias y ausencias, tanto físicas como emocionales, que habéis tenido que sufrir y entender durante estos cuatro años. Si pensamos en las presencias que os he podido dedicar, y las que habéis podido esperar, lo primero es pedir perdón por mi baja prevalencia. Gracias por aceptar y respetar todas mis ausencias a lo largo de estos años. Gracias por disfrutar de todos los Hits, o veces que me esperabais y he podido corresponderos. Gracias por valorar todos mis False alarms, o veces que no me esperabais y he sacado un momento para vosotros. Pero sobre todo, perdón y gracias por ser tan comprensivos con mis Misses, o veces que me necesitabais y no he podido estar.

Y ya para terminar, gracias a mi familia, por darle sentido a todo lo que hago. A Íñigo, que me has apoyado y acompañado de forma incondicional, aunque yo no haya podido estar para ti. A mis padres, por el esfuerzo de intentar entenderme y escucharme, y de hacer mis alegrías vuestras. A mi abuela, por rezar tanto por mí. Todo lo que hago, incluido esto, lo hago por y para ti. Estoy seguro de que recordaré estos años como los años en los que he podido disfrutar contigo, y no como los años que hice la tesis.

Funding

This dissertation has been funded by the Ministry of Science and Innovation of Spain through project TED2021-132544B-I00. The research has also been supported by the Doctoral School of Navarra (EDONA) and the Public University of Navarra (UPNA). This support has been essential for the completion of this dissertation.

*Ahora sí,
esto ya es lo último.*

Abstract

This dissertation examines the evaluation of binary classification models by analyzing current metrics and practices, proposing a novel validation framework, and exploring their practical implications for land change science and environmental evaluation. Motivated by structural issues in land change model (LCM) validation, the manuscript focuses on applications where the event of interest is rare.

The first Chapter introduces the general problem of evaluating binary classifications and situates it within a broader interdisciplinary context, with particular reference to LCM. It provides a review of a literature that is dispersed across multiple disciplines, bringing together and systematizing different approaches to the evaluation of binary events. The chapter characterizes the main strands of this literature, summarizes their key critiques, and highlights the challenges arising from both the complexity of binary event evaluation and the fragmentation of existing research.

The second Chapter applies concepts from classification evaluation literature to propose and explore the CSI framework as an alternative suitable for any application where the event of interest is rare. In addition, it identifies and corrects an inconsistency in the formulation of the Gilbert Skill Score (GSS).

The third Chapter applies the findings and the interdisciplinary literature review to LCM validation practices. It examines how validation is currently conducted, highlighting key methodological limitations, particularly in the presence of spatial dependence and the rarity of land change. The chapter shows how the proposed CSI framework can address strongly rooted blunders in the field.

The fourth Chapter presents an empirical application based on LCM, in which the proposed methodological framework is used to estimate the economic value of soil erosion. The approach combines LCM, the revised universal soil loss equation (RUSLE), and replacement cost methods to provide a dynamic and spatially explicit valuation of erosion. The chapter demonstrates how LCM can be effectively integrated into an economic valuation framework.

Resumen

Esta tesis examina la evaluación de modelos de clasificación binaria mediante el análisis de las métricas y prácticas existentes, la propuesta de un nuevo marco de validación y la exploración de sus implicaciones prácticas en la ciencia del cambio de uso del suelo y en la valoración ambiental. Motivada por problemas estructurales en la validación de los modelos de cambio de uso del suelo (LCM), la investigación se centra en aplicaciones en las que el evento de interés es poco frecuente.

El primer capítulo introduce el problema general de la evaluación de clasificaciones binarias y lo sitúa en un contexto interdisciplinar más amplio, con especial atención a los LCM. Para ello, revisa una literatura dispersa en múltiples disciplinas, integrando y sistematizando los distintos enfoques existentes para la evaluación de eventos binarios. El capítulo identifica las principales corrientes de esta literatura, resume sus críticas fundamentales y pone de manifiesto los desafíos derivados tanto de la complejidad inherente a la evaluación de estos eventos como de la fragmentación del conocimiento disponible.

El segundo capítulo aplica conceptos de la literatura de evaluación de clasificación para proponer y analizar el marco CSI como una alternativa adecuada en contextos donde el evento de interés es poco frecuente. Asimismo, identifica y corrige una inconsistencia en la formulación del Gilbert Skill Score (GSS), contribuyendo a una interpretación más coherente de esta métrica.

El tercer capítulo traslada los resultados obtenidos y la revisión interdisciplinar a las prácticas de validación de los LCM. En él se analiza cómo se lleva a cabo actualmente la validación, poniendo el foco en limitaciones metodológicas clave, especialmente en presencia de dependencia espacial y en contextos caracterizados por la baja frecuencia de los cambios de uso del suelo. El capítulo muestra cómo el marco

CSI propuesto permite abordar errores profundamente arraigados en este campo de estudio.

El cuarto capítulo presenta una aplicación empírica basada en LCM, en la que el marco metodológico propuesto se emplea para estimar el valor económico de la erosión del suelo. El enfoque combina los LCMs, la Revised Universal Soil Loss Equation (RUSLE) y métodos de costes de reposición, con el fin de proporcionar una valoración dinámica y espacialmente explícita de la erosión. Este capítulo demuestra cómo los LCM pueden integrarse de manera efectiva en un marco de valoración económica.

Contents

Acknowledgments	i
Funding	iii
Abstract	v
Resumen	vi
Table of Contents	viii
List of Figures	xii
List of Tables	xvi
Introduction	1
1 Chapter 1: Land Change Models and Binary Classification Evaluation: A Literature Review	4
1.1 Land change and its modeling	5
1.1.1 Land change	5
1.1.2 Land change models	6
1.1.2.1 Land change models and their applications	6
1.1.2.2 Historical overview of LCM	9
1.1.2.3 Structure of pattern-based LCM	13
1.1.3 Challenges in land change science	16
1.2 Evaluation metrics for binary classification	26
1.2.1 Single threshold metrics	30
1.2.1.1 Quantity and Allocation differences	30
1.2.1.2 First order single threshold metrics	34
1.2.1.3 Second order single threshold metrics	38
1.2.2 Multiple threshold metrics	41
1.2.3 Comparability of single and multiple threshold metrics	45

1.2.4	Agreement vs Skill	47
1.2.4.1	Definition of Agreement and Skill	47
1.2.4.2	Overview of Skill metrics	49
1.2.5	Limits and domain of application of classification evaluation metrics	54
1.2.6	Current LCM validation practices	57
1.2.6.1	Hard validation practices	58
1.2.6.2	Soft validation practices	60
1.2.7	Binary classifications in Economics	61
2	Chapter 2: Introducing the CSI Framework as a Novel Methodology for Binary Classification Evaluation with Rare Events	88
2.1	Introduction	89
2.2	Methods	91
2.2.1	Construction of TOC and CSI plots	91
2.2.2	AUCSI and AUCSIS: new metrics to evaluate the Agreement and Skill of multiple thresholds	97
2.2.3	Distribution of Total and Reference areas in CSI and TOC plots	99
2.3	Analyzing the properties of the CSI framework	102
2.3.1	Revisiting metrics of multiple thresholds	102
2.3.1.1	AUCSI as a metric of Agreement of the rare event at multiple thresholds	102
2.3.1.2	AUCSI Score as a metric of Skill of the rare event at multiple thresholds	108
2.3.2	Revisiting metrics of single thresholds	112
2.3.2.1	CSI as a metric of Agreement of the rare event at a single threshold	112
2.3.2.2	Gilbert Skill Score as a metric of Skill of the rare event at a single threshold	116
2.3.3	Simulation-Based Assessment of the CSI Framework	125
2.3.3.1	Comparison between two simulated classifiers	126
2.3.3.2	Comparison of multiple concave classifiers	129
2.4	Discussion	133
2.5	Conclusion	136
3	Chapter 3: The CSI Framework in Land Change Model Validation	143
3.1	Introduction	145
3.2	Current LCM validation approaches and their limitations	147
3.3	Methods	149
3.4	Results	151

3.4.1	Validation with conventional practices	151
3.4.2	Validation with the FOM framework	157
3.4.3	Insights on the current use of FOM in LCM	160
3.4.4	Considering other alternative baselines	164
3.5	Discussion	166
4	Chapter 4: The economic value of erosion: Integrating land changes and nutrient prices	177
4.1	Introduction	180
4.2	Methods	181
4.3	Case study	184
4.3.1	Data	184
4.3.2	Results	185
4.4	Discussion	191
	General Conclusions and Further Research Lines	197
4.5	General conclusions	198
4.6	Conclusiones generales	203
4.7	Further research lines	208
A	Appendix Chapter 1	216
B	Appendix Chapter 2	218
B.1	The math behind CSI curves	218
B.1.1	Reconstruction of Contingency Table from CSI curves	218
B.1.2	Slopes of CSI curves	219
B.1.3	Equation of the baseline CSI curve	220
B.2	Construction of AUC and AUCS	221
B.3	Construction of AUCSI and AUCSIS	223
B.4	Question 1	226
B.4.1	AUC	226
B.4.2	AUCSI	226
B.4.3	AUCS	227
B.4.4	AUCSIS	227
B.5	Question 2	228
B.5.1	AUC	228
B.5.2	AUCSI	228
B.5.3	AUCS	228
B.5.4	AUCSIS	229
B.6	Question 3	230

B.6.1	AUC	230
B.6.2	AUCSI	231
B.6.3	AUCS	233
B.6.4	AUCSIS	234
B.7	Design of the simulated classifiers	236
B.7.1	Calculation of the upper and lower boundaries of Figure 2.12 .	237
B.7.2	Simulated TOC Curves	237
C	Appendix Chapter 3	240
C.1	Data sources and data processing for the case study	240
	Index of Definitions	241
	Index of Equations	243

List of Figures

1.1	Representation of a pattern-based model structure. Bold box frames indicate input data, red box frames indicate hard classified outputs, and green box frames indicate soft classified outputs. <i>Source: Adapted from Ren et al. (2019)</i>	14
1.2	Toy examples of three map comparisons and their corresponding contingency tables used to derive Quantity and Allocation differences. P stands for presence and A for absence.	32
1.3	Toy example of a multiple threshold evaluation of a classifier. <i>Source: Adapted from Pontius (2022)</i>	42
1.4	Representation of the ROC (a) and Precision-Recall (b) curves. The green line represents the random classification baseline. <i>Source: Self elaboration</i>	43
1.5	Illustration of two classifications of the same observations, showing different levels of Quantity disagreement. <i>Source: Self elaboration</i> . . .	47
2.1	Representation of the TOC (a) and CSI (b) plots. <i>Source: Self elaboration</i>	94
2.2	Representation of the areal comparison of the three analyses. The green and red areas are compared for each of the questions. The first and second columns correspond to the CSI and TOC graphs, respectively. Panels a) and b) address Question (1), panels c) and d) address Question (2), and panels e) and f) address Question (3). <i>Source: Self elaboration</i>	103
2.3	Proportion of the Total area covered at each threshold by TOC (blue lines) and CSI (orange line) plots. The solid lines correspond to a population with 5% prevalence. The dotted lines correspond to a population with 25% prevalence. The triangular markers on the lines indicate the proportion of the Total area that is covered at P . The circular markers indicate the points where the slope is equal to the slope of the diagonal line. <i>Source: Self elaboration</i>	104

2.4	Representation of the sensitivity of equations (2.6) and (2.7) to prevalence at $x = P$. Panel (a) shows the sensitivity of the elasticities; Panel (b) shows the sensitivity of the asymmetries between the right and left of P ; Panel (c) shows the sensitivity of the accumulated proportion of the Total area at P . The triangles correspond to the same markers as in Figure 2.3. <i>Source: Self elaboration.</i>	106
2.5	Proportion of the Reference area covered at each threshold by TOC (blue lines) and CSI (orange line) plots. The solid lines correspond to a population with 5% prevalence. The dotted lines correspond to a population with 25% prevalence. The triangular markers on the lines indicate the proportion of the Reference area that is covered at P . The circular markers indicate the points where the slope is equal to the slope of the diagonal line. <i>Source: Self elaboration.</i>	109
2.6	Representation of the sensitivity of equations (2.8) and (2.9) to prevalence at $x = P$. Panel (a) shows the sensitivity of the elasticities; Panel (b) shows the sensitivity of the asymmetries between the right and left of P ; Panel (c) shows the sensitivity of the accumulated proportion of the Reference area at P . The triangles correspond to the same markers as in Figure 2.5. <i>Source: Self elaboration.</i>	110
2.7	Representation of the CSI plots with a prevalence of 10% and 50%. The colored areas represent the space where classifications can get a CSI above 60%. <i>Source: Self elaboration.</i>	114
2.8	Representation of the two CSI plots with prevalences of 20% (a) and 50% (b). The dashed lines represent thresholds of a CSI of 30% and 60%. <i>Source: Self elaboration.</i>	115
2.9	Representation of multiple thresholds for the illustrative classifier using OA (a) and CSI (b). Points correspond to contingency tables of a perfect classifier, a baseline classifier, and a selected reference classifier under different definitions of Skill. <i>Source: Self elaboration.</i>	119
2.10	Venn diagram of the contingency table of the Reference Classification and its corresponding random classification (b). <i>Source: Self elaboration.</i>	122
2.11	TOC (a) and CSI (b) plots of two of the classifiers of the simulation.	127
2.12	Relationships between single-threshold and multi-threshold metrics. Panels (a) and (b) show Agreement-based metrics, while panels (c) and (d) show Skill-based metrics. Color indicates the skewness of the curve, determined by the point at which it converges toward the baseline. Labeled points correspond to the same classifiers shown in Figure 2.11. .	130
3.1	Observed Forest and Non-Forest presence at 2005 (a) and 2023 (b). Forest presence in (b) is decomposed into Forest persistence and Forest gains.	150

3.2	Simulated soft and hard maps for Forest presence (a)–(b) and Forest gains (c)–(d). Panels (a)–(b) show the suitability map and corresponding hard-classified map for Forest presence in 2023. Panels (c)–(d) show the suitability map and corresponding hard-classified map for Forest gains during 2005–2023. In the lower suitability map, locations with Forest presence in 2005 are excluded. The hard-classified maps are constrained to the same total quantity of simulated Forest presence to facilitate comparison.	152
3.3	ROC curves corresponding to the validation of Forest presence (a) and Forest gain (b) of the suitability and categorical maps.	153
3.4	Agreement and disagreement between observed and simulated Forest presence (a) and Forest gain (b). In Panel (b) forest in 2005 is excluded from the extent as it is not eligible for Forest gain Source: Self elaboration.	154
3.5	Graphical representation of the FOM framework applied to our suitability and categorical maps.	158
3.6	TOC (a) and FOM (b) plots showing model performance across different prevalence levels. <i>Source: Self elaboration.</i>	162
3.7	Representation of the TOC (a) and FOM (b) plots. <i>Source: Self elaboration.</i>	166
4.1	Graphical summary of the steps involved in the methodology. 1 application of CLUE-S to estimate a land use time-series. 2 application of the RUSLE equation to estimate erosion rates. 3 application of the replacement cost method to estimate the economic valuation of erosion. Source: Self elaboration.	182
4.2	Spatial distribution of (a) Nitrogen, (b) Phosphorus and (c) Potassium densities (normalized colour scales). North, Transition and South areas are marked by a blue line. Source: Self elaboration based on ESDAC.	185
4.3	Evolution of land uses (column 1), soil erosion (column 2), and valuation of erosion (column 3), in years 2005 (row 1), 2023 (row 2) and 2030 (row 3). North, Transition and South areas are marked by a coloured line. Source: Self elaboration.	186
4.4	Evolution of (a) land uses, (b) soil erosion, (c) prices and (d) erosion valuation for the whole region. Panel (b) expresses erosion as an index 2005 = 100. The vertical line delimits the observation (2005–2023) and projection periods (2024–2030). Source: (a), (b) and (d), Self elaboration based on the results; (c) adapted from Campión et al (2024).	189
4.5	Changes in valuation of erosion between 2005 and 2030. Map of the region and two zoomed areas. North, Transition and South areas are marked by a coloured line.	190

B.1	TOC (a) and CSI (b) plots of the most and less efficient possible concave TOC curve. <i>Source: Own elaboration.</i>	238
-----	--	-----

List of Tables

1.1	Contingency table for binary classification showing the relationship between observed and classified outcomes.	29
1.2	Finley's contingency table for tornado forecast	30
1.3	Summary of binary classification performance metrics derived from the contingency table, including formulas and commonly used alternative terminology.	35
1.4	Agreement and skill metrics applied to Finley's data.	56
1.5	Contingency table for land changes	58
2.1	Contingency table for binary classification with formal notation, where outcomes are defined conditional on a classification threshold.	92
2.2	Table summarizing Agreement and Skill across questions.	112
2.3	Contingency tables for illustrative classification cases, including the perfect classifier, a baseline classifier, and intermediate cases used in the evaluation of skill metrics. These tables correspond to the points represented in Figure 2.9.	118
2.4	Example of classification, random classification, and numerators for the illustration in Figure 2.10.	124
2.5	Comparison of Agreement and Skill results between Classifiers A and B. We compare metrics that consider both classes (OA, HSS, AUC) against those of only the minority class (CSI, GSS, AUCSI).	127
3.1	Contingency table comparing the observed land presence in 2005 and 2023 in km ²	150
3.2	Contingency table of observed and simulated Forest presence and Forest gain in km ²	153
4.1	Temporal evolution (2005, 2023 and 2030) of land use (percentage of the total area for each land use type within a specific region); erosion (in tonnes per hectare in a specific area); and valuation (in millions € in a specific area).	187

Introduction

Binary classification models are widely used across disciplines to support decision-making in contexts where the outcome of interest has two possible states, such as the occurrence or absence of an event. They are applied in fields as diverse as medical diagnosis, meteorology, machine learning, economics, and environmental modeling, including land change analysis, all of which have, to varying degrees, shaped the development and perspective of this dissertation. In all these cases, the evaluation of model performance plays a central role, as it determines how results are interpreted and ultimately whether they can be trusted for practical use. However, despite the extensive use of binary classifiers, the methodologies employed to assess their performance remain fragmented, often inconsistent, and frequently misaligned with the specific characteristics of the problems to which they are applied.

Within this setting, it is important to clarify the context in which this dissertation has been developed, as this helps guide its interpretation. The initial motivation for this research arises from a series of methodological challenges encountered in the validation of Land Change Models (LCM). In particular, many of these challenges derive from a defining characteristic of land change processes: changes are typically rare events. This low prevalence has important methodological, practical, and even ethical implications for how models are evaluated and how their results are interpreted and communicated.

As is to be expected, this issue is not exclusive to LCMs but is shared across a wide range of disciplines in which the event of interest is rare. Various disciplines have developed parallel evaluation frameworks for such cases, often using distinct terminology and emphasizing different aspects of performance. This has led to a decentralized body of knowledge in which similar concepts are repeatedly redefined, hindering both accessibility for end users and the transfer of methodological advances

across fields. Consequently, the difficulties, limitations, and insights identified in this dissertation extend well beyond the specific application that motivates it. In this sense, LCMs serve both as a concrete case study and as a unifying context through which broader challenges in binary classification can be illustrated and addressed.

Against this background, the main objective of this dissertation is to review, systematize, and advance the evaluation of binary classification models, with particular attention to their application in LCMs. This objective is pursued through two complementary research goals.

The first goal is to make the concepts and methodologies of binary classification evaluation more accessible, coherent, and useful for end users. To this end, the dissertation provides a structured synthesis of existing evaluation metrics and frameworks, highlighting their relationships, limitations, and areas of applicability. Building on this foundation, it introduces a new methodological framework designed to improve interpretability and facilitate the joint analysis of different types of classification outputs. Throughout the manuscript, a central motivation is the development of tools that enable users to better understand model behavior beyond single summary statistics.

The second goal is to explore how these methodological advances can improve validation practices in LCM. LCMs represent a particularly relevant application domain, as they typically involve low-prevalence events and strong links to policy-making. Despite this, their validation practices often rely on metrics that are not well-suited to these characteristics. By incorporating insights from the broader literature on binary classification and applying the proposed framework to LCM contexts, this dissertation aims to refine and modernize existing validation approaches, providing more informative and reliable assessments of model performance.

Beyond these two core objectives, the dissertation also seeks to bridge the gap between methodological development and empirical application. While advances in evaluation metrics are often discussed at a theoretical level, their practical implications become fully apparent only when they are applied to real-world problems. In this regard, the dissertation includes an empirical application that links land use modeling with the economic valuation of environmental processes. This application serves to illustrate how differences in model performance and error structures can translate

into differences in economic outcomes and policy-relevant conclusions, highlighting the broader importance of appropriate evaluation practices.

The structure of the dissertation reflects these objectives. Chapter 1 provides a comprehensive review and organization of binary classification evaluation methods. Chapter 2 introduces a new framework for the analysis of classification performance, emphasizing interpretability and the distinction between different types of errors. Chapter 3 applies these insights to the validation of LCMs, critically assessing existing practices and proposing methodological improvements. Finally, Chapter 4 extends the analysis to an empirical setting, demonstrating how model outputs can be integrated into an economic valuation framework to support policy-relevant analysis.

Chapter 1

Land Change Models and Binary Classification Evaluation: A Literature Review

1.1 Land change and its modeling

1.1.1 Land change

Land use refers to the human activities and management practices that take place on the earth’s surface (Bakker and Veldkamp 2008; FAO and UNSD 2020). Land use refers to the way land is used for human activities, such as urban, agricultural, or forestry uses. The dynamics of land use and land change are pivotal in shaping the economic, environmental, and social dimensions of sustainable development (Lambin 2006).

Economically, land use directly affects and is affected by economic development (Duke and Wu 2014), as it is tightly connected with patterns of economic growth and resource allocation. For instance, the conversion of agricultural land to urban areas can drive economic growth through infrastructure development and increased housing supply. However, this conversion often comes at the cost of agricultural productivity, rural livelihoods, or landscape transformations, illustrating tradeoffs between land use and economic outcomes (Turner et al. 2007).

Environmentally, land use is a critical factor influencing several key processes, including biodiversity, soil health, and climate stability. Deforestation for agricultural expansion, for example, has led to large biodiversity losses in tropical regions, threatening countless species and disrupting ecosystems (Meyfroidt et al. 2013). Similarly, unsustainable agricultural practices can accelerate soil erosion, leading to the degradation of arable land and compromising food security (Borrelli et al. 2017). Additionally, land changes such as urban sprawl and deforestation are major contributors to greenhouse gas emissions, contributing to climate change and its associated impacts (Foley et al. 2005; Turner et al. 2007).

Socially, land use plays a fundamental role in shaping communities and influencing social equity. Urbanization, while often a driver of economic opportunity, can lead to social disparities, such as the displacement of low-income populations and the loss of communal lands vital for rural communities (Smith 1996; Borras Jr. et al. 2011; UN-Habitat 2020). Urbanization is the main cause of rural depopulation and land abandonment. The transition from traditional land uses to more commercial or industrial purposes can disrupt social structures, erode cultural heritage, and reduce

access to essential resources like clean water and arable land (Borras Jr. et al. 2011). Moreover, the social implications of land use are evident in issues such as food security, where the shift from subsistence agriculture to cash crops can influence local food systems and populations' livelihoods (UN-Habitat 2020).

In most regions of the world, land uses tend to remain rather static over time (Whittle et al. 2012; Pontius, Boersma, et al. 2008). This means that land changes are generally infrequent phenomena. However, the relative infrequency of land changes compared to land use persistence does not diminish their significance. Land changes are often reflections of economic and social dynamics within regions. Thus, they tend to signal broader tendencies that can influence even those areas where land use remains unchanged. For example, urban expansion in one region can alter economic pressures, resource availability, and environmental conditions in neighboring areas, leading to indirect spillover effects across the landscape (Lambin et al. 2000).

1.1.2 Land change models

1.1.2.1 Land change models and their applications

Land change models are tools for the conceptual and quantitative representation of land use dynamics and patterns. These models serve as means to identify the factors that drive changes in land use, simulate the underlying processes that generate these changes, and project potential future trajectories of land use (Vliet 2016; Lambin et al. 2000). The utility of LCM extends far beyond predictions. They provide a framework for exploring the complex interactions and feedback loops that shape land use over time. Moreover, LCMs facilitate communication among stakeholders by providing a common platform to visualize and discuss potential land use scenarios (Noszczyk 2019; Verburg, Alexander, et al. 2019). This makes them useful not only in academic research but also in practical applications like policy formulation, urban planning, and environmental conservation.

LCMs are often categorized into predictive and explanatory models, each serving distinct but complementary purposes (Verburg, Neumann, et al. 2011). Explanatory models, also known as process-based models, emphasize understanding the underlying processes and interactions that shape land changes (Overmars et al. 2007). By

providing insights into historical and current land use dynamics, these models help stakeholders identify key drivers of change and design interventions that align with socio-economic and environmental objectives. These models are effectively being implemented as a tool to enhance the conversation between social, economic, and environmental actors.

An example of an explanatory model application can be found in agricultural land use analysis. Explanatory LCMs have been employed to reconstruct historical land use patterns and understand how socio-economic shifts, technological advancements, and policy changes have influenced agricultural expansion or contraction over time (Lambin et al. 2000; Verburg, Schot, et al. 2004). By uncovering these patterns, researchers and policymakers can identify the drivers that have historically led to land degradation or productivity increases, enabling them to craft targeted policies that encourage sustainable agricultural practices while mitigating adverse environmental impacts (Geist and Lambin 2002).

Predictive models, also known as pattern-based models, focus on simulating future land changes based on observed trends, drivers, and policy interventions (Verburg, Schot, et al. 2004; Clarke and Gaydos 1998). These models are crucial for scenario analysis, enabling policymakers to assess the potential impacts of various strategies on land use patterns, infrastructure development, and ecosystem services (Veldkamp and Verburg 2004).

For instance, in REDD+ (Reducing Emissions from Deforestation and Forest Degradation) projects, predictive LCMs are essential tools. REDD+ projects aim to mitigate climate change by preserving forests in developing countries, thus reducing CO₂ emissions (Angelsen 2009). LCMs are employed to model land use scenarios that identify areas at high risk of deforestation, allowing stakeholders to assess the potential impacts of various conservation strategies (Busch et al. 2012). LCMs are also used to estimate the carbon savings that could result from avoided deforestation, ensuring that REDD+ projects contribute effectively to climate change mitigation (Houghton 2005). Moreover, LCMs assist in verifying that emission reductions are genuinely achieved, meeting international standards and supporting the validation of REDD+ initiatives (UNFCCC 2014).

In urban planning, LCMs are used to manage the complex challenges caused by

rapid urbanization (Clarke and Gaydos 1998; Batty 2007). One of these applications is the establishment and monitoring of Urban Growth Boundaries (UGB). UGBs are designed to limit urban sprawl and protect surrounding agricultural and natural lands (Nelson and Dawkins 2004). LCMs are used to create scenarios and predict the potential effects of UGBs on land use patterns, infrastructure demands, and ecosystem services (Verburg, Schot, et al. 2004). LCMs provide critical insights for balancing the need for urban development with the preservation of open spaces and farmland. Through these models, planners can refine UGBs to balance economic growth and environmental sustainability (Batty 2007).

Payment for Ecosystem Services (PES) programs also benefit from the application of predictive LCMs. PES programs compensate landowners for managing their land in ways that provide ecological benefits, such as water purification, carbon sequestration, or biodiversity conservation (Wunder 2005). LCMs support the design and evaluation of these programs by modeling the effects of different land management practices on ecosystem services (Verburg, Schot, et al. 2004; Verburg, Alexander, et al. 2019). By identifying priority areas for PES, LCMs enable stakeholders to forecast the long-term outcomes of these programs on both land use and ecosystem health, ensuring that PES initiatives are strategically implemented to maximize environmental benefits (Engel et al. 2008; Duke and Wu 2014).

The inductive nature of pattern-based models makes them particularly useful in applications where data availability and quality are high. These characteristics enable the identification of meaningful patterns that can be projected into future extrapolations. Despite their inductive focus, pattern-based models often incorporate prior theories to enhance their predictive accuracy. However, the primary strength of pattern-based models lies in their capacity to leverage large datasets, making them highly applicable in real-world scenarios where timely and reliable predictions are critical. In fact, most of the practical applications of LCMs use on pattern-based models.

Although the models used as illustrations in this dissertation are pattern-based, the methods developed in it apply to the outputs of any LCM. These methods compare simulated categories with observed categories. In the case of pattern-based models, the main output they generate are simulated land change maps, usually composed of

pixels. In contrast, for agent-based models, maps of land changes are only one of the many outputs generated by the models. Other outputs may include variations in the behavior of the model’s agents or variations in economic and environmental impacts. As the methods developed in this dissertation can evaluate both the presence and absence of changes, they are perfectly extrapolated to the vast majority of LCMs.

1.1.2.2 Historical overview of LCM

The evolution of LCM is closely linked to technical and scientific advancements in other research areas, such as economics, ecology, remote sensing, geographic information systems, or computational science. Briassoulis (2000) provides a well-documented historical perspective on the development of LCM. Throughout their history, LCMs have been predominantly empirical and statistical, concentrating on urban expansion and agricultural land use. According to Briassoulis, the origins of land change science can be traced back to the 19th century, with the contributions from pioneers like Marsh (1965) and Thünen (1966). Marsh presents a view of humans as integral components of ecosystems, emphasizing the nature-society relationship as a key determinant of the impacts of human intervention. In contrast, Von Thünen adopts a utilitarian perspective on land management, where the expected economic return from land use dictates its transformation. These opposing viewpoints laid the foundation for the two principal approaches taken by subsequent studies of land change.

The 20th century is marked by the gradual development and proliferation of the first LCMs across various fields of study. The main characteristic of this first batch of models is their spatial aggregation (Hall et al. 1995a). These models, primarily based on theoretical principles, were constructed using a wide range of scientific approaches. In economics, many of the econometric tools and models developed in the second half of the 20th century were integrated into LCM, including input-output models, spatial equilibrium models, CGE models, and diffusion models, with a particular focus on urban and agricultural land changes (Duke and Wu 2014).

Following the integration of economic models into LCM, the sociology-oriented fields also made significant contributions to understanding land change, particularly through the lens of human ecology. During the 1960s and 1970s, sociological studies

of urban spatial and social structures began to adopt quantitative, empirical approaches, initially focusing on social area analysis and later moving towards more sophisticated inductive techniques such as factorial ecology (Sweetser 1965). These studies concentrated on variables like socio-economic status, family dynamics, and ethnic composition to explain variations in land use, particularly residential patterns (Briassoulis 2000).

In the subsequent years, ecosystem-based approaches emerged as a response to the limitations of earlier land change models. These models aimed to integrate ecological, economic, social, and spatial dimensions of land change (Turner et al. 2007; Verburg, Kok, et al. 2006). Unlike the more uni-dimensional approaches discussed previously, which often viewed land primarily through an economic or socio-cultural perspective, ecosystem-based models treat land and land use as dynamic components of a larger environmental system (Liu, Dietz, et al. 2007). These models recognize that land has intrinsic ecological properties and functions that interact with human activities in complex ways. The focus shifts from merely understanding spatial patterns or economic outputs to analyzing the broader interactions between human actions and natural systems. This holistic perspective allows for a more integrated analysis of land change, taking into account environmental sustainability alongside economic and social factors (Verburg, Kok, et al. 2006).

The fast adoption of these models responded to the emergence of environmental and climatic concerns of the last decades of the 20th century (Turner et al. 2007). In fact, these integrated approaches are the origin of a large share of the current LCM, particularly pattern-based models. The late 1990s and early 2000s marked an abrupt proliferation of some of the most popular LCM used today, including CLUES (Verburg, Soepboer, et al. 2002), SLEUTH (Clarke and Gaydos 1998), GEOMOD (Hall et al. 1995b; Pontius 1994), LCM (Eastman 2006), or Dinamica-EGO (Soares-Filho et al. 2002), among others.

Within this context, Baker (1989) distinguished one of the earlier taxonomies of LCM, depending on their level of aggregation, which continues to influence the field today. He identified three types of models based on their spatial approaches. Whole landscape models capture phenomena in aggregate for an entire region, ignoring the

spatial distribution of these within the region. Distributional landscape models focus on how land area is distributed among different subregions. They can locate the changes in certain demarked areas, such as municipalities, provinces, or states. Finally, spatial landscape models, currently known as spatially explicit models, emphasize the location and configuration of these phenomena. Baker also discussed various mathematical approaches used in the models, like Markov chain models, projection models, and differential equations, all contributing to the evolution of spatial landscape modeling.

Simultaneously, concerns about the impacts of land change on ecosystems, climate change, and the environment at the end of the 20th century intensified, leading to increased focus on land change science (Lambin 2006). This trend culminated in the establishment of the land use and Land-Cover Change (LUCC) project in 1994, a collaborative initiative by the International Geosphere-Biosphere Programme (IGBP) and the International Human Dimensions Programme on Global Environmental Change (IHDP). The LUCC aimed to explore the potential for creating a unified core research program at the international level (Gutman et al. 2004). In recent decades, the field of land use and land cover change research has matured, becoming more integrative by focusing on both the drivers and impacts of land change. This evolution has led to the establishment of the Global Land Programme (GLP) as a successor to the LUCC, reflecting a broader understanding of the complex interactions involved in land change (Global Land Programme 2024).

Building on the advancements in ecosystem-based and integrated modeling approaches, the latter part of the 20th century also saw significant technological progress that transformed how land changes were studied and monitored. The development of spatial data management systems, such as Geographic Information Systems (GIS), and the increasing accessibility of remote sensing technologies allowed researchers to observe and record land cover changes with unprecedented detail and accuracy (Longley et al. 2005; Foody 2002a). These tools facilitated large-scale projects focused on land change, often coordinated at the international level, which provided valuable insights into both global and regional environmental shifts (Turner et al. 2007; Strahler et al. 2006).

However, while these technological tools revolutionized the way data on land cover

were collected and processed, it became clear that technology alone was not sufficient. This proliferation of datasets highlighted the importance of theoretically informed and methodologically rigorous analyses (Turner et al. 2007). On the one hand, without the integration of robust theories regarding nature–society interactions, technological outputs risk being reduced to visually impressive but shallow representations of change (Briassoulis 2000; Verburg, Neumann, et al. 2011). On the other hand, the lack of fundamental methodological tools to analyze these new types of datasets has challenged the interpretation, control, and use of LCM outputs (Pontius, Huffaker, et al. 2004; Foody 2002a). Consequently, the success of land change research in this period depended not only on technological advancements but also on the continued development of comprehensive frameworks capable of interpreting and applying data to real-world policy and environmental challenges (Turner et al. 2007).

Recently, the trajectory of LCM has been characterized by the integration of fast technological advancements, primarily through three action lines. First, the refinement of spatial data and its increased frequency have enabled the use of much higher spatial and temporal resolutions than in the past. This has also highlighted the influence of input and output resolutions on model performance (Camacho Olmedo et al. 2018; Pontius, Castella, et al. 2018; Baker, Warner, et al. 2013; Castilla et al. 2009). Second, improved computational capacity has allowed for the consideration of more complex models, incorporating a greater number of variables with enhanced spatial and temporal resolution (Verburg, Schot, et al. 2004). Finally, the emergence of machine learning and artificial intelligence techniques has largely influenced LCM, especially in the 2010s. Today, it is common to find models using neural networks, multilayer perceptrons, support vector machines, random forests, or others (Maxwell et al. 2018). However, while these machine learning models can capture nonlinear dynamics and enhance predictive capabilities, they often operate as “black boxes”, providing little insight into the underlying mechanisms driving their predictions. The combination of machine learning techniques with high-resolution and high-dimensional data has also resulted in models that are less interpretable and controllable (Sohl and Claggett 2013; Viana et al. 2023; Rudin 2019).

The rapid increase in data availability and computational capacity has not been

matched by equivalent advances in fundamental methods for analyzing this information. Despite significant technological and technical progress, the accuracy of LCM outputs appears to have reached a plateau. In recent years, several studies suggest that model selection or parametrization may be less critical sources of error than issues related to data quality and model uncertainty (Pontius, Castella, et al. 2018; Verburg, Neumann, et al. 2011). Consequently, research topics such as data quality, model uncertainty, and model validation have become central in land change science. These lines of research address the inherent complexity of land changes and their modeling, raising important concerns about the validity and interpretation of LCM outputs (Pontius, Castella, et al. 2018; Stehman and Foody 2019; Shao et al. 2019; Vliet et al. 2016; Olofsson, Foody, Herold, et al. 2014; Tong and Feng 2020).

Within this new line of research, the work of Clark University stands out. Led by Professor Robert Gilmore Pontius Jr., they have published a wide range of articles analyzing the current state of land change modeling and proposing new methods for evaluating their validity. Among other contributions, they have made notable advances in areas such as the analysis of time series of categorical and continuous maps (Bilintoh 2023; Pontius, Krithivasan, et al. 2017; Pontius and Parmentier 2014), LCM validation (Pontius, Boersma, et al. 2008; Pontius and Millones 2011; Pontius, Peethambaram, et al. 2011), and signal detection (Pontius and Parmentier 2014; Pontius and Si 2014). The concepts and methods described in this body of literature will be one of the primary sources of information used in this dissertation.

1.1.2.3 Structure of pattern-based LCM

The typical structure of a pattern-based LCM algorithm follows a similar logical progression, represented in Figure 1.1. These models consist of two main components: one determines the quantity of land use or land change, while the other independently establishes the spatial location criteria. The only input data used by any of them are land use maps and maps of explanatory variables. From the overlay of land use maps from two time points, we can get the resulting map of land changes. Thus, in an initial step, we have the information about land uses, land changes, and the values of a set of explanatory variables for each individual pixel.

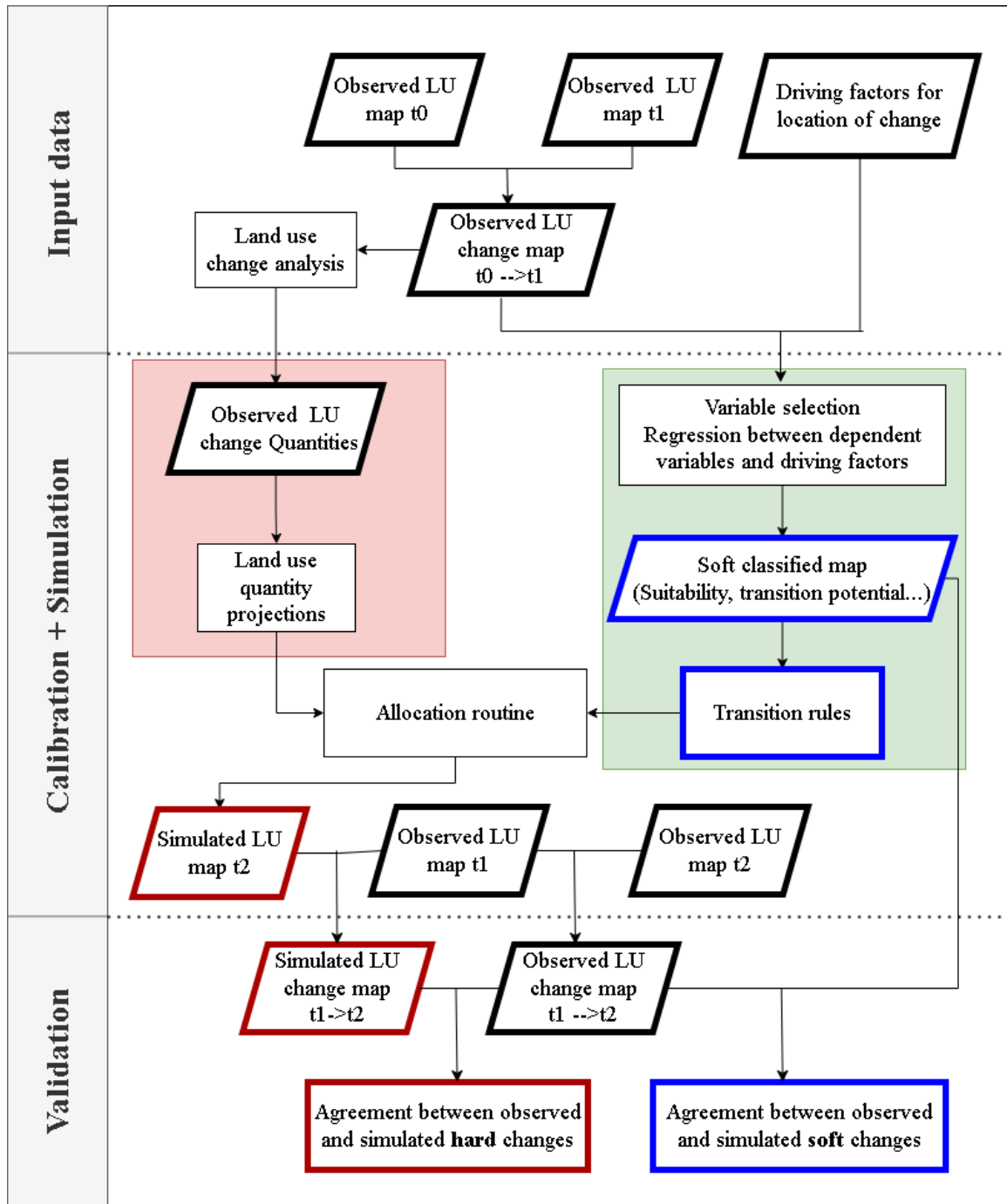


FIGURE 1.1: Representation of a pattern-based model structure. Bold box frames indicate input data, red box frames indicate hard classified outputs, and green box frames indicate soft classified outputs. *Source:* Adapted from Ren et al. (2019)

Regarding the algorithms that determine the quantity, these modules extrapolate the observed quantity, of either land use or land change, between t_0 and t_1 . This component of the model constrains the extension of each land use or land change that has to be achieved in each simulated time step. The method for quantity extrapolation can be either determined by the model or decided by the modeler. The most popular procedures are Markov chains, linear extrapolations, econometric methods, or scenario designs.

As for the component that dictates the location, the first step consists of a regression between dependent and independent variables, using each pixel as an observation. The dependent variable is the study event that, depending on the model, is the presence or absence of a land use category, a change in land use, or a transition from one land use to another. In any case, the independent variables are the driving factors determined by the modeler. The variable selection is not determined by the model, so it is usually decided through statistical methods, literature review, or expert knowledge. The regression method is also open for the modeler, but the most frequent choices are logistic regression, neural networks, multilayer perceptrons, support vector machines, or random forests.

The result from this regression is a suitability value for each pixel and each event, which indicates how adequate the particular event is for that location. The aggregation of the output from all the pixels results in suitability maps if the binary variable is land use, and transition potential maps if it is land change. These maps are commonly referred to as soft maps or soft classified maps.

In subsequent modules, this initial soft map is modified using a set of prior assumptions, which vary from model to model. These theoretical frameworks, often based on preestablished land use principles, modify the soft initial maps to consider previous land uses and their predisposition to change. For example, the CLUE-S model (Verburg, Soepboer, et al. 2002) only allows a certain land use to gain within an arbitrary distance from a pixel with the same use.

Once the quantity and the location are simulated, the model spatially allocates the extrapolated quantity into a categorical map of land uses in t_2 . These maps are commonly known as hard maps or hard classified maps. This spatial allocation is done by allocating the land use or land change to the category with the highest value within

each pixel. As this hardened map is unlikely to fulfill the quantity constraint, the models iterate over the soft maps until their hardened version matches the required quantities. This approach prioritizes matching quantities over their location, which is one of the reasons why LUCM tends to have larger errors in the location of land uses than in their quantity.

The resulting hard classification from this iteration is the final output of the simulation model. Now, the outputs of the models are validated against the observed data. Two outputs are traditionally validated in LCM: the initial soft classified map resulting from the regression and the final hard classified map resulting from the allocation routine. The overlay of the observed land use map of t_1 and t_2 results in the map of observed changes. The overlay of the observed land use map of t_1 and the simulated one in t_2 results in the map of simulated changes. This three-map comparison is the source of any validation procedure applied in LCM.

1.1.3 Challenges in land change science

The study and modeling of land change is a challenging task, primarily due to the complexity land changes entail (Sohl and Claggett 2013; Rindfuss et al. 2004). Pontius, Castella, et al. (2018) summarized several structural, ontological, and epistemological challenges that land change science faces, each hindering the scientific progress and practical advancements in the field.

Structural challenges involve practical and social aspects of land use science, such as integrating diverse data sources and methods, but also the social dynamics influencing researchers. Pontius (2022) highlights how social settings can shape scientists' behaviors and decisions, affecting the quality of their research. In his book, Pontius highlights the social reasons behind researchers' and modelers' decisions about metrics used and results reported, which involve inadequate methodological choices, bad research practices, and "publish or perish" type of pressures. Land change science has been particularly affected by these social constraints, which is a big part of this dissertation's motivation. In many cases, these factors can have substantially more influence on the scientific progress than the other two challenges.

Ontological challenges refer to issues related to the nature of land use and land cover themselves. The fundamental problem lies in defining and classifying the problem of study in a way that captures their real-world complexity (Rindfuss et al. 2004). Land use categories can be fluid and context-dependent, making it difficult to create stable, universally applicable definitions (Mushtaq et al. 2024). This variability can affect how changes are detected and interpreted, as the categories used in modeling may not fully align with actual land use patterns (Ahlqvist et al. 2016).

Epistemological challenges concern the nature and scope of the knowledge about land changes. These challenges involve understanding the limitations of existing data, theories, and models, considering the extent to which they can accurately represent real-world dynamics (García-Álvarez 2018). The main contributions of this dissertation directly address some of the primary epistemological challenges of land change science.

On one side, technical and practical limitations of remote sensing procedures largely influence the quality and consistency of the datasets used by LCM (Foody 2024; Pontius and Spencer 2005). Verburg, Neumann, et al. (2011) describes the thematic, temporal, and spatial inconsistencies between and within land use datasets. The technical issues related to land use datasets are one of the main sources of uncertainty in the LCM process (García-Álvarez et al. 2019).

On the other hand, the complexity and multidimensionality of the time series of categorical maps hinder the pattern recognition process, and thus, pattern projection. Pattern-based LCM tries to identify past land change patterns and project them into the future. However, the detection of these patterns is highly dependent on the dimensions and resolutions selected for our analysis. For example, if we use annual maps to extrapolate agricultural expansion, we might be missing relevant information about crop rotation, seasonal trends, or others.

In these lines, land change processes and patterns tend to be temporally and spatially non-stationary (Pontius, Castella, et al. 2018). The environmental, economic, and social drivers of land changes (processes) are unlikely to be constant over time or among regions. Similarly, the rate and location of land change (patterns) also vary with time and place. This challenges the projection of observed tendencies into future scenarios (Aldwaik and Pontius 2012; Vliet et al. 2016; Chen and Pontius 2010).

Uncertainty in LCM

Uncertainty is a central factor within most of the mentioned challenges of land change science. In this dissertation, we define uncertainty as the degree of imprecision or unpredictability in representing and forecasting events. Uncertainty entails various factors that introduce a gap between real-world phenomena and their modeled representations (García-Álvarez 2018). In LCMs, uncertainty refers to the elements within the modeling process that limit our ability to accurately simulate or project observed trends of land change.

Numerous studies have identified uncertainty analysis as a priority research area within the field of LCM (Pontius, Boersma, et al. 2008; Sohl and Claggett 2013). Pattern-based models risk propagating uncertainty from observed data into their projections, which can have substantial effects on model outputs (Verburg, Tabeau, et al. 2013; Pontius and Spencer 2005). Authors emphasize the need to validate LCM as a way to understand the uncertainties and errors associated with each application (Verburg, Tabeau, et al. 2013; Verburg, Schot, et al. 2004; Pontius, Castella, et al. 2018; Brown et al. 2013). García-Álvarez (2018) identifies five main sources of uncertainty in LCMs: model conceptualization, data sources and processing, technical issues, model parametrization, validation, and process variability. Within these categories, the literature highlights three key factors as primary contributors to uncertainty in LCMs: data quality (Pontius, Boersma, et al. 2008; Foody 2024), multidimensionality of land changes (Sohl and Claggett 2013; Rindfuss et al. 2004), and lack of stationarity in the modeled event (Brown et al. 2013; Vliet et al. 2016; Aldwaik and Pontius 2012).

Data quality

The uncertainty associated with map generation and processing is a critical issue in land change science. Researchers often treat maps as a perfect representation of reality. Yet, it is common for ground datasets to contain errors, sometimes larger than the observed changes themselves (Foody 2013). Land use and land cover mapping typically catalogs maps as accurate when the overall accuracy exceeds 80-90% (Olofsson, Foody, Stehman, et al. 2013; Pontius and Spencer 2005; Stehman and Foody 2019). Plus, observed land change is not directly measured, but it is the result of

the overlay of two land use maps at different time points, with their overlaid errors. However, actual changes often occur in only a small fraction of the total area, which is substantially smaller than the map errors. As a result, a significant proportion of the land changes used by LCM are often map errors.

The literature underscores map error as a primary source of uncertainty in land change science. Pontius and Lippitt (2006) explored whether map errors could explain differences between land cover maps from two different points in time. Foody (2013) demonstrates that land cover change accuracy depends on observed data error and the prevalence of change. Olofsson, Foody, Herold, et al. (2014) highlights the challenges remote sensing faces in generating accurate land change datasets, stressing the importance of accounting for map errors when interpreting changes over time.

In a field where the prevalence of change is so low, the assessment of uncertainty becomes a crucial element. For example, LCMs often propagate the uncertainty in the maps to the projected quantities and locations of land changes (García-Álvarez 2018). Still, despite the established importance of accuracy assessment within the remote sensing community (Foody 2002b; Strahler et al. 2006), land change studies often fail to adequately evaluate the accuracy of land change maps. Many studies overlook the information obtained from accuracy assessments, leading to incomplete or misleading conclusions from their results (Olofsson, Foody, Herold, et al. 2014; Tang et al. 2021). Errors in ground observed data can contribute significantly to misinterpretation, potentially leading to over- or underestimation of accuracy measures and inaccurate assessments of change extent (Foody 2010).

Land use map accuracy assessments are closely influenced by the prevalence of the classified categories. Foody (2010) notes that accuracy and area estimates are intimately linked, meaning that area estimation quality varies based on class prevalence. Specifically, variation in the prevalence of the target class can significantly impact both their estimated quantity and location (Foody 2013). Despite this, land change science, and LCM in particular, often overlooks prevalence as a critical factor. Researchers assume that results from applications with divergent change rates are fully comparable, ignoring the level of uncertainty of their databases.

Additionally, modelers often overlook the fact that most data are typically produced for specific purposes unrelated to LCM. Consequently, using such data in applications for which they were not designed can introduce new uncertainties (García-Álvarez 2018). For instance, some datasets are created for agricultural control and monitoring (Gobierno de Navarra 2025), where the thematic definition and classification of agricultural land uses are more precise than those for urban land uses. Similarly, datasets intended for urban planning are constructed using cadastral information and therefore give less attention to other land uses. Modelers must ensure that their data sources align with the needs and requirements of their specific applications.

Data quality issues in LCM have become an increasing focus of research. The literature highlights the impact of data uncertainty, classification errors, and scale effects on LCM outcomes (Stehman and Foody 2019; Foody 2013). Burnicki (2012) note that the reliability of LCMs depends largely on the certainty of observed changes, emphasizing the need to incorporate uncertainty analysis into LCM studies as a critical research priority. Inaccuracies in land cover data can propagate through models, affecting both calibration and validation processes, and ultimately influencing the interpretation of model results (Stehman and Foody 2019; Foody 2013). Addressing these issues is therefore essential to improve the robustness and credibility of land change projections.

Multidimensionality

Pattern-based LCMs rely on identifying and extrapolating observed land change patterns into the future. However, land changes are inherently multidimensional, which implies that information contained in a time series of maps must be reduced in order to be analyzed, inevitably leading to a loss of information regarding when, where, and how changes occur. This multidimensionality arises because land change processes involve several interrelated dimensions, including spatial location, temporal dynamics, transition types, and the intensity of changes (Rindfuss et al. 2004).

For example, while an overall increase in forest land use may be observed over a given time interval, projecting this change requires a deeper understanding of its underlying pattern: it is necessary to consider when and where the change occurred, whether forest areas are simultaneously experiencing gains and losses, and whether

the expansion is uniform or driven by specific types of forest transitions. Thus, the multidimensionality of land change patterns refers to the complexity introduced by the need to reduce high-dimensional spatio-temporal information into simplified representations, which can obscure relevant dynamics and affect both the detection of patterns and the outcomes of LCMs.

Drawn from the works by Verburg, Neumann, et al. (2011) and Pontius and Santacruz (2014), we distinguish four primary dimensions in land changes, each with its own range of resolutions:

- **Thematic Dimension:** This dimension refers to the number and types of land use classes considered in the analysis. When abstracting reality into a model, the modeler must make decisions about the level of detail required for categorization. Often, this decision is influenced by the original classification of the data sources, as well as the specific research questions being addressed. The choice of thematic resolution can substantially alter the results of a study (Aldwaik, Onsted, et al. 2015; Camacho Olmedo et al. 2018; García-Álvarez et al. 2019; Varga et al. 2019; Pontius, Castella, et al. 2018).

Land use classes can range from a simple binary classification (e.g., presence or absence of a single category) to a classification where each location is treated as a unique category. Additionally, the types of land changes, which only result from the comparison of at least two maps, depend on the land use classes considered. With a binary classification of land uses, we have four types of events: persistence of presence, change from presence to absence (loss), change from absence to presence (gain), and persistence of absence. The higher the thematic resolution, the more land use classes we consider, leading to a quadratic increase in the number of possible transitions (n^2 , for $n \in \mathbb{N}, n > 1$).

Focusing on the resolution of the events, we can further differentiate changes at the extent, category, and transition levels. The transition level refers to changes from one land use category in origin to another in destination (e.g., transitioning from agriculture to urban use). The category level focuses on changes within a specific category, either as an origin or destination (e.g., changes from agriculture). The extent level aggregates changes across all categories.

- **Temporal Dimension:** This dimension addresses the timing of land changes, either by the temporal extension of the analysis or the frequency of the maps. Thus, the temporal frequency ranges from changes observed between two points in time (initial and final maps) to more complex analyses involving multiple time points. Certain patterns, such as alternation (Bilintoh 2023), are only detectable when more than two observed maps are available. Additionally, some land change phenomena, like crop rotations, may require specific temporal resolutions to be accurately captured.

While data availability often limits temporal resolution, the choice of temporal frequency can also be driven by the needs of the modeler. The most common temporal resolution in LCM includes two or three time points, dividing the calibration and validation intervals. However, with the increasing availability of time series of spatial data, tools for analyzing patterns in multiple time points have gained importance.

- **Spatial Dimension:** This dimension involves the choice of the minimum mapping unit (MMU) and its size. The MMU refers to the kind of geometry considered within maps, whether it is pixels or polygons. Pixels are grid-based units, often used in raster data structures where space is divided into uniform cells. Each pixel represents a single value (e.g., land cover type). Pixels work well for high-resolution remote sensing, where smaller cell sizes can capture detailed variations. However, because each pixel has a fixed resolution, the "mixed pixel problem" can arise when multiple land cover types are present within a single pixel, potentially reducing classification accuracy in heterogeneous landscapes. Polygons, on the other hand, are vector-based units that represent spatial features with boundaries, allowing them to vary in shape and size. Polygons can capture irregularly shaped areas like forest boundaries, fields, or urban zones, making them useful for delineating areas with clear boundaries. They offer flexibility in size and shape, which is advantageous for representing specific features accurately. However, they are less comparable across maps and may be computationally intensive in complex landscapes.

The spatial resolution of this dimension refers to the size of the MMU. The spatial dimension ranges from considering the territory as a single unit to using

the highest resolution allowed by remote sensing technology. The choice of spatial resolution affects the detail of the analysis and can influence the detected patterns of land change. There is an extensive body of literature on the influence of MMU in the analysis of land change and the LCM process (Camacho Olmedo et al. 2018; Pontius, Castella, et al. 2018).

- **Components of Change:** This dimension distinguishes between the net and total quantities of land changes. Land changes can be categorized into Quantity and Allocation (Pontius and Santacruz 2014). Changes in Quantity refer to the net changes in the size of a category, i.e., the difference between the initial and final extent of each land use. Changes in Allocation, on the other hand, involve shifts that do not affect the net extension of land uses, as they represent simultaneous losses and gains within a category (Pontius and Millones 2011). This distinction is somewhat artificial, as it does not observe specific pixels but rather an accounting of changes. However, its utility lies in the fact that Allocation changes can constitute a large portion of total changes. Later sections delve deeper into the concepts of Quantity and Allocation, two key analysis tools that will be frequently employed throughout this dissertation.

Allocation changes may occur through Exchange or Shift (Pontius and Santacruz 2014)1.2.1.1. Exchange represents trade-offs between two categories, where, for example, a transition ($A \rightarrow B$) in one pixel occurs simultaneously with the reverse transition ($B \rightarrow A$) in another pixel, resulting in no change in quantity of either of the categories. Shift, in contrast, involves transitions across three or more categories without altering the net quantity of at least one of them. This could include scenarios like $A \rightarrow B \rightarrow C$, where A gains Quantity, C loses Quantity, and B undergoes a Shift; or closed loops such as $A \rightarrow B \rightarrow C \rightarrow A$, which indicate a Shift across all three categories.

Any analysis and interpretation of these dimensions requires choosing an appropriate resolution. For instance, if we want to analyze the spatial dimension of the changes in a certain land use, we must select a resolution that is suitable for our analysis. A resolution that is too large may add unnecessary complexity, making it difficult to interpret patterns of interest. Similarly, a resolution that is too small may cause patterns of interest to become spatially diluted and thus, undetectable.

Let's think of an example of a LCM whose goal is to predict crop rotation. In its temporal dimension, the ideal resolution would be the cycle of the objective crop, let's say half a year. Considering more frequent maps would still allow the detection of the crop rotation patterns, but it would add unnecessary noise to the process, and more computational power would be required. The use of annual maps would have much less noise, but it would only capture a partial image of the patterns of rotation. Moreover, any of the selected resolutions would hinder the detection of patterns in other relevant crops with different rotation cycles.

The specific combinations of dimensions and their resolutions considered in our analysis have a substantial influence on the patterns we are able to detect (Baker, Warner, et al. 2013; García-Álvarez 2018; García-Álvarez et al. 2019; Pontius and Millones 2011; Roy et al. 2016). In our example with crop rotation, the detection of relevant patterns depends not only on the temporal resolution but also on the remaining resolutions selected for our analysis. Even if we use a half-year temporal resolution, crop rotation patterns will be undetectable if we don't use appropriate land use categories or spatial detail. Crop rotation might be increasing for the whole region, but this growth can be spatially heterogeneous. Similarly, crop rotation might be increasing for the overall temporal extent, but there might be some intervals with increases and others with decreases. All the decisions regarding resolutions are likely to affect the detection of any other pattern relevant to the modeling process.

These modeling decisions also have direct implications for the prevalence and characterization of observed changes. As temporal resolution increases (i.e., more frequent observations), the likelihood of detecting changes also increases, since more intermediate transitions become observable. Conversely, coarser temporal resolutions may aggregate multiple transitions into a single observed change or even mask changes entirely, leading to an underestimation of prevalence. Similarly, a more detailed spatial resolution necessarily represents the same or higher prevalence of changes.

This has important consequences for the analysis and evaluation of LCMs. Differences in any resolution can alter not only the number of observed changes but also their apparent timing, location, and persistence, which in turn affects model calibration and validation. As a result, models built under different resolution choices may not be directly comparable, even when they are applied to the same raw data.

The choice of resolution not only determines the level of detail of the analysis but also shapes the statistical structure of the observed events. As will be discussed throughout this dissertation, these differences play a key role in understanding model behavior and the performance of evaluation metrics.

Stationarity

In the context of LCMs, stationarity refers to the assumption that the relationships between drivers of land change and land use patterns remain constant over time and across space. This means that the processes influencing land change do not vary across different time periods or geographic regions, allowing the model to apply historical trends and relationships to predict future changes.

The stationarity assumption adopted by LCM refers to both temporal and spatial stationarity. Temporal stationarity means that the relationships between land use variables and their influencing factors remain consistent over time, allowing models to use past patterns and trends as reliable indicators of future land changes. Spatial stationarity, on the other hand, assumes that these relationships are uniform across different geographic locations within the region, implying that similar environmental and socioeconomic factors will drive land changes in the same way across space.

Stationarity is a major assumption in LCMs (Vliet et al. 2016; Bakker and Veldkamp 2012a; Aldwaik and Pontius 2012). Models are calibrated on historical data, relying on observed patterns and correlations to forecast land changes. Thus, models operate under the assumption that the underlying drivers of land change will remain constant over time (Brown et al. 2013). However, the socioeconomic and biophysical factors influencing land system dynamics are continually evolving, making the assumption of stationarity questionable. For example, urbanization trends, agricultural practices, or conservation policies may evolve in ways that disrupt previously observed relationships. Similarly, focusing on pattern-based models, the quantities and locations of land changes are neither constant over time nor in space. Chen and Pontius (2010) emphasize that accurate predictions depend on stable relationships between land use and explanatory variables. The lack of stationarity of land change patterns makes LCM propagate large shares of uncertainty.

In discussing the presumed stationarity of land change for pattern analysis, research debates between inferring future land changes from current change patterns or from historical land use patterns (Bakker and Veldkamp 2012a). Some researchers support inferring future changes from existing land use patterns, arguing that this approach captures the effects of a long historical process, thus offering more stable insights. This viewpoint assumes that fluctuations in land productivity, demographic pressure, or economic development cancel each other out over time. Conversely, others believe these changes are structural and require predictions based on recent patterns. According to these authors, recent processes are more reflective of the immediate past than of long-term averages (Bakker and Veldkamp 2012a; Lambin et al. 2000; Bakker and Veldkamp 2012b).

Explicitly recognizing stationarity assumptions and examining data for signs of non-stationarity are critical steps in addressing and understanding model uncertainty (Brown et al. 2013). Several studies illustrate how the number and separation of time-points included in the calibration of LCMs model significantly influenced the calibration result (Blecic et al. 2015; Paegelow 2018; Rosa et al. 2015; Versteegen et al. 2016). By recognizing and incorporating the dynamic nature of land use drivers, researchers can enhance the reliability of their models, leading to valuable insights into land change processes.

1.2 Evaluation metrics for binary classification

The complexity of land use changes introduces important challenges for LCM validation (Rindfuss et al. 2004; Brown et al. 2013). Land changes are often rare, spatially heterogeneous, and subject to multiple sources of uncertainty. These features have direct methodological, practical, and interpretative implications for model evaluation. However, existing methods for model validation and map comparison do not always fully address these challenges (Bilintoh et al. 2024; Vliet et al. 2016; Viana et al. 2023), and advancements in validation have not kept pace with the rapid development of models (Rindfuss et al. 2004; Foody 2024; Foody 2023).

Many of these difficulties are not unique to land change science, but arise more generally in contexts where the event of interest is rare and unevenly distributed.

This observation motivates a broader perspective within this dissertation. It becomes necessary to examine how similar problems have been addressed in other fields. In this sense, LCM validation can be framed within the wider domain of binary classification evaluation, a well-established area that has developed a rich set of concepts, metrics, and frameworks across disciplines.

Evaluation metrics for binary classification measure the relationship between a binary outcome and an index variable. These metrics are widely used across various research fields, including meteorology (Mason 1982; Murphy 1996; Wilks 2006), medical diagnostics (Hanley and McNeil 1982), economics (Lahiri and Wang 2013; Berge and Jorda 2011), and machine learning (Fawcett 2006). In LCM, the binary outcome typically represents the presence or absence of change, while the index variable is usually derived from a regression between observed changes and their driving factors (Verburg, Schot, et al. 2004).

Due to this wide range of applications, the literature on classification evaluation metrics has developed across disciplines and remains largely fragmented. Different fields have introduced related concepts, metrics, and evaluation frameworks, often using distinct terminology and focusing on specific types of problems. As a result, despite the existence of a rich and well-developed body of work, its structure is not always transparent to end users.

This Section aims to consolidate and organize this dispersed literature by providing a structured and coherent overview of classification evaluation metrics relevant to the key topics addressed in this dissertation. It clarifies the relationships between existing metrics and approaches, and shows how land change science can integrate these tools into its validation process in a consistent and interpretable way.

One significant challenge in evaluating classification accuracy is the presence of imbalanced datasets, where one class is significantly rarer than the other. Class imbalance hinders the evaluation of agreement for each class, because one class is underrepresented while the other is overrepresented (He and Garcia 2009). Thus, applications with imbalanced classes require special attention in the choice of evaluation metrics (Saito and Rehmsmeier 2015). In such cases, metrics that consider agreement for both presence and absence, such as OA, tend to reflect the agreement of the majority class. Consequently, these metrics may report high values simply

because it is easier for observations and predictions to coincide when the majority class dominates (Pontius and Parmentier 2014).

Imbalanced datasets are common across various fields, such as economics (Aldasoro et al. 2018; Granger and Pesaran 2000; Lahiri and Yang 2013), credit scoring (Abdou and Pointon 2011; Engelmann 2011; Liu, Liu, et al. 2024), species distribution (Phillips et al. 2006; Hanberry and He 2013), rare diseases (Garon et al. 2015; Movahedi et al. 2023), medicine (Gao et al. 2018), or meteorology (Wilks 2006; Gilbert 1884; Ebert and McBride 2000; Hamill and Juras 2006). In many of these applications, the prevalence of the binary variable can vary substantially over space and time. For example, the frequency of economic crisis might vary over time, or species spatial distributions may change rapidly (Chen, Hill, et al. 2011). In the specific case of land changes, they are often rare within a region, but their prevalence varies with regions and time.

Within this context, the literature underscores some common concerns for the evaluation of accuracy in applications with low and varying prevalence. How should we measure the contribution to the accuracy of each of the classes? (Gilbert 1884; Wilks 2006; Pontius, Boersma, et al. 2008) Are metrics insensitive to prevalence (equitable), and thus, comparable across applications with different prevalence? (Hogan et al. 2010; Hoffman 2020; Chicco and Jurman 2020) Do metrics weigh equally Misses and False alarms? (Ebert and Milne 2022)

Many of these concerns cannot be addressed separately from the specific research questions of their applications. For example, applications where the only focus is on one of the classes are not affected by the accuracy of the other classes. Similarly, applications where Misses and False alarms have different practical costs don't need them to be equally weighted. The link between evaluation metrics and research questions is a critical and influential element in the analysis of any metric (Pontius 2022; Hand 2012a).

Binary classifications can be either deterministic or probabilistic (Chicco and Jurman 2020; Hossin and Sulaiman 2015; Hoffman 2020). Deterministic classifications assign each observation to a single class based on specific criteria or thresholds. These classifications operate under the assumption that each input belongs precisely to

one class, something necessary for many applications¹. These classifications are also known as hard classifications. In contrast, probabilistic classifications associate each observation with a set of probabilities, indicating the likelihood of belonging to each category. Probabilistic classifications are particularly useful in situations where data overlap between classes, enabling more flexible decision-making in contexts where absolute certainty is difficult to achieve. These classifications are also known as soft classifications.

Deterministic classifications are evaluated through single threshold metrics because the classifications use a single threshold of the index variable to classify the observations. Observations with values of the index variable above that threshold are classified as presence, and values below as absence (see Fig.1.3 for visual representation). The overlay of the observed and classified outcomes results in a 2x2 matrix, usually called a contingency table. Table 1.1 shows the entries of the contingency table. Hits (H) are observed presences correctly classified; Correct Rejections (C) are observed absences correctly classified; Misses (M) are observed presences classified as absences; and False Alarms (F) are observed absences classified as presences.

		Observed		Total Classified
		Presence	Absence	
Classified	Presence	H	F	Classified Presence (H+F)
	Absence	M	C	Classified Absence (M+C)
Total Observed		Observed Presence (H+M)	Observed Absence (C+F)	Population (H+C+M+F)

TABLE 1.1: Contingency table for binary classification showing the relationship between observed and classified outcomes.

In contrast, probabilistic classifications are evaluated through multiple threshold metrics. These metrics generate hard classifications of multiple thresholds of the index variable, and evaluate them all together. Thus, in practice, multiple threshold

¹This distinction is homologous to the soft and hard maps of LCM previously defined.

metrics are built under the same principles as single threshold metrics, which are drawn from contingency tables.

As an illustrative example of the variety of classification evaluation metrics and their properties, in the upcoming sections, we will calculate several metrics on the tornado forecast data in Finley (1884) (Tab. 1.2). Finley’s forecast is a classic example of an imbalanced dataset, which has inspired much of the literature in binary classification evaluation (see Section 1.2.4).

		Observed		Total
		Presence	Absence	
Predicted	Presence	28	72	100
	Absence	23	2680	2703
Total		51	2752	2803

TABLE 1.2: Finley’s contingency table for tornado forecast

1.2.1 Single threshold metrics

1.2.1.1 Quantity and Allocation differences

Quantity, Allocation, and Total differences (QD, AD, and TD) are mathematical representations that arise from the comparison of two datasets defined over populations of equal size. This can result from the comparison of two populations at different time points, two data sources of the same population, or a prediction against an observation, among others.

Quantity difference indicates the absolute difference between the size of a category between the two datasets (Pontius and Si 2014). For example, if we compare the unemployment of a country between two time points, the Quantity difference is the net difference between the two unemployment rates. In contrast, Allocation difference reflects the extent of discrepancy between the two datasets that is not reflected in net differences. In the previous example, we could have no change in the unemployment rate, but observe differences because some individuals have lost their jobs and others have found new ones. Total differences represent all the observations that differ in the comparison, which is the sum of Quantity and Allocation differences.

Figure 1.2 shows an illustration of three comparisons of maps between two time points (t_0 and t_1). As we are comparing the same population at different time points, the differences are referred to as changes. The figure illustrates the contingency tables of each comparison, as well as their changes in Quantity and Allocation, using presence as the reference class. In panel a), only the two upper locations change over time, resulting in two units of total change (TD=2). However, the net count of locations with the presence of the binary variable remains constant (QD=0). The difference comes because one observation with presence has changed to absence, while another absence has moved to presence (AD=2). Thus, the differences in this illustration are only due to Allocation.

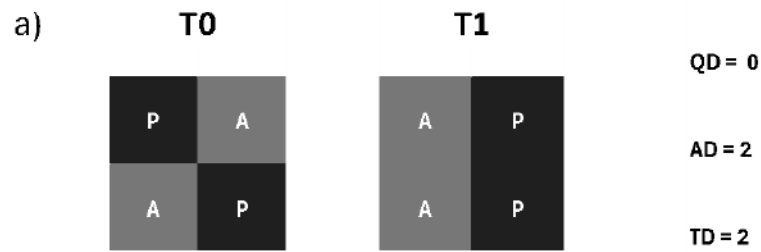
The corresponding contingency table shows how the marginal totals for each category remain constant, which means there is no change in quantity of either presence or absence. However, there is one Miss (absence changed to presence) and one False alarm (presence changed to absence), which don't affect the Quantity but affect Allocation and Total change.

In panel b), the only location that changes is the upper right one, which changes from absence to presence (TD=1). In this case, there are no pairs of observations changing from absence to presence and from presence to absence, so the Allocation change is 0. Therefore, all the changes are due to Quantity. In the contingency table, the totals are no longer constant, because t_1 has one more location with presence.

Finally, panel c) shows both Quantity and Allocation differences. All the observations have changed, but the lower right one (TD=3). There is one change from absence to presence that can be paired with another from presence to absence, and thus, not affecting Quantity (AD=2). Additionally, there is one extra gain of presence which can't be paired with a loss (QD=1). Once again, the totals of the contingency table indicate a gain in presence (or loss in absence). However, in contrast to panel b), now there are more observations off the main diagonal of the contingency table.

This illustration sets the basis for the mathematical representation of Quantity and Allocation differences. Quantity, Allocation, and Total differences are defined in equations 1.1 to 1.3 (Pontius 2022), using presence as the reference category.

$$\mathbf{QD} = |\text{Observed Presence} - \text{Classified Presence}|$$



		T1		
		P	A	Total
T0	P	1	1	2
	A	1	1	2
Total		2	2	4



		T1		
		P	A	Total
T0	P	2	0	2
	A	1	1	2
Total		3	1	4



		T1		
		P	A	Total
T0	P	1	1	2
	A	2	0	2
Total		3	1	4

FIGURE 1.2: Toy examples of three map comparisons and their corresponding contingency tables used to derive Quantity and Allocation differences. P stands for presence and A for absence.

$$\begin{aligned}
&= |(H + M) - (H + F)| \\
&= |M - F|
\end{aligned} \tag{1.1}$$

$$\mathbf{AD} = 2 \min(M, F) \tag{1.2}$$

$$\begin{aligned}
\mathbf{TD} &= \mathbf{QD} + \mathbf{AD} \\
&= |M - F| + 2 \min(F, M) = M + F
\end{aligned} \tag{1.3}$$

In Finley’s forecast (Tab. 1.4), 95 predictions were incorrect ($\mathbf{TD} = 95$). However, a large share of this error is due to an overestimation of tornado occurrences. In particular, the Quantity difference amounts to 49, which is nearly as large as the observed number of tornado events (51). By contrast, the Allocation difference is 46. This indicates that Finley’s errors do not arise solely from an inability to distinguish between the presence and absence of tornadoes, but largely from a systematic tendency to forecast too many tornado events.

The distinction between Quantity and Allocation differences is essential for interpreting comparison results properly. In diagnostic testing, for example, it is important to know whether errors arise because a test systematically overestimates or underestimates the number of disease cases, or because it fails to distinguish correctly between diseased and non-diseased individuals. In meteorology, it is important to determine whether forecasts overestimate or underestimate the number of extreme events, since these two types of error have very different implications for the population. Similarly, in LCM, it is a key issue whether the model fails to estimate the total area of change or its spatial allocation correctly. This information about the sources of error is crucial for error assessment, because it directly shapes how results are interpreted and, consequently, the practical decisions based on them.

In this dissertation, we adopt the methods, concepts, and terminology described by Pontius (2022), as these were specifically designed with applications in land change science in mind. However, the concepts are not limited to this field and have been described and applied in other disciplines, highlighting their broad relevance and utility. For instance, in meteorology, Ebert and McBride (2000) decomposes total disagreement into displacement, volume, and pattern errors, which parallel the notions of

Quantity and Allocation differences. Similarly, in traditional statistics involving estimations of continuous values, bias, measured by the mean error (ME), corresponds to whether an estimation systematically overestimates or underestimates the observed value, akin to the concept of Quantity difference. Likewise, mean absolute error (MAE) aligns with the Total difference, while the term $MAE - ME$ is equivalent to Allocation difference (Pontius 2022).

1.2.1.2 First order single threshold metrics

First-order binary classification evaluation metrics describe proportions derived directly from the entries and marginal totals of a contingency table. These metrics quantify specific aspects of the relationship between observed and classified outcomes. For example, User’s Accuracy (UA) (Eq.1.6) is the proportion of classified presence that is correctly classified.

First-order metrics are straightforward to interpret because they offer clear information about the outcome of the classification. However, the use of only one threshold metric will probably miss the complexity of contingency tables and class distribution. Wilks (2006) points to the dimensionality of contingency tables as a challenge for their summarization. 2x2 contingency tables have $2 \cdot 2 - 1 = 3$ degrees of freedom. Yet, any scalar verification measure, regardless of the number of entries considered, reduces this three-dimensional data to a single value, inevitably resulting in a loss of information. In other words, classifications can succeed or fail in various ways, and each metric combines these different outcomes in its own unique manner. For instance, UA (Eq.1.6) informs about how well presence is classified, but gives no information about absence classification. Similarly, Overall Agreement (OA, Eq.1.4) gives information about the Agreement in both presences and absences, but doesn’t inform about the contribution of each of them to Agreement.

Table 1.3 compiles some of the most popular first-order metrics. Among them, we can highlight popular metrics such as OA (Eq.1.4), Producer’s Accuracy (PA) (Eq.1.5), UA (Eq.1.6), and Critical Success Index (CSI) (Eq.1.7).

OA, interpreted in terms of contingency tables, refers to the proportion of all the classifications that agree with the observed data. The following equation shows its mathematical expression:

Observed				
		P	A	
Classified	P	H	F	$UA = \frac{H}{H + F}$
	A	M	C	$FOR = \frac{M}{C + M}$
		$PA = \frac{H}{H+M}$	$SP = \frac{C}{C+F}$	$CSI = \frac{H}{H + M + F}$
$Pr = \frac{H + M}{H + M + C + F}$		$OA = \frac{H + C}{H + C + M + F}$		

Acronym	Name	Also known as
PA	Producer's Accuracy	True positive rate, recall, sensitivity, hit rate
SP	Specificity	True negative rate
FAR	False Alarm Rate	False positive rate, 1 – Specificity
Pr^2	prevalence	Event rate, prior presence probability
UA	User's Accuracy	Positive predictive value, precision
FOR	False omission rate	Miss rate
CSI	Critical Success Index	Threat index, Jaccardi Index, Figure of merit
OA	Overall Agreement	Percent correct, Accuracy

TABLE 1.3: Summary of binary classification performance metrics derived from the contingency table, including formulas and commonly used alternative terminology.

$$OA = \frac{H + C}{H + C + M + F} = \frac{H + C}{\text{Population Size}} \quad (1.4)$$

OA is one of the most widely used single-threshold metrics because it summarizes correct classifications for both presence and absence in a single, easily interpretable value. However, its interpretability comes at the cost of information loss. OA becomes less informative when the class of interest is underrepresented, as it is strongly influenced by the majority class. In such cases, high agreement can be achieved by trivial predictions that simply reproduce the dominant class (Hand 2012a).

A simple analogy is forecasting the weather in a desert climate. A model that always predicts sunshine can achieve high agreement because sunny days dominate, even if the model provides no information about rainfall and never predicts it. Similarly, in LCMs, OA often reports high values because land persistence is much more frequent than change, meaning that Correct Rejections dominate the contingency table.

This limitation is clearly illustrated in Finley’s tornado forecasts, where OA reaches 96.6%, indicating that most predictions are correct. However, this high value largely reflects the rarity of tornado events rather than genuine predictive ability, as Finley’s forecast had large errors in tornadoes. The strong dependence of OA on class frequency, as shown in Finley’s forecast, was a key motivation for the development of alternative evaluation metrics derived from his work.

PA or Precision represents the proportion of observed events correctly classified. UA or Recall is the proportion of classified events correctly classified. Their mathematical definition is given by:

$$PA = \frac{H}{H + M} \quad (1.5)$$

$$UA = \frac{H}{H + F} \quad (1.6)$$

In applications to categorical variables, where more than two categories are considered, PA and UA are often used as category-based Agreement metrics. This is often the case in land use science and remote sensing (Congalton and Green 2009; Stehman 1997). These metrics offer valuable and easy-to-interpret information, as

they only combine two entries of the contingency tables.

PA measures the Agreement in observed presence. It represents the proportion of observed presence that is correctly classified, since the sum of Hits and Misses corresponds to the total observed presence. In this sense, PA takes the observed Quantity as given and does not depend on prevalence (Pontius 2022). UA, in contrast, measures the Agreement in classified presence. It represents the proportion of classified presence that is correct, since the sum of Hits and False Alarms corresponds to the total classified presence. UA depends on how much presence is predicted by the model and on prevalence, as the likelihood of correctly identifying presence increases when presence is more common in the population (Pontius 2022). Once again, we observe how the relationship between prevalence and evaluation metrics is a key element in their proper interpretation.

In Finley’s forecast, PA takes a value of 55%, meaning that 55% of the actual tornadoes were correctly forecasted. Meanwhile, UA gets a substantially lower value (28%), meaning that only 28% of the predicted tornadoes were right. A part of the difference between PA and UA can be explained by the fact that Finley overestimated the quantity of tornadoes. When Quantity is overestimated, PA will tend to be higher because it is more likely to correctly classify an actual tornado. Similarly, UA will tend to be smaller because a part of the forecasts will necessarily be wrong.

Each PA and UA only considers one type of error (either M or F), omitting the information from two entries of the contingency tables. CSI (Eq.1.7), also known as Threat score, Tanimoto similarity coefficient, Jaccard index, or figure of merit, combines the entries of UA and PA as it accounts for both Misses and False alarms. CSI evaluates the Agreement between observed and classified presence, ignoring the contribution of Correct Rejections to classification Agreement. This characteristic makes it particularly useful in cases where events of interest are rare. Using the previous example of weather forecasting in a desert climate, only forecasts that focus on rainfall events and correctly identify when they occur will get high CSI values.

CSI is given by the expression:

$$CSI = \frac{H}{H + M + F} = \frac{|\text{Observed Presence} \cap \text{Classified Presence}|}{|\text{Observed Presence} \cup \text{Classified Presence}|} \quad (1.7)$$

CSI ranges from 0 to 1. A CSI value of 1 indicates perfect Agreement, where all classified presence matches observed presence. A value of 0 means a perfect disagreement between the classified and observed presence.

The literature describes some of the properties of CSI. First, CSI tends to yield higher values for commonly occurring events compared to rare events. This characteristic is also explained by the previous argument: when an event is more prevalent, even uninformative classifiers can achieve higher Agreement of presence (Hand 2012b). Second, papers describe how CSI is often maximized when there is a Quantity overestimation (Baldwin and Kain 2006). The literature suggests that classifications maximizing CSI can result in a systematic overestimation of the quantity of presence. In this line, Pontius and Millones (2011) argues how CSI combines both Quantity and Allocation disagreements in one single metric, which he points to as one of the main limitations of the metric. Consistent with the previous property of CSI, its inability to distinguish between Quantity and Allocation disagreement hinders the interpretation of the metric, as it misses critical information about the sources of error of our classification.

In Finley’s forecast, CSI gets a value of 23%. CSI is always smaller than or equal to the smallest of PA and UA, as it combines both of their errors. A part of the reason why this value is low is that the prevalence of tornadoes is only 2%. The amount of tornadoes is overestimated, so according to what we just discussed, CSI would probably be smaller if there were no Quantity error. Note also that if we only report CSI, we can’t tell how much of the error is due to the overestimation of the quantity of tornadoes.

1.2.1.3 Second order single threshold metrics

Second-order classification evaluation metrics evaluate the relationships between multiple first-order metrics. Their added complexity tries to account for factors such as class imbalances or error costs. Thus, second-order metrics often incorporate more entries from the contingency table, weighting them to address some of the information lost in first-order metrics. However, this added complexity comes at the cost of interpretability, as the influence of the contingency table’s entries becomes harder to trace.

Some second-order metrics evaluate the performance of a classifier relative to a reference baseline, such as random guessing. This distinction allows us to differentiate between Agreement and Skill. An agreement measures the degree of coincidence between observed and classified outcomes, without considering how that level of agreement compares to a reference. In contrast, Skill measures the improvement of the classification over a baseline level of performance. For this reason, these metrics are more appropriately interpreted as measures of Skill rather than Agreement, as will be discussed in the following sections (See section 1.2.4).

The most popular second-order metrics include metrics such as F1-score, Matthews Correlation Coefficient (MCC), Geometric Mean, or likelihood ratios (Tharwat 2021). The F-measures (Rijsbergen 1979) (Eq. 1.8) are metrics similar to CSI in that they do not consider Correct rejections. Like CSI, F-measures focus on the classification ability of presence, which makes them particularly useful when addressing class imbalances in applications where the class of interest is the rare one. F-measures are calculated as the weighted harmonic mean between user accuracy (UA) and producer accuracy (PA). The β allows the metric to emphasize the user's ($\beta < 1$) or producer's accuracy ($\beta > 1$), depending on the relative cost of False alarms to Misses.

$$F_{\beta} = \frac{(1 + \beta^2) \times \text{UA} \times \text{PA}}{\beta^2 \times \text{UA} + \text{PA}} = \frac{(1 + \beta^2)H}{\beta^2(H + M) + H + F} \quad (1.8)$$

In practice, the most popular F-measure is F-1 (Eq. 1.9), which corresponds to the F-measure with $\beta = 1$. The F1 score is a monotonic transformation of the CSI, taking values that are always equal to or higher than those of the CSI, and therefore leading to a closely related interpretation. Thus, the direction of any comparison between models based on CSI is preserved when using F1 (Christen et al. 2023). The relationship between both metrics can be directly observed from Eq. 1.9. F1 places relatively greater emphasis on Hits, which results in systematically higher values. F-measures offer a way to weight the relative costs of False alarms and Misses, but tend to be more complex to interpret. On the other hand, CSI offers a straightforward interpretation but gives equal weight to False alarms and Misses and tend to report low values. As was the case with CSI, F-measures range from 1 to 0. When applied to Finley's data, F1 yields a value of 37%, compared to 23% for CSI.

$$F_1 = \frac{2 \times \text{UA} \times \text{PA}}{\text{UA} + \text{PA}} = \frac{2H}{2H + M + F} \quad (1.9)$$

Over the last few years, MCC (Matthews 1975) has become one of the most popular second-order metrics. MCC is equivalent to Pearson’s correlation applied to binary classifications (See Appendix A). We can express the MCC in terms of classification outcomes:

$$\text{MCC} = \frac{\text{Cov}(r, c)}{\sigma_r \sigma_c} = \frac{H \cdot C - M \cdot F}{\sqrt{(H + F)(H + M)(C + F)(C + M)}} \quad (1.10)$$

The MCC value ranges from -1 to +1, where +1 indicates a perfect prediction, and -1 indicates total disagreement between prediction and observation. A value of 0 indicates performance no better than random, as a random prediction would classify C at the same rate as F, and H at the same rate as M (see Section 1.2.4). As MCC entails a comparison against a random performance, it is a metric of Skill rather than Agreement³. This is a relevant observation, as the use of MCC is often promoted over other Agreement metrics (Chicco, Tötsch, et al. 2021; Zhu 2020; Chicco and Jurman 2020), when Agreement and Skill metrics measure different concepts.

MCC evaluates both presence and absence by incorporating all four entries of the contingency table, which allows it to account for the balance between classes. MCC gets high values only when the classifier performs well across both classes, effectively weighting performance according to class prevalence (Chicco 2017). For this reason, it is often regarded as a comprehensive metric for binary classification, particularly in the presence of class imbalance (Chicco and Jurman 2020). However, this apparent robustness is a direct consequence of its nature as a Skill metric, since MCC implicitly compares observed performance against a reference defined by the marginal distributions.

Despite its increasing popularity and proposals to adopt it as a standard metric (Chicco and Jurman 2023), several studies have pointed out limitations of correlation-based measures in classification evaluation, particularly regarding their interpretability and sensitivity to prevalence and marginal distributions (Powers 2011; Flach 2015).

³As a reference, note the similarity between Equation 1.10 and those in Section 1.2.4.

Applied to Finley’s forecasts, MCC gets a value of 38%, which is almost identical to F1’s value. However, this score comes from the evaluation of the forecasts of both the presence and absence of tornadoes. Thus, the low value is explained because, even if the absences of tornadoes are accurately predicted, a large share of the Agreement is due to their dominance in the dataset. MCC subtracts the effect that the prevalence of no tornadoes has on the overall agreement.

1.2.2 Multiple threshold metrics

Metrics based on multiple thresholds are used to evaluate probabilistic or soft classifications, where each observation is assigned a probability rather than a binary output. Unlike binary classifications, soft classifications do not specify a fixed quantity of presence or absence. To evaluate their performance, soft classifications must be hardened into binary outcomes, as the ultimate purpose of soft classifications is to guide binary decision-making. Since there are multiple ways to harden soft classifications (e.g., using different probability thresholds), evaluation metrics for these cases focus on the overall performance across all possible thresholds.

Every multiple threshold method and metric is built under the same logic. All the observations that have a value of the soft classifier above a given threshold are classified as presence, and those below as absence. This operation is repeated iteratively throughout the multiple thresholds of the soft classification. Figure 1.3 illustrates this logic applied to two thresholds. Since the highest threshold classifies all the observations as absence, and the lowest threshold does so as presence, multiple threshold curves evaluate the classifier at multiple quantities of classified presence. Each of the evaluated thresholds produces a contingency table, from which single threshold accuracy metrics are calculated. Multiple threshold curves represent the aggregation of those single threshold metrics at multiple evaluations.

The two most widely used methodologies for multiple-threshold evaluation are the Receiver Operating Characteristic (ROC) curve and the Precision-Recall Curve (PRC)(Green and Swets 1966; Saito and Rehmsmeier 2015; Davis and Goadrich 2006; Fawcett 2006). These two curves follow a fundamentally different approach to the measurement of Agreement. ROC curves (Fig. 1.4a) represent Hit rate (or Sensitivity, PA of presence, Tab.1.3) against False Alarm rate. Hit rate or PA is

Classifier's value	90	65	50	45	40	30	30	10	10
Observed	P	A	A	P	P	P	A	A	A
Predicted	P	P	A	A	A	A	A	A	A
Threshold	60								

(a)

Classifier's value	90	65	50	45	40	30	30	10	10
Observed	P	A	A	P	P	P	A	A	A
Predicted	P	P	P	P	P	A	A	A	A
Threshold	35								

(b)

FIGURE 1.3: Toy example of a multiple threshold evaluation of a classifier. *Source: Adapted from Pontius (2022).*

the proportion of observed presence correctly classified, and False Alarm Rate is the proportion of observed absence wrongly classified. ROC curves consider all four entries of contingency tables, as they evaluate the classification of both presence and absence. By visualizing these rates across multiple thresholds, ROC curves provide insights into a model's ability to distinguish between presence and absence classes at the whole set of potential classifications. Furthermore, the slope of the ROC curve can be interpreted as the trade-off between correctly classifying observed presences and wrongly classifying observed absences.

Note that ROC curves are designed with metrics that use the observed events as reference. This means that the classification doesn't influence the size of the denominator of the metrics. This is the reason why ROC curves can't have negative

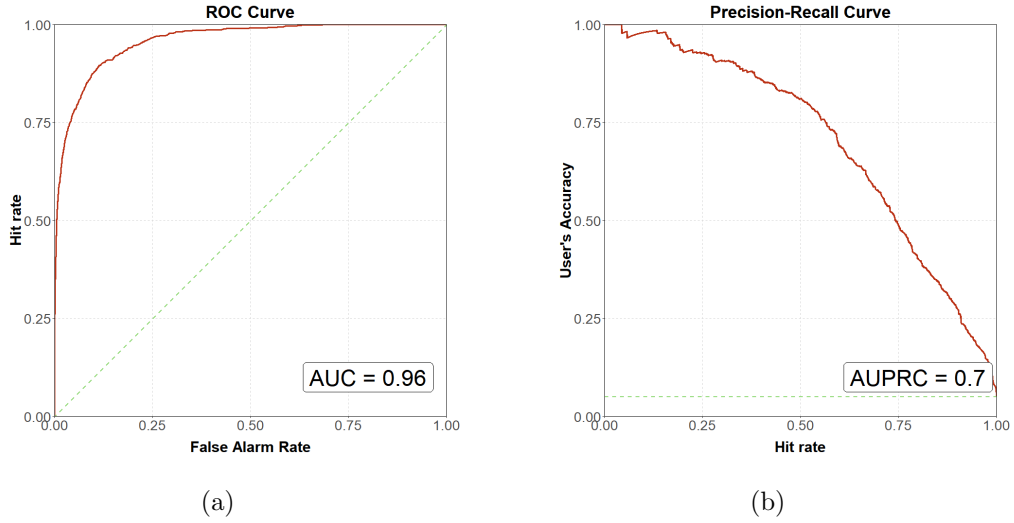


FIGURE 1.4: Representation of the ROC (a) and Precision-Recall (b) curves. The green line represents the random classification baseline. *Source: Self elaboration.*

slopes, as neither Hits or False alarms can decrease with smaller thresholds. The goal of this design is a standardized comparison of classification results in populations with different prevalence. However, as further discussed in different sections of this dissertation, the sensitivity of the ROC framework to prevalence is one of its biggest criticisms (Lobo et al. 2008).

Pontius and Si (2014) designed Total Operating Characteristic (TOC) curves as an alternative to overcome some of the limitations of ROC curves. TOC curves offer a more interpretable representation of the information shown in ROC curves (Pontius 2022; Pontius and Si 2014) because they present information in absolute terms rather than ratios. The primary advantage of this is that it allows for the reconstruction of the contingency table associated with each threshold, along with Quantity and Allocation disagreement and prevalence. The horizontal axis in TOC curves represent classified presence. This enhances the interpretation of the curves by enabling the differentiation between quantity and allocation disagreement for each point on the curve. Such differentiation is vital for many research questions, as it visualizes whether classifications overestimate or underestimate the quantity of classified presence. A more extensive description and interpretation of TOC curves can be found in Chapter 2.

On the other hand, PRC curves (Fig. 1.4b) plot Precision or User’s Accuracy (Eq.1.6) against Hit rate or Producer’s Accuracy (Eq.1.5). Precision represents the proportion of classified presence correctly classified. Hit rate or PA is the proportion of observed presence correctly classified. Thus, the slope of PRC can be interpreted as the trade-off between the Agreement in classified and observed presence, i.e., how much our correctly classified presence changes to classify more of the observed presence. Unlike ROC, PRC curves don’t consider Correct rejections on its accuracy metrics. Therefore, it is a framework that focuses exclusively on the Agreement of presence, as they were CSI and F-measures.

The baseline considered in both methods (Fig.1.4, green lines) is a random allocation, i.e, a classifier that classifies presence uniformly at the rate of prevalence. In ROC, the baseline takes the form of a diagonal line, as the rate at which the random classifier classifies Hits is the same as that of False alarms. In PRC, the baseline is a horizontal line located at $y = \text{prevalence}$, as the random classifier has a constant UA.

The Area Under the ROC Curve (AUC) and the Area under the Precision-Recall Curve (AUPRC) are popular metrics that summarize the Agreement of multiple thresholds (Davis and Goadrich 2006; Tharwat 2021). These metrics they are measures of Allocation Agreement because they evaluate the Agreement of all the possible classified quantities of classified presence. The AUC’s values range between 0 and 1, with 1 being a perfect classification, 0.5 equal to random classification, and 0 the worst possible classification. As the standard multiple-threshold metric, AUC provides an overview of a classifier’s ability to differentiate between positive and negative classes. AUC measures the degree of monotonic relationship between the index and the binary variable because ROC curves can’t have negative slopes (Pontius 2022).

AUC considers both H and C. Thus, parallel to metrics with the same construction, such as OA, AUC tends to get high values when there is a substantial class imbalance, influenced by the Agreement in the majority class. Therefore, one of the main criticisms of AUC is that it provides no information in applications with class imbalances (Lobo et al. 2008; Hand 2009). Despite an extensive body of literature underscoring this property of the metric, AUC is still often used as an accuracy metric in applications where the event of interest is rare, as is the case in land change

science.

AUPRC has the same range of values as AUC. However, 0.5 does not represent a random Agreement, as the baseline of PRC changes with prevalence. AUPRC shares the same definition of Agreement as PRC, which does not consider C. This focus makes AUPRC appropriate when the class of interest is rare. AUPRC is becoming increasingly adopted in applications with this characteristic (Blockeel et al. 2013; Davis and Goadrich 2006; Saito and Rehmsmeier 2015).

Other metrics have been developed to address the distortions caused by class imbalance in AUC. The Partial Area Under the ROC Curve (pAUC) measures the AUC on specific ranges of threshold values, providing an evaluation of the most relevant thresholds of the classifier. pAUC has applications in various fields, including species distribution (Peterson et al. 2008), screening research (Baker and Pinsky 2001), and medicine (Malik et al. 2016; Nahm 2022). pAUC and other partial area metrics face the challenge of selecting the cutoff criterion. pAUC’s cutoff criterion can be determined using utility functions, statistical criteria, or decision analysis (Dodd and Pepe 2003; McClish 1989). In practice, the most used criterion depends on Agreement metrics such as PA or specificity (Carrington et al. 2023; Mas et al. 2013). However, these pAUCs have some issues regarding the arbitrariness of specifying the cutoff range (Ma et al. 2013). For example, if the cutoff is a PA range of 0.4-0.6, we will evaluate our classifier when it detects around half of the observed presence. Yet, our classifier might require a large overestimation of quantity to achieve those PA values, meaning that we would still be evaluating unrealistic classifications. This arbitrariness hinders the comparison and interpretation of pAUC values.

1.2.3 Comparability of single and multiple threshold metrics

Following the logic underlying ROC and PRC curves (Fig. 1.3), any single-threshold metric can be extended into a multiple-threshold representation by evaluating it across a sequence of contingency tables defined by varying thresholds. This approach consists of representing the chosen single-threshold metric on the y-axis, while the classified quantity (i.e., $H + F$) or, equivalently, the decision threshold, is represented on the x-axis.

The potential of this representation is particularly relevant, as it allows for a unified analysis of soft and hard validation using consistent elements of the contingency table. This is essential for interpretation, since it enables distinguishing whether classification errors arise from the estimation of the continuous (soft) output or from the selection of the threshold used to obtain the binary (hard) classification.

In practice, single-threshold and multiple-threshold metrics are typically applied using different configurations of the contingency table. As a result, their outcomes are not directly comparable. For example, in LCM applications, soft validation is commonly assessed using AUC, while hard validation relies on metrics such as OA or CSI. These metrics do not measure Agreement in the same way, which complicates the interpretation of results and prevents identifying whether discrepancies arise from the model output or from the choice of threshold.

To illustrate this issue, consider the evaluation of GDP growth as an indicator of economic recession. Given that recessions are rare events, one might use PRC and AUPRC for multiple-threshold evaluation. However, when selecting a specific threshold, the evaluation would typically rely on metrics such as CSI, F-measures, or PA/UA. Since these metrics are based on different arrangements of the entries of the contingency table, their results are not directly comparable to AUPRC. Consequently, one might incorrectly conclude that GDP growth is not an appropriate indicator of economic recession, when in fact the issue lies in the choice of threshold. Conversely, one might attribute poor performance to an inadequate threshold when the underlying problem is the lack of predictive power of GDP growth itself.

The representation of single-threshold metrics across multiple thresholds constitutes a central methodological approach in this dissertation (see Fig. 2.5). Specifically, we propose a framework that, first, represents metrics as a function of the classified quantity ($H + F$), enabling the visualization of Quantity and Allocation disagreement, as well as the role of prevalence. Second, we compute the area under these curves as a metric of soft validation, ensuring comparability between soft and hard evaluation. Although the dissertation focuses on the case of CSI, this methodology can be extended to any single-threshold metric.

Despite its potential, this approach remains underexplored in the literature. For instance, Hamill and Juras (2006) uses this idea to relate the Critical Success Index

(CSI) to bias, while Pontius, Castella, et al. (2018) extends the structure of the TOC curve by linking CSI to $H + F$. Similarly, such representations of OA are common in the literature (Pontius and Millones 2011; Hogan et al. 2010), yet these contributions do not generalize the approach to establish a consistent framework connecting soft and hard validation.

1.2.4 Agreement vs Skill

1.2.4.1 Definition of Agreement and Skill

All the metrics mentioned in Section 1.2 fall under the category of **Agreement** metrics. These metrics measure the Agreement between classifications and actual observed data. However, having a large Agreement doesn't necessarily mean that the classification is actually skillful.

Figure 1.5 illustrates two different classifications of the same population. Both classifications in the figure have the same value of OA (80%) with different types of errors. While both classifications have only Quantity disagreement, the first classification underestimates the Quantity by 20 observations, whereas the second overestimates it by the same amount.

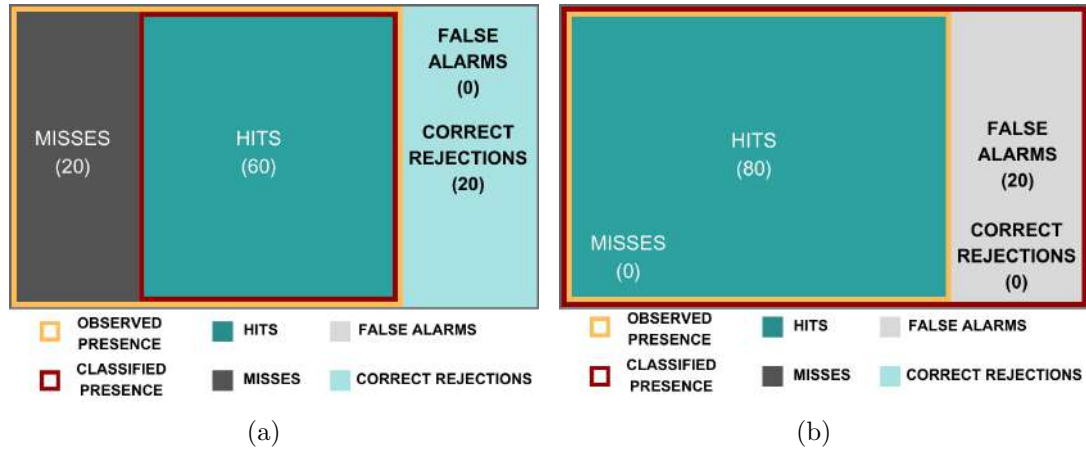


FIGURE 1.5: Illustration of two classifications of the same observations, showing different levels of Quantity disagreement. *Source: Self elaboration.*

It is straightforward to think that even if both classifications have the same Agreement they are not equally good. The first classifier correctly allocates all the presence

it simulates. Thus, we could think that if the classifier increased the quantity of presence it simulates, it would get a close to perfect classification. Meanwhile, the second classifier only generates Agreement because it overestimates the quantity of presence, which is the majority event. As presence takes most of the observations, panel b is simply classifying everything as presence, which is clearly an unskillful classification.

Our illustration shows how the Agreement of our classification doesn't imply a good performance. Metrics that assess model performance are known as **Skill** metrics (Doolittle 1888; Gilbert 1884). According to Briggs and Ruppert (2005), a collection of predictions is skillful if their expected Agreement is larger than the expected Agreement of a naive prediction. In various fields, the most used naive prediction is randomness (Mason 1982; Hand 2009; Fawcett 2006). Thus, the Skill in many metrics and models is measured as how much randomness is improved.

The definition of a random classification is the one that correctly classifies the presence at the uniform rate of prevalence (see Section 1.2.4.2). This is so because the probability of randomly getting a correct presence is equal to the proportion of presence in the population. Going back to Figure 1.5, in panel (a), a random classifier would be expected to correctly classify $60 \times 0.8 = 48$ presences (H) and 12 correct absences (C), resulting in an OA of 54%. In contrast, the classification in panel (b) is identical to a random classification. Therefore, it is clear that classification A has more merit than classification B, as it is the only one that can outperform a random classification.

Our illustration represents a common scenario observed in many research fields where the event of interest is rare. For instance, in land change science, any model will score high OA regardless of its performance. This is simply due to the low prevalence of changes, which causes observed and classified persistence overlay. However, this high OA does not imply that the model is performing well, as any more sophisticated model is likely to outperform it. Therefore, to assess a classifier's performance, it is necessary to compare it to a baseline and measure how much the baseline is outperformed.

This illustration underscores the relevance of the conceptual differentiation between the Agreement and Skill of classifications. Yet, these two concepts are often overlooked in the literature. It is common to find claims about the quality of models'

performance based on metrics of Agreement, mainly OA. However, Agreement metrics only inform about the similarity between observations and classifications, and not about how much of that Agreement is due to the quality of the model. Similarly, even if Agreement and Skill metrics have different interpretations, some Skill metrics, such as MCC, are employed and interpreted as metrics of Agreement. Again, this could lead to potentially flawed conclusions. A classification can outperform a certain baseline, but still doesn't get values of Agreement.

It is also noteworthy to mention that not all the baselines necessarily need to be based on a random assumption. Most phenomena in the real world are not random, so a randomness baseline can be insufficiently ambitious in many applications. For example, research such as Hamill and Juras (2006) illustrates how in spatial data, the spatial heterogeneity of the binary variable makes randomness a poor baseline. This is because the prevalence can be low for the whole region, but high in areas where the presence concentrates. If unambitious baselines are used to assess model performance, models will tend to get high values of Skill metrics without actually being skillful.

1.2.4.2 Overview of Skill metrics

The “Finley’s affair”, spanning from 1884 to 1893, marks a pivotal era in the evolution of prediction verification methods, setting the foundation for substantive advancements in the field (Murphy 1996). This period was triggered by Finley (1884) publication of binary tornado prediction experiments. Gilbert (1884) response to Finley’s publication paved the way for many popular binary classification metrics. This is one of the earliest papers that differentiates the concepts of Agreement and Skill in binary forecasts. Some of the metrics developed in response to these two works remain widely used in modern binary classification evaluation. Among them, we can highlight the Gilbert Skill score (GSS) (Gilbert 1884), the Heidke Skill score (HSS) (Heidke 1926), which was already defined by Doolittle (1888), the Peirce Skill score (PSS) (Pierce 1884), and the Clayton Skill score (CSS) (Clayton 1934).

Many of these metrics have been repeatedly reinvented and redefined over time, often taking on different names depending on the field of study (Mbizvo and Larner 2024). For instance, the Gilbert Score has been reintroduced as the CSI (Schaefer

1990), Threat Score (Palmer and Allen 1949), and Taimoto Index (Donaldson et al. 1975) in meteorology, the Jaccard Index (Jaccard 1908) in environmental sciences, F^* (Hand, Christen, et al. 2021) in machine learning, or Figure of Merit (Pontius, Boersma, et al. 2008) in land change. The frequency with which different disciplines have required the fundamental concepts behind these metrics highlights their persistent relevance in evaluating binary classifications. The issues and arguments derived from these metrics address the main concerns regarding binary classification evaluation.

Classification Skill is typically expressed as the percentage improvement relative to the baseline classification. Skill metrics are the result of the comparison of a specific Agreement measure A to the Agreement A_{baseline} of baseline classification relative to a perfect classification, calculated as follows (Wilks 2006; Woodcock 1976):

$$\text{Skill Score} = \frac{A - A_{\text{baseline}}}{A_{\text{perf}} - A_{\text{baseline}}} \quad (1.11)$$

Where A_{perf} refers to the score of a perfect classification.

GSS is the first Skill scored derived from “Finley’s affair”. GSS is a metric that uses CSI’s definition of Agreement. Gilbert tried to adapt CSI to account for the Hits that a random classification would generate. In practice, GSS intends to measure the difference in Agreement, measured by CSI, between the actual and random classifications. Using the Skill definition of Equation 1.11, where $A_{\text{perf}} = 1$, Gilbert defined his metric as:

$$GSS = \frac{H - H^{\text{rand}}}{H - H^{\text{rand}} + F + M} \quad (1.12)$$

An equivalent expression can be written in terms of CSI:

$$GSS = \frac{\frac{H}{H + F + M} - \frac{H^{\text{rand}}}{H + F + M}}{1 - \frac{H^{\text{rand}}}{H + F + M}} \quad (1.13)$$

Where H^{rand} is the Hits that a random forecast would score with the same quantity of classified presence, given by:

$$H^{\text{rand}} = (H + F) \frac{H + M}{E} = (H + F) \cdot \text{prevalence} \quad (1.14)$$

Similarly, the remaining entries of the contingency table for the random classification are:

$$F^{\text{rand}} = (H + F) \frac{F + C}{E} \quad (1.15)$$

$$M^{\text{rand}} = (H + M) \frac{M + C}{E} \quad (1.16)$$

$$C^{\text{rand}} = (C + F) \frac{M + C}{E} \quad (1.17)$$

Going back to Finley's forecasts, GSS scores almost the same value as CIS (22%). The interpretation of these results is that not only observed and forecasted tornadoes match 23% of the time, but the forecast also achieved 22% of the agreement that could not have occurred by chance.

Heidke Skill Score (HSS) is a Skill metric based on OA, accounting for both the Hits and Correct rejections that are due to chance. HSS accounts for the probability of a correct observation, regardless it is a correct presence or absence. The baseline in HSS is the OA obtained by a random classifier that classifies the same quantity of presence as the original classifier, i.e., the proportion correct that would be achieved by random classifications that are statistically independent of the observations (Wilks 2006). As a random forecast would predict both events correctly, the baseline considers both of them:

$$OA^{\text{rand}} = \frac{H^{\text{rand}} + C^{\text{rand}}}{E} \quad (1.18)$$

$$= \frac{(H + F)(H + M) + (C + M)(C + F)}{E^2} \quad (1.19)$$

From the application of Equation 1.11, we get the Heidke Skill Score (HSS), given by:

$$HSS = \frac{OA - OA^{\text{rand}}}{1 - OA^{\text{rand}}} = \frac{2(HC - FM)}{(H + F)(F + C) + (H + M)(M + C)} \quad (1.20)$$

HSS can take on values within the range $\{-\infty, 1\}$. An HSS of 1 indicates a perfect Skill, an HSS of 0 indicates no Skill beyond random chance, and a negative HSS indicates worse than random chance.

Peirce Skill Score (PSS) (Eq.1.26) shares a similar formulation with the HSS, as they both share OA’s definition of Agreement and follow Equation 1.11. However, HSS and PSS differ in how they define the random classification that corresponds to a perfect classification. They both define a random classifier as one that generates Hits at the rate of prevalence. In HSS, the random classification that corresponds to a perfect classification classifies the same quantity as the original classifier (H+F) (Eq.1.14). In PSS, the random classification that corresponds to a perfect classification classifies the same quantity as the observed (H+M) (Ebert and Milne 2022; Wilks 2006). This is called an unbiased random classification, as the random classifier has no Quantity disagreement ($H + F = H + M$). Equation 1.21 shows how HSS and PSS share the same numerator, but differ in their denominator.

$$PSS = \frac{OA - OA^{\text{rand}}}{1 - OA^{\text{rand,u}}} \quad (1.21)$$

Where $OA^{\text{rand,u}}$ is the OA of the unbiased random classification. Ebert and Milne (2022) explain the reasoning behind this adjustment of PSS. He claims that a perfect classification can’t be obtained by classifying the same Quantity as the observed classification (H+F), because it would have Quantity disagreement, and wouldn’t be able to get an $OA = 1$. Thus, as the perfect classifier is one with no Quantity disagreement, the corresponding random classification shouldn’t have any either. As PSS considers a random classification with a different Quantity than HSS, it uses $OA^{\text{rand,u}}$ instead of OA^{rand} . Ebert and Milne (2022) claim that PSS is the most appropriate Skill metric because it considers the most realistic baseline for a perfect classifier.

$$OA^{\text{rand,u}} = \frac{H^{\text{rand,u}} + C^{\text{rand,u}}}{E} = \frac{(H + M)^2 + (C + F)^2}{E^2} \quad (1.22)$$

where $H^{\text{rand,u}}$ and $C^{\text{rand,u}}$ denote the expected number of Hits and Correct Rejections under an unbiased random classification, defined as:

$$H^{\text{rand,u}} = (H + M) \frac{H + M}{E} \quad (1.23)$$

$$C^{\text{rand,u}} = (C + F) \frac{C + F}{E} \quad (1.24)$$

The remaining entries of the contingency table are:

$$F^{\text{rand,u}} = M^{\text{rand,u}} = \frac{(H + M)(C + F)}{E} \quad (1.25)$$

Thus, the resulting PSS can be written as:

$$PSS = \frac{\frac{H + C}{E} - \frac{(H + M)^2 + (C + F)^2}{E^2}}{1 - \frac{(H + M)^2 + (C + F)^2}{E^2}} = \frac{HC - MF}{(H + M)(C + F)} \quad (1.26)$$

A perfect classification yields a score of 1 (when $M = F = 0$), random classifications score 0 ($HC = MF$), and the worst classification gets a value of -1 ($H = C = 0$). Unlike the Heidke score, the Peirce Skill Score increases the value of correct classification for rare events as the likelihood of the event varies, thereby encouraging the classification of rare events rather than discouraging them due to their low frequency (Ebert and Milne 2022).

Equation 1.27 represents PSS as the difference between the PA and the false alarm rate, which are the same two accuracy metrics considered by ROC curves. In fact, PSS is the vertical distance between a point in an ROC curve and the uniform line (Manzato 2005; Manzato 2007). In diagnostic testing, this is a popular metric known as the Youden Index (Youden 1950). Note also the similarities between PSS (Eq.1.26), HSS (Eq.1.20), GSS (Eq.1.12) and MCC (Eq.1.10), which all share the same concept in their numerator of skill.

$$PSS = \frac{H}{H + M} - \frac{F}{C + F} \quad (1.27)$$

Applied to Finley’s data, HSS scores a value of 36% against the 55% of PSS. MCC, which also considers the agreement of both presence and absence, gets a value of 38%. This divergence in Skill scores highlights the need to understand the metrics, their properties, and to get coherent interpretations and conclusions for them.

1.2.5 Limits and domain of application of classification evaluation metrics

Despite the usefulness of each of the metrics, no classification evaluation metric is universally applicable, nor can we state that one is better than the other. The immense variety of Agreement and Skill metrics described in these sections raises the question of whether a standard can be designed to tackle all the research questions and data characteristics. Foody (2023) refers to the unrealistic desire to have a single optimal accuracy value as the reason for the popularity of some binary classification metrics. Many researchers emphasize the importance of matching accuracy metrics to research-specific questions and data characteristics, arguing against the use of standard accuracy metrics because of their popularity (Pontius 2022; Hand 2012b; Hand, Christen, et al. 2021). The knowledge about the properties and interpretation of these metrics is crucial for their proper application. The election of an inappropriate evaluation metric can lead to incorrect conclusions from research questions (Hand 2012b).

Different metrics reflect different perspectives on the classification ability derived from the same contingency tables. Wilks (2006) argues that any evaluation approach based on contingency tables summarizes a three-dimensional problem into a single metric, because the four entries of the contingency table have at least one degree of freedom. The reduction of three-dimensional data into a single value necessarily results in a loss of information. There are multiple ways in which the entries of contingency tables can be combined to assess classification Agreement and Skill. Pontius and Si (2014) underscores the importance of reporting the metrics along with the

corresponding contingency table. Contingency tables allow the calculation of any binary classification evaluation metric, while specific evaluation metrics don't allow the reconstruction of contingency tables.

Table 1.4 summarizes all the values of the agreement and skill metrics described in this chapter, applied to Finley's data. The large variety of Agreement and Skill values illustrates the complexity of the analysis of the same contingency table. It would be a bold statement to claim any forecast quality without the specific forecast goal. If the objective is to achieve a high number of correct forecasts, the OA shows that the forecast meets this goal. Yet, if the goal is to get a model that outperforms randomness in the forecast of the two events, MCC, HSS, and PSS suggest different conclusions.

If the goal is to accurately predict tornadoes, both the CSI and the F1 score indicate that our forecasts fall short of this objective. However, F1 suggests a slightly better outcome, as it assigns greater importance to Misses compared to False Alarms, something that may be desirable in tornado forecasting due to the difference in the costs associated with each type of error. Finally, the GSS reveals that not only are we unable to forecast many tornadoes correctly, but even a random forecast would yield similar results.

Finley's forecast also illustrates the relevance of interpreting the metrics along with their contingency table. If we only had the values of the metrics shown in Table 1.4, we would miss crucial information to interpret the results and assess the errors in our forecast. From the contingency table (Tab. 1.2), we can know that prevalence is extremely low (1.8%), which contextualizes the metrics that consider the agreement of both tornadoes and no tornadoes. The table also shows that our forecast overestimates the quantity of tornadoes, something impossible to deduce from the metrics. This information reveals that to improve our forecast, we should not only improve our ability to discriminate between tornadoes and no tornadoes (Allocation), but also we should forecast fewer tornadoes overall (Quantity).

The literature review carried out in this Section highlights the lack of practical sense of the quest for a single standard accuracy metric. Every metric provides a different perspective of contingency tables, and it is their proper application and

TABLE 1.4: Agreement and skill metrics applied to Finley’s data.

Metric Type	Metric Name	Value
Agreement	Overall Agreement (OA)	97%
	User’s Accuracy (UA)	Tornado: 28%, No Tornado: 99%
	Producer’s Accuracy (PA)	Tornado: 55%, No Tornado: 97%
	Critical Success Index (CSI)	23%
	F1 Score	37%
Skill	Matthews Correlation Coefficient (MCC)	38%
	Gilbert Skill Score (GSS)	22%
	Heidke Skill Score (HSS)	36%
	Peirce Skill Score (PSS)	52%

interpretation, rather than their properties, that make them good metrics. For example, lots of criticism has been made of the dependency of metrics such as OA on prevalence. Yet, these metrics are adequate for applications where the classification of presence is equally relevant to that of absence, which requires the consideration of Correct Rejections and Agreement. In applications where the focus is exclusively on the correct classification of presence, metrics such as CSI and F1 are more useful because they don’t consider Correct Rejections. In cases where the research question requires the evaluation of a model’s performance with respect to a baseline, Skill metrics need to be used. As an illustrative example, following the aim for a standard evaluation metric, claims have been made about the adequacy of MCC over F1. Nonetheless, each metric represents different information about contingency tables. In fact, literature compares MCC with F1 when one is a metric of Skill of both presence and absence, and the other is a metric of Agreement of only presence.

A similar reflection can be made in the comparison of multiple and single threshold metrics. Multiple threshold metrics respond to the need to simultaneously evaluate several quantities of classified presence. Criticism of multiple thresholds metrics often points to the fact that unrealistic quantities of classified presence are evaluated, and thus, considered relevant. However, this is the main purpose of these metrics. Thus,

what is often referred to as limitations are rather misapplications or misinterpretations. A similar tendency can be observed in the criticism of AUC. AUC is criticized for being sensitive to imbalanced data and potentially misleading. Nevertheless, while imbalanced datasets shouldn't be in the domain of applications of AUC, AUC still reflects the information it is designed for. AUC measures the degree of a monotonic relationship between an index and a binary variable, regardless of prevalence (Pontius 2022). The wrong practical use and interpretation of AUC reflects the lack of understanding of the metric.

Single threshold metrics, on the other hand, require an a priori choice of the cutoff value, which can be arbitrary and may not reflect the classifier's overall performance. However, this requirement is not inherently a limitation but rather a response to specific research needs. The selection of the specific threshold should respond to the research question and not the optimization of arbitrary evaluation metrics. Some applications might require a limited number of False Alarms, needing to focus on Quantity disagreement. Some applications might require the correct diagnosis of as much observed presence as possible, needing to focus on the reduction of Misses and Allocation disagreement. Therefore, the selection of the threshold should be guided by the research question and data characteristics rather than aiming for the highest score of a standard metric.

1.2.6 Current LCM validation practices

LCM validation methods are a subset of map comparison methods. Map comparison methods analyze the difference between two or more maps. These maps can belong to the same time point but from different sources, different time points but same source, or have a simulated and observed origin, among others. LCM validation methods specifically compare at least three maps: an observed land use map in t_0 , an observed land use map in t_1 , and a simulated land use map in t_1 . The overlay of these maps provides the observed and simulated change, which are the observed and classified binary outcomes we need to validate. The presence or absence of the binary variable is now either change or persistence of land uses. Thus, the research and advancements in binary classification evaluation offer a suite of tools to land change science. The

adoption and development of binary classification metrics for LCM validation metrics is essential to assess many of the challenges of land change science (Section 1.1.3).

1.2.6.1 Hard validation practices

Hard-classified maps are the most frequent LCM output. The literature presents a wide range of validation metrics used for hard validations (Tong and Feng 2020). The metric selection criteria vary depending on whether the goal is to validate land uses versus land use changes, patch-based versus pixel-based configurations, or Allocation versus Quantity differences, among other factors. This dissertation focuses on metrics that compare Allocation and Quantity changes pixel by pixel.

Pixel-by-pixel validation is primarily based on contingency tables, which now show the presence or absence of land change (see Table 1.5). Therefore, Hits are now observed changes are correctly simulated; Correct rejections are observed persistence are correctly simulated; Misses are observed changes are simulated as persistence; and False alarms are observed persistence are simulated as changes (Pontius and Millones 2011; Pontius and Parmentier 2014; Pontius 2022). If the source map includes more than two categories, there is an additional type of error in the location of the change, where the change location is correctly identified but assigned to an incorrect category (Pontius and Parmentier 2014).

		Observed		
		Change	Persistence	Total
Simulated	Change	H	F	H + F
	Persistence	M	C	C + M
Total		H + M	C + F	Extent

TABLE 1.5: Contingency table for land changes

The most frequently used metric for hard validation is OA. However, as extensively discussed in this Section, OA considers the Agreement of both presence and absence of the binary variable. This means that, applied to LCM, it measures the Agreement of changes and persistence. Land changes are usually rare events, with

common prevalence rates lying under 15%. This means that OA is mainly showing the Agreement due to correct persistence, being persistence substantially easier to simulate. As a result, modelers tend to report high accuracy values and claim good model performance without actually evaluating the predictive ability of changes.

It is also common to see modelers report PA and UA values. These metrics are more adequate for LCM validation because they don't consider correct rejections in their definition of Agreement, and thus, address class imbalance. However, these metrics show limited information about simulations because they only consider two entries of the contingency tables. Applied to LCM, PA represents the proportion of observed changes the model detects, but it doesn't inform about how much change it has to simulate to detect them. UA shows the proportion of the simulated changes that are correct, but it says nothing about how much of the observed change is detected.

In recent years, modelers have started incorporating CSI as a metric that doesn't consider correct rejections and combines PA and UA. In the field of LCM, CSI is commonly known as Figure of Merit (Pontius, Boersma, et al. 2008). CSI is an appropriate metric for LCM because it considers Hits, False alarms, and Misses exclusively in a single metric. Yet, CSI doesn't allow the distinction of Quantity and Allocation disagreements, something critical for the interpretation of LCM results (Pontius, Castella, et al. 2018).

In addition to the pixel-by-pixel analysis, more recent methods have been developed that account for near-hits, addressing both categorical (Hagen 2003; Hagen-Zanker 2009) and spatial fuzziness (Pontius and Millones 2011; Vliet 2016). Categorical fuzzy methods acknowledge that land changes are not always distinctly defined and that the boundaries between different land use categories can be blurred. Rather than evaluating whether a pixel has been correctly classified into a specific category, fuzzy methods allow for a gradation of Agreement, taking into account the degree to which a pixel belongs to a particular category. This means that a partially correct classification can be recognized as a "near-hit," which is particularly useful in areas where land changes are subtle or gradual. The multi-resolution approach allows for the evaluation of land change models at different spatial scales. This method is especially valuable for capturing change patterns that may be apparent at larger

scales but not as clear on a pixel-by-pixel basis. By integrating multiple levels of resolution, the method distinguishes between minor and major errors, depending on the closeness of False Alarms and Misses. Fuzzy evaluation methods are also based on contingency tables, and as such, can be measured by Quantity and Allocation disagreement-based methods. As such, they offer a necessary complement to traditional pixel-based methods, as they offer a nuanced perspective on the types of errors the model makes.

The concept of Skill is yet to be formalized into the field of land change science. The literature does not show any proper introduction of the concept, nor a distinction between the Agreement and Skill metrics used. The only Skill metric we identify in the land change field is the Kappa index, which is the specific name used in the field for Heidke Skill Score (HSS)(Eq.1.20). Pontius and Millones (2011) provides a variety of reasons why Kappa is not an adequate metric for model validation, most of them related to the interpretability of the metric. In Chapter 4, we delve into and criticize the arguments given by Pontius and Millones (2011).

The wide variety of validation metrics illustrates the importance of basing the choice of evaluation metrics on specific research questions (Stehman 1997). In the field of LCM, researchers often draw unrealistic conclusions due to the use of metrics that are not well-suited to their research objectives. For example, Tang et al. (2021) found that nearly 30% of the reviewed Cellular Automata applications reported Agreement, which is a metric of land use Agreement rather than land change Agreement. Similarly, 37% of the LCM analyzed by Vliet et al. (2016) used validation metrics such as OA, Kappa, and/or AUC/ROC, which evaluate the Agreement of both change and persistence. Moreover, 23% of the studies did not report any validation.

1.2.6.2 Soft validation practices

Soft validation of LCM models is often seen as a complementary process to hard validation. While hard-classified outputs address both the Quantity and spatial allocation of land uses in simulations, soft validation focuses exclusively on the allocation aspect, offering a probabilistic representation of land changes. As a result, hard outputs tend to receive more attention, as they provide clearer categorical outcomes.

Nevertheless, soft outputs are equally important for a comprehensive understanding of model behavior and performance. Soft validation allows researchers to trace sources of error in their models, as it evaluates intermediate outputs of the model instead of only the final result. In fact, most models incorporate both hard and soft outputs within their algorithms.

In LCM, soft validation consists of the evaluation of the ability of a soft classified map to correctly classify land changes. In contrast to the large variety of hard validation metrics used for hard LCM validation, soft validation approaches are rather limited. The main and almost only method used for LCM soft validation is the ROC curves and their corresponding AUC [1.2.2](#). The ROC curve provides a graphical representation of the trade-off between how much erroneous change our model commits to correctly simulate changes. ROC curves assess the model’s ability to distinguish between land change and persistence. The AUC, which measures the overall area under the ROC curve, serves as a summary statistic of the model’s discriminatory ability.

Applied to LCM, AUC offers a single value reflecting the model’s ability to distinguish between change and persistence. However, AUC has its limitations in the context of LCM soft validation. In [Chapter 4](#), we will delve into the reasons why AUC fails to account for the model’s ability to accurately simulate changes.

The inadequacy of AUC for LCM validation contrasts with its popularity. AUC is the predominant soft LCM validation methodology ([Camacho Olmedo et al. 2018](#)), without any adopted methodological alternative. In practice, soft LCM validation is often overseen, but if performed, it is primarily done with the ROC/AUC paradigm. The majority of LCM applications report high AUC values, from which modelers claim good model performance. However, they don’t specify the prevalence of their case study, nor the Agreement due to correct changes, which are critical elements of LCM evaluation. Additionally, in LCM literature, there is no applied methodology that evaluates the Skill of soft maps instead of their Agreement.

1.2.7 Binary classifications in Economics

The need to predict events with binary outcomes arises frequently in economics [Lahiri and Yang \(2013\)](#). Many economic problems can be framed as the relationship between

an underlying index variable and a binary outcome. Questions such as whether a borrower will default, whether a firm will exit the market, or whether an economy will enter a recession are inherently binary. Binary classification models provide a framework to represent and analyze these problems by mapping complex economic processes into two possible states. As such, they play a central role in both research and policy, where decisions often depend on whether an event occurs.

These models are used across a wide range of applications. In credit risk analysis, they are employed to estimate the likelihood of default or credit approval (Abdou and Pointon 2011; Engelmann 2011; Liu, Liu, et al. 2024). At the firm level, they are used to study decisions such as export market entry (Costa et al. 2019; Costa et al. 2022) or business exit (Paroissien et al. 2021). In macroeconomics, binary classification supports the identification of business cycle turning points and the prediction of recessions or financial stress episodes (Berge and Jordà 2011; Aldasoro et al. 2018; Duprey et al. 2022). Since the early adoption of ROC/AUC methods in economics (Granger and Pesaran 2000), these tools have become standard for evaluating predictive performance in such contexts. For instance, Lahiri and Wang (2013) uses ROC analysis to assess predictions of GDP decline, while Berge and Jorda (2011) evaluates how accurately economic activity is classified into expansions and recessions, including the performance of the NBER Dating Committee.

The evaluation of these models is critical due to the economic consequences of classification errors. Misclassifying default risk may lead financial institutions to extend credit that later becomes non-performing, while overly conservative models may restrict credit to solvent borrowers, limiting investment and growth. More generally, different types of classification errors have asymmetric economic implications. A missed event (e.g., failing to identify a profitable investment opportunity) may imply forgone gains, whereas a false alarm (e.g., investing in an unprofitable project) may lead to direct financial losses. As a result, the relative importance of errors depends on the decision context, and a model that minimizes one type of error may not be optimal from an economic perspective.

This consideration naturally leads to the role of loss functions in economic evaluation (Elliott and Lieli 2013). Standard econometric models for binary outcomes, such

as logit or probit, are typically estimated by maximizing likelihood functions, without explicitly incorporating the loss function associated with the underlying decision problem. However, in many applications, the costs and benefits associated with Hits, False alarms, and other outcomes differ substantially. Moreover, forecasts are often intended for heterogeneous users, each with their own loss function, which makes it difficult to embed a single, universally relevant loss structure into model estimation or evaluation (Elliott and Lieli 2013). In this context, evaluation tools such as ROC curves are particularly useful, as they provide a framework that allows different loss functions to be implicitly considered through the trade-off between hit rates and false alarm rates.

The most common approach to evaluating binary classification models is the AUC (Fawcett 2006; Hanley and McNeil 1982; Hand and Anagnostopoulos 2013). Its widespread adoption has made it a standard benchmark across disciplines, including economics. However, as we have previously discussed, a substantial body of literature shows that AUC can provide misleadingly optimistic assessments when the event of interest is rare (Hand and Anagnostopoulos 2013; Fawcett 2006; Davis and Goadrich 2006; Ozenne et al. 2015; Sofaer et al. 2018). This limitation is particularly relevant in economics, where many events of interest, such as recessions or financial crises, are infrequent.

As emphasized by Lahiri and Yang (2013), in the context of rare events, evaluation should not be driven by metrics that reward correct classification of the frequent outcome, such as overall accuracy. Instead, greater attention should be given to the correct identification of the rare event, focusing on measures such as hit rates and false alarm rates. Despite this, such considerations are often overlooked in applied economic work (Lahiri and Yang 2023). At the same time, the increasing use of machine learning methods, often characterized by limited interpretability, has further increased the reliance on evaluation metrics. This reinforces the need for a careful and context-aware assessment of classification performance, both in methodological and economic terms.

References

- Abdou, H.A. and J. Pointon (2011). “Credit Scoring, Statistical Techniques and Evaluation Criteria: A Review of the Literature”. In: *Intelligent Systems in Accounting, Finance and Management* 18, pp. 59–88. DOI: [10.1002/isaf.325](https://doi.org/10.1002/isaf.325).
- Ahlqvist, Ola et al., eds. (2016). *Land Use and Land Cover Semantics: Principles, Best Practices, and Prospects*. 1st. CRC Press, p. 360. ISBN: 9781351228596. DOI: [10.1201/9781351228596](https://doi.org/10.1201/9781351228596).
- Aldasoro, I., C. Borio, and M. Drehmann (2018). “Early Warning Indicators of Banking Crises: Expanding the Family”. In: *Journal of Financial Transformation* 48, pp. 142–155.
- Aldwaik, Safaa Zakaria, Jeffrey A. Onsted, and Robert Gilmore Pontius (Mar. 2015). “Behavior-based aggregation of land categories for temporal change analysis”. en. In: *International Journal of Applied Earth Observation and Geoinformation* 35, pp. 229–238. ISSN: 1569-8432. DOI: [10.1016/j.jag.2014.09.007](https://doi.org/10.1016/j.jag.2014.09.007). URL: <https://www.sciencedirect.com/science/article/pii/S0303243414002013> (visited on 04/18/2023).
- Aldwaik, Safaa Zakaria and Robert Gilmore Pontius (2012). “Intensity Analysis to Unify Measurements of Size and Stationarity of Land Changes by Interval, Category, and Transition”. In: *Landscape and Urban Planning* 106.1, pp. 103–114. DOI: [10.1016/j.landurbplan.2012.02.010](https://doi.org/10.1016/j.landurbplan.2012.02.010).
- Angelsen, Arild (2009). *Realising REDD+: National Strategy and Policy Options*. Bogor, Indonesia: Center for International Forestry Research (CIFOR).
- Baker, Benjamin A., Timothy A. Warner, et al. (Mar. 2013). “Does spatial resolution matter? A multi-scale comparison of object-based and pixel-based methods for detecting change associated with gas well drilling operations”. In: *International Journal of Remote Sensing* 34.5, pp. 1633–1651. ISSN: 0143-1161. DOI: [10.1080/01431161.2012.724540](https://doi.org/10.1080/01431161.2012.724540). URL: <https://doi.org/10.1080/01431161.2012.724540> (visited on 10/21/2024).

- Baker, Stuart G and Paul F Pinsky (June 2001). “A Proposed Design and Analysis for Comparing Digital and Analog Mammography: Special Receiver Operating Characteristic Methods for Cancer Screening”. In: *Journal of the American Statistical Association* 96.454, pp. 421–428. ISSN: 0162-1459. DOI: [10.1198/016214501753168136](https://doi.org/10.1198/016214501753168136). URL: <https://doi.org/10.1198/016214501753168136> (visited on 08/15/2024).
- Baker, William L. (Mar. 1989). “A review of models of landscape change”. en. In: *Landscape Ecology* 2.2, pp. 111–133. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/BF00137155](http://link.springer.com/10.1007/BF00137155). URL: <http://link.springer.com/10.1007/BF00137155> (visited on 10/22/2024).
- Bakker, M. and A. Veldkamp (Nov. 2008). “modeling land change: the issue of use and cover in wide-scale applications”. In: *Journal of Land Use Science* 3.4, pp. 203–213. ISSN: 1747-423X. DOI: [10.1080/17474230802465181](https://doi.org/10.1080/17474230802465181). URL: <https://doi.org/10.1080/17474230802465181> (visited on 07/17/2024).
- Bakker, Martha and Antonie Veldkamp (Dec. 2012a). “Changing relationships between land use and environmental characteristics and their consequences for spatially explicit land-use change prediction”. In: *Journal of Land Use Science* 7.4, pp. 407–424. ISSN: 1747-423X. DOI: [10.1080/1747423X.2011.595833](https://doi.org/10.1080/1747423X.2011.595833). URL: <https://doi.org/10.1080/1747423X.2011.595833> (visited on 07/23/2024).
- (Dec. 2012b). “Changing relationships between land use and environmental characteristics and their consequences for spatially explicit land-use change prediction”. In: *Journal of Land Use Science* 7.4, pp. 407–424. ISSN: 1747-423X. DOI: [10.1080/1747423X.2011.595833](https://doi.org/10.1080/1747423X.2011.595833). URL: <https://doi.org/10.1080/1747423X.2011.595833> (visited on 07/23/2024).
- Baldwin, Michael E. and John S. Kain (Aug. 2006). “Sensitivity of Several Performance Measures to Displacement Error, Bias, and Event Frequency.” en. In: *Weather and Forecasting* 21.4. DOI: <https://doi.org/10.1175/WAF933.1>. URL: https://journals.ametsoc.org/view/journals/wefo/21/4/waf933_1.xml (visited on 08/01/2024).
- Batty, Michael (2007). *Cities and Complexity: Understanding Cities with Cellular Automata, Agent-Based Models, and Fractals*. Cambridge, MA: MIT Press.

- Berge, T. J. and Ò. Jordà (2011). “Evaluating the Classification of Economic Activity into Recessions and Expansions”. In: *American Economic Journal: Macroeconomics* 3.2, pp. 246–277.
- Berge, Travis J. and Oscar Jorda (2011). “Evaluating the Classification of Economic Activity into Recessions and Expansions”. In: *American Economic Journal: Macroeconomics* 3.2. Publisher: American Economic Association, pp. 246–277. ISSN: 1945-7707. URL: <https://www.jstor.org/stable/41237149> (visited on 10/23/2024).
- Bilintoh, Thomas Mumuni (Dec. 2023). “Methods to Analyze Land Transitions during Multiple Time Intervals”. en. PhD thesis. Clark University.
- Bilintoh, Thomas Mumuni, Robert Gilmore Pontius, and Zhen Liu (Aug. 2024). “Analyzing the Losses and Gains of a Land Category: Insights from the Total Operating Characteristic”. en. In: *Land* 13.8. Number: 8 Publisher: Multidisciplinary Digital Publishing Institute, p. 1177. ISSN: 2073-445X. DOI: [10.3390/land13081177](https://doi.org/10.3390/land13081177). URL: <https://www.mdpi.com/2073-445X/13/8/1177> (visited on 08/13/2024).
- Blecic, Ivan, Arnaldo Cecchini, and Giuseppe A. Trunfio (Mar. 2015). “How much past to see the future: a computational study in calibrating urban cellular automata”. In: *International Journal of Geographical Information Science* 29.3, pp. 349–374. ISSN: 1365-8816. DOI: [10.1080/13658816.2014.970190](https://doi.org/10.1080/13658816.2014.970190). URL: <https://doi.org/10.1080/13658816.2014.970190> (visited on 05/02/2023).
- Blockeel, Hendrik et al., eds. (2013). *Machine Learning and Knowledge Discovery in Databases*. en. 1st ed. Vol. 8190. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer Berlin Heidelberg. ISBN: 978-3-642-40993-6. DOI: [10.1007/978-3-642-40994-3](https://doi.org/10.1007/978-3-642-40994-3). URL: <http://link.springer.com/10.1007/978-3-642-40994-3> (visited on 08/14/2024).
- Borras Jr., Saturnino M. et al. (2011). “Towards a Better Understanding of Global Land Grabbing: An Editorial Introduction”. In: *The Journal of Peasant Studies* 38.2, pp. 209–216. DOI: [10.1080/03066150.2011.559005](https://doi.org/10.1080/03066150.2011.559005).
- Borrelli, P., D. A. Robinson, L. R. Fleischer, et al. (2017). “An Assessment of the Global Impact of 21st Century Land Use Change on Soil Erosion”. In: *Nature Communications* 8.1, p. 2013. DOI: [10.1038/s41467-017-02142-7](https://doi.org/10.1038/s41467-017-02142-7).

- Briassoulis, Helen (2000). “Analysis of Land Use Change: Theoretical and Modeling Approaches”. en. PhD thesis. Lesvos, Greece: University of the Aegean.
- Briggs, William and David Ruppert (2005). “Assessing the Skill of Yes/No Predictions”. In: *Biometrics* 61.3. Publisher: [Wiley, International Biometric Society], pp. 799–807. ISSN: 0006-341X. URL: <https://www.jstor.org/stable/3695609> (visited on 08/13/2024).
- Brown, Daniel G. et al. (Oct. 2013). “Opportunities to improve impact, integration, and evaluation of land change models”. In: *Current opinion in environmental sustainability* 5.5, pp. 452–457. DOI: [10.1016/j.cosust.2013.07.012](https://doi.org/10.1016/j.cosust.2013.07.012). URL: <https://doi.org/10.1016/j.cosust.2013.07.012>.
- Burnicki, Amy C. (May 2012). “Impact of error on landscape pattern analyses performed on land-cover change maps”. en. In: *Landscape Ecology* 27.5, pp. 713–729. ISSN: 1572-9761. DOI: [10.1007/s10980-012-9719-2](https://doi.org/10.1007/s10980-012-9719-2). URL: <https://doi.org/10.1007/s10980-012-9719-2> (visited on 10/21/2024).
- Busch, Jonah et al. (2012). “Structuring Economic Incentives to Reduce Emissions from Deforestation within Indonesia”. In: *Proceedings of the National Academy of Sciences* 109.4, pp. 1062–1067. DOI: [10.1073/pnas.1106612109](https://doi.org/10.1073/pnas.1106612109).
- Camacho Olmedo, María Teresa et al., eds. (2018). *Geomatic Approaches for Modeling Land Change Scenarios*. en. Lecture Notes in Geoinformation and Cartography. Cham: Springer International Publishing. ISBN: 978-3-319-60800-6. DOI: [10.1007/978-3-319-60801-3](https://doi.org/10.1007/978-3-319-60801-3). URL: <http://link.springer.com/10.1007/978-3-319-60801-3> (visited on 07/26/2024).
- Carrington, André M. et al. (Jan. 2023). “Deep ROC Analysis and AUC as Balanced Average Accuracy, for Improved Classifier Selection, Audit and Explanation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45.1. Conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence, pp. 329–341. ISSN: 1939-3539. DOI: [10.1109/TPAMI.2022.3145392](https://doi.org/10.1109/TPAMI.2022.3145392). URL: <https://ieeexplore.ieee.org/document/9693294> (visited on 07/19/2024).
- Castilla, Guillermo et al. (Jan. 2009). “The impact of thematic resolution on the patch-mosaic model of natural landscapes”. en. In: *Landscape Ecology* 24.1, pp. 15–23. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/s10980-008-9310-z](https://doi.org/10.1007/s10980-008-9310-z). URL: <http://link.springer.com/10.1007/s10980-008-9310-z> (visited on 04/10/2023).

- Chen, Hao and Robert Gilmore Pontius (Nov. 2010). “Diagnostic tools to evaluate a spatial land change projection along a gradient of an explanatory variable”. en. In: *Landscape Ecology* 25.9, pp. 1319–1331. ISSN: 1572-9761. DOI: [10.1007/s10980-010-9519-5](https://doi.org/10.1007/s10980-010-9519-5). URL: <https://doi.org/10.1007/s10980-010-9519-5> (visited on 07/23/2024).
- Chen, I-Ching, Jane K. Hill, et al. (2011). “Rapid range shifts of species associated with high levels of climate warming”. In: *Science* 333.6045, pp. 1024–1026. DOI: [10.1126/science.1206432](https://doi.org/10.1126/science.1206432).
- Chicco, Davide (Dec. 2017). “Ten quick tips for machine learning in computational biology”. In: *BioData Mining* 10.1, p. 35. ISSN: 1756-0381. DOI: [10.1186/s13040-017-0155-3](https://doi.org/10.1186/s13040-017-0155-3). URL: <https://doi.org/10.1186/s13040-017-0155-3> (visited on 08/15/2024).
- Chicco, Davide and Giuseppe Jurman (Jan. 2020). “The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation”. In: *BMC Genomics* 21, p. 6. ISSN: 1471-2164. DOI: [10.1186/s12864-019-6413-7](https://doi.org/10.1186/s12864-019-6413-7). URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6941312/> (visited on 07/19/2024).
- (Feb. 2023). “The Matthews correlation coefficient (MCC) should replace the ROC AUC as the standard metric for assessing binary classification”. In: *BioData Mining* 16, p. 4. DOI: [10.1186/s13040-023-00308-w](https://doi.org/10.1186/s13040-023-00308-w). URL: <https://biodatamining.biomedcentral.com/articles/10.1186/s13040-023-00308-w>.
- Chicco, Davide, Niklas Tötsch, and Giuseppe Jurman (Feb. 2021). “The Matthews correlation coefficient (MCC) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation”. In: *BioData Mining* 14, p. 13. DOI: [10.1186/s13040-021-00244-z](https://doi.org/10.1186/s13040-021-00244-z).
- Christen, Peter, David J. Hand, and Nishadi Kirielle (Oct. 2023). “A Review of the F-Measure: Its History, Properties, Criticism, and Alternatives”. In: *ACM Comput. Surv.* 56.3, 73:1–73:24. ISSN: 0360-0300. DOI: [10.1145/3606367](https://doi.org/10.1145/3606367). URL: <https://dl.acm.org/doi/10.1145/3606367> (visited on 11/18/2024).
- Clarke, Keith C. and Leonard J. Gaydos (1998). “Loose-coupling a Cellular Automaton Model and GIS: Long-term Urban Growth Prediction for San Francisco

- and Washington/Baltimore”. In: *International Journal of Geographical Information Science* 12.7, pp. 699–714. ISSN: 1365-8816. DOI: [10.1080/136588198241617](https://doi.org/10.1080/136588198241617).
- Clayton, H.H (1934). “Rating weather forecasts”. In: *Bulletin of the American Meteorological Society* 15, pp. 279–283.
- Congalton, Russell G. and Kass Green (2009). *Assessing the Accuracy of Remotely Sensed Data: Principles and Practices*. CRC Press.
- Costa, S., E. Diniz, and B. Araujo (2019). “Firm Heterogeneity and Export Market Entry: A Binary Response Approach”. In: *Economic modeling* 79, pp. 163–174.
- (2022). “Threshold Effects in Export Market Decisions: Evidence from a Panel Binary Model”. In: *Applied Economics* 54.12, pp. 1393–1408.
- Davis, Jesse and Mark Goadrich (2006). “The relationship between Precision-Recall and ROC curves”. en. In: *Proceedings of the 23rd international conference on Machine learning - ICML '06*. Pittsburgh, Pennsylvania: ACM Press, pp. 233–240. ISBN: 978-1-59593-383-6. DOI: [10.1145/1143844.1143874](https://doi.org/10.1145/1143844.1143874). URL: <http://portal.acm.org/citation.cfm?doid=1143844.1143874> (visited on 08/14/2024).
- Dodd, Lori E. and Margaret S. Pepe (Sept. 2003). “Partial AUC Estimation and Regression”. en. In: *Biometrics* 59.3, pp. 614–623. ISSN: 0006-341X, 1541-0420. DOI: [10.1111/1541-0420.00071](https://doi.org/10.1111/1541-0420.00071). URL: <https://academic.oup.com/biometrics/article/59/3/614-623/7286731> (visited on 02/01/2024).
- Donaldson, R.J, R.M Dyer, and M.J Kraus (1975). “An objective evaluator of techniques for predicting severe weather events.” In: *Preprints*. Springfield, VI, USA: National Technical Information Service, pp. 312–326.
- Doolittle, M.H (1888). “Association ratios”. In: *Bulletin of the Philosophical Society of Washington* 10, pp. 83–87, 94–96.
- Duke, Joshua M. and JunJie Wu, eds. (2014). *The Oxford handbook of land economics*. en. New York: Oxford University Press. ISBN: 978-0-19-976374-0.
- Duprey, T., B. Klaus, and T. A. Peltonen (2022). “Dating Financial Stress Episodes Using a Systemic Risk Indicator”. In: *Journal of Financial Stability* 58, p. 100952.
- Eastman, J. R. (2006). *Idrisi Andes: Guide to GIS and Image Processing*. USA: Clark University.
- Ebert, E. E and J. L McBride (Dec. 2000). “Verification of precipitation in weather systems: determination of systematic errors”. In: *Journal of Hydrology* 239.1, pp. 179–

202. ISSN: 0022-1694. DOI: [10.1016/S0022-1694\(00\)00343-7](https://doi.org/10.1016/S0022-1694(00)00343-7). URL: <https://www.sciencedirect.com/science/article/pii/S0022169400003437> (visited on 10/02/2024).
- Ebert, Philip A. and Peter Milne (Feb. 2022). “Methodological and conceptual challenges in rare and severe event forecast verification”. English. In: *Natural Hazards and Earth System Sciences* 22.2. Publisher: Copernicus GmbH, pp. 539–557. ISSN: 1561-8633. DOI: [10.5194/nhess-22-539-2022](https://doi.org/10.5194/nhess-22-539-2022). URL: <https://nhess.copernicus.org/articles/22/539/2022/> (visited on 09/10/2024).
- Elliott, Graham and Robert P. Lieli (May 2013). “Predicting binary outcomes”. en. In: *Journal of Econometrics* 174.1, pp. 15–26. ISSN: 03044076. DOI: [10.1016/j.jeconom.2013.01.003](https://doi.org/10.1016/j.jeconom.2013.01.003). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0304407613000171> (visited on 10/23/2024).
- Engel, Stefanie, Stefano Pagiola, and Sven Wunder (2008). “Designing Payments for Environmental Services in Theory and Practice: An Overview of the Issues”. In: *Ecological Economics* 65.4, pp. 663–674. DOI: [10.1016/j.ecolecon.2008.03.011](https://doi.org/10.1016/j.ecolecon.2008.03.011).
- Engelmann, B. (2011). “The Basel II Risk Parameters: Estimation, Validation, and Backtesting”. In: *Journal of Banking Regulation* 12.3, pp. 195–212.
- FAO and UNSD (2020). *System of Environmental-Economic Accounting for Agriculture, Forestry and Fisheries (SEEA AFF)*. en. 1st ed. Rome, Italy: FAO and United Nations Statistical Division. ISBN: 978-92-5-132207-9. DOI: [10.4060/ca7735en](https://doi.org/10.4060/ca7735en). URL: <http://www.fao.org/documents/card/en/c/ca7735en> (visited on 09/12/2024).
- Fawcett, Tom (June 2006). “An introduction to ROC analysis”. en. In: *Pattern Recognition Letters* 27.8, pp. 861–874. ISSN: 01678655. DOI: [10.1016/j.patrec.2005.10.010](https://doi.org/10.1016/j.patrec.2005.10.010). URL: <https://linkinghub.elsevier.com/retrieve/pii/S016786550500303X> (visited on 01/08/2024).
- Finley, J.P (1884). “Tornado predictions”. In: *American Meteorological Journal* 1, pp. 85–88.
- Flach, Peter (2015). *Machine Learning: The Art and Science of Algorithms that Make Sense of Data*. Cambridge University Press.
- Foley, Jonathan A. et al. (July 2005). “Global Consequences of Land Use”. en. In: *Science* 309.5734, pp. 570–574. ISSN: 0036-8075, 1095-9203. DOI: [10.1126/science](https://doi.org/10.1126/science).

1111772. URL: <https://www.science.org/doi/10.1126/science.1111772> (visited on 01/28/2023).
- Foody, Giles M. (2002a). “Status of Land Cover Classification Accuracy Assessment”. In: *Remote Sensing of Environment* 80.1, pp. 185–201. DOI: [10.1016/S0034-4257\(01\)00295-4](https://doi.org/10.1016/S0034-4257(01)00295-4).
- (Apr. 2002b). “Status of land cover classification accuracy assessment”. en. In: *Remote Sensing of Environment* 80.1, pp. 185–201. ISSN: 00344257. DOI: [10.1016/S0034-4257\(01\)00295-4](https://doi.org/10.1016/S0034-4257(01)00295-4). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0034425701002954> (visited on 09/09/2024).
- (Oct. 2010). “Assessing the accuracy of land cover change with imperfect ground reference data”. In: *Remote Sensing of Environment* 114.10, pp. 2271–2285. ISSN: 0034-4257. DOI: [10.1016/j.rse.2010.05.003](https://doi.org/10.1016/j.rse.2010.05.003). URL: <https://www.sciencedirect.com/science/article/pii/S0034425710001434> (visited on 07/19/2024).
- (Aug. 2013). “Ground reference data error and the mis-estimation of the area of land cover change as a function of its abundance”. In: *Remote Sensing Letters* 4.8, pp. 783–792. ISSN: 2150-704X. DOI: [10.1080/2150704X.2013.798708](https://doi.org/10.1080/2150704X.2013.798708). URL: <https://doi.org/10.1080/2150704X.2013.798708> (visited on 07/19/2024).
- (Oct. 2023). “Challenges in the real world use of classification accuracy metrics: From recall and precision to the Matthews correlation coefficient”. en. In: *PLOS ONE* 18.10. Publisher: Public Library of Science, e0291908. ISSN: 1932-6203. DOI: [10.1371/journal.pone.0291908](https://doi.org/10.1371/journal.pone.0291908). URL: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0291908> (visited on 07/19/2024).
- (Feb. 2024). “Ground Truth in Classification Accuracy Assessment: Myth and Reality”. In: *Geomatics* 4.1, pp. 81–90. ISSN: 2673-7418. DOI: [10.3390/geomatics4010005](https://doi.org/10.3390/geomatics4010005). URL: <https://www.mdpi.com/2673-7418/4/1/5> (visited on 02/23/2024).
- Gao, Ting et al. (Oct. 2018). “Predicting pathological response to neoadjuvant chemotherapy in breast cancer patients based on imbalanced clinical data”. en. In: *Personal and Ubiquitous Computing* 22.5, pp. 1039–1047. ISSN: 1617-4917. DOI: [10.1007/s00779-018-1144-3](https://doi.org/10.1007/s00779-018-1144-3). URL: <https://doi.org/10.1007/s00779-018-1144-3> (visited on 06/07/2024).

- García-Álvarez, David (2018). “Aproximación al estudio de la incertidumbre en la modelización del cambio de usos y coberturas del suelo”. PhD thesis. Universidad de Granada.
- García-Álvarez, David et al. (Nov. 2019). “Thematic resolution influence in spatial analysis. An application to Land Use Cover Change (LUCC) modeling calibration”. en. In: *Computers, Environment and Urban Systems* 78, p. 101375. ISSN: 01989715. DOI: [10.1016/j.compenvurbsys.2019.101375](https://doi.org/10.1016/j.compenvurbsys.2019.101375). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0198971519302315> (visited on 07/28/2023).
- Garon, Edward B. et al. (2015). “Pembrolizumab for the Treatment of Non-Small-Cell Lung Cancer”. In: *New England Journal of Medicine* 372.21, pp. 2018–2028. DOI: [10.1056/NEJMoa1501824](https://doi.org/10.1056/NEJMoa1501824). URL: <https://www.nejm.org/doi/full/10.1056/NEJMoa1501824> (visited on 06/06/2024).
- Geist, Helmut J. and Eric F. Lambin (2002). “Proximate Causes and Underlying Driving Forces of Tropical Deforestation”. In: *BioScience* 52.2, pp. 143–150. DOI: [10.1641/0006-3568\(2002\)052\[0143:PCAUDF\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2002)052[0143:PCAUDF]2.0.CO;2).
- Gilbert, G.K. (1884). “Finley’s Tornado Predictions”. In: *American Meteorological Journal* 1.5, pp. 166–172. URL: https://archive.org/details/sim_american-meteorological-journal_1884-09_1_5/page/168/mode/2up?view=theater.
- Global Land Programme (2024). *Global Land Programme*. <https://glp.earth/>. Accessed: 2024-10-23.
- Gobierno de Navarra (2025). *Sistema de Información Geográfica de Parcelas Agrícolas (SIGPAC) - Navarra*. <https://sigpac.navarra.es>. Accessed: 2025-01-14.
- Granger, C.W.J. and M.H. Pesaran (2000). “Economic and statistical measures of forecast accuracy”. In: *Journal of Forecasting* 19, pp. 537–560. DOI: [10.1002/1099-131X\(200012\)19:7<537::AID-FOR769>3.0.CO;2-G](https://doi.org/10.1002/1099-131X(200012)19:7<537::AID-FOR769>3.0.CO;2-G).
- Green, D.M. and John A. Swets (1966). *Signal detection theory and psychophysics*. New York: Wiley Press.
- Gutman, Garik et al., eds. (2004). *Land Change Science: Observing, Monitoring and Understanding Trajectories of Change on the Earth’s Surface*. en. 1st ed. 2004. Remote Sensing and Digital Image Processing 6. Dordrecht: Imprint: Springer. ISBN: 978-1-4020-2561-7. DOI: [10.1007/978-1-4020-2562-4](https://doi.org/10.1007/978-1-4020-2562-4).

- UN-Habitat (2020). *World Cities Report 2020: The Value of Sustainable Urbanization*. Nairobi: United Nations Human Settlements Programme.
- Hagen, Alex (Apr. 2003). “Fuzzy set approach to assessing similarity of categorical maps”. In: *International Journal of Geographical Information Science* 17.3, pp. 235–249. ISSN: 1365-8816. DOI: [10.1080/13658810210157822](https://doi.org/10.1080/13658810210157822). URL: <https://doi.org/10.1080/13658810210157822> (visited on 07/30/2024).
- Hagen-Zanker, Alex (Jan. 2009). “An improved Fuzzy Kappa statistic that accounts for spatial autocorrelation”. In: *International Journal of Geographical Information Science* 23.1, pp. 61–73. ISSN: 1365-8816. DOI: [10.1080/13658810802570317](https://doi.org/10.1080/13658810802570317). URL: <https://doi.org/10.1080/13658810802570317> (visited on 07/30/2024).
- Hall, C. A. S. et al. (July 1995a). “modeling Spatial and Temporal Patterns of Tropical Land Use Change”. en. In: *Journal of Biogeography* 22.4/5, p. 753. ISSN: 03050270. DOI: [10.2307/2845977](https://www.jstor.org/stable/2845977?origin=crossref). URL: <https://www.jstor.org/stable/2845977?origin=crossref> (visited on 10/22/2024).
- (1995b). “modeling Spatial and Temporal Patterns of Tropical Land Use Change”. In: *Journal of Biogeography* 22.4-5, pp. 753–757. ISSN: 0305-0270. DOI: [10.2307/2845977](https://www.jstor.org/stable/2845977).
- Hamill, Thomas M. and Josip Juras (2006). “Measuring forecast skill: is it real skill or is it the varying climatology?” en. In: *Quarterly Journal of the Royal Meteorological Society* 132.621C, pp. 2905–2923. ISSN: 1477-870X. DOI: [10.1256/qj.06.25](https://onlinelibrary.wiley.com/doi/abs/10.1256/qj.06.25). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1256/qj.06.25> (visited on 08/16/2024).
- Hanberry, B. B. and H. S. He (May 2013). “Prevalence, statistical thresholds, and accuracy assessment for species distribution models”. en. In: *Web Ecology* 13.1, pp. 13–19. ISSN: 1399-1183. DOI: [10.5194/we-13-13-2013](https://we.copernicus.org/articles/13/13/2013/). URL: <https://we.copernicus.org/articles/13/13/2013/> (visited on 06/07/2024).
- Hand, D. J. and C. Anagnostopoulos (Apr. 2013). “When is the area under the receiver operating characteristic curve an appropriate measure of classifier performance?” In: *Pattern Recognition Letters* 34.5, pp. 492–495. DOI: [10.1016/j.patrec.2012.12.004](https://doi.org/10.1016/j.patrec.2012.12.004). URL: <https://doi.org/10.1016/j.patrec.2012.12.004>.
- Hand, David J. (Oct. 2009). “Measuring classifier performance: a coherent alternative to the area under the ROC curve”. en. In: *Machine Learning* 77.1, pp. 103–123.

- ISSN: 1573-0565. DOI: [10.1007/s10994-009-5119-5](https://doi.org/10.1007/s10994-009-5119-5). URL: <https://doi.org/10.1007/s10994-009-5119-5> (visited on 02/18/2026).
- Hand, David J. (2012a). “Assessing the Performance of Classification Methods”. en. In: *International Statistical Review* 80.3, pp. 400–414. ISSN: 1751-5823. DOI: [10.1111/j.1751-5823.2012.00183.x](https://doi.org/10.1111/j.1751-5823.2012.00183.x). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1751-5823.2012.00183.x> (visited on 09/04/2024).
- (2012b). “Assessing the Performance of Classification Methods”. en. In: *International Statistical Review* 80.3, pp. 400–414. ISSN: 1751-5823. DOI: [10.1111/j.1751-5823.2012.00183.x](https://doi.org/10.1111/j.1751-5823.2012.00183.x). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1751-5823.2012.00183.x> (visited on 09/04/2024).
- Hand, David J., Peter Christen, and Nishadi Kirielle (Mar. 2021). “F*: an interpretable transformation of the F-measure”. en. In: *Machine Learning* 110.3, pp. 451–456. ISSN: 1573-0565. DOI: [10.1007/s10994-021-05964-1](https://doi.org/10.1007/s10994-021-05964-1). URL: <https://doi.org/10.1007/s10994-021-05964-1> (visited on 09/06/2024).
- Hanley, J. A. and B. J. McNeil (Apr. 1982). “The meaning and use of the area under a receiver operating characteristic (ROC) curve.” en. In: *Radiology* 143.1, pp. 29–36. ISSN: 0033-8419, 1527-1315. DOI: [10.1148/radiology.143.1.7063747](https://doi.org/10.1148/radiology.143.1.7063747). URL: <http://pubs.rsna.org/doi/10.1148/radiology.143.1.7063747> (visited on 06/06/2024).
- He, Haibo and Edwardo A. Garcia (2009). “Learning from Imbalanced Data”. In: *IEEE Transactions on Knowledge and Data Engineering* 21.9, pp. 1263–1284. DOI: [10.1109/TKDE.2008.239](https://doi.org/10.1109/TKDE.2008.239).
- Heidke, P (1926). “Berechnung des erfolges und der güte der windstärkvorhersagen im sturmwarnungsdienst”. In: *Geografika Annaler* 8, pp. 301–349.
- Hoffman, Michael M (2020). “The MCC-F1 curve: a performance evaluation technique for binary classification”. en. In: *arXiv*. DOI: <https://doi.org/10.48550/arXiv.2006.11278>.
- Hogan, Robin J. et al. (Apr. 2010). “Equitability Revisited: Why the “Equitable Threat Score” Is Not Equitable”. en. In: *Weather and Forecasting*. Section: Weather and Forecasting. DOI: [10.1175/2009WAF2222350.1](https://doi.org/10.1175/2009WAF2222350.1). URL: https://journals.ametsoc.org/view/journals/wefo/25/2/2009waf2222350_1.xml (visited on 08/14/2024).

- Hossin, M. and M.N. Sulaiman (Mar. 2015). “A Review on Evaluation Metrics for Data Classification Evaluations”. en. In: *International Journal of Data Mining & Knowledge Management Process* 5.2, pp. 01–11. ISSN: 2231007X, 22309608. DOI: [10.5121/ijdkp.2015.5201](https://doi.org/10.5121/ijdkp.2015.5201). URL: <http://www.aircconline.com/ijdkp/V5N2/5215ijdkp01.pdf> (visited on 08/14/2024).
- Houghton, Richard A. (2005). “Tropical Deforestation as a Source of Greenhouse Gas Emissions”. In: *Tropical Deforestation and Climate Change*, pp. 13–21.
- Jaccard, Paul (1908). “Nouvelles recherches sur la distribution florale”. fr. In: *Bulletin de la Société Vaudoise des Sciences Naturelles* 44, pp. 223–270. DOI: [10.5169/SEALS-268384](https://doi.org/10.5169/SEALS-268384). URL: <https://www.e-periodica.ch/digbib/view?pid=bsv-002:1908:44::485> (visited on 09/06/2024).
- Lahiri, K. and C. Yang (2023). “ROC and PRC Approaches to Evaluate Recession Forecasts”. In: *Journal of Business Cycle Research* 19.2, pp. 119–148. DOI: [10.1007/s41549-023-00082-4](https://doi.org/10.1007/s41549-023-00082-4).
- Lahiri, Kajal and J. George Wang (Jan. 2013). “Evaluating probability forecasts for GDP declines using alternative methodologies”. en. In: *International Journal of Forecasting* 29.1, pp. 175–190. ISSN: 01692070. DOI: [10.1016/j.ijforecast.2012.07.004](https://doi.org/10.1016/j.ijforecast.2012.07.004). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0169207012001094> (visited on 10/23/2024).
- Lahiri, Kajal and Liu Yang (Jan. 2013). “Chapter 19 - Forecasting Binary Outcomes”. In: *Handbook of Economic Forecasting*. Ed. by Graham Elliott and Allan Timmermann. Vol. 2. Handbook of Economic Forecasting. Elsevier, pp. 1025–1106. DOI: [10.1016/B978-0-444-62731-5.00019-1](https://doi.org/10.1016/B978-0-444-62731-5.00019-1). URL: <https://www.sciencedirect.com/science/article/pii/B9780444627315000191> (visited on 10/23/2024).
- Lambin, E F et al. (2000). *A core project of the International Geosphere-Biosphere Programme and the International Human Dimensions Programme on Global Environmental Change*. en. Tech. rep. 48. Stockholm: International Geosphere-Biosphere Programme, p. 125.
- Lambin, Eric F. (2006). *Land-Use and Land-Cover Change: Local Processes and Global Impacts*. eng. Global Change - the IGBP Ser. Berlin, Heidelberg: Springer Berlin / Heidelberg. ISBN: 978-3-540-32201-6.

- Liu, J., J. Liu, et al. (2024). “Enhancing credit risk prediction based on ensemble tree-based feature transformation and logistic regression”. In: *Journal of Forecasting* 43.2, pp. 429–455. DOI: [10.1002/for.3040](https://doi.org/10.1002/for.3040).
- Liu, Jianguo, Thomas Dietz, et al. (2007). “Complexity of Coupled Human and Natural Systems”. In: *Science* 317.5844, pp. 1513–1516. DOI: [10.1126/science.1144004](https://doi.org/10.1126/science.1144004).
- Lobo, Jorge M., Alberto Jiménez-Valverde, and Raimundo Real (Mar. 2008). “AUC: a misleading measure of the performance of predictive distribution models”. en. In: *Global Ecology and Biogeography* 17.2, pp. 145–151. ISSN: 1466-822X, 1466-8238. DOI: [10.1111/j.1466-8238.2007.00358.x](https://doi.org/10.1111/j.1466-8238.2007.00358.x). URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1466-8238.2007.00358.x> (visited on 04/02/2024).
- Longley, Paul A. et al. (2005). *Geographic Information Systems and Science*. 2nd ed. Chichester: Wiley.
- Ma, Hua et al. (2013). “On use of partial area under the ROC curve for evaluation of diagnostic performance”. In: *Statistics in Medicine* 32.6. Received 19 June 2012, Accepted 11 2 2013, Published online 18 March 2013, pp. 1102–1117. DOI: [10.1002/sim.5777](https://doi.org/10.1002/sim.5777). URL: <https://wileyonlinelibrary.com/doi/abs/10.1002/sim.5777>.
- Malik, Rizwan et al. (June 2016). “Diagnostic Accuracy of Optical Coherence Tomography and Scanning Laser Tomography for Identifying Glaucoma in Myopic Eyes”. English. In: *Ophthalmology* 123.6. Publisher: Elsevier, pp. 1181–1189. ISSN: 0161-6420, 1549-4713. DOI: [10.1016/j.ophttha.2016.01.052](https://doi.org/10.1016/j.ophttha.2016.01.052). URL: [https://www.aaojournal.org/article/S0161-6420\(16\)00158-5/abstract](https://www.aaojournal.org/article/S0161-6420(16)00158-5/abstract) (visited on 08/15/2024).
- Manzato, Agostino (Dec. 2005). “An Odds Ratio Parameterization for ROC Diagram and Skill Score Indices”. en. In: *Weather and Forecasting* 20.6. Section: Weather and Forecasting, pp. 918–930. DOI: [10.1175/WAF899.1](https://doi.org/10.1175/WAF899.1). URL: https://journals.ametsoc.org/view/journals/wefo/20/6/waf899_1.xml (visited on 09/10/2024).
- (Oct. 2007). “A Note On the Maximum Peirce Skill Score”. en. In: *Weather and Forecasting* 22.5. Section: Weather and Forecasting, pp. 1148–1154. DOI: [10.1175/WAF1041.1](https://doi.org/10.1175/WAF1041.1). URL: https://journals.ametsoc.org/view/journals/wefo/22/5/waf1041_1.xml (visited on 09/10/2024).

- Marsh, G. P. (1965). *Man and Nature; Or, the Earth as Modified by Human Action*. Cambridge, MA: Belknap Press of Harvard University Press.
- Mas, Jean-François et al. (Sept. 2013). “A Suite of Tools for ROC Analysis of Spatial Models”. en. In: *ISPRS International Journal of Geo-Information* 2.3. Number: 3 Publisher: Multidisciplinary Digital Publishing Institute, pp. 869–887. ISSN: 2220-9964. DOI: [10.3390/ijgi2030869](https://doi.org/10.3390/ijgi2030869). URL: <https://www.mdpi.com/2220-9964/2/3/869> (visited on 07/24/2024).
- Mason, I. B. (1982). “A Model for Assessment of Weather Forecasts”. In: *Australian Meteorological Magazine* 30, pp. 291–303.
- Matthews, B. W. (Oct. 1975). “Comparison of the predicted and observed secondary structure of T4 phage lysozyme”. In: *Biochimica et Biophysica Acta (BBA) - Protein Structure* 405.2, pp. 442–451. ISSN: 0005-2795. DOI: [10.1016/0005-2795\(75\)90109-9](https://doi.org/10.1016/0005-2795(75)90109-9). URL: <https://www.sciencedirect.com/science/article/pii/0005279575901099> (visited on 08/15/2024).
- Maxwell, Aaron E., Timothy A. Warner, and Fang Fang (2018). “Implementation of Machine-Learning Classification in Remote Sensing: An Applied Review”. In: *International Journal of Remote Sensing* 39.9, pp. 2784–2817. DOI: [10.1080/01431161.2018.1433343](https://doi.org/10.1080/01431161.2018.1433343).
- Mbizvo, Gashirai K. and Andrew J. Larner (Jan. 2024). “On the Dependence of the Critical Success Index (CSI) on Prevalence”. en. In: *Diagnostics* 14.5. Number: 5 Publisher: Multidisciplinary Digital Publishing Institute, p. 545. ISSN: 2075-4418. DOI: [10.3390/diagnostics14050545](https://doi.org/10.3390/diagnostics14050545). URL: <https://www.mdpi.com/2075-4418/14/5/545> (visited on 08/01/2024).
- McClish, Donna Katzman (Aug. 1989). “Analyzing a Portion of the ROC Curve”. en. In: *Medical Decision Making* 9.3, pp. 190–195. ISSN: 0272-989X, 1552-681X. DOI: [10.1177/0272989X8900900307](https://doi.org/10.1177/0272989X8900900307). URL: <http://journals.sagepub.com/doi/10.1177/0272989X8900900307> (visited on 08/14/2024).
- Meyfroidt, Patrick et al. (Oct. 2013). “Globalization of land use: distant drivers of land change and geographic displacement of land use”. en. In: *Current Opinion in Environmental Sustainability* 5.5, pp. 438–444. ISSN: 18773435. DOI: [10.1016/j.cosust.2013.04.003](https://doi.org/10.1016/j.cosust.2013.04.003). URL: <https://linkinghub.elsevier.com/retrieve/pii/S1877343513000353> (visited on 01/28/2023).

- Movahedi, Faezeh, Rema Padman, and James F Antaki (Apr. 2023). “Limitations of receiver operating characteristic curve on imbalanced data: Assist device mortality risk scores”. In: *The Journal of thoracic and cardiovascular surgery* 165.4, 1433–1442.e2. ISSN: 0022-5223. DOI: [10.1016/j.jtcvs.2021.07.041](https://doi.org/10.1016/j.jtcvs.2021.07.041). URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8800945/> (visited on 06/06/2024).
- Murphy, Allan H. (Mar. 1996). “The Finley Affair: A Signal Event in the History of Forecast Verification”. en. In: *Weather and Forecasting* 11.1. Section: Weather and Forecasting. ISSN: 1520-0434. URL: https://journals.ametsoc.org/view/journals/wefo/11/1/1520-0434_1996_011_0003_tfaase_2_0_co_2.xml (visited on 09/05/2024).
- Mushtaq, Fatima et al. (Aug. 2024). “Land Cover and Land Use Ontology—Evolution of International Standards, Challenges, and Opportunities”. en. In: *Land* 13.8, p. 1202. ISSN: 2073-445X. DOI: [10.3390/land13081202](https://doi.org/10.3390/land13081202). URL: <https://www.mdpi.com/2073-445X/13/8/1202> (visited on 10/25/2024).
- Nahm, Francis Sahngun (Jan. 2022). “Receiver operating characteristic curve: overview and practical use for clinicians”. English. In: *Korean Journal of Anesthesiology* 75.1. Publisher: Korean Society of Anesthesiologists, pp. 25–36. ISSN: 2005-6419, 2005-7563. DOI: [10.4097/kja.21209](https://doi.org/10.4097/kja.21209). URL: <http://ekja.org/journal/view.php?doi=10.4097/kja.21209> (visited on 08/15/2024).
- Nelson, Arthur C. and Casey J. Dawkins (2004). *Urban Containment in the United States: History, Models, and Techniques for Regional and Metropolitan Growth Management*. Chicago: American Planning Association Press.
- Noszczyk, Tomasz (Aug. 2019). “A review of approaches to land use changes modeling”. In: *Human and Ecological Risk Assessment: An International Journal* 25.6, pp. 1377–1405. ISSN: 1080-7039. DOI: [10.1080/10807039.2018.1468994](https://doi.org/10.1080/10807039.2018.1468994). URL: <https://doi.org/10.1080/10807039.2018.1468994> (visited on 03/03/2023).
- Olofsson, Pontus, Giles M. Foody, Martin Herold, et al. (May 2014). “Good practices for estimating area and assessing accuracy of land change”. en. In: *Remote Sensing of Environment* 148, pp. 42–57. ISSN: 0034-4257. DOI: [10.1016/j.rse.2014.02.015](https://doi.org/10.1016/j.rse.2014.02.015). URL: <https://www.sciencedirect.com/science/article/pii/S0034425714000704> (visited on 04/19/2023).

- Olofsson, Pontus, Giles M. Foody, Stephen V. Stehman, et al. (Feb. 2013). “Making better use of accuracy data in land change studies: Estimating accuracy and area and quantifying uncertainty using stratified estimation”. en. In: *Remote Sensing of Environment* 129, pp. 122–131. ISSN: 0034-4257. DOI: [10.1016/j.rse.2012.10.031](https://doi.org/10.1016/j.rse.2012.10.031). URL: <https://www.sciencedirect.com/science/article/pii/S0034425712004191> (visited on 05/10/2023).
- Overmars, Koen P., Peter H. Verburg, and Tom (A.) Veldkamp (2007). “Comparison of a deductive and an inductive approach to specify land suitability in a spatially explicit land use model”. In: *Land Use Policy* 24.3. Integrated Assessment of the Land System: The Future of Land Use, pp. 584–599. ISSN: 0264-8377. DOI: <https://doi.org/10.1016/j.landusepol.2005.09.008>. URL: <https://www.sciencedirect.com/science/article/pii/S0264837706000688>.
- Ozenne, Brice, Fabien Subtil, and Delphine Maucourt-Boulch (2015). “The precision–recall curve overcame the optimism of the receiver operating characteristic curve in rare diseases”. In: *Journal of Clinical Epidemiology* 68.8, pp. 855–859. DOI: [10.1016/j.jclinepi.2015.02.010](https://doi.org/10.1016/j.jclinepi.2015.02.010). URL: <https://doi.org/10.1016/j.jclinepi.2015.02.010>.
- Paegelow, M. (2018). “Impact and Integration of Multiple Training Dates for Markov Based Land Change Modeling”. en. In: *Geomatic Approaches for Modeling Land Change Scenarios*. Ed. by María Teresa Camacho Olmedo et al. Cham: Springer International Publishing, pp. 121–138. ISBN: 978-3-319-60801-3. DOI: [10.1007/978-3-319-60801-3_7](https://doi.org/10.1007/978-3-319-60801-3_7). URL: https://doi.org/10.1007/978-3-319-60801-3_7 (visited on 07/23/2024).
- Palmer, W.C and R.A Allen (1949). “Note on the Accuracy of Forecasts Concerning the Rain Problem”. In: *U.S. Weather Bureau manuscript*.
- Paroissien, E., J. Peña, and E. Tello (2021). “Business Discontinuance and Informality: A Binary Event Study”. In: *Small Business Economics* 57, pp. 233–254.
- Peterson, A. Townsend, Monica Papeş, and Jorge Soberón (Apr. 2008). “Rethinking receiver operating characteristic analysis applications in ecological niche modeling”. In: *Ecological modeling* 213.1, pp. 63–72. ISSN: 0304-3800. DOI: [10.1016/j.ecolmodel.2007.11.008](https://doi.org/10.1016/j.ecolmodel.2007.11.008). URL: <https://www.sciencedirect.com/science/article/pii/S0304380007006163> (visited on 08/15/2024).

- Phillips, Steven J., Robert P. Anderson, and Robert E. Schapire (Jan. 2006). “Maximum entropy modeling of species geographic distributions”. In: *Ecological modeling* 190.3, pp. 231–259. ISSN: 0304-3800. DOI: [10.1016/j.ecolmodel.2005.03.026](https://doi.org/10.1016/j.ecolmodel.2005.03.026). URL: <https://www.sciencedirect.com/science/article/pii/S030438000500267X> (visited on 06/06/2024).
- Pierce, C.S (1884). “The numerical measure of the success of predictions”. In: *Science* 4, pp. 453–454.
- Pontius, Robert G. Jr., Diane Huffaker, and Kevin Denman (2004). “Useful Techniques of Validation for Spatially Explicit Land-Change Models”. In: *Ecological modeling* 179.4, pp. 445–461. DOI: [10.1016/j.ecolmodel.2004.05.010](https://doi.org/10.1016/j.ecolmodel.2004.05.010).
- Pontius, Robert Gilmore (1994). “Modeling Tropical Land Use Change and Assessing Policies to Reduce Carbon Dioxide Release from Africa”. Ph.D. Dissertation. Syracuse, New York: State University of New York, College of Environmental Science and Forestry.
- (2022). *Metrics That Make a Difference: How to Analyze Change and Error*. en. Advances in Geographic Information Science. Cham: Springer International Publishing. ISBN: 978-3-030-70764-4. DOI: [10.1007/978-3-030-70765-1](https://doi.org/10.1007/978-3-030-70765-1). URL: <https://link.springer.com/10.1007/978-3-030-70765-1> (visited on 01/08/2024).
- Pontius, Robert Gilmore, Wideke Boersma, et al. (Mar. 2008). “Comparing the input, output, and validation maps for several models of land change”. en. In: *The Annals of Regional Science* 42.1, pp. 11–37. ISSN: 0570-1864, 1432-0592. DOI: [10.1007/s00168-007-0138-2](https://doi.org/10.1007/s00168-007-0138-2). URL: <http://link.springer.com/10.1007/s00168-007-0138-2> (visited on 06/04/2024).
- Pontius, Robert Gilmore, Jean-Christophe Castella, et al. (2018). “Lessons and Challenges in Land Change Modeling Derived from Synthesis of Cross-Case Comparisons”. en. In: *Trends in Spatial Analysis and modeling*. Ed. by Martin Behnisch and Gotthard Meinel. Vol. 19. Series Title: Geotechnologies and the Environment. Cham: Springer International Publishing, pp. 143–164. ISBN: 978-3-319-52520-4. DOI: [10.1007/978-3-319-52522-8_8](https://doi.org/10.1007/978-3-319-52522-8_8). URL: http://link.springer.com/10.1007/978-3-319-52522-8_8 (visited on 12/13/2022).
- Pontius, Robert Gilmore, Roopa Krithivasan, et al. (July 2017). “Methods to summarize change among land categories across time intervals”. en. In: *Journal of*

- Land Use Science* 12.4, pp. 218–230. ISSN: 1747-423X, 1747-4248. DOI: [10.1080/1747423X.2017.1338768](https://doi.org/10.1080/1747423X.2017.1338768). URL: <https://www.tandfonline.com/doi/full/10.1080/1747423X.2017.1338768> (visited on 01/08/2024).
- Pontius, Robert Gilmore and Christopher D Lippitt (Jan. 2006). “Can Error Explain Map Differences Over Time?” en. In: *Cartography and Geographic Information Science* 33.2, pp. 159–171. ISSN: 1523-0406, 1545-0465. DOI: [10.1559/152304006777681706](https://doi.org/10.1559/152304006777681706). URL: <http://www.tandfonline.com/doi/abs/10.1559/152304006777681706> (visited on 09/09/2024).
- Pontius, Robert Gilmore and Marco Millones (Aug. 2011). “Death to Kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment”. en. In: *International Journal of Remote Sensing* 32.15, pp. 4407–4429. ISSN: 0143-1161, 1366-5901. DOI: [10.1080/01431161.2011.552923](https://doi.org/10.1080/01431161.2011.552923). URL: <https://www.tandfonline.com/doi/full/10.1080/01431161.2011.552923> (visited on 01/08/2024).
- Pontius, Robert Gilmore and Benoit Parmentier (Mar. 2014). “Recommendations for using the relative operating characteristic (ROC)”. en. In: *Landscape Ecology* 29.3, pp. 367–382. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/s10980-013-9984-8](https://doi.org/10.1007/s10980-013-9984-8). URL: <http://link.springer.com/10.1007/s10980-013-9984-8> (visited on 01/08/2024).
- Pontius, Robert Gilmore, Smitha Peethambaram, and Jean-Christophe Castella (Jan. 2011). “Comparison of Three Maps at Multiple Resolutions: A Case Study of Land Change Simulation in Cho Don District, Vietnam”. en. In: *Annals of the Association of American Geographers* 101.1, pp. 45–62. ISSN: 0004-5608, 1467-8306. DOI: [10.1080/00045608.2010.517742](https://doi.org/10.1080/00045608.2010.517742). URL: <http://www.tandfonline.com/doi/abs/10.1080/00045608.2010.517742> (visited on 01/31/2023).
- Pontius, Robert Gilmore and Alí Santacruz (Nov. 2014). “Quantity, exchange, and shift components of difference in a square contingency table”. en. In: *International Journal of Remote Sensing* 35.21, pp. 7543–7554. ISSN: 0143-1161, 1366-5901. DOI: [10.1080/2150704X.2014.969814](https://doi.org/10.1080/2150704X.2014.969814). URL: <https://www.tandfonline.com/doi/full/10.1080/2150704X.2014.969814> (visited on 01/08/2024).
- Pontius, Robert Gilmore and Kangping Si (Mar. 2014). “The total operating characteristic to measure diagnostic ability for multiple thresholds”. In: *International*

- Journal of Geographical Information Science* 28.3, pp. 570–583. ISSN: 1365-8816. DOI: [10.1080/13658816.2013.862623](https://doi.org/10.1080/13658816.2013.862623). URL: <https://doi.org/10.1080/13658816.2013.862623> (visited on 10/09/2023).
- Pontius, Robert Gilmore and Joseph Spencer (Apr. 2005). “Uncertainty in Extrapolations of Predictive Land-Change Models”. en. In: *Environment and Planning B: Planning and Design* 32.2, pp. 211–230. ISSN: 0265-8135, 1472-3417. DOI: [10.1068/b31152](https://doi.org/10.1068/b31152). URL: <http://journals.sagepub.com/doi/10.1068/b31152> (visited on 09/12/2024).
- Powers, David M. W. (2011). “Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness and Correlation”. In: *Journal of Machine Learning Technologies* 2.1, pp. 37–63.
- Ren, Yanjiao et al. (2019). “Spatially Explicit Simulation of Land Use/Land Cover Changes: Current Coverage and Future Prospects”. In: *Earth-Science Reviews* 190, pp. 398–415. DOI: [10.1016/j.earscirev.2019.01.001](https://doi.org/10.1016/j.earscirev.2019.01.001).
- Rijsbergen, Cornelius J. Van (1979). *Information Retrieval*. London: Butterworth and Co.
- Rindfuss, Ronald R. et al. (2004). “Developing a Science of Land Change: Challenges and Methodological Issues”. In: *Proceedings of the National Academy of Sciences* 101.39, pp. 13976–13981. DOI: [10.1073/pnas.0401545101](https://doi.org/10.1073/pnas.0401545101).
- Rosa, Isabel M. D. et al. (Jan. 2015). “modeling land cover change in the Brazilian Amazon: temporal changes in drivers and calibration issues”. en. In: *Regional Environmental Change* 15.1, pp. 123–137. ISSN: 1436-378X. DOI: [10.1007/s10113-014-0614-z](https://doi.org/10.1007/s10113-014-0614-z). URL: <https://doi.org/10.1007/s10113-014-0614-z> (visited on 07/23/2024).
- Roy, Hari G., Dennis M. Fox, and Karine Emsellem (Sept. 2016). “Dormant categories and spatial resolution affect the perception of land cover change model performance”. en. In: *Cybergeo: European Journal of Geography*. Publisher: CNRS-UMR Géographie-cités 8504. ISSN: 1278-3366. DOI: [10.4000/cybergeo.27794](https://doi.org/10.4000/cybergeo.27794). URL: <https://journals.openedition.org/cybergeo/27794> (visited on 06/10/2024).
- Rudin, Cynthia (2019). “Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead”. In: *Nature Machine Intelligence* 1, pp. 206–215. DOI: [10.1038/s42256-019-0048-x](https://doi.org/10.1038/s42256-019-0048-x).

- Saito, Takaya and Marc Rehmsmeier (Mar. 2015). “The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets”. en. In: *PLOS ONE* 10.3. Ed. by Guy Brock, e0118432. ISSN: 1932-6203. DOI: [10.1371/journal.pone.0118432](https://doi.org/10.1371/journal.pone.0118432). URL: <https://dx.plos.org/10.1371/journal.pone.0118432> (visited on 08/14/2024).
- Schaefer, Joseph T. (1990). “The Critical Success Index as an Indicator of Warning Skill”. en. In: *Weather and Forecasting* 5.4. DOI: [https://doi.org/10.1175/1520-0434\(1990\)005<0570:TCSIAA>2.0.CO;2](https://doi.org/10.1175/1520-0434(1990)005<0570:TCSIAA>2.0.CO;2). URL: https://journals.ametsoc.org/view/journals/wefo/5/4/1520-0434_1990_005_0570_tcsiaa_2_0_co_2.xml (visited on 08/01/2024).
- Shao, Guofan, Lina Tang, and Jiangfu Liao (Nov. 2019). “Overselling overall map accuracy misinforms about research reliability”. en. In: *Landscape Ecology* 34.11, pp. 2487–2492. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/s10980-019-00916-6](https://doi.org/10.1007/s10980-019-00916-6). URL: <http://link.springer.com/10.1007/s10980-019-00916-6> (visited on 11/09/2024).
- Smith, Neil (1996). *The New Urban Frontier: Gentrification and the Revanchist City*. London: Routledge.
- Soares-Filho, Britaldo Silveira, Gustavo Coutinho Cerqueira, and Cássio Lopes Penachin (2002). “Dinamica—A Stochastic Cellular Automata Model Designed to Simulate the Landscape Dynamics in an Amazonian Colonization Frontier”. In: *Ecological modeling* 154.3, pp. 217–235. ISSN: 0304-3800. DOI: [10.1016/S0304-3800\(02\)00059-5](https://doi.org/10.1016/S0304-3800(02)00059-5).
- Sofaer, Helen R., Jennifer A. Hoeting, and Catherine S. Jarnevich (2018). “The area under the precision-recall curve as a performance metric for rare binary events”. In: *Methods in Ecology and Evolution* 9.4, pp. 678–685. DOI: [10.1111/2041-210X.13140](https://doi.org/10.1111/2041-210X.13140). URL: <https://doi.org/10.1111/2041-210X.13140>.
- Sohl, Terry L. and Peter R. Claggett (Nov. 2013). “Clarity versus complexity: Land-use modeling as a practical tool for decision-makers”. In: *Journal of Environmental Management* 129, pp. 235–243. ISSN: 0301-4797. DOI: [10.1016/j.jenvman.2013.07.027](https://doi.org/10.1016/j.jenvman.2013.07.027). URL: <https://www.sciencedirect.com/science/article/pii/S0301479713005069> (visited on 07/18/2024).

- Stehman, Stephen V. (Oct. 1997). “Selecting and interpreting measures of thematic classification accuracy”. In: *Remote Sensing of Environment* 62.1, pp. 77–89. ISSN: 0034-4257. DOI: [10.1016/S0034-4257\(97\)00083-7](https://doi.org/10.1016/S0034-4257(97)00083-7).
- Stehman, Stephen V. and Giles M. Foody (Sept. 2019). “Key issues in rigorous accuracy assessment of land cover products”. In: *Remote Sensing of Environment* 231, p. 111199. ISSN: 0034-4257. DOI: [10.1016/j.rse.2019.05.018](https://doi.org/10.1016/j.rse.2019.05.018). URL: <https://www.sciencedirect.com/science/article/pii/S0034425719302111> (visited on 09/12/2024).
- Strahler, Alan H et al. (2006). “Global land cover validation: Recommendations for evaluation and accuracy assessment of global land cover maps”. In: *European Communities, Luxembourg* 51.4, pp. 1–60.
- Sweetser, Frank L. (Mar. 1965). “Factorial ecology”. en. In: *Demography* 2.1, pp. 372–385. ISSN: 0070-3370, 1533-7790. DOI: [10.2307/2060125](https://doi.org/10.2307/2060125). URL: <https://read.dukeupress.edu/demography/article/2/1/372/172597/Factorial-ecology-Helsinki-1960> (visited on 10/24/2024).
- Tang, Xiaojing et al. (Apr. 2021). “Spatiotemporal assessment of land use/land cover change and associated carbon emissions and uptake in the Mekong River Basin”. In: *Remote Sensing of Environment* 256, p. 112336. ISSN: 0034-4257. DOI: [10.1016/j.rse.2021.112336](https://doi.org/10.1016/j.rse.2021.112336). URL: <https://www.sciencedirect.com/science/article/pii/S0034425721000547> (visited on 10/21/2024).
- Tharwat, Alaa (Jan. 2021). “Classification assessment methods”. en. In: *Applied Computing and Informatics* 17.1, pp. 168–192. ISSN: 2634-1964, 2210-8327. DOI: [10.1016/j.aci.2018.08.003](https://doi.org/10.1016/j.aci.2018.08.003). URL: <https://www.emerald.com/insight/content/doi/10.1016/j.aci.2018.08.003/full/html> (visited on 08/13/2024).
- Thüinen, J. H. von (1966). *Isolated State*. Trans. by C. M. Wartenberg. Originally published in 1826. Oxford: Pergamon Press.
- Tong, Xiaohua and Yongjiu Feng (May 2020). “A review of assessment methods for cellular automata models of land-use change and urban growth”. In: *International Journal of Geographical Information Science* 34.5, pp. 866–898. ISSN: 1365-8816. DOI: [10.1080/13658816.2019.1684499](https://doi.org/10.1080/13658816.2019.1684499). URL: <https://doi.org/10.1080/13658816.2019.1684499> (visited on 07/29/2024).

- Turner, B. L., Eric F. Lambin, and Anette Reenberg (Dec. 2007). “The emergence of land change science for global environmental change and sustainability”. en. In: *Proceedings of the National Academy of Sciences* 104.52, pp. 20666–20671. ISSN: 0027-8424, 1091-6490. DOI: [10.1073/pnas.0704119104](https://doi.org/10.1073/pnas.0704119104). URL: <https://pnas.org/doi/full/10.1073/pnas.0704119104> (visited on 01/28/2023).
- UNFCCC (2014). *Warsaw Framework for REDD+*. <https://unfccc.int/topics/land-use/workstreams/reddplus>.
- Varga, Orsolya Gyöngyi et al. (June 2019). “Intensity Analysis and the Figure of Merit’s components for assessment of a Cellular Automata – Markov simulation model”. en. In: *Ecological Indicators* 101, pp. 933–942. ISSN: 1470160X. DOI: [10.1016/j.ecolind.2019.01.057](https://doi.org/10.1016/j.ecolind.2019.01.057). URL: <https://linkinghub.elsevier.com/retrieve/pii/S1470160X19300743> (visited on 12/13/2022).
- Veldkamp, A and P. H Verburg (Aug. 2004). “modeling land use change and environmental impact”. In: *Journal of Environmental Management*. modeling land use change and environmental impact 72.1, pp. 1–3. ISSN: 0301-4797. DOI: [10.1016/j.jenvman.2004.04.004](https://doi.org/10.1016/j.jenvman.2004.04.004). URL: <https://www.sciencedirect.com/science/article/pii/S0301479704000684> (visited on 03/19/2026).
- Verburg, P. H., P. P. Schot, et al. (2004). “Land use change modeling: Current practice and research priorities”. In: *GeoJournal* 61.4, pp. 309–324. DOI: [10.1007/s10708-004-4946-y](https://doi.org/10.1007/s10708-004-4946-y). URL: <https://doi.org/10.1007/s10708-004-4946-y>.
- Verburg, Peter H, Peter Alexander, et al. (June 2019). “Beyond land cover change: towards a new generation of land use models”. en. In: *Current Opinion in Environmental Sustainability*. Sustainability governance and transformation 38, pp. 77–85. ISSN: 1877-3435. DOI: [10.1016/j.cosust.2019.05.002](https://doi.org/10.1016/j.cosust.2019.05.002). URL: <https://www.sciencedirect.com/science/article/pii/S1877343518301362> (visited on 05/02/2023).
- Verburg, Peter H., Kasper Kok, et al. (2006). “Modeling Land-Use and Land-Cover Change”. en. In: *Land-Use and Land-Cover Change*. Ed. by Eric F. Lambin and Helmut Geist. Series Title: Global Change - The IGBP Series. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 117–135. ISBN: 978-3-540-32201-6. DOI: [10.1007/3-540-32202-7_5](https://doi.org/10.1007/3-540-32202-7_5). URL: http://link.springer.com/10.1007/3-540-32202-7_5 (visited on 04/28/2022).

- Verburg, Peter H., Katharina Neumann, and Louisa Nol (2011). “Challenges in Using Land Use and Land Cover Data for Global Change Studies”. In: *Global Change Biology* 17.2, pp. 974–989. DOI: [10.1111/j.1365-2486.2010.02307.x](https://doi.org/10.1111/j.1365-2486.2010.02307.x).
- Verburg, Peter H., Welmoed Soepboer, et al. (2002). “Modeling the Spatial Dynamics of Regional Land Use: The CLUE-S Model”. In: *Environmental Management* 30.3, pp. 391–405. ISSN: 0364-152X. DOI: [10.1007/s00267-002-2630-x](https://doi.org/10.1007/s00267-002-2630-x).
- Verburg, Peter H., Andrzej Tabeau, and Erez Hatna (2013). “Assessing spatial uncertainties of land allocation using a scenario approach and sensitivity analysis: a study for land use in Europe”. In: *Journal of Environmental Management* 127. Epub 2012 Sep 29. PMID: 23026356, S132–S144. DOI: [10.1016/j.jenvman.2012.08.038](https://doi.org/10.1016/j.jenvman.2012.08.038). URL: <https://doi.org/10.1016/j.jenvman.2012.08.038>.
- Verstegen, Judith A. et al. (Jan. 2016). “Detecting systemic change in a land use system by Bayesian data assimilation”. In: *Environmental modeling & Software* 75, pp. 424–438. ISSN: 1364-8152. DOI: [10.1016/j.envsoft.2015.02.013](https://doi.org/10.1016/j.envsoft.2015.02.013). URL: <https://www.sciencedirect.com/science/article/pii/S1364815215000626> (visited on 07/23/2024).
- Viana, Cláudia M., Robert Gilmore Pontius, and Jorge Rocha (Aug. 2023). “Four Fundamental Questions to Evaluate Land Change Models with an Illustration of a Cellular Automata–Markov Model”. en. In: *Annals of the American Association of Geographers*, pp. 1–15. ISSN: 2469-4452, 2469-4460. DOI: [10.1080/24694452.2023.2232435](https://doi.org/10.1080/24694452.2023.2232435). URL: <https://www.tandfonline.com/doi/full/10.1080/24694452.2023.2232435> (visited on 09/07/2023).
- Vliet, Jasper van (2016). “Calibration and validation of land use models”. PhD thesis. Wageningen University.
- Vliet, Jasper van et al. (Aug. 2016). “A review of current calibration and validation practices in land-change modeling”. In: *Environmental modeling & Software* 82, pp. 174–182. ISSN: 1364-8152. DOI: [10.1016/j.envsoft.2016.04.017](https://doi.org/10.1016/j.envsoft.2016.04.017). URL: <https://www.sciencedirect.com/science/article/pii/S1364815216301128> (visited on 07/23/2024).
- Whittle, Martin et al. (Sept. 2012). “Detection of tropical deforestation using ALOS-PALSAR: A Sumatran case study”. In: *Remote Sensing of Environment* 124, pp. 83–98. ISSN: 0034-4257. DOI: [10.1016/j.rse.2012.04.027](https://doi.org/10.1016/j.rse.2012.04.027). URL: <https://www.sciencedirect.com/science/article/pii/S0034425712000838>.

- [sciencedirect.com/science/article/pii/S0034425712002039](https://www.sciencedirect.com/science/article/pii/S0034425712002039) (visited on 07/19/2024).
- Wilks, Daniel S. (2006). *Statistical methods in the atmospheric sciences*. en. 2nd ed. International geophysics series volume 91. Amsterdam Paris: Elsevier. ISBN: 978-0-12-751966-1.
- Woodcock, Frank (Oct. 1976). “The Evaluation of Yes/No Forecasts for Scientific and Administrative Purposes”. en. In: *Monthly Weather Review* 104.40. Section: Monthly Weather Review, pp. 1209–1214. ISSN: 1520-0493. URL: https://journals.ametsoc.org/view/journals/mwre/104/10/1520-0493_1976_104_1209_teoyff_2_0_co_2.xml (visited on 09/11/2024).
- Wunder, Sven (2005). “Payments for Environmental Services: Some Nuts and Bolts”. In: *CIFOR Occasional Paper* 42.
- Youden, W. J. (1950). “Index for rating diagnostic tests”. en. In: *Cancer* 3.1, pp. 32–35. ISSN: 1097-0142. DOI: [10.1002/1097-0142\(1950\)3:1<32::AID-CNCR2820030106>3.0.CO;2-3](https://doi.org/10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.0.CO;2-3). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/1097-0142%281950%293%3A1%3C32%3A%3AAID-CNCR2820030106%3E3.0.CO%3B2-3> (visited on 09/10/2024).
- Zhu, Qiuming (2020). “On the performance of Matthews correlation coefficient (MCC) for imbalanced dataset”. In: *Pattern Recognition Letters*. Available online: <https://doi.org/10.1016/j.patrec.2020.03.030>. DOI: [10.1016/j.patrec.2020.03.030](https://doi.org/10.1016/j.patrec.2020.03.030).

Chapter 2

Introducing the CSI Framework as a Novel Methodology for Binary Classification Evaluation with Rare Events

2.1 Introduction

This chapter applies concepts from classification evaluation literature (Chapter 1) to propose novel alternative metrics that are suitable not only for LCM but also for various other research fields. The goal of this chapter is to contribute fundamentally to the understanding and development of classification Agreement and Skill metrics, with a particular focus on their applicability to rare events.

As reviewed in Chapter 1, classification evaluation involves assessing an index variable’s ability to accurately classify the presence or absence of a binary variable. Widely used tools in soft classification evaluation include the Receiver Operating Characteristic (ROC) curves and the Total Operating Characteristic (TOC) curves, along with their Area Under the Curve (AUC) metrics (Fawcett 2006; Green and Swets 1966; Pontius and Si 2014). These methodologies evaluate the capacity of various thresholds of the index variable to correctly classify the presence of a binary variable. Such applications have been extensively employed across a diverse range of fields, including health (Kessler, Berglund, et al. 2005), disease detection (Garon et al. 2015), psychology (Kessler, Andrews, et al. 2002), machine learning (Provost and Fawcett 1997), species distribution (Phillips et al. 2006), and land change science (Pontius and Si 2014).

Despite its popularity, the AUC has several limitations widely acknowledged in the literature. The ROC curve can present an overly optimistic performance when applied to imbalanced data (Lobo et al. 2008; Movahedi et al. 2023). The AUC tends to evaluate thresholds that may not be practically meaningful or realistic in clinical or practical settings, particularly when the prevalence is low (Parodi et al. 2022). This issue arises because AUC considers all possible thresholds across the entire range of the index variable, including those that may not have practical relevance or utility. Consequently, the AUC may overestimate the discriminatory ability of an index variable by including thresholds unlikely to be used in real-world scenarios, leading to an inflated perception of the variable’s diagnostic performance.

In the context of classification evaluation, Skill refers to the ability of a model to perform better than random chance or a simple baseline, measuring the model’s effective classification ability. Many research fields fail to distinguish between Agreement

and Skill metrics, leading to misinterpretations of model performance. Researchers may erroneously report Agreement results as definitive indicators of their classification models' effectiveness, resulting in overly optimistic conclusions about their applicability and utility. In this context, AUC, which measures Agreement, is frequently interpreted as a metric of Skill. To enhance the validation of classification evaluations, it is essential for researchers to interpret these metrics correctly, ensuring they select the appropriate evaluation tools based on the specific characteristics of their datasets and research objectives.

Critical Success Index (CSI), also known as Threat score, Gilbert score, Jaccard Index, or Figure of Merit, is a single threshold metric that focuses on the Agreement of the presence of the binary variable (Gilbert 1884). It is a widely used metric for applications with imbalanced data, such as meteorology (Baldwin and Kain 2006; Hamill and Juras 2006; Schaefer 1990; Wang 2014), disease detection (Mbizvo et al. 2023), land change (Pontius, Peethambaram, et al. 2011; Varga et al. 2019), or economics (Sitthiyot and Holasut 2022). CSI is designed as a single threshold metric, but it can be applied to multiple thresholds. Similar to AUC, CSI has also been misused as a metric of Skill.

In this chapter, we revisit a range of soft and hard classification evaluation metrics, intending to develop a unified framework better suited to rare events. The proposed framework contributes to the evaluation of binary classifications in several ways. First, it introduces a new set of soft validation metrics that ignore the influence of the majority class. Second, it provides a common basis for soft and hard validation metrics by defining them in relation to contingency table entries in a consistent way, which improves their comparability. Third, it offers a unified graphical representation that facilitates the interpretation of classification results by making prevalence, as well as Quantity and Allocation disagreement, more explicit. To achieve this, the chapter reconstructs the framework step by step, examines its properties critically, and compares it with the predominant ROC/AUC paradigm.

For soft metrics of Agreement, we introduce the area under the CSI curve (AUCSI) as a metric that addresses some of the main limitations of AUC. We also introduce AUCSI Skill as a measure of Skill of the rare class at multiple thresholds. Both metrics provide an alternative for evaluating the classification ability of multiple thresholds,

giving greater weight to the most relevant thresholds. We demonstrate that AUCSI and AUCSI Skill are less dependent on prevalence than ROC-based metrics, which is particularly beneficial for applications with low prevalence.

For hard metrics, we analyze the CSI as an Agreement metric and the GSS as its corresponding Skill metric. We compare and contextualize GSS with respect to other popular Skill metrics. We argue that the GSS is conceptually flawed due to the baseline it considers. We draw on the principles of other similar metrics to propose a revised Skill metric associated with CSI.

After presenting and justifying the design and properties of the CSI framework, we develop a set of simulations to evaluate its behavior and compare it with the ROC/AUC paradigm. The goal is to examine the practical implications of adopting this framework.

2.2 Methods

2.2.1 Construction of TOC and CSI plots

The only input for constructing ROC and CSI plots is the contingency tables at multiple thresholds. Each threshold of the index variable generates a contingency table. Contingency tables are created by classifying as presence of all the observations with values of the index variable higher than the threshold. Values of the index variable under the threshold are classified as absent. This results in a threshold-dependent 2x2 contingency table shown in Table 2.1, with the following entries: correctly classified presences are Hits (H), observed absences classified as presences are False Alarms (F), observed presences classified as absences are Misses (M), and correctly classified absences are Correct Rejections (C). Prevalence (Pr) refers to the proportion of the population that contains the presence of the binary variable or $(H + M)/(H + M + F + C)$.

The ROC curve is a commonly utilized method in binary classifications. ROC is constructed by plotting $H/(H + M)$ or Sensitivity on the y-axis and $F/(F + C)$ or $1 - \text{Specificity}$ on the x-axis (Fig.1.4). Meanwhile, TOC curves contain two additional bits of information compared to ROC. They show the sizes of observed presence (P) and absence (Q), from which we deduce the population size and prevalence (Pontius and Si 2014). For this reason, we rely on TOC curves in our analysis rather than the

TABLE 2.1: Contingency table for binary classification with formal notation, where outcomes are defined conditional on a classification threshold.

		Reference		
		Presence	Absence	Total classified
Classification	Presence	H_t	F_t	$H_t + F_t$
	Absence	M_t	C_t	$M_t + C_t$
	Total Reference	$H_t + M_t = P$	$F_t + C_t = Q$	$P + Q = E$

Symbol	Meaning
t	Denotation for a quantity of classified presence
H_t	Hits: number of observations that are observed presence and have an index value greater than or equal to threshold t
M_t	Misses: number of observations that are observed presence and have an index value less than threshold t
F_t	False Alarms: number of observations that are observed absence and have an index value greater than or equal to threshold t
C_t	Correct Rejections: number of observations that are observed absence and have an index value less than threshold t
P	Number of observations that are observed presence
Q	Number of observations that are observed absence
E	Population size or Extent

^a Source: Adapted from Pontius Jr, R. G., & Si, K. (2014).

more conventional ROC representation. Throughout the chapter, references to the properties and limitations of TOC curves also apply to ROC curves, as both share the same AUC.

The TOC curve represents $H + F$ on the x-axis and H on the y-axis. The x-axis of the TOC curve is equivalent to the quantity of classified presence. In Chapter 1, we defined Quantity disagreement as the net difference between the amount of observed and classified presence. Allocation disagreement is the difference between the observed and classified presence due to the less-than-optimal assignment (Pontius, Peethambaram, et al. 2011). Thus, TOC allows the differentiation between Quantity and Allocation disagreement, something ROC can't do. This differentiation is

crucial to properly interpret the performance of classifiers. Both TOC and ROC are constructed by representing the contingency tables of successive thresholds. The interpretation of the shape of the TOC curve is equivalent to that of the ROC curve.

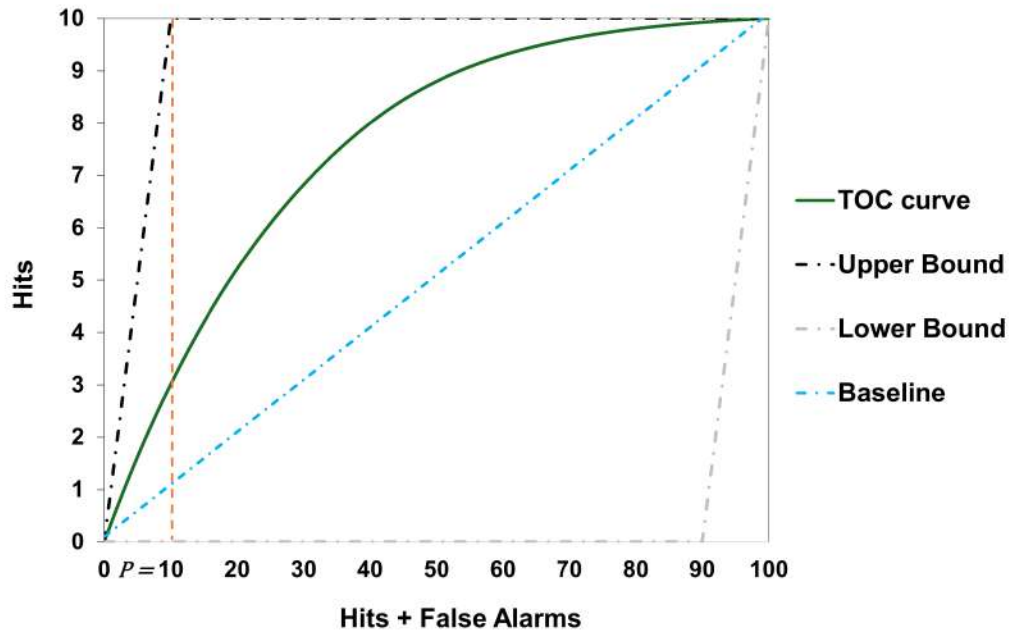
Figure 2.1a shows the TOC plot. The green curve shows the TOC curve of a given classifier. The height of the green curve represents the Hits that the classifier generates at a given threshold. The x-axis shows $H + F$, i.e., the quantity of classified presence. The slope of TOC curves shows the intensity at which Hits are classified in each threshold. The upper bound represents the best possible classifier, which can only have Quantity disagreement but no Allocation disagreement. The lower bound of the TOC plot recreates the worst possible classifier, which is the one with the maximum possible Allocation disagreement. The baseline represents a classifier that generates Hits at a uniform rate equal to prevalence. This classifier is considered to have no classification ability because it does not distinguish the presence and absence of the binary variable between observations (Pontius and Parmentier 2014). Pontius and Si (2014) provides further description and interpretation of the TOC plot.

CSI (Eq.1.7) is a metric that measures the Agreement between the observed and classified presence of a given threshold (Gilbert 1884; Schaefer 1990; Pontius, Peethambaram, et al. 2011). Popular single-threshold binary classification metrics, such as F1 (Eq.1.9) or the Matthews Correlation Coefficient (Eq.1.10), rely on more complex combinations of the entries of the contingency table, which limits their interpretability. CSI has a more straightforward interpretation because it represents proportions derived from the contingency table.

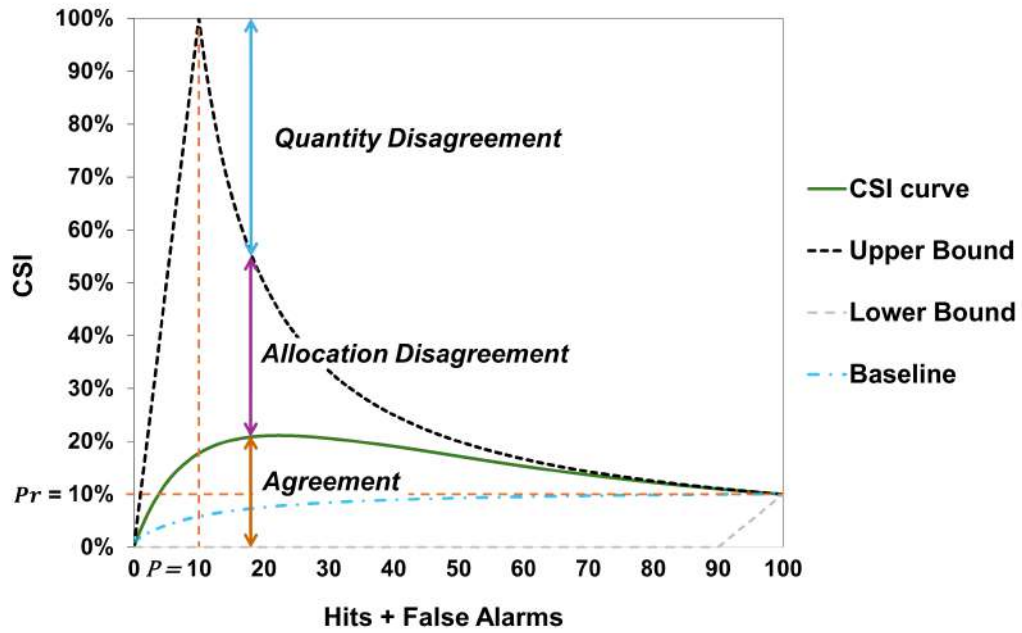
It is calculated as the intersection between the classified and the observed presence as a proportion of their union. The intersection between observed and classified presence is H . The classified presence is $H + F$, while the observed presence is $H + M$. Thus, CSI represents the proportion of classified and observed presence that is correctly classified. Equation (1.7) shows the mathematical definition of CSI. CSI ranges from 1, which means a perfect classification, to 0, which means no Agreement between the classified and the observed presence.

$$\text{CSI} = \frac{H}{H + F + M}$$

CSI is designed to be applied to individual contingency tables. However, following



(a)



(b)

FIGURE 2.1: Representation of the TOC (a) and CSI (b) plots. *Source: Self elaboration.*

the same logic that builds the ROC and TOC curves, we can apply Equation (1.7) to multiple thresholds of the index variable (Pontius 2018; Wang 2014). This way, we can visualize the CSI associated with each of the contingency tables at every threshold. To separate Quantity and Allocation disagreement, we can represent the CSI as a function of $H + F$, which is the same configuration used in TOC. From now on, we will differentiate the CSI value, which is the value that CSI takes at a given threshold, and the CSI curve, which is the set of CSI of multiple thresholds of an index variable.

Figure 2.1b shows the CSI plot of the same classification of Panel (b). The green curve shows the CSI curve of the same classifier used in Figure 2.1a. The height of the curve represents the CSI value of a given threshold, which is the height of the TOC divided by $H + F + M$. The vertical axis of the plot ranges from 0 to 100%, which is the range of values that the CSI can have. The horizontal axis represents $H + F$, which is the same axis as the TOC plot. The representation of $H + F$ or the classified quantity of presence in the horizontal axis allows the visualization of Quantity and Allocation disagreement. The plot's horizontal axis ranges from 0 to E because that is the maximum quantity of presence that can be classified. The classifiers used to build the upper and lower bounds and the baseline of TOC are the same as in CSI, which means the three lines derive from the same contingency tables.

Equation (2.1) and Equation (2.2) represent the upper and lower bounds of CSI as a function of Hits + False Alarms (x), respectively:

$$\text{Upper bound} = \begin{cases} \frac{x}{P} & \text{if } x \leq P; \text{ where } H(x) = x \text{ and } F(x) = 0 \\ \frac{P}{x} & \text{if } x > P; \text{ where } H(x) = P \text{ and } F(x) = x - P \end{cases} \quad (2.1)$$

$$\text{Lower bound} = \begin{cases} 0 & \text{if } x \leq Q; \text{ where } H(x) = 0 \text{ and } F(x) = x \\ \frac{x - Q}{E} & \text{if } x > Q; \text{ where } H(x) = x - Q \text{ and } F(x) = Q \end{cases} \quad (2.2)$$

The upper bound has a different shape to the left and the right of the observed quantity of presence (P). To the left of P , F is 0 because the observed quantity of presence is not yet classified. Here, every increase in classified quantity of presence

means an increase in H and a decrease of the same amount in M . Meanwhile, the upper bound to the right of P changes its shape because it generates F without any additional H . The upper bound peaks at $(P, 1)$. This point is the only possible combination of $H + F$ that generates a CSI of 1 because here Quantity disagreement is 0. As for the lower bound, in the interval $[0, Q]$ no H are generated, and only F increases, resulting in CSI values of 0%. To the right of Q , all the absences are classified as presence, which means that the classifier can only classify H . This creates the triangle at the bottom right corner of the graph.

While the representation of the baseline in TOC is always the diagonal line, in CSI, the baseline is a strictly concave curve that approaches the point (E, Pr) . Figure 2.1b shows how all the CSI curves must converge to this point because here all the population is classified as presence and the CSI value is equal to prevalence. Thus, any CSI curve, including the baseline, has a maximum CSI value of at least Pr .

The CSI plot allows the visualization of the Quantity and Allocation disagreement composition of CSI values. As shown in Figure 2.1b, the vertical distance between the upper boundary of CSI and 1 represents Quantity disagreement; the vertical distance between the green curve and the upper boundary of CSI represents Allocation disagreement, and the distance between the horizontal axis and the curve represents the Agreement measured by CSI. The decomposition of these three components takes the following mathematical form:

$$H + F + M = H + |M - F| + 2 \cdot \min(M, F) = \text{Agreement} + QD + AD;$$

$$\frac{H + F + M}{H + F + M} = \frac{H + |M - F| + 2 \cdot \min(M, F)}{H + F + M};$$

$$1 = \frac{H}{H + F + M} + \frac{|M - F|}{H + F + M} + \frac{2 \cdot \min(M, F)}{H + F + M}$$

Where QD and AD stand for Quantity and Allocation disagreement.

CSI curves have all the information necessary to reconstruct the contingency tables of each of the thresholds. The peak of the upper CSI bound is located at

$x = P$, from which we can know the presence of the population. All the lines converge to $CSI = Pr$ as x approaches E , from which we can know the size of the population and the prevalence. With these bits of information, we can calculate H , F , M , and C of any point of the CSI curves. The reconstruction of contingency tables is a critical issue because it informs about the population size and prevalence, and it allows practitioners to compute a wide variety of measures for each threshold (Pontius and Si 2014).

2.2.2 AUCSI and AUCSIS: new metrics to evaluate the Agreement and Skill of multiple thresholds

The area under the ROC curve (AUC) is a summary metric used to measure the degree of association between the index and a binary variable. TOC curves reveal the same AUC as ROC curves. Despite the popularity of its use, AUC has several well-known limitations (Lobo et al. 2008; Movahedi et al. 2023; Pontius and Si 2014). In this section, we focus on two closely related issues.

First, AUC evaluates thresholds that may not be meaningful in practical settings, particularly when prevalence is low. This issue arises because AUC integrates performance over all possible thresholds of the index variable, including those that would never be used in practice. Using the example of rainfall in a desert, a classifier that frequently predicts rain would be evaluated across thresholds where rain is almost always predicted. These thresholds contribute to the AUC because the few sunny days predicted are correct, even if the overall prediction is useless. As a result, AUC can overestimate the discriminatory ability of an index variable by assigning weight to unrealistic thresholds, leading to an inflated assessment of classification performance. From now on, realistic thresholds or classifications refer to those with small Quantity differences, and thus, are more likely to be used in practical settings.

Second, as a consequence of the former, AUC depends on prevalence in practical applications, even if this dependence is not explicit in its mathematical formulation. As shown by Lobo et al. (2008), applying the same classifier to the same phenomenon can yield substantially different AUC values if the size of the populations is modified

by adding absences. For example, consider a species whose distribution is concentrated in a specific area. Expanding the study region beyond this area reduces prevalence and can artificially increase the apparent classification performance reported by AUC, even though the classifier itself remains unchanged. Related to the first limitation, we would be artificially including more of the easy-to-predict sunny days. This makes direct comparisons across classifiers or applications difficult and potentially misleading.

We propose the area under the CSI curve (AUCSI) and the corresponding AUCSI Skill (AUCSIS) as alternative metrics to address these limitations of AUC. The key advancement of these metrics is that they do not account for Correct Rejections and therefore focus exclusively on the event of interest. As a result, they are not driven by the Agreement in the majority class and portray a better classification assessment when the event of interest is rare. In the upcoming sections, we will study the properties and characteristics of AUCSI to define in which applications it complements or outperforms AUC.

First, we define AUCSI and describe how it is computed. Analogous to AUC, AUCSI is defined as the proportion of the total feasible area of the CSI plot that is covered by a given CSI curve. The total feasible area corresponds to the region between the upper and lower bounds of CSI. Intuitively, AUCSI measures how much of the Allocation disagreement associated with the rare event is avoided by a classifier across multiple thresholds.

By construction, AUCSI ranges from 0 to 1. A value of 1 corresponds to the area covered by a perfect classifier, while a value of 0 corresponds to the worst possible classifier. This normalization ensures that AUCSI is comparable across populations with different sizes and prevalence. Moreover, expressing AUCSI on the same scale as AUC, and as a proportion of a total feasible area, facilitates direct comparison and interpretation between the two metrics.

Equation (2.3) defines the calculation of AUCSI. The denominator represents the total feasible area, which we refer to as the Total Area. The numerator corresponds to the area between the CSI curve and the lower bound.

$$\text{AUCSI} = \frac{\text{Area between the CSI curve and the lower CSI bound}}{\text{Area between the upper and lower CSI bound}} \quad (2.3)$$

Both AUC and ASCI measure Allocation Agreement, not Allocation Skill. As defined in Chapter 1, Agreement refers to the similarity of a classification with the observation; Skill refers to the difference in the Agreement of a classifier against a baseline, i.e., how much they outperform the baseline. Therefore, to measure Skill we need to discount the corresponding baselines to the metrics.

The same baseline classifier gets a different value of AUC and AUCSI. This is due to the difference in the definition of Agreement that these metrics have. In TOC plots, the baseline always covers 50% of the feasible area. In CSI plots, the AUCSI of the baseline depends positively on prevalence. In AUC, the Skill of a classifier is measured as how much it exceeds 0.5; in AUCSI, the baseline to beat changes with prevalence.

As a result, we need to normalize AUC and AUCSI as the differences from their baseline to compare how each metric ranks the Skill of a classifier. We will do so by calculating the proportion of the area between the upper bound and the baseline (Reference area) that a TOC or CSI curve covers. Equations (2.4) and (2.5) calculate the difference of the metrics from their baseline.

$$\text{AUCS} = \frac{\text{AUC} - 0.5}{1 - 0.5} = 2 \cdot \text{AUC} - 1 \quad (2.4)$$

$$\text{AUCSIS} = \frac{\text{AUCSI} - \text{AUCSI}^{\text{Baseline}}}{1 - \text{AUCSI}^{\text{Baseline}}} \quad (2.5)$$

Where $\text{AUCSI}^{\text{Baseline}}$ is the AUCSI of the baseline.

2.2.3 Distribution of Total and Reference areas in CSI and TOC plots

Once AUCSI and AUCSIS are defined, we can study their properties and evaluate whether they overcome any of the limitations of AUC. The two limitations of the AUC of the study, which are the evaluation of unrealistic thresholds and the dependence on prevalence, are closely related to the shape of TOC plots and to where they

concentrate their area. For example, TOC curves get a share of their area from the right region of the graph, which are those irrelevant fully rainy predictions in the desert.

Thresholds with a larger share of the Total area in TOC and CSI plots have a greater contribution to AUC and AUCSI, respectively. Similarly, thresholds with a larger share of the Reference area in TOC and CSI plots have a greater contribution to AUCS and AUCSIS. Therefore, we need to identify and compare critical regions of CSI and TOC plots to assess whether AUCSI and AUCSIS have desirable properties for applications with rare events.

To identify these critical regions, we will analyze how the areas are distributed along the horizontal axis in the TOC and CSI plots. We express the proportion of the Total and Reference area covered as a function of $H + F$. We are not evaluating specific CSI curves, but how the Total and Reference areas are distributed along the graph. The function informs about the proportion of the area accumulated at any $H + F$, that is, how much of the AUC or AUCSI is explained. Thus, the derivative of this function with respect to $H + F$ represents the contribution of those particular thresholds or classifications to the metrics.

Let $x = H + F$ denote the quantity of classified presence. Equations (2.6) and (2.7) express the proportion of the *Total Area* as a function of x for AUC and AUCSI, respectively. The denominator represents the Total Area, defined as the area between the upper and lower bounds, while the numerator corresponds to the area accumulated up to x .

$$\text{Covered AUC}(x) = \frac{\text{AUC}^{\text{Upper}}(x)}{\text{AUC}^{\text{Upper}}(E)} \quad (2.6)$$

$$\text{Covered AUCSI}(x) = \frac{\text{AUCSI}^{\text{Upper}}(x)}{\text{AUCSI}^{\text{Upper}}(E)} \quad (2.7)$$

where $\text{AUC}^{\text{Upper}}$ and $\text{AUCSI}^{\text{Upper}}$ denote the AUC and AUCSI of the upper bound, respectively.

Equations 2.8 and 2.9 define the proportion of the *Reference Area* as a function of x . The denominator corresponds to the difference between the upper bound and the baseline, while the numerator represents the accumulated Reference Area up to

x .

$$\frac{\text{AUC}^{\text{Upper}}(x) - \text{AUC}^{\text{Baseline}}(x)}{\text{AUC}^{\text{Upper}}(E) - \text{AUC}^{\text{Baseline}}(E)} = \frac{\text{AUC}^{\text{Upper}}(x) - \text{AUC}^{\text{Baseline}}(x)}{0.5} \quad (2.8)$$

$$\frac{\text{AUCSI}^{\text{Upper}}(x) - \text{AUCSI}^{\text{Baseline}}(x)}{\text{AUCSI}^{\text{Upper}}(E) - \text{AUCSI}^{\text{Baseline}}(E)} = \frac{\text{AUCSI}^{\text{Upper}}(x) - \text{AUCSI}^{\text{Baseline}}(x)}{1 - \text{AUCSI}^{\text{Baseline}}(E)} \quad (2.9)$$

where $\text{AUC}^{\text{Baseline}}$ and $\text{AUCSI}^{\text{Baseline}}$ denote the AUC and AUCSI of the baseline, respectively.

Equations (2.6), (2.7), (2.8) and (2.9) show that the proportion of Total and Reference area covered by TOC and CSI plots only depends on the proportion of the population classified as presence (x/E) and prevalence (see Appendix). Therefore, all the populations with the same prevalence have the same distribution of areas in their plots regardless of their size. However, we need to analyze how the properties of AUC and AUCSI change with prevalence. We will study these properties through the application of equations (2.6), (2.7), (2.8) and (2.9). The analysis will address the two mentioned limitations of AUC, focusing on three fundamental questions.

Question 1. *Does CSI concentrate more area around P than TOC?*

P is a critical threshold in the analysis of the metrics because it is the only point where Quantity disagreement is 0, and thus, where complete Agreement can be achieved. Quantities lower than P mean under-prediction, and vice versa. If CSI has a higher contribution to the areas around P , AUCSI and AUCSIS are a metric that favors Quantity Agreement more than AUC and AUCS, i.e., they get higher values when classified quantities of presence are closer to the observed ones. This is a desired property when the research question requires small deviations in classified Quantities from the observed ones, being particularly important in cases where the prevalence is low. Question (1) is illustrated in Figure 2.2a and b.

Question 2. *Is there an asymmetry between the contribution to the Total (and Reference) area in the right and left neighborhoods of P in any of the metrics?*

If Question (1) addresses Quantity Agreement, Question (2) searches for directions of the potential Quantity biases of the metrics. Imbalances in the contribution

to the areas between the right and left regions of P imply a Quantity disagreement bias of the metric. If the right region has more area than the left region, the metric favors curves skewed to the right, and thus, it can result in a systematic Quantity overestimation. Similarly, when the left region has more contribution to the Reference area than the right region, the metric is biased towards Quantity underestimation. If CSI has less asymmetry around P than TOC, CSI-derived metrics are more adequate than TOC-derived metrics for research questions requiring the classification of small Quantity overestimation or underestimation. Question (2) is illustrated in Figure 2.2c and d.

Question 3. *Do AUCSI (and AUCSIS) weigh the last thresholds of the index variable less than AUC (and AUCS)?*

This question addresses whether the metrics evaluate unrealistic thresholds. Many applications of these metrics are not interested in the far-right region of TOC curves, because these are thresholds where False Alarms are too high. If CSI-derived metrics weigh far regions of the graph less than TOC-derived metrics, then it is a more adequate metric for such applications. Question (3) is illustrated in Figure 2.2e and f.

2.3 Analyzing the properties of the CSI framework

2.3.1 Revisiting metrics of multiple thresholds

2.3.1.1 AUCSI as a metric of Agreement of the rare event at multiple thresholds

Figure 2.3 evaluates two populations with different prevalence. The figure plots the proportion of the Total area of CSI and TOC as a function of $H + F$ (Eq.2.6, 2.7). The horizontal axis is the quantity of classified presence of the rare event expressed as a proportion of the population, which is $H + F$ divided by E . The vertical axis is the proportion of the Total area covered, as a percentage. The dotted lines represent the proportion of the Total areas covered by CSI and TOC in a population with a 5% prevalence. The solid lines show the proportion of the areas covered in a

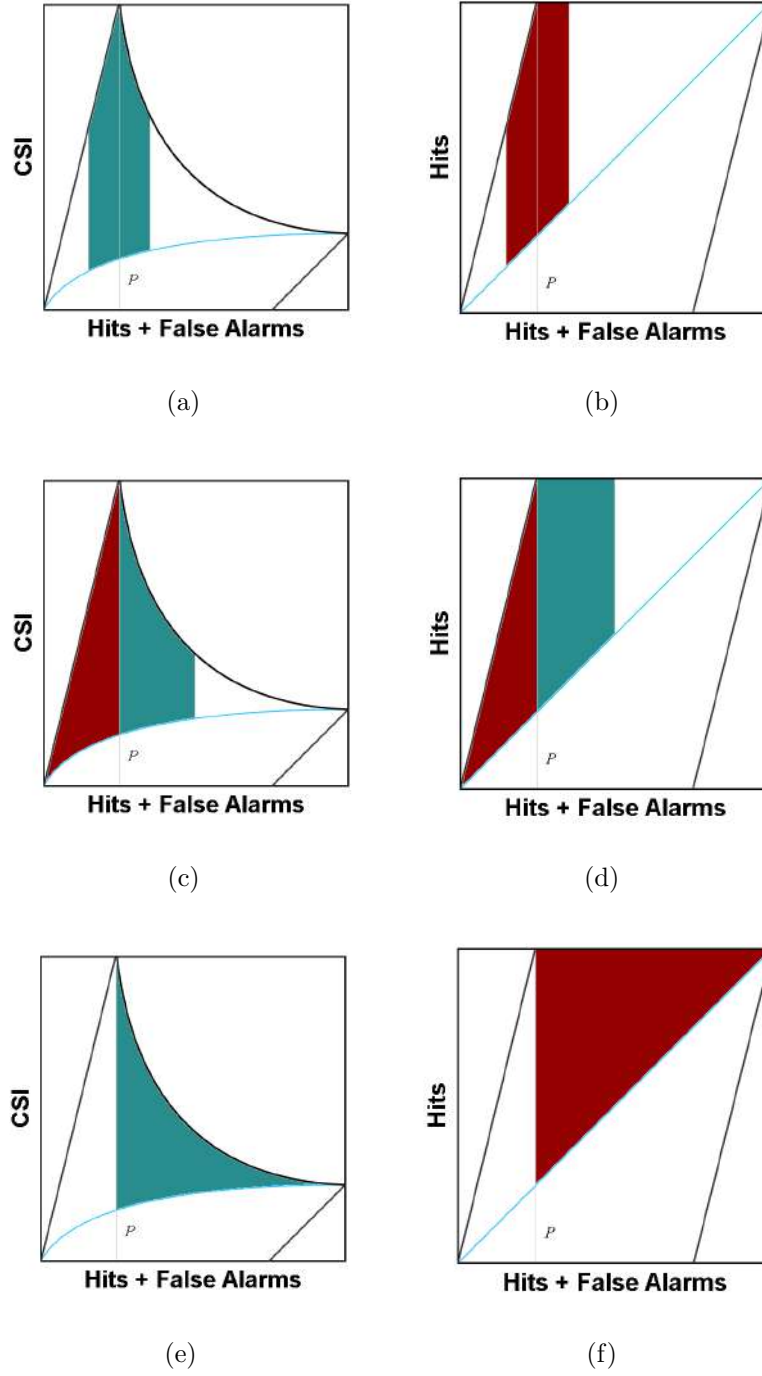


FIGURE 2.2: Representation of the areal comparison of the three analyses. The green and red areas are compared for each of the questions. The first and second columns correspond to the CSI and TOC graphs, respectively. Panels a) and b) address Question (1), panels c) and d) address Question (2), and panels e) and f) address Question (3). *Source: Self elaboration.*

population with a 25% of prevalence. The triangular markers on the lines indicate the proportion of the Total area that is covered at P . The diagonal line represents a uniform distribution of the Total area along the x -axis, i.e., a hypothetical plot where all the classified Quantities have the same contribution to the Total area. The circular markers indicate the points where the slope is equal to the slope of the diagonal line.

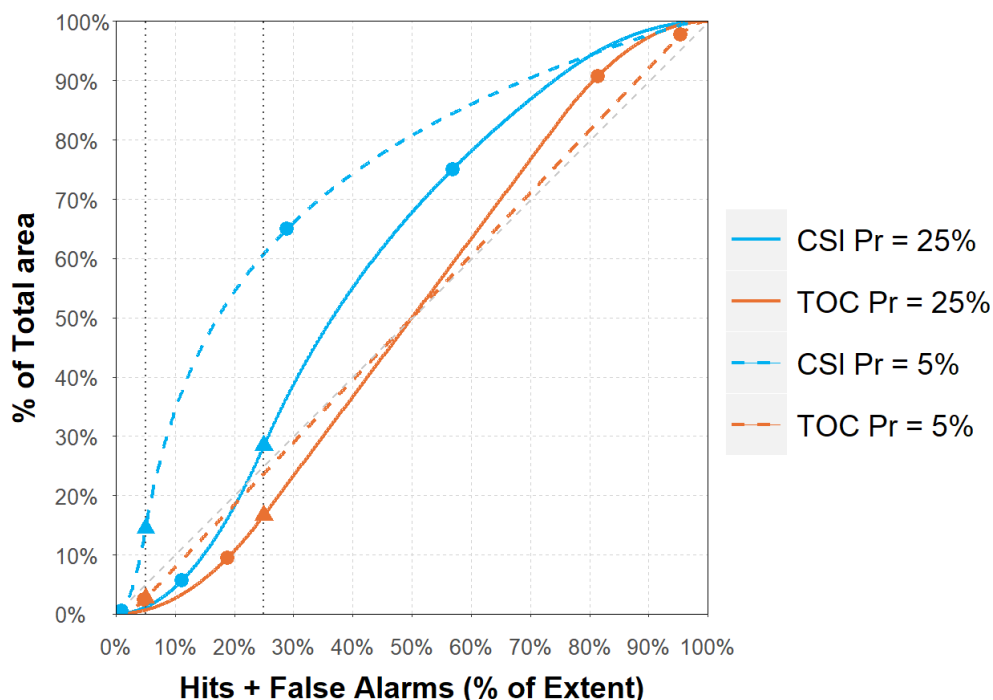


FIGURE 2.3: Proportion of the Total area covered at each threshold by TOC (blue lines) and CSI (orange line) plots. The solid lines correspond to a population with 5% prevalence. The dotted lines correspond to a population with 25% prevalence. The triangular markers on the lines indicate the proportion of the Total area that is covered at P . The circular markers indicate the points where the slope is equal to the slope of the diagonal line. *Source: Self elaboration.*

Figure 2.3 provides two key elements for the analysis: triangular markers and slopes. The triangular markers indicate how much area is accumulated up to P . If the marker of one curve lies above another, that curve accumulates more area before P , and therefore less area beyond it. This implies that the corresponding metric places less weight on classifications with large Quantity overestimation, which are typically less realistic.

The slopes of the curves represent the marginal contribution of each level of classified Quantity to the Total Area, and thus to the value of the metric. A slope

greater than one indicates that the corresponding classified Quantity contributes more than under a uniform distribution. In other words, that range of classified Quantities has a stronger influence on the metric. The circular markers identify the interval of classified Quantities for which this contribution is above the average.

The figure shows that the slope of CSI at P is substantially larger than one, whereas the slope of TOC is close to one. This difference holds across the prevalences considered, with CSI consistently exhibiting a steeper slope at P than TOC. In addition, the range of classified Quantities with above-average contribution, indicated by the circular markers, is shorter for CSI. This implies that CSI concentrates a larger share of its Total Area around P , whereas AUC distributes its Total Area more evenly along the horizontal axis, assigning relatively less weight to values near P . These observations provide an affirmative answer to Question (1) for the prevalences shown.

The triangular markers provide a clear answer to Question (3). In all cases, the markers for CSI lie above those for TOC, indicating that CSI accumulates more Total Area at lower levels of classified presence. As a result, the far-right region, associated with unrealistically high Quantities, represents a smaller share of the Total Area under CSI. This again supports an affirmative answer to Question (3) for the prevalences considered.

By contrast, Question (2) cannot be resolved through visual inspection alone. Comparing the differences in slopes on either side of the triangular markers is not straightforward. This question is therefore addressed in the subsequent analysis.

Although Figure 2.3 answers the questions for specific prevalences, it is necessary to examine how these results generalize. Figure 2.4 analyzes the sensitivity of Eqs. (2.6) and (2.7) to prevalence at $H + F = P$. Each panel addresses one of the three questions: panel (a) examines how the contribution to Total Area varies with prevalence (Question 1); panel (b) evaluates asymmetries around P (Question 2); and panel (c) shows how the accumulated proportion of Total Area at P changes with prevalence (Question 3).

To assess Question (1), we examine the slopes of Eqs. (2.6) and (2.7) at P . Since these expressions depend on x/E , elasticities can be used to generalize the results across populations. Figure 2.4a shows that CSI consistently exhibits higher elasticities

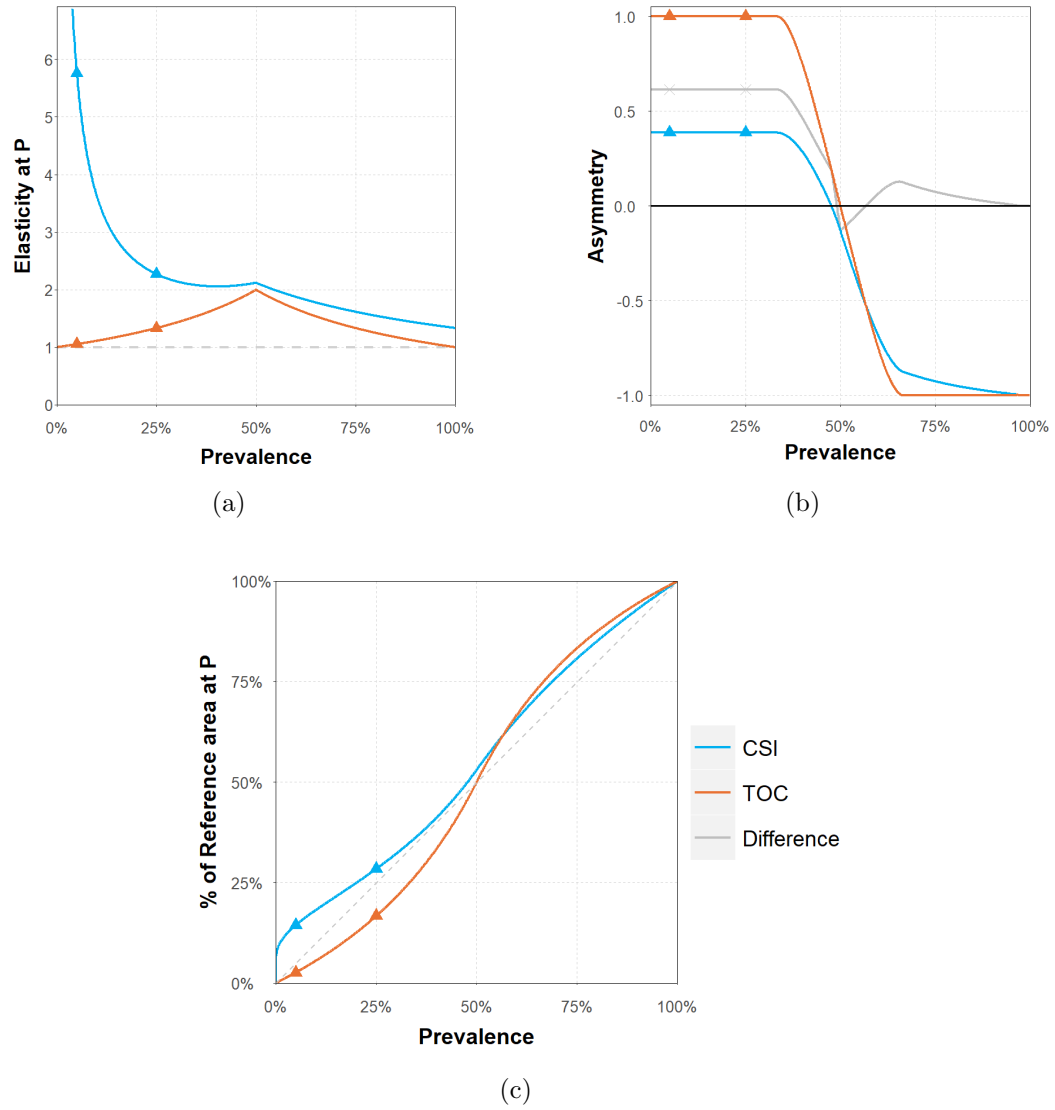


FIGURE 2.4: Representation of the sensitivity of equations (2.6) and (2.7) to prevalence at $x = P$. Panel (a) shows the sensitivity of the elasticities; Panel (b) shows the sensitivity of the asymmetries between the right and left of P ; Panel (c) shows the sensitivity of the accumulated proportion of the Total area at P . The triangles correspond to the same markers as in Figure 2.3. *Source: Self elaboration.*

than TOC. This confirms that the contribution to Total Area at $H + F = P$ is always larger for CSI, regardless of prevalence.

The magnitude of this difference varies with prevalence. As prevalence approaches zero, the elasticity of CSI at P tends to infinity, indicating that CSI increasingly concentrates its Total Area around P . In contrast, the contribution of TOC at P

converges to one. Thus, the lower the prevalence, the greater the difference between AUCSI and AUC in terms of the contribution around P , and this relationship is non-linear. These results show that AUCSI prioritizes correct classifications near the observed quantity of presence even when the event is rare, whereas AUC does not.

Answer 1. *Affirmative regardless of prevalence.*

We now turn to Question (2). Figure 2.4b compares the regions to the left and right of P to assess asymmetry. Asymmetry is defined as the ratio between the larger and smaller areas, minus one. Positive values indicate asymmetry toward the right, while negative values indicate asymmetry toward the left. The grey line represents the difference in asymmetry between AUC and CSI. The analysis considers symmetric windows of size $\min(P, Q)$ around P , which is the largest feasible window for each prevalence.¹

Both metrics exhibit a similar pattern: low prevalences lead to greater contribution from the right side of P , while higher prevalences reduce this asymmetry and eventually reverse it. However, they differ in magnitude. For prevalences below 50%, TOC exhibits substantially higher asymmetry than CSI. CSI's asymmetry is smaller and less sensitive to changes in prevalence, except in a narrow range around 50%, where differences are minor.

Overall, CSI displays consistently lower asymmetry than TOC. While both AUCSI and AUC are biased toward Quantity overestimation for rare events and toward underestimation for common events, this bias is weaker in AUCSI.

Answer 2. *Affirmative for both metrics and most prevalences. AUCSI favors Quantity Agreement more than AUC.*

Finally, Figure 2.4c addresses Question (3). For prevalences below 50%, TOC lies below the diagonal at P , indicating that it accumulates area at a rate lower than uniform up to P . In contrast, CSI lies above the diagonal for all prevalences, meaning that it consistently accumulates area more intensively around P . CSI remains above TOC for prevalences below approximately 57%, while beyond this threshold, both metrics converge.

¹The window size is an arbitrary choice that affects quantitative results but not qualitative conclusions.

Answer 3. *Affirmative for prevalences below approximately 57%. For higher prevalences, the difference becomes negligible.*

These results confirm that AUCSI assigns more weight than AUC to classified Quantities close to the observed ones. For rare events, AUC derives a substantial share of its value from large and unrealistic Quantities, whereas AUCSI concentrates its contribution on more plausible classifications. This reinforces the conclusion that AUCSI mitigates one of the main limitations of AUC.

2.3.1.2 AUCSI Score as a metric of Skill of the rare event at multiple thresholds

Once we have evaluated AUCSI and AUC as metrics of Agreement, we can extend the analysis to AUCSIS and AUC as metrics of Skill. To do so, we focus on Eqs. (2.8) and (2.9), shifting attention from the Total Area to the Reference Area.

As in the previous section, Figures 2.5 and 2.6 provide graphical representations of these equations. Figure 2.5 considers fixed prevalences and varying $H + F$, while Figure 2.6 considers varying prevalences and fixed $H + F = P$. The curves and markers follow the same construction as in Figures 2.3 and 2.4. Their interpretation is also analogous, with the difference that the analysis now refers to the Reference Area, and therefore to Skill.

Figure 2.5 shows patterns similar to those observed for Agreement. In contrast to the Agreement case, both TOC and CSI exhibit slopes at P that are above the uniform benchmark. However, CSI's slopes are consistently larger across prevalences. Moreover, CSI displays a narrower range of classified Quantities with above-average contribution. As a result, AUC distributes its Reference Area more evenly along the horizontal axis, whereas AUCSIS concentrates it around P . For the prevalences shown, this provides an affirmative answer to Question (1).

Regarding Question (3), the triangular markers for CSI lie above those for TOC in both cases. This indicates that CSI accumulates a larger share of the Reference Area at lower levels of classified presence. Consequently, unrealistic thresholds contribute less to AUCSIS than to AUC. For the displayed prevalences, the answer to Question (3) is again affirmative.

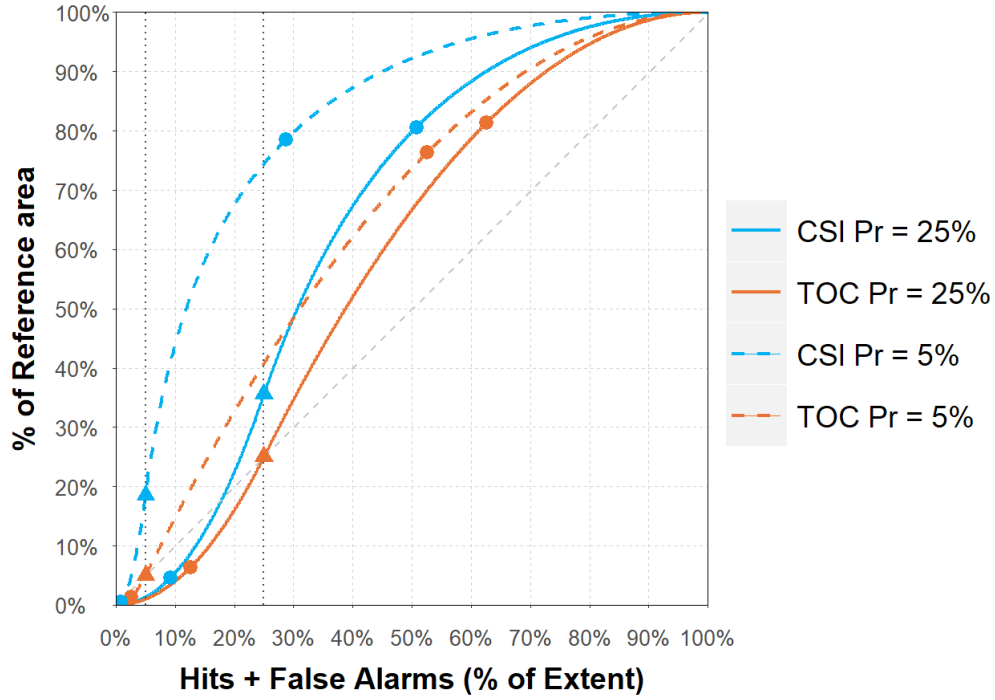


FIGURE 2.5: Proportion of the Reference area covered at each threshold by TOC (blue lines) and CSI (orange line) plots. The solid lines correspond to a population with 5% prevalence. The dotted lines correspond to a population with 25% prevalence. The triangular markers on the lines indicate the proportion of the Reference area that is covered at P . The circular markers indicate the points where the slope is equal to the slope of the diagonal line. *Source: Self elaboration.*

As in the previous section, Question (2) cannot be resolved through visual inspection alone. We therefore turn to Figure 2.6.

Figure 2.6 generalizes these results across prevalences. Panel (a) addresses Question (1) by showing the elasticities of Eqs. (2.8) and (2.9) at P . CSI consistently exhibits higher elasticities than TOC, confirming that the contribution to the Reference Area at $H + F = P$ is always larger for CSI, regardless of prevalence.

The magnitude of this difference depends on prevalence. As prevalence approaches zero, the elasticity of CSI at P tends to infinity, indicating an increasing concentration of Reference Area around P . In contrast, the contribution of TOC remains constant at 2. This highlights a key limitation of AUC: although it gives weight to classifications near the observed Quantity, it does not adapt to rarity. As a result, AUC fails to adequately measure Skill when the event is rare. The non-linear behavior of AUCSIS

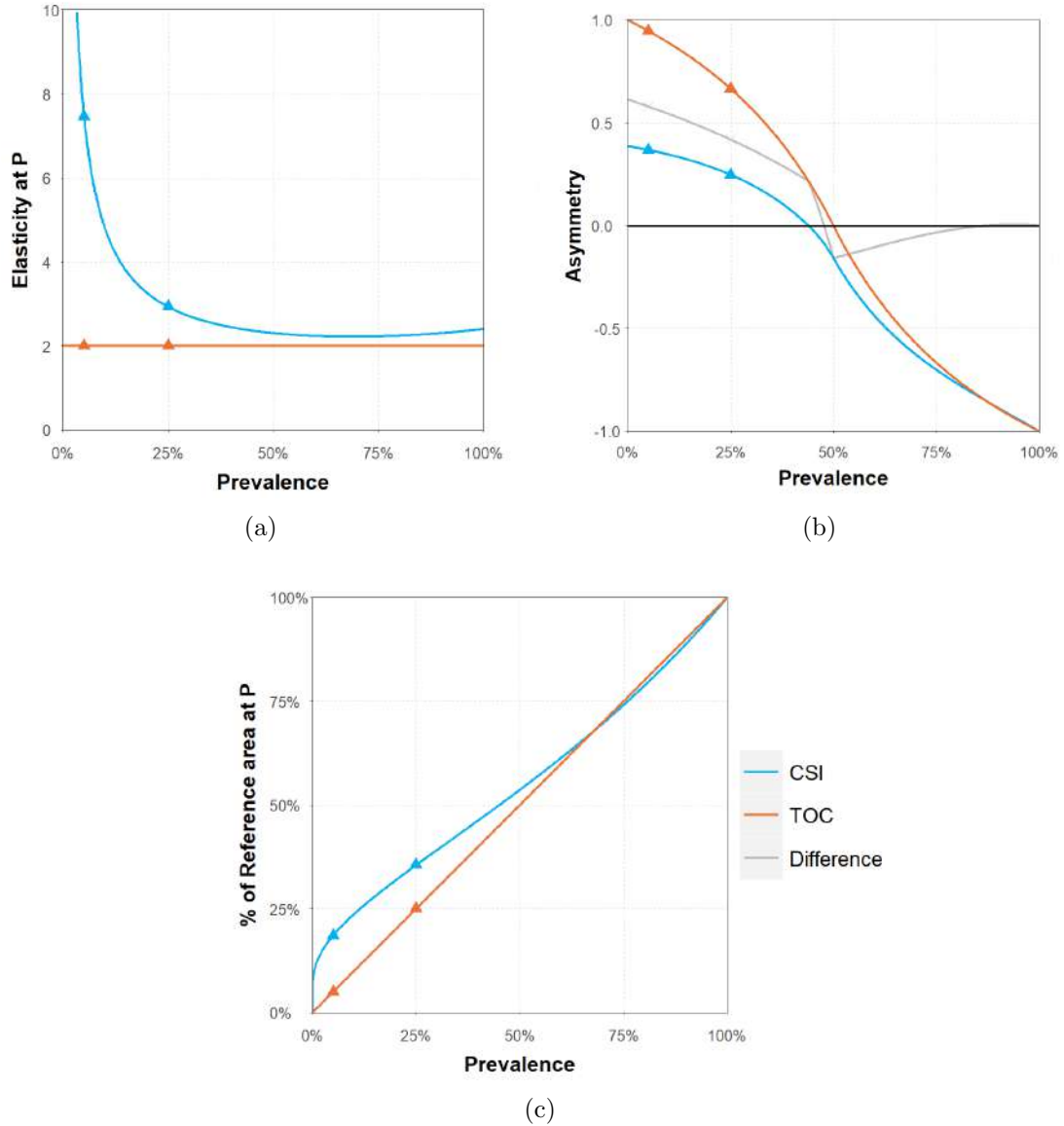


FIGURE 2.6: Representation of the sensitivity of equations (2.8) and (2.9) to prevalence at $x = P$. Panel (a) shows the sensitivity of the elasticities; Panel (b) shows the sensitivity of the asymmetries between the right and left of P ; Panel (c) shows the sensitivity of the accumulated proportion of the Reference area at P . The triangles correspond to the same markers as in Figure 2.5. *Source: Self elaboration.*

at low prevalences shows that it overcomes this limitation, with differences between the two metrics becoming larger as prevalence decreases.

The conclusions drawn from Question (1) are consistent for both the Agreements

and Skill metrics. They both show that CSI-derived metrics have a greater contribution from thresholds with small Quantity differences. They both show non-linear differences for low prevalences. TOC-derived metrics do not adapt their measurement of Agreement and Skill to the scarcity of the event of study.

Answer 4. *Affirmative regardless of prevalence.*

We now consider Question (2). Figure 2.6b evaluates asymmetry around P , defined as the ratio between the larger and smaller areas minus one. The analysis follows the same procedure as in Section 2.3.1.1, using symmetric windows of size $\min(P, Q)$.²

Both metrics exhibit similar qualitative patterns: low prevalences generate asymmetry toward the right of P , while higher prevalences reduce and eventually reverse this asymmetry. However, they differ in magnitude and in the location of the tipping point. For TOC, the tipping point occurs at $Pr = 50\%$, whereas for CSI it lies within the interval $5\% < Pr < 35\%$, depending on the window size.

Overall, CSI exhibits lower asymmetry for most prevalences. For low prevalences, both metrics favor the right region, but CSI's asymmetry is substantially smaller and less sensitive to prevalence. Between the tipping points, CSI shifts toward left asymmetry earlier than TOC. For $Pr > 50\%$, both metrics exhibit asymmetry toward the left, with CSI showing slightly higher values. These results indicate that, although both metrics are biased toward Quantity overestimation for rare events, this bias is weaker for CSI.

Answer 5. *Affirmative for both metrics and most prevalences. CSI exhibits lower asymmetry, particularly when prevalence is low.*

Finally, Figure 2.6c addresses Question (3). For TOC, the proportion of Reference Area accumulated at P equals prevalence, implying a uniform accumulation up to P . In contrast, CSI accumulates more area than TOC for prevalences below approximately 68%. Beyond this threshold, both metrics converge.

The relationship is non-linear. As prevalence decreases, the difference between CSI and TOC increases. For example, at 25% prevalence, CSI covers 35% of its Reference Area at P , compared to 25% for TOC. At 5% prevalence, these values are 20% and 5%, respectively. Thus, the gap widens as the event becomes rarer.

²The window size is an arbitrary choice that affects quantitative results but not qualitative conclusions.

These results confirm that AUCSIS assigns more weight than AUC to classified Quantities close to the observed ones, particularly when prevalence is low. Consequently, AUCSIS reduces the influence of unrealistic thresholds and provides a more informative measure of Skill in rare event settings.

Answer 6. *Affirmative for all the cases where prevalence is below 68%.*

TABLE 2.2: Table summarizing Agreement and Skill across questions.

AUC vs AUCSI		
	Agreement	Skill
Question (1)	Affirmative at any prevalence. AUCSI concentrates the area around P more intensively than uniform; AUC doesn't.	Affirmative at any prevalence. Both AUCSI and AUC concentrate the area around P more intensively than uniformly.
Question (2)	Affirmative at most prevalences	Affirmative at most prevalences
Question (3)	Affirmative at prevalences $< 57\%$	Affirmative at prevalences $< 68\%$

2.3.2 Revisiting metrics of single thresholds

2.3.2.1 CSI as a metric of Agreement of the rare event at a single threshold

CSI was originally introduced by Gilbert (1884) as a measure of agreement in tornado forecasts. Although it has been predominantly used in meteorology (Wang 2014), it is increasingly applied in fields where the event of interest is rare, including epidemiology, ecology, remote sensing, and machine learning (Mbizvo et al. 2023; Pontius, Boersma, et al. 2008; Powers 2011; Saito and Rehmsmeier 2015; Sofaer et al. 2018).

CSI (Eq. 1.7) measures the Agreement between observed and classified presence. Two features of this definition are central to its interpretation. First, CSI is a metric of Agreement, not Skill. It captures similarity between observed and classified rare events, regardless of whether this agreement arises from model ability or from the abundance of the rare event in the sample. As a result, CSI values are not directly

comparable across datasets with different prevalences. In practice, however, CSI is often used to evaluate model performance without accounting for this distinction, which can lead to misleading conclusions.

Second, CSI focuses exclusively on the rare event. Correct Rejections are not included in Eq. (1.7), so the absence class does not contribute to Agreement. This is desirable when the rare event is relevant and scarce, but it limits the informativeness of CSI in more balanced settings.

Although widely used, CSI has two well-known shortcomings (Baldwin and Kain 2006; Wang 2014). First, CSI depends strongly on prevalence. When prevalence is low, high CSI values are harder to achieve, not because good classifications are impossible, but because only a narrow range of classified Quantities can yield high scores. Second, CSI is often maximized under Quantity overestimation (Baldwin and Kain 2006), meaning that it tends to favor classifications that overpredict the rare event.

To understand these issues, it is useful to extend CSI to multiple thresholds, as in the previous section. Representing CSI as a curve allows us to analyze how Agreement is distributed across different classified Quantities and to distinguish between Allocation and Quantity disagreement.

Figure 2.7 illustrates the effect of prevalence on CSI. The figure compares two populations (10% and 50% prevalence) under a perfect classifier. The horizontal line marks a CSI value of 60%, and the shaded regions indicate the classified Quantities that achieve values above this threshold.

The range of Quantities that yield high CSI values is substantially larger at higher prevalence. At 10% prevalence, $\text{CSI} > 60\%$ is achieved only for a narrow interval of classified presence, whereas at 50% prevalence this interval is much wider. Moreover, the shaded area is larger at higher prevalence, indicating that not only do more Quantities exceed the threshold, but they do so more easily. This confirms that the difficulty of obtaining high CSI values increases as prevalence decreases.

The second limitation, the Quantity overestimation bias, can also be understood through CSI curves. For a given CSI level, the set of classified Quantities that achieve that value is typically asymmetric around P . In low-prevalence settings, there are more Quantities to the right of P than to the left, meaning that the same CSI can

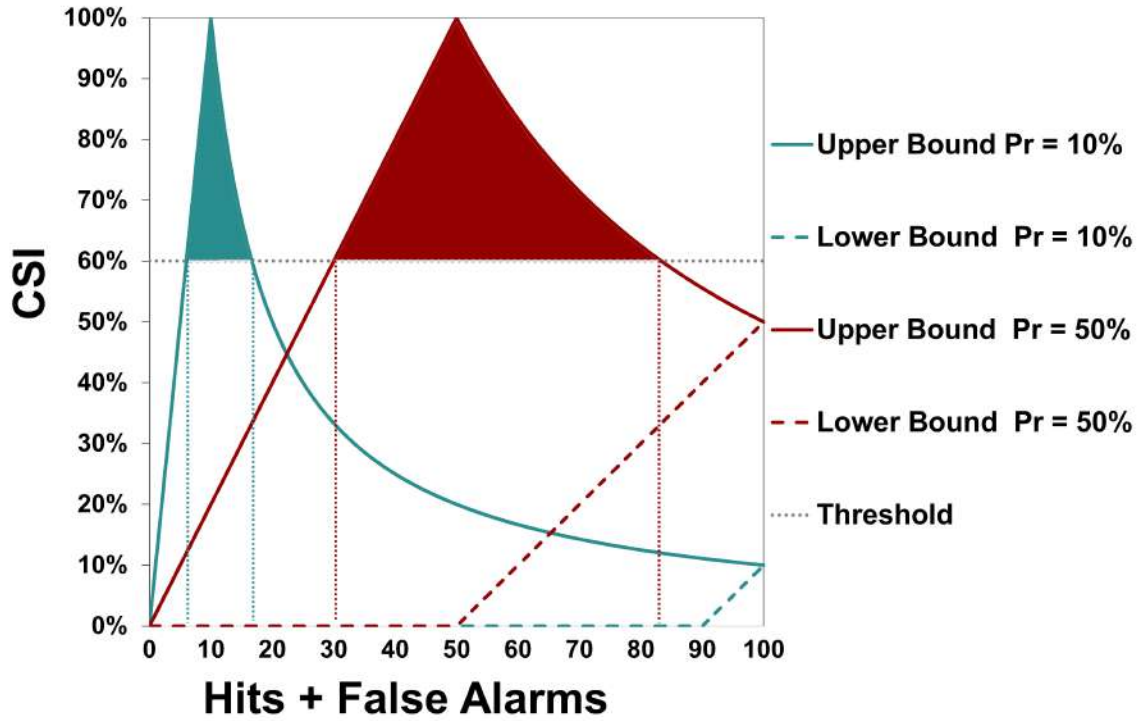


FIGURE 2.7: Representation of the CSI plots with a prevalence of 10% and 50%. The colored areas represent the space where classifications can get a CSI above 60%. *Source: Self elaboration.*

be obtained with larger overestimation than underestimation.

Figure 2.8 illustrates this asymmetry. For example, a CSI of 30% can be achieved over a wide range of classified Quantities, with substantially more room for overestimation than underestimation. For example, in panel (a), a CSI of 30% can be achieved with classified Quantities ranging from 7% to 68% ($Q_2 = 61\%$). This implies that underestimation is limited to 13 percentage points, whereas overestimation can reach up to 45 percentage points. However, this asymmetry is not constant. As the CSI threshold increases, the range of feasible Quantities becomes narrower, and the bias decreases. For instance, a CSI of 60% can only be obtained within a smaller interval ($Q_1 = 20\%$), with at most 7 percentage points of underestimation and 13 of overestimation. In addition, the direction and magnitude of the bias depend on prevalence. At higher prevalences, the asymmetry becomes smaller and may even reverse for certain CSI levels.

The comparison between Figures 2.8a and 2.8b provides additional insight into

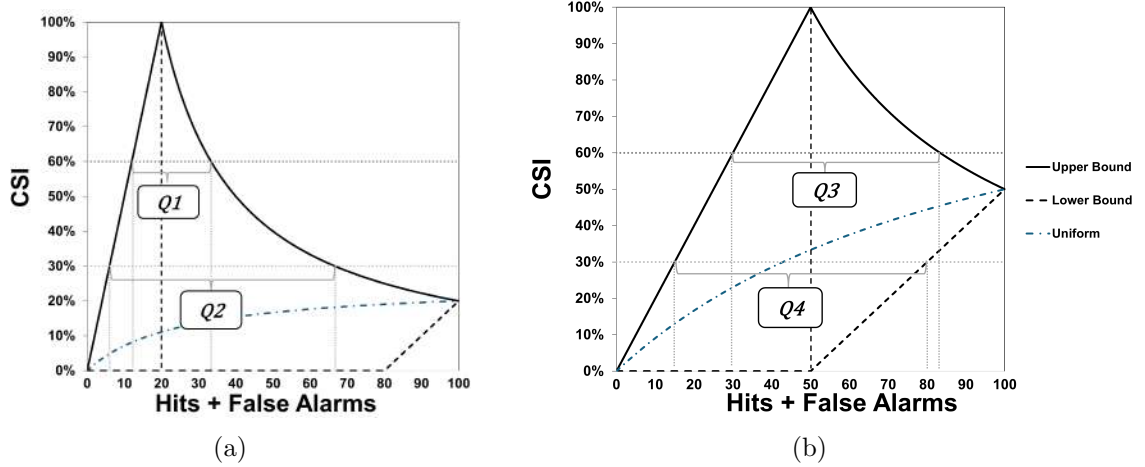


FIGURE 2.8: Representation of the two CSI plots with prevalences of 20% (a) and 50% (b). The dashed lines represent thresholds of a CSI of 30% and 60%. *Source: Self elaboration.*

the Quantity bias of CSI. For a prevalence of 60%, the curve still favors Quantity overestimation to achieve a CSI of 60% (Q_3). However, although Q_3 is substantially larger than Q_1 , the magnitude of the bias does not increase. In fact, Q_4 is smaller than Q_2 , and the direction of the bias shifts at higher prevalence.

Several conclusions follow from this comparison. First, the magnitude of the bias decreases as the CSI value increases. Second, the bias does not increase with prevalence; instead, it may decrease and even reverse, leading to Quantity underestimation when the CSI is below prevalence.

This suggests that the bias of CSI is not strictly toward presence overestimation, but more generally toward overestimating the less frequent class. Consequently, the Quantity overestimation bias is primarily relevant in situations where both CSI and prevalence are low.

As a closing remark, these findings should be interpreted in light of the definition of CSI. As a metric of Agreement, CSI reflects the opportunities for agreement in the data. When the event is rare, there are fewer opportunities for agreement, and therefore fewer classifications can attain high CSI values. This is not a limitation of the metric, but a direct consequence of its definition.

The main issue lies in its interpretation. CSI is often used as a proxy for Skill, that is, to evaluate the performance of a classifier relative to a baseline. However,

Agreement and Skill are fundamentally different concepts: Agreement measures similarity, whereas Skill measures improvement over a reference. Using CSI as a Skill metric without accounting for prevalence can therefore lead to misleading conclusions. In the following section, we examine the corresponding Skill formulation of CSI and its implications.

2.3.2.2 Gilbert Skill Score as a metric of Skill of the rare event at a single threshold

Gilbert (1884) introduced the Gilbert Skill Score (GSS) as a measure of classifier performance. His work represents one of the earliest attempts to incorporate the concept of Skill into binary classification metrics (Murphy 1996). In particular, Gilbert showed how randomness can be explicitly accounted for by discounting the proportion of Agreement that is due to chance. Although GSS was originally proposed as a Skill counterpart to CSI, its formulation suggests a more general principle: each Agreement metric can be associated with a corresponding Skill metric.

GSS is one of the Skill metrics derived from CSI and is commonly referred to in the literature as the Equitable Threat Score (ETS) (Schaefer 1990; Mesinger and Brill 2004). It evaluates classification Skill by discounting the Hits that are expected under a random classifier. More broadly, GSS follows the logic of Skill scores, which compare the Agreement achieved by a classifier to that of a baseline, relative to a perfect classification (Eq. 1.12). In this section, we place GSS in the context of other Skill metrics to assess its logical foundation.

The conceptual basis of Skill metrics has been widely discussed in the literature. Ebert and Milne (2022) revisits the definition of Skill for several widely used metrics, including the Pierce Skill Score (PSS) (Eq. 1.26), the Heidke Skill Score (HSS) (Eq. 1.20), and the Relative Improvement over Chance (RIOCI) (Farrington and Loeber 1989; Loeber and Dishion 1983). They conclude that only the formulation underlying PSS provides a logically consistent definition of Skill. The multiple-threshold representation introduced in this thesis allows these concepts to be interpreted visually, with the classified quantity of the rare event on the horizontal axis playing a key role.

Figure 2.9 provides an illustrative example based on multiple contingency tables for a fixed population. The baseline curve represents a random classifier, i.e., a classifier that correctly detects the rare event at the prevalence rate. Although different Skill metrics rely on different definitions of Agreement, all of them are constructed from contingency tables and can therefore be represented across thresholds. Figure 2.9 illustrates this using OA (panel a, Eq. 1.4) and CSI (panel b) as Agreement metrics. The figure is not tied to a specific metric, but serves as a general tool to visualize the baselines and perfect classifiers underlying different Skill definitions.

Consider a population with a prevalence of 30%. Suppose we evaluate the Skill of classification A (the Reference Classification). This classification generates the contingency table in Table 2.3a, with a CSI of 45.5% and an OA of 70%. It classifies 50% of the population as presence, implying a Quantity disagreement of 20%.

Table 2.3b and point B in Figure 2.9 represent the baseline used by HSS. This baseline assumes that a random classifier produces the same quantity of classified presence as the Reference Classification. Point F represents the perfect classifier under HSS. By contrast, Table 2.3d and point D correspond to the baseline used by PSS, which assumes no Quantity disagreement. Table 2.3e and point E represent the corresponding perfect classifier for PSS.

As shown by Ebert and Milne (2022), these differences can be interpreted geometrically. In the case of HSS, Skill can be represented as the ratio AB/BF , i.e., the proportion of the distance between the baseline (B) and the perfect classifier (F) that is covered by the observed classification (A). Note that A, B, and F share the same value of $H + F$, and therefore the same Quantity disagreement. We refer to this definition as *Type 1 Skill*. However, Ebert and Milne (2022) argues that the perfect classifier in HSS is not feasible without altering prevalence. This is illustrated in Figure 2.9, where point F lies outside the feasible bounds of the metric.

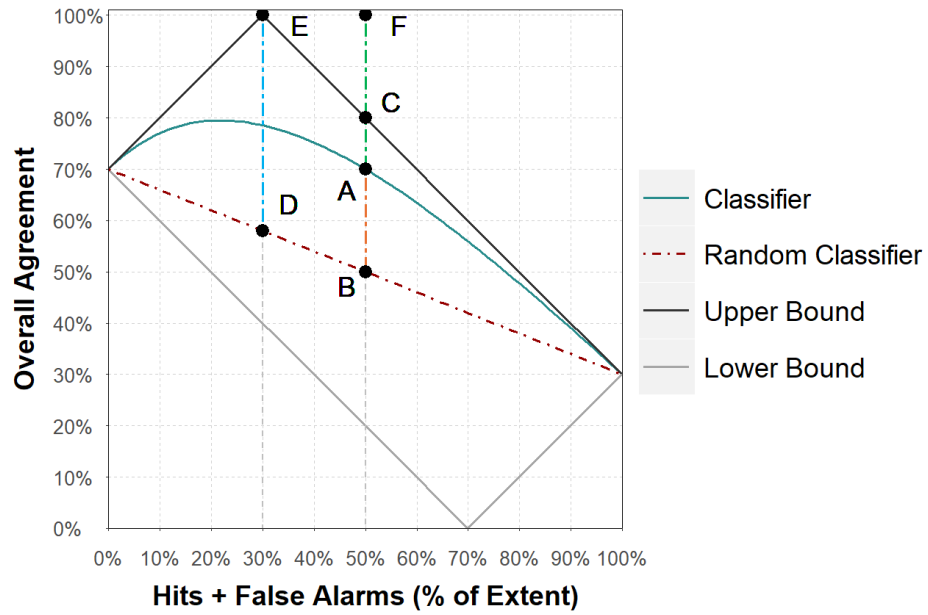
Similarly, RIOC is defined in equation (2.10). RIOC is graphically defined as AB/BC in Figure 2.9a.

$$\text{RIOC} = \frac{\text{OA} - \text{OA}^{\text{random}}}{\text{OA}^{\text{Allocation}} - \text{OA}^{\text{random}}} \quad (2.10)$$

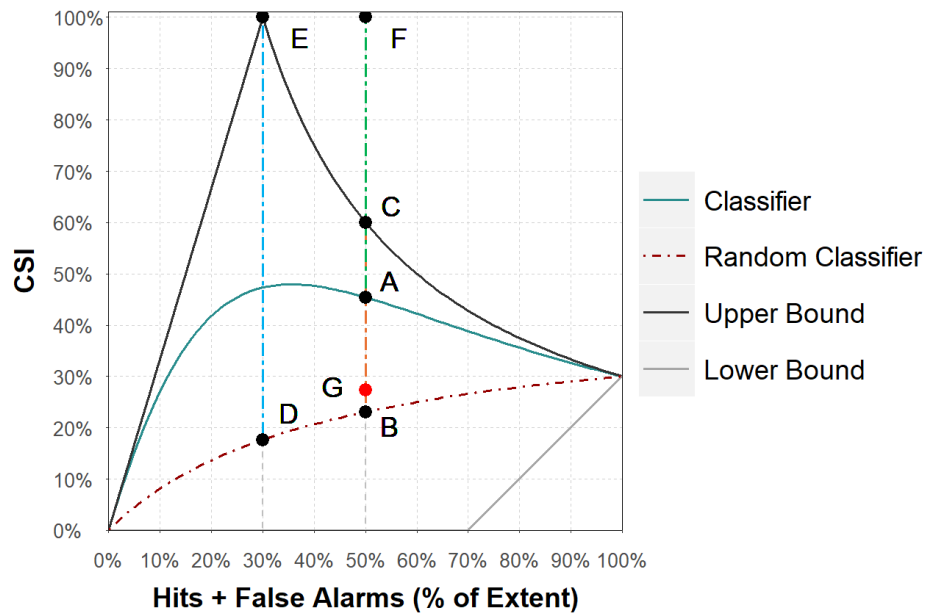
Where $\text{OA}^{\text{random}}$ is defined in Equation 1.2.4.2, and $\text{OA}^{\text{Allocation}}$ is the OA that a classifier with only Quantity disagreement would achieve at the same quantity of

TABLE 2.3: Contingency tables for illustrative classification cases, including the perfect classifier, a baseline classifier, and intermediate cases used in the evaluation of skill metrics. These tables correspond to the points represented in Figure 2.9.

(a)				
CSI = 45.5%		Reference		
OA = 70%		Presence	Absence	Total classified
Classification	Presence	25	25	50
	Absence	5	45	50
	Total Reference	30	70	100
(b)				
CSI = 23%		Reference		
OA = 50%		Presence	Absence	Total classified
Classification	Presence	15	35	50
	Absence	15	35	50
	Total Reference	30	70	100
(c)				
CSI = 60%		Reference		
OA = 80%		Presence	Absence	Total classified
Classification	Presence	30	20	50
	Absence	0	50	50
	Total Reference	30	70	100
(d)				
CSI = 17.6%		Reference		
OA = 58%		Presence	Absence	Total classified
Classification	Presence	9	21	30
	Absence	21	49	70
	Total Reference	30	70	100
(e)				
CSI = 100%		Reference		
OA = 100%		Presence	Absence	Total classified
Classification	Presence	30	0	30
	Absence	0	70	70
	Total Reference	30	70	100



(a)



(b)

FIGURE 2.9: Representation of multiple thresholds for the illustrative classifier using OA (a) and CSI (b). Points correspond to contingency tables of a perfect classifier, a baseline classifier, and a selected reference classifier under different definitions of Skill. *Source: Self elaboration.*

classified presence, calculated as:

$$\text{OA}^{\text{Allocation}} = \frac{2 \cdot \min(H + F, H + M)}{H + F + M + C} \quad (2.11)$$

RIOC shares the same definition of baseline with HSS, i.e., a random classifier that classifies the same quantity of presence. As in Type 1 Skill, all A, B, and C have the same $H + F$. However, RIOC's perfect classifier differs from that of HSS. RIOC's perfect classifier is one with no Allocation disagreement but Quantity disagreement. Note how point C has an OA of 80%, far from the 100% of an actual perfect classification. This 80% OA represents the highest score we can get with an optimal Allocation of the classified Quantity. RIOC measures the extent to which the classifier reduces the Allocation disagreement compared with a random classifier. Thus, we can define RIOC as a metric of Allocation Skill instead of overall Skill. This is a subset of Type 1 Skill.

As a metric of Allocation Skill, RIOC, along with other metrics with the same definition of a perfect classifier, provides valuable information but requires an adequate interpretation. In the Reference classification in our particular example, RIOC captures some of the overall Skill, because its main contribution to disagreement is Allocation and not Quantity. However, we can also think of a case in which we overestimate the quantity of the rare event so much that our classification is located in the further right extreme of Figure 2.9. In that case, the RIOC will approach 1 because the classification covers much of the distance between the Random classifier and the Upper bound. This high score does not indicate overall Skill but rather a high Allocation Skill. Yet, this Allocation Skill is meaningless when compared to the substantial Quantity disagreement.

The perfect classifier considered in the denominator by PSS is a classification with no Quantity and Allocation disagreement (point *E*). PSS is no longer a Type 1 Skill because its observed, perfect, and baseline classification don't have the same Quantity disagreement. This definition of Skill will be referred to as Type 2 Skill. The numerator in PSS is identical to the one in HSS. The numerator corresponds to *AB* in Figure 2.9, and it measures how much the classifier exceeds the random classifier at the same quantity of classified presence. However, their denominators differ in their definition of a perfect classifier. The denominator in PSS is the segment *DE*. This is

the difference between a classifier with no Quantity and Allocation disagreement and a random classifier with no Quantity disagreement.

But, where does GSS stand in this regard? What Type of Skill does it entail? Can we graphically represent it along with these metrics? The answer to these questions is linked to the logical foundation of GSS.

First of all, to evaluate GSS's definitions of Skill, we need to analyze Equations (1.12) and (1.14). Equation (1.14) refers to the Hits obtained by the random classifier defined in GSS. The baseline in Equation (1.12) is the same for the numerator and the denominator. It considers the Agreement that the random classifier generates the same quantity of classified presence of the classifier. This indicates that GSS is a Type 1 Skill metric, as it shares the definition of the baseline with HSS and RIOCI. The perfect classifier in GSS has a CSI score of 1, so it is a metric of overall Skill, not only Allocation Skill. Therefore, we can define GSS as a version of HSS that only focuses on the Skill in the rare event classification.

However, when we graphically represent the GSS in Figure 2.9b, we observe a discrepancy with HSS. HSS's projection into the CSI plot should be equal to GSS. Nevertheless, the projection of the baseline of HSS in the CSI plot corresponds to point *B*, while GSS's baseline is at point *G*. Graphically, GSS is defined as AG/FG instead of AB/BF . This discrepancy reveals a fundamental mathematical error in GSS.

Equation 1.13 is a mathematical derivation of the original proposal by Gilbert (1884). As a refresher, Equation Eq.1.12 was defined as:

$$GSS = \frac{H - H^{\text{random}}}{H + F + M - H^{\text{random}}}$$

Which can also be expressed in terms of CSI in Equation 1.13:

$$GSS = \frac{\frac{H}{H + F + M} - \frac{H^{\text{random}}}{H + F + M}}{1 - \frac{H^{\text{random}}}{H + F + M}}$$

Gilbert's proposal subtracts the Hits due to chance in both the numerator and the denominator of CSI (Eq.1.12). This proposal is similar to how other Skill metrics

are mathematically built. Even if the motivation is correct, the logic behind this transformation is flawed.

Figure 2.10 represents the Venn's diagrams of the classifications of our example. Panel (a) corresponds to the Reference Classification. Panel (b) to the random classification at the same quantity of presence, or the random classification of Skill Type 1 (Tab. 2.4b). OA is defined as the ratio between the intersection of classifications and references, which include both presence and absence ($H + C$), and their union ($H + F + M + C$). By definition, the union is constant for any classification that classifies the whole population. In Figure 2.10, the union is $25 + 25 + 5 + 45 = 100$ for the Reference Classification and $35 + 15 + 15 + 35 = 100$ for the random classification. The intersection is $25 + 45 = 70$ for the Reference Classification and $15 + 35 = 50$ for the random classification. Thus, the union remains constant even if the intersection changes.

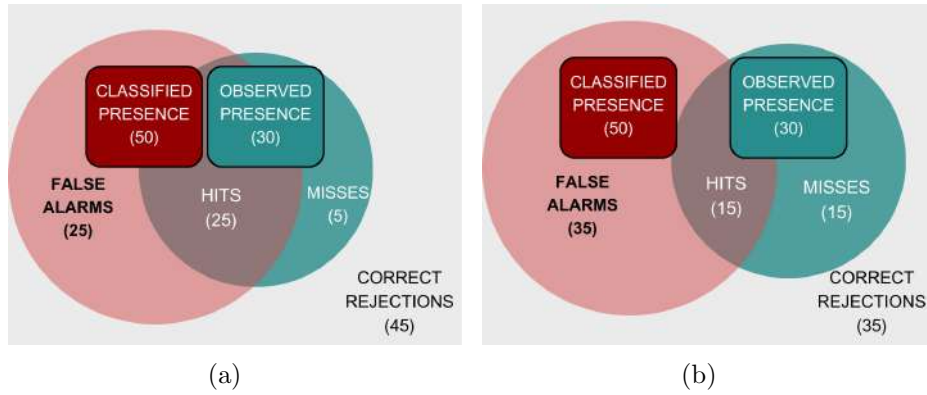


FIGURE 2.10: Venn diagram of the contingency table of the Reference Classification and its corresponding random classification (b). *Source: Self elaboration.*

Nevertheless, this is not the case in GSS. GSS's Agreement metric is CSI. CSI is the ratio between the intersection of classified and observed presence (H) and their union ($H + F + M$). In Figure 2.10, the size of the classified and observed presence is constant. We can look at the union as $(H + F) + (H + M) - H$. In this case, the difference in the intersections between the two classifications affects the unions. In our example, the unions are $25 + 25 + 5 = 55$ and $35 + 15 + 15 = 75$, and the intersections are 25 and 15. The union is no longer constant in CSI.

Equation (1.13) does not account for the difference in the size of the unions between the Reference Classification and the random classification. In particular, the numerator is written as

$$\frac{H}{H + F + M} - \frac{H^{\text{random}}}{H + F + M},$$

which implicitly assumes that both classifications share the same union, namely $H + F + M$. However, the union of the random classifier is in general different and given by $H^{\text{random}} + F^{\text{random}} + M^{\text{random}}$ (Eqs.1.14-1.16). Consequently, its CSI should be defined as

$$\text{CSI}^{\text{random}} = \frac{H^{\text{random}}}{H^{\text{random}} + F^{\text{random}} + M^{\text{random}}}.$$

Therefore, a Skill metric consistent with Eq. (1.11) must take the form:

$$\text{Revised GSS} = \frac{\text{CSI} - \text{CSI}^{\text{random}}}{1 - \text{CSI}^{\text{random}}} \quad (2.12)$$

Note that in Eq. (2.12) the size of the union is $H + F + M$ for the Reference Classification and $H^{\text{random}} + F^{\text{random}} + M^{\text{random}}$ for the random classification. The Revised GSS expressed in terms of the entries of the contingency table can be found in Appendix B.3. Equation (2.13) shows how Gilbert's original proposal must be adjusted to account for these differences in union size:

$$\text{Revised GSS} = \frac{H - H^{\text{random}} \cdot \frac{H + F + M}{H^{\text{random}} + F^{\text{random}} + M^{\text{random}}}}{H + F + M - H^{\text{random}} \cdot \frac{H + F + M}{H^{\text{random}} + F^{\text{random}} + M^{\text{random}}}} \quad (2.13)$$

The Revised GSS ranges from 1 to $-\infty$. A perfect classification attains a CSI of 1 and therefore a Revised GSS of 1. A random classification yields a value of 0, while classifications performing worse than random take negative values.

This mathematical correction affects the Skill score of the classifications. Table 2.4 compares the CSIs defined by GSS (Eq.1.12) and the Revised GSS (Eq.2.12), as well as their Skills. In our example, the CSI of the random classification is 23% according to GSS and 27% according to the Revised GSS. This creates a 4-point gap between

the two baselines. The graphical representation of the discrepancy between GSS and Revised GSS is also shown in Figure 2.9. Point F represents the current definition of the baseline of GSS, and point B is the baseline considered by Revised GSS. The figure shows how the baseline in GSS deviates from the random classification, affecting the metric’s numerator (CG) and denominator (FG). The Revised GSS is BC/BF .

TABLE 2.4: Example of classification, random classification, and numerators for the illustration in Figure 2.10.

		Classification	Random Classification	Skill
GSS (Eq.1.12)	Union	55	55	
	Intersection	25	$\frac{30 \times 50}{100} = 15$	25%
	CSI	45.5%	27%	
Revised GSS (Eq.2.12)	Union	55	65	
	Intersection	25	$\frac{30 \times 50}{100} = 15$	29%
	CSI	45.5%	23%	

If we compare the points in Figure 2.9, we see that the Revised GSS has the same definition of a perfect and random classification as HSS, i.e., it is a Type 1 overall Skill metric. Nonetheless, we could also consider other CSI-derived Skill metrics with alternative Skill definitions.

First, we can define a Type 1 Allocation Skill metric, which is a version of RIOG that only focuses on the rare event. In our example, this would consider Table 2.4a, b, and c as observed, baseline, and perfect classifications, respectively. However, instead of using OA, the new metric uses CSI. This Allocation GSS is graphically represented in Figure 2.9b as AB/BC . Allocation GSS is mathematically defined as:

$$\text{Allocation GSS} = \frac{\frac{H}{H + M + F} - \frac{H^{\text{random}}}{H^{\text{random}} + F^{\text{random}} + M^{\text{random}}}}{\frac{H + \min(M + F)}{H + 2\min(M + F)} - \frac{H^{\text{random}}}{H^{\text{random}} + F^{\text{random}} + M^{\text{random}}}} \quad (2.14)$$

The Allocation GSS also ranges between 1 and ∞ . A perfect Allocation gets a value of 1, a random allocation gets a value of 0, and a worse-than-random allocation gets negative values.

Similarly, we can also create a CSI-derived metric with a Type 2 Skill definition. This would be the PSS version that only considers observed and classified rare events. Again, this metric differs from PSS in that it uses CSI instead of OA. This Type 2 GSS is graphically represented in Figure 2.9b as AB/DE , and is mathematically defined by:

$$\text{Revised GSS2} = \frac{\frac{H}{H + M + F} - \frac{H^{\text{random}}}{H^{\text{random}} + F^{\text{random}} + M^{\text{random}}}}{1 - \frac{H^{\text{random,u}}}{H^{\text{random,u}} + F^{\text{random,u}} + M^{\text{random,u}}}} \quad (2.15)$$

Where H_{ru} , M_{ru} and F_{ru} are the H, M, and F of a random classification with no Quantity disagreement (Eq.1.23, 1.25).

2.3.3 Simulation-Based Assessment of the CSI Framework

This section evaluates the properties of the CSI framework and illustrates the practical implications of Sections 2.3.1 and 2.3.2 through a series of simulations. The aim is to show how similar AUC or AUC Skill values can correspond to very different classification abilities in the context of rare events, while AUCSI and AUCSI Skill provide a more accurate representation of performance for the rare class.

We simulate classifiers that produce strictly concave TOC curves over the interval $[0, E]$, covering different regions of the plot. Concave TOC curves ensure that AUC reflects the strength of the monotonic relationship between the index variable and the presence of the binary variable, as the curves do not cross the baseline (Pontius 2022).

These curves represent desirable classifiers, since they concentrate classification ability at higher thresholds of the index variable. The simulated curves differ in the rate at which Hits are generated, leading to different combinations of AUC and AUCSI values.

Details of the simulation design are provided in Appendix B.7. To illustrate the behavior of the metrics, the classifiers are applied to a population with low prevalence. The population consists of 10,000 synthetic observations, of which 1,000 correspond to the rare event (a prevalence of 10%). As shown in Equations (2.6), (2.7), (2.8), and (2.9), the areal distributions of CSI and TOC do not depend on population size, so the results generalize to any population with similar prevalence.

2.3.3.1 Comparison between two simulated classifiers

Figure 2.11 presents two of the simulated classifiers. Panels (a) and (b) display their TOC and CSI curves, respectively. Suppose the objective is to correctly classify the rare event, and the choice is between these two classifiers. Visual inspection suggests that Classifier A is preferable to Classifier B in most practical settings. For a wide range of realistic thresholds³, Classifier A yields higher Agreement and Skill. Thus, if the goal is to select a single threshold, Classifier A is generally preferred because it produces more Hits for realistic quantities of classified rare events.

This advantage extends to comparisons across multiple thresholds. Classifier A generates more Hits whenever the classified presence is less than 2.5 times the observed presence, and therefore outperforms Classifier B over this range. Classifier B is only preferable in situations where minimizing Misses is the primary objective, as it reaches the upper bounds earlier.

However, AUC fails to rank Classifier A above Classifier B. Table 2.5 summarizes the Agreement and Skill metrics for both classifiers. Each classifier attains an AUC of 0.9, which could be interpreted as a strong classification ability.

The difference lies in how Hits are distributed across thresholds. Classifier A concentrates Hits in the left region of the plot, while Classifier B distributes them more evenly. The areas between the two TOC curves in the left and right regions are approximately equal, which leads to identical AUC values. As a result, Classifier

³Remember that realistic thresholds or classifications refer to those with small Quantity disagreement, and thus, are more likely to be used in practical settings.

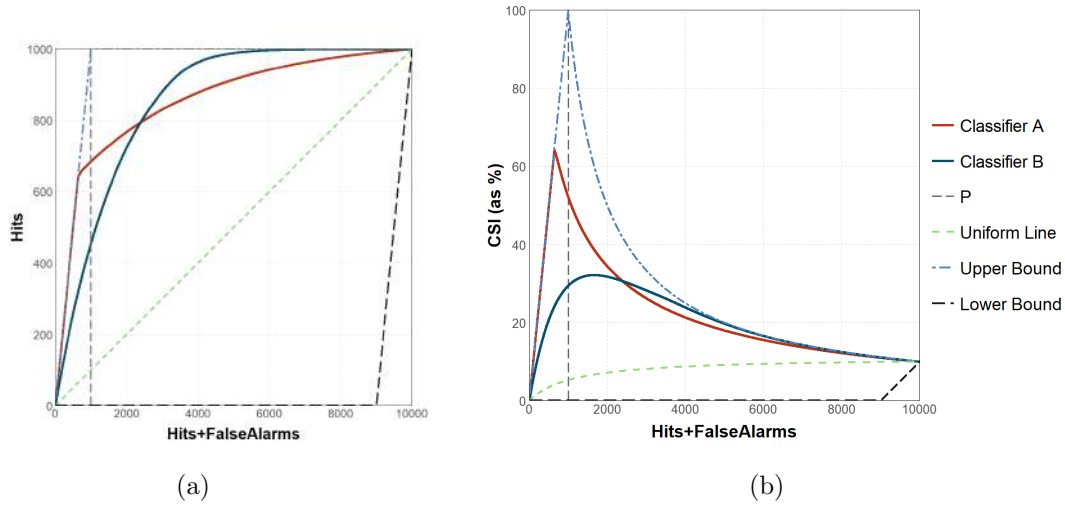


FIGURE 2.11: TOC (a) and CSI (b) plots of two of the classifiers of the simulation.

B derives a substantial portion of its AUC from thresholds that are not relevant for practical decision-making.

TABLE 2.5: Comparison of Agreement and Skill results between Classifiers A and B. We compare metrics that consider both classes (OA, HSS, AUC) against those of only the minority class (CSI, GSS, AUCSI).

		Hard classification		Soft classification	
		CSI	OA	AUCSI	AUC
Agreement	Classifier A	[10, 60%]	[10%, 97%]	79%	90%
	Classifier B	[10, 31%]	[10%, 92%]	69%	90%
		RGSS	HSS	AUCSIS	AUCS
Skill	Classifier A	[0, 57%]	[0, 67%]	71%	80%
	Classifier B	[0, 23%]	[0, 29%]	57%	80%

In contrast, AUCSI ranks Classifier A above Classifier B. In Figure 2.11, the slope of the TOC curve depends only on the rate at which Hits are generated, whereas the slope of the CSI curve depends on both the rate of Hits and the accumulation of Hits and False Alarms. That is, CSI penalizes Hits if they imply large Quantity disagreements.

For Classifier A, Hits are concentrated at the first thresholds, where Misses are high but False Alarms remain close to zero. This results in steep initial segments of the CSI curve, which contribute substantially to the Total area. In contrast, Classifier B requires a larger number of False Alarms to generate Hits, which limits the steepness of its CSI curve and reduces the accumulated area.

All CSI curves converge to the point (E, Pr) , which reduces the contribution of the right-hand side of the graph to differences in area. Consequently, differences in performance are driven primarily by the left region, where Classifier A clearly dominates. This results in an AUCSI of 79% for Classifier A compared to 69% for Classifier B.

The same pattern appears in the corresponding Skill measures. Although both classifiers achieve identical AUC Skill values (80%), AUCSI Skill assigns 71% to Classifier A and 57% to Classifier B. Therefore, CSI-based metrics are able to distinguish between the two classifiers in both Agreement and Skill, whereas AUC-based metrics are not.

Both classifiers attain lower values under AUCSI than under AUC, reflecting a more restrictive notion of Agreement that penalizes False Alarms more directly. While AUC-based metrics suggest near-perfect performance, CSI-based metrics reveal more moderate and differentiated classification ability.

We now evaluate which assessment better serves the objective of accurately classifying the rare event. In most applications, multiple-threshold analyses are used to guide the selection of a single threshold. Metrics such as AUC summarize performance across thresholds, but the final purpose of a classifier is to support a hard classification. For this reason, comparability between single- and multiple-threshold metrics is essential, as soft metrics should guide the final threshold selection.

To assess whether AUC and AUCSI are aligned with hard classification results, we compare Agreement and Skill across soft and hard metrics. Figure 2.11b and Table 2.5 show that both classifiers are bounded by a maximum CSI value, corresponding to the peak of their CSI curves. Classifier A reaches a maximum CSI of 60%, implying that any threshold entails at least 40% disagreement in the rare class. In contrast, Classifier B reaches at most 31% Agreement (or 69% disagreement). Similarly, Classifier B gets substantially lower Skill values of either RGSS or HSS.

This evidence allows us to answer the central question: which classifier should be selected to accurately classify the rare event? Visual inspection already suggested that Classifier A is preferable. CSI-based metrics support this conclusion, as both multiple-threshold and realistic single-threshold evaluations indicate higher Agreement and Skill for Classifier A. In contrast, AUC would treat both classifiers as equivalent. In this example, only CSI-derived metrics are consistent with the ranking implied by hard classification results.

Regardless of whether AUC or AUCSI is used, soft metrics differ in magnitude from single-threshold metrics. While CSI and RGSS values are generally below 50%, soft Agreement and Skill metrics exceed this level. This discrepancy is particularly evident for Classifier B, where hard Agreement and Skill are low (around 30% and 23%), yet soft metrics still exceed 56%.

This difference arises from two factors. First, AUC and AUCSI are primarily related to Allocation disagreement, whereas CSI and RGSS incorporate both Allocation and Quantity disagreement. Since hard metrics account for an additional source of disagreement, they tend to yield lower Agreement values. Second, soft metrics aggregate performance across multiple thresholds, while CSI and GSS correspond to specific threshold choices. Soft metrics can therefore be interpreted as an average over threshold-specific performance, which smooths extreme values.

Applying this interpretation to Classifier B clarifies the discrepancy. At the threshold where maximum CSI is achieved, disagreement is largely driven by Quantity disagreement. A maximum CSI of 33% corresponds to approximately 34% Quantity disagreement and 33% Allocation disagreement. In contrast, AUCSI isolates the Allocation component, measuring the proportion of feasible Allocation disagreement avoided by the classifier. Furthermore, although Classifier B achieves a relatively low peak CSI, it maintains similar CSI values across a wide range of thresholds. This leads to slightly higher AUCSI values despite weak performance at any specific threshold.

2.3.3.2 Comparison of multiple concave classifiers

Figure 2.12 shows the relationship between single- and multiple-threshold metrics of Agreement and Skill across the full set of simulated classifiers. Panels (a) and (b) present Agreement metrics, while panels (c) and (d) show the corresponding Skill

metrics. The color scale reflects the skewness of the curves, measured by the point at which the slope of the TOC and CSI curves equals that of the baseline, i.e., where marginal contributions to area begin to diminish. Green indicates curves skewed to the left of P , while red indicates curves skewed to the right. Bold lines represent the theoretical bounds of the metrics under concave TOC curves. Labeled points correspond to the same classifiers shown in Figure 2.11

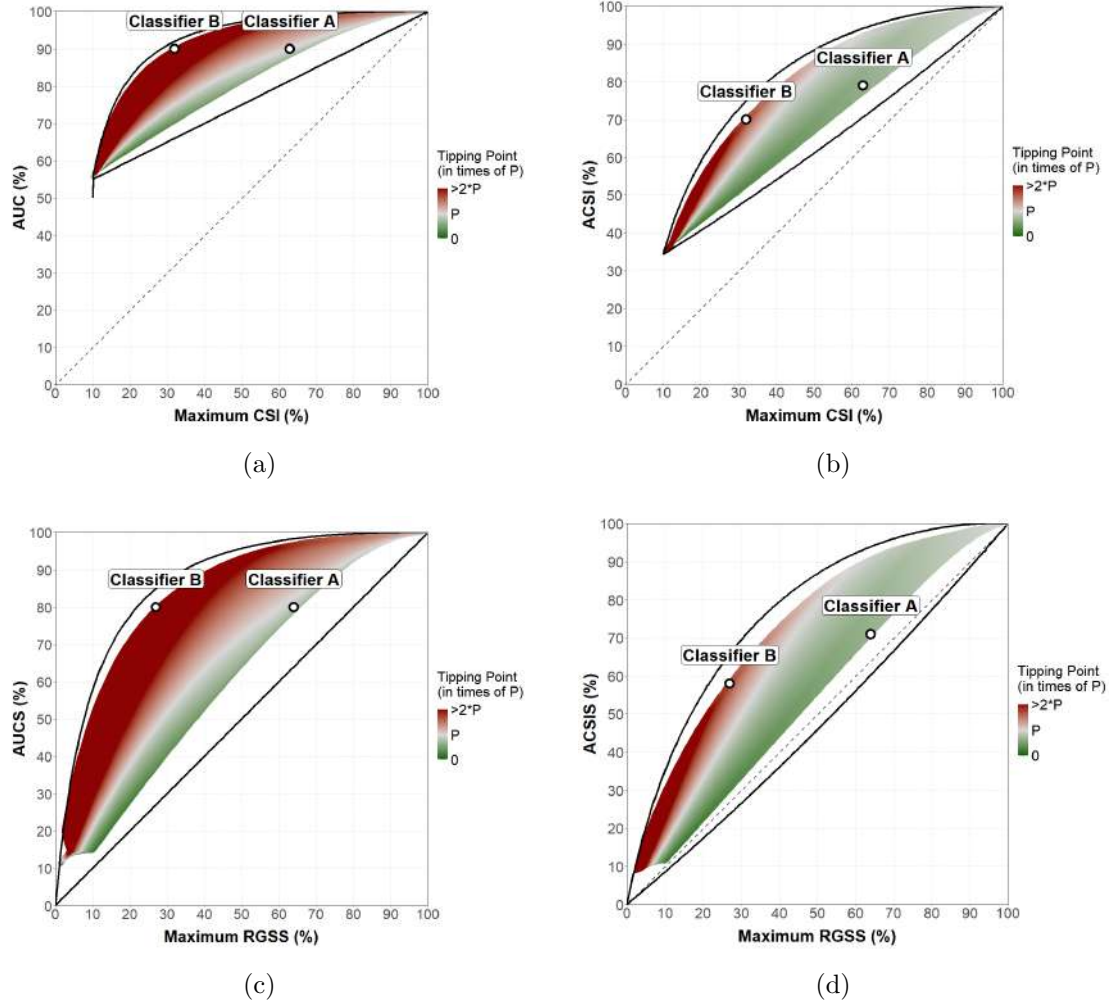


FIGURE 2.12: Relationships between single-threshold and multi-threshold metrics. Panels (a) and (b) show Agreement-based metrics, while panels (c) and (d) show Skill-based metrics. Color indicates the skewness of the curve, determined by the point at which it converges toward the baseline. Labeled points correspond to the same classifiers shown in Figure 2.11.

The figure shows that soft Agreement metrics consistently exceed their hard counterparts, for the reasons discussed before. This relationship between soft and hard metrics follows a similar pattern for both AUC and AUCSI. At low levels of Agreement, all metrics take similar values. As TOC curves separate from the baseline, a wider set of curve shapes can produce the same maximum CSI, causing the upper and lower bounds in Figure 2.12 to diverge. As classifiers approach perfection, these bounds converge to 100%, since only a limited set of curves can achieve high CSI values. For example, in Figure 2.11b, many curve configurations can yield a CSI of 50%, while far fewer can achieve 90%.

Despite these similarities, important differences emerge between AUC and AUCSI. Comparing panels (a) and (b) shows that, for a given AUCSI value, the range of possible maximum CSI values is substantially narrower than for AUC. This has direct implications for interpretation, because it means that two identical AUC values can correspond to hard classification outcomes that classify the rare event substantially differently. For instance, an AUC of 90% may be associated with CSI values ranging from approximately 29% to 80%, spanning cases of both high and low Agreement. Notably, this uncertainty is greatest for AUC values between 75% and 95%, which are often considered “good” in the literature but are, in fact, the least informative for the rare event.

In contrast, AUCSI exhibits a much tighter relationship with hard Agreement. For example, an AUCSI of 90% corresponds to CSI values ranging from approximately 53% to 88%. The slope of the boundary region in panel (b) is also closer to one, indicating a stronger alignment between AUCSI and maximum CSI. This suggests that AUCSI is more sensitive to differences in threshold-specific performance and better reflects disagreement in the rare event.

The color patterns provide additional insight. Curves with tipping points near P (light colors) achieve some of the highest AUCSI values but only moderate AUC values, confirming that AUCSI emphasizes classification ability with smaller Quantity disagreement, while AUC does not. Curves skewed to the right of P (red tones), which tend to overestimate Quantity in hard classification, achieve high values under both metrics, which is consistent with the findings of the previous sections. However, the red band is much wider under AUC, indicating a stronger Quantity bias. This shows

further evidence that AUC can assign high scores to classifiers that perform poorly in terms of rare event classification.

The figure also clarifies the behavior of Classifiers A and B. Although both achieve the same AUC, Classifier B derives most of its area from regions to the right of P , leading to low maximum CSI. It lies close to the upper bound, representing one of the worst classifiers compatible with that AUC value. In contrast, Classifier A concentrates area around P , achieving higher CSI while maintaining the same AUC. AUCSI correctly distinguishes between the two classifiers, reflecting their differences in hard classification performance.

Panels (c) and (d) present the corresponding analysis for Skill metrics. Unlike Agreement metrics, Skill metrics are normalized relative to a baseline and therefore range from 0 to 1. As in the Agreement case, soft Skill metrics tend to exceed hard Skill (RGSS). However, AUCSI Skill remains more closely aligned with RGSS than AUC Skill. For example, an AUC Skill of 80% may correspond to GSS values ranging from 22% to 80%, whereas an AUCSI Skill of 80% corresponds to a narrower range of approximately 40% to 83%.

Finally, the bounds of concave TOC curves yield an important theoretical result. For such curves, the maximum achievable CSI is bounded by the corresponding AUC, implying that:

$$\text{CSI}_{\max} \leq 2 \cdot \text{AUC} - 1.$$

This corresponds to the lower black bound in Panel (a). Equivalently, this implies a minimum disagreement of $2 - 2 \cdot \text{AUC}$, regardless of prevalence. For example, an AUC of 70% implies, even in the best case, at least 60% disagreement in one of the classes. In particular, achieving more Agreement than disagreement in one of the classes requires an AUC greater than 0.75.

The upper bound of Agreement associated with a given AUC, represented by the upper black bound in Panel (a), does depend on prevalence. This relationship is again nonlinear. As prevalence decreases, this bound expands, meaning that the same AUC value can correspond to a wider range of classification outcomes. Consequently, AUC becomes less informative in low-prevalence settings, as it fails to accurately reflect performance in the rare event classification.

2.4 Discussion

A central limitation of AUC is that it assigns similar importance to all thresholds, regardless of their relevance for the classification of the rare event. In applications where Quantity disagreement matters, this is problematic: not all correct Allocations contribute equally to the Agreement of the rare event. Correct classifications associated with small Quantity disagreements are more informative than those obtained at extreme and unrealistic thresholds. AUCSI addresses this issue by assigning greater weight to these regions, while AUC distributes weight uniformly and remains insensitive to the relative size of the rare event and the majority class, i.e., to the balance between Misses and False Alarms.

A second limitation of AUC is its insensitivity to prevalence. As prevalence decreases, the ratio P/Q becomes increasingly non-linear, yet AUC assigns a similar contribution to the area around P . Although this invariance is often considered a desirable property, it becomes a limitation when the objective is to evaluate the classification of the rare event. AUCSI adapts to this non-linearity, and the divergence between AUC and AUCSI increases as prevalence approaches zero.

These two limitations explain the differences observed between the metrics throughout this chapter. AUC measures Hits and Correct rejections relative to the size of the classes, trying to be insensitive to prevalence. In contrast, AUCSI weights Allocation Agreement according to the Quantity disagreement it entails. When prevalence is low, Quantity disagreement becomes the dominant source of error, while the relative importance of Misses decreases. In this setting, AUC tends to overestimate classification performance, as it does not account for the imbalance between Misses and False Alarms. In contrast, AUCSI remains informative and favors classifiers that concentrate their predictive ability with smaller Quantity disagreements.

Although AUCSI still incorporates unrealistic thresholds, it assigns them less weight. Partial AUC measures attempt to address this issue by restricting the range of integration using performance-based cutoffs, such as sensitivity (Peterson et al. 2008) or true positive rate (Dodd and Pepe 2003). However, these approaches introduce arbitrariness and hinder comparability across studies. In contrast, AUCSI distributes weight based solely on prevalence, providing a consistent and non-arbitrary evaluation

across thresholds.

Both AUC and AUCSI exhibit a bias toward right-skewed curves under low prevalence, which corresponds to Quantity overestimation in hard classification of the rare event. However, this bias is substantially smaller for AUCSI. This difference is particularly relevant in applications where threshold selection is required, as AUC may systematically favor classifiers that overestimate the quantity of the rare event.

Prevalence emerges as the main determinant of the differences between AUC and AUCSI. These differences are non-linear and become more pronounced as prevalence decreases. The results suggest a threshold around 25% prevalence, below which the divergence between the metrics becomes substantial. In this regime, AUCSI assigns greater relative importance to correct classifications with small Quantity disagreements, amplifying differences in classifier behavior that AUC fails to capture.

This insight connects with a broader limitation of evaluation metrics. Prevalence is not only relevant for AUC and AUCSI, but for any agreement metric, including OA, CSI, or any other studied in Chapter 1. When prevalence is low, Overall Agreement can be maximized by underpredicting the rare event, leading to misleadingly high scores. In contrast, CSI is maximized under overestimation of the rare event. This behavior highlights the importance of selecting metrics aligned with the research objective.

The extensive analysis of this chapter underscores some practical implications of the misuse of AUC, and the amendment of using the CSI framework. It becomes clear that when the objective is to rank classifiers in low-prevalence settings, AUC often fails to discriminate between models with different classification ability at relevant thresholds. Two classifiers may achieve identical AUC values while exhibiting substantial differences in their ability to correctly identify the rare event. In contrast, AUCSI more effectively distinguishes classifiers that generate Hits in realistic regions of the threshold space.

This limitation of AUC is particularly relevant in fields where events are intrinsically rare, such as species distribution modeling (Phillips et al. 2006), rare disease detection (Garon et al. 2015; Movahedi et al. 2023), and medical diagnostics (Gao et al. 2018). Despite this, AUC remains widely used in these contexts, often leading

to overestimation of performance and suboptimal threshold selection. AUCSI provides a more appropriate alternative by aligning Allocation Agreement with Quantity disagreement.

AUC also struggles to compare classifiers across populations with different prevalences. Small changes in prevalence can produce large changes in the ratio P/Q , especially below the 25% threshold. This limits the comparability of AUC across studies. In contrast, AUCSI provides a more consistent basis for comparison, as it adapts to prevalence.

This issue is especially relevant in applications where prevalence varies across space, time, or data partitions. For example, in epidemiology (Reid et al. 2012) and ecology (Hanberry and He 2013), prevalence can differ substantially between regions or periods, as well as between training and validation datasets. In such cases, AUC may yield inconsistent evaluations, whereas AUCSI maintains interpretability.

Finally, CSI curves offer additional analytical advantages beyond ROC and TOC curves. They provide a direct visualization of the contributions of Allocation and Quantity disagreement across thresholds, allowing researchers to identify thresholds that best align with their objectives and to assess improvements relative to a baseline.

More importantly, CSI curves provide a unified framework for the joint interpretation of hard and soft validation. Any hard classification corresponds to a single point on the CSI curve, while the full curve summarizes performance across all possible thresholds. This allows both evaluations to be interpreted using the same definition of Agreement and within the same conceptual space.

This unified perspective informs about key elements for the evaluation of binary classifications: It makes prevalence and Quantity disagreement explicit and provides directly comparable hard metrics (CSI and its Skill) and soft metrics (AUCSI and its Skill) within a single analytical structure. By doing so, it moves beyond the use of isolated summary metrics and instead frames validation as the analysis of how classification performance evolves across thresholds. As a result, it offers a tool for more more consistent and transparent classification evaluation and communication.

2.5 Conclusion

In this chapter, we propose AUCSI as an alternative metric to AUC for the assessment of binary classifiers for rare events. Our findings illustrate how the two metrics differ in their definition of Agreement: AUC considers all the thresholds as similar contributors to Agreement, which distorts its properties when prevalence is low; meanwhile, AUCSI ranks the Agreement relative to the Quantity disagreement they entail, thus prioritizing classifiers with classification in realistic classifications.

The findings suggest that AUCSI is better suited than AUC for applications where prevalence is low, a common characteristic in many research fields. The introduction of CSI graphs as complements to TOC graphs presents a valuable tool for researchers. Both graphs provide more useful information than ROC because they allow the visualization of Allocation and Quantity disagreements of each of the thresholds. This is a crucial advancement in the evaluation of the classifier’s performance across different thresholds, as Allocation and Quantity disagreement are particularly relevant for studies where the type of disagreement needs to be carefully considered. Our proposal contributes to both multiple and single threshold evaluation.

This research aligns with previous work emphasizing that the choice of evaluation metrics should depend on the research question and the structural characteristics of the problem (Hand 2012; Pontius 2022). In this context, we recommend the use of AUCSI over AUC in applications with low prevalence where Quantity disagreement is relevant. We also recommend AUCSI for comparing classifiers across populations with different prevalences when the rare event is scarce. More broadly, we recommend interpreting the four metrics proposed in this framework (CSI, RGSS, AUCSI, and AUCSIS) within their graphical representation. This joint interpretation of curves and summary metrics enables a clearer communication of results, links threshold-specific behavior with aggregated performance, and moves beyond the limitations of isolated summary metrics.

References

- Baldwin, Michael E. and John S. Kain (Aug. 2006). “Sensitivity of Several Performance Measures to Displacement Error, Bias, and Event Frequency.” en. In: *Weather and Forecasting* 21.4. DOI: <https://doi.org/10.1175/WAF933.1>. URL: https://journals.ametsoc.org/view/journals/wefo/21/4/waf933_1.xml (visited on 08/01/2024).
- Dodd, Lori E. and Margaret S. Pepe (Sept. 2003). “Partial AUC Estimation and Regression”. en. In: *Biometrics* 59.3, pp. 614–623. ISSN: 0006-341X, 1541-0420. DOI: [10.1111/1541-0420.00071](https://doi.org/10.1111/1541-0420.00071). URL: <https://academic.oup.com/biometrics/article/59/3/614-623/7286731> (visited on 02/01/2024).
- Ebert, Philip A. and Peter Milne (Feb. 2022). “Methodological and conceptual challenges in rare and severe event forecast verification”. English. In: *Natural Hazards and Earth System Sciences* 22.2. Publisher: Copernicus GmbH, pp. 539–557. ISSN: 1561-8633. DOI: [10.5194/nhess-22-539-2022](https://doi.org/10.5194/nhess-22-539-2022). URL: <https://nhess.copernicus.org/articles/22/539/2022/> (visited on 09/10/2024).
- Farrington, David P. and Rolf Loeber (Sept. 1989). “Relative improvement over chance (RIOCI) and phi as measures of predictive efficiency and strength of association in 2x2 tables”. en. In: *Journal of Quantitative Criminology* 5.3, pp. 201–213. ISSN: 0748-4518, 1573-7799. DOI: [10.1007/BF01062737](https://doi.org/10.1007/BF01062737). URL: <http://link.springer.com/10.1007/BF01062737> (visited on 09/12/2024).
- Fawcett, Tom (June 2006). “An introduction to ROC analysis”. en. In: *Pattern Recognition Letters* 27.8, pp. 861–874. ISSN: 01678655. DOI: [10.1016/j.patrec.2005.10.010](https://doi.org/10.1016/j.patrec.2005.10.010). URL: <https://linkinghub.elsevier.com/retrieve/pii/S016786550500303X> (visited on 01/08/2024).
- Gao, Ting et al. (Oct. 2018). “Predicting pathological response to neoadjuvant chemotherapy in breast cancer patients based on imbalanced clinical data”. en. In: *Personal and Ubiquitous Computing* 22.5, pp. 1039–1047. ISSN: 1617-4917. DOI: [10.1007/s00779-018-1144-3](https://doi.org/10.1007/s00779-018-1144-3). URL: <https://doi.org/10.1007/s00779-018-1144-3> (visited on 06/07/2024).
- Garon, Edward B. et al. (2015). “Pembrolizumab for the Treatment of Non-Small-Cell Lung Cancer”. In: *New England Journal of Medicine* 372.21, pp. 2018–2028.

- DOI: [10.1056/NEJMoa1501824](https://doi.org/10.1056/NEJMoa1501824). URL: <https://www.nejm.org/doi/full/10.1056/NEJMoa1501824> (visited on 06/06/2024).
- Gilbert, G.K. (1884). “Finley’s Tornado Predictions”. In: *American Meteorological Journal* 1.5, pp. 166–172. URL: https://archive.org/details/sim-american-meteorological-journal_1884-09_1_5/page/168/mode/2up?view=theater.
- Green, D.M. and John A. Swets (1966). *Signal detection theory and psychophysics*. New York: Wiley Press.
- Hamill, Thomas M. and Josip Juras (2006). “Measuring forecast skill: is it real skill or is it the varying climatology?” en. In: *Quarterly Journal of the Royal Meteorological Society* 132.621C, pp. 2905–2923. ISSN: 1477-870X. DOI: [10.1256/qj.06.25](https://doi.org/10.1256/qj.06.25). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1256/qj.06.25> (visited on 08/16/2024).
- Hanberry, B. B. and H. S. He (May 2013). “Prevalence, statistical thresholds, and accuracy assessment for species distribution models”. en. In: *Web Ecology* 13.1, pp. 13–19. ISSN: 1399-1183. DOI: [10.5194/we-13-13-2013](https://doi.org/10.5194/we-13-13-2013). URL: <https://web.copernicus.org/articles/13/13/2013/> (visited on 06/07/2024).
- Hand, David J. (2012). “Assessing the Performance of Classification Methods”. en. In: *International Statistical Review* 80.3, pp. 400–414. ISSN: 1751-5823. DOI: [10.1111/j.1751-5823.2012.00183.x](https://doi.org/10.1111/j.1751-5823.2012.00183.x). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1751-5823.2012.00183.x> (visited on 09/04/2024).
- Kessler, R. C., G. Andrews, et al. (Aug. 2002). “Short screening scales to monitor population prevalences and trends in non-specific psychological distress”. en. In: *Psychological Medicine* 32.6, pp. 959–976. ISSN: 1469-8978, 0033-2917. DOI: [10.1017/S0033291702006074](https://doi.org/10.1017/S0033291702006074). URL: <https://www.cambridge.org/core/journals/psychological-medicine/article/short-screening-scales-to-monitor-population-prevalences-and-trends-in-nonspecific-psychological-distress/F141675CCD0E08C0FB98E01C006B4E0D> (visited on 06/06/2024).
- Kessler, R.C., P. Berglund, et al. (2005). “Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the national comorbidity survey replication”. English. In: *Archives of General Psychiatry* 62.6, pp. 593–602. ISSN: 1538-3636. DOI: [10.1001/archpsyc.62.6.593](https://doi.org/10.1001/archpsyc.62.6.593).

- Lobo, Jorge M., Alberto Jiménez-Valverde, and Raimundo Real (Mar. 2008). “AUC: a misleading measure of the performance of predictive distribution models”. en. In: *Global Ecology and Biogeography* 17.2, pp. 145–151. ISSN: 1466-822X, 1466-8238. DOI: [10.1111/j.1466-8238.2007.00358.x](https://doi.org/10.1111/j.1466-8238.2007.00358.x). URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1466-8238.2007.00358.x> (visited on 04/02/2024).
- Loeber, Rolf and Thomas J. Dishion (1983). “Early predictors of male delinquency: A review”. In: *Psychological Bulletin* 94.1, pp. 68–99. DOI: [10.1037/0033-2909.94.1.68](https://doi.org/10.1037/0033-2909.94.1.68). URL: <https://doi.org/10.1037/0033-2909.94.1.68>.
- Mbizvo, Gashirai K. et al. (2023). “Using Critical Success Index or Gilbert Skill Score as composite measures of positive predictive value and sensitivity in diagnostic accuracy studies: Weather forecasting informing epilepsy research”. en. In: *Epilepsia* 64.6, pp. 1466–1468. ISSN: 1528-1167. DOI: [10.1111/epi.17537](https://doi.org/10.1111/epi.17537). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/epi.17537> (visited on 07/19/2024).
- Mesinger, Fedor and Keith Brill (2004). “Bias normalized precipitation scores”. en. In: *17th Conf. on Probability and Statistics*. Vol. CD-ROM, J12.6. Seattle, WA, USA: Preprints.
- Movahedi, Faezeh, Rema Padman, and James F Antaki (Apr. 2023). “Limitations of receiver operating characteristic curve on imbalanced data: Assist device mortality risk scores”. In: *The Journal of thoracic and cardiovascular surgery* 165.4, 1433–1442.e2. ISSN: 0022-5223. DOI: [10.1016/j.jtcvs.2021.07.041](https://doi.org/10.1016/j.jtcvs.2021.07.041). URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8800945/> (visited on 06/06/2024).
- Murphy, Allan H. (Mar. 1996). “The Finley Affair: A Signal Event in the History of Forecast Verification”. en. In: *Weather and Forecasting* 11.1. Section: Weather and Forecasting. ISSN: 1520-0434. URL: https://journals.ametsoc.org/view/journals/wefo/11/1/1520-0434_1996_011_0003_tfaase_2_0_co_2.xml (visited on 09/05/2024).
- Parodi, Stefano et al. (Oct. 2022). “The clinical meaning of the area under a receiver operating characteristic curve for the evaluation of the performance of disease markers”. In: *Epidemiology and Health* 44, e2022088. ISSN: 2092-7193. DOI: [10.4178/epih.e2022088](https://doi.org/10.4178/epih.e2022088). URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10089712/> (visited on 06/06/2024).

- Peterson, A. Townsend, Monica Papeş, and Jorge Soberón (Apr. 2008). “Rethinking receiver operating characteristic analysis applications in ecological niche modeling”. In: *Ecological modeling* 213.1, pp. 63–72. ISSN: 0304-3800. DOI: [10.1016/j.ecolmodel.2007.11.008](https://doi.org/10.1016/j.ecolmodel.2007.11.008). URL: <https://www.sciencedirect.com/science/article/pii/S0304380007006163> (visited on 08/15/2024).
- Phillips, Steven J., Robert P. Anderson, and Robert E. Schapire (Jan. 2006). “Maximum entropy modeling of species geographic distributions”. In: *Ecological modeling* 190.3, pp. 231–259. ISSN: 0304-3800. DOI: [10.1016/j.ecolmodel.2005.03.026](https://doi.org/10.1016/j.ecolmodel.2005.03.026). URL: <https://www.sciencedirect.com/science/article/pii/S030438000500267X> (visited on 06/06/2024).
- Pontius, Robert Gilmore (Sept. 2018). “Criteria to Confirm Models that Simulate Deforestation and Carbon Disturbance”. en. In: *Land* 7.3, p. 105. ISSN: 2073-445X. DOI: [10.3390/land7030105](https://doi.org/10.3390/land7030105). URL: <http://www.mdpi.com/2073-445X/7/3/105> (visited on 09/02/2023).
- (2022). *Metrics That Make a Difference: How to Analyze Change and Error*. en. Advances in Geographic Information Science. Cham: Springer International Publishing. ISBN: 978-3-030-70764-4. DOI: [10.1007/978-3-030-70765-1](https://doi.org/10.1007/978-3-030-70765-1). URL: <https://link.springer.com/10.1007/978-3-030-70765-1> (visited on 01/08/2024).
- Pontius, Robert Gilmore, Wideke Boersma, et al. (Mar. 2008). “Comparing the input, output, and validation maps for several models of land change”. en. In: *The Annals of Regional Science* 42.1, pp. 11–37. ISSN: 0570-1864, 1432-0592. DOI: [10.1007/s00168-007-0138-2](https://doi.org/10.1007/s00168-007-0138-2). URL: <http://link.springer.com/10.1007/s00168-007-0138-2> (visited on 06/04/2024).
- Pontius, Robert Gilmore and Benoit Parmentier (Mar. 2014). “Recommendations for using the relative operating characteristic (ROC)”. en. In: *Landscape Ecology* 29.3, pp. 367–382. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/s10980-013-9984-8](https://doi.org/10.1007/s10980-013-9984-8). URL: <http://link.springer.com/10.1007/s10980-013-9984-8> (visited on 01/08/2024).
- Pontius, Robert Gilmore, Smitha Peethambaram, and Jean-Christophe Castella (Jan. 2011). “Comparison of Three Maps at Multiple Resolutions: A Case Study of Land Change Simulation in Cho Don District, Vietnam”. en. In: *Annals of the Association of American Geographers* 101.1, pp. 45–62. ISSN: 0004-5608, 1467-8306. DOI:

- 10.1080/00045608.2010.517742. URL: <http://www.tandfonline.com/doi/abs/10.1080/00045608.2010.517742> (visited on 01/31/2023).
- Pontius, Robert Gilmore and Kangping Si (Mar. 2014). “The total operating characteristic to measure diagnostic ability for multiple thresholds”. In: *International Journal of Geographical Information Science* 28.3, pp. 570–583. ISSN: 1365-8816. DOI: 10.1080/13658816.2013.862623. URL: <https://doi.org/10.1080/13658816.2013.862623> (visited on 10/09/2023).
- Powers, David M. W. (2011). “Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness and Correlation”. In: *Journal of Machine Learning Technologies* 2.1, pp. 37–63.
- Provost, Foster and Tom Fawcett (1997). “Analysis and visualization of classifier performance: comparison under imprecise class and cost distributions”. In: *Proceedings of the Third International Conference on Knowledge Discovery and Data Mining. KDD’97*. Place: Newport Beach, CA. AAAI Press, pp. 43–48.
- Reid, Heidi L. et al. (May 2012). “Characterizing the spatial and temporal variation of malaria incidence in Bangladesh, 2007”. In: *Malaria Journal* 11.1, p. 170. ISSN: 1475-2875. DOI: 10.1186/1475-2875-11-170. URL: <https://doi.org/10.1186/1475-2875-11-170> (visited on 06/07/2024).
- Saito, Takaya and Marc Rehmsmeier (Mar. 2015). “The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets”. en. In: *PLOS ONE* 10.3. Ed. by Guy Brock, e0118432. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0118432. URL: <https://dx.plos.org/10.1371/journal.pone.0118432> (visited on 08/14/2024).
- Schaefer, Joseph T. (1990). “The Critical Success Index as an Indicator of Warning Skill”. en. In: *Weather and Forecasting* 5.4. DOI: [https://doi.org/10.1175/1520-0434\(1990\)005<0570:TCSIAA>2.0.CO;2](https://doi.org/10.1175/1520-0434(1990)005<0570:TCSIAA>2.0.CO;2). URL: https://journals.ametsoc.org/view/journals/wefo/5/4/1520-0434_1990_005_0570_tcsiaa_2_0_co_2.xml (visited on 08/01/2024).
- Sitthiyot, Thitithep and Kanyarat Holasut (2022). “On the Evaluation of Skill in Binary Forecast”. en. In: *Thailand and The World Economy* 40.3.
- Sofaer, Helen R., Jennifer A. Hoeting, and Catherine S. Jarnevich (2018). “The area under the precision-recall curve as a performance metric for rare binary events”.

- In: *Methods in Ecology and Evolution* 9.4, pp. 678–685. DOI: [10.1111/2041-210X.13140](https://doi.org/10.1111/2041-210X.13140). URL: <https://doi.org/10.1111/2041-210X.13140>.
- Varga, Orsolya Gyöngyi et al. (June 2019). “Intensity Analysis and the Figure of Merit’s components for assessment of a Cellular Automata – Markov simulation model”. en. In: *Ecological Indicators* 101, pp. 933–942. ISSN: 1470160X. DOI: [10.1016/j.ecolind.2019.01.057](https://doi.org/10.1016/j.ecolind.2019.01.057). URL: <https://linkinghub.elsevier.com/retrieve/pii/S1470160X19300743> (visited on 12/13/2022).
- Wang, Chung-Chieh (Aug. 2014). “On the Calculation and Correction of Equitable Threat Score for Model Quantitative Precipitation Forecasts for Small Verification Areas: The Example of Taiwan”. en. In: *Weather and Forecasting* 29.4. Section: Weather and Forecasting. DOI: [10.1175/WAF-D-13-00087.1](https://doi.org/10.1175/WAF-D-13-00087.1). URL: https://journals.ametsoc.org/view/journals/wefo/29/4/waf-d-13-00087_1.xml (visited on 08/14/2024).

Chapter 3

The CSI Framework to Address Misconceptions and Improve Practices in Land Change Model Validation

Preface

This chapter follows directly from the framework developed in Chapter 2. While the previous chapter establishes a general approach to the evaluation of binary classifications, the purpose of this chapter is to apply that framework to the specific case of LCM validation.

In LCMs, the problem of validation can be understood as a binary classification task, where the events of interest are the presence or absence of land changes, and model outputs are compared with observed outcomes. Moreover, as discussed in Chapter 1, land change is an inherently rare phenomenon. This connection makes LCMs a natural setting for the application of the tools and concepts introduced in Chapter 2.

The development of validation practices in LCM has largely taken place within its own field. In contrast to other disciplines, these practices have remained relatively stable over time and have not fully incorporated the debates, innovations, and methodological advances developed elsewhere. Regardless of the reasons behind this, this gap is one of the main motivations of this thesis. This chapter, therefore, represents a central component of the dissertation, aiming to provide clear and appropriate tools to support the adaptation of LCM validation practices to current standards in binary classification evaluation.

Within the structure of the thesis, this chapter acts as the link between methodological development and practical application. First, it brings together advances from different disciplines in the evaluation of binary classifications, discussed in Chapter 1, and presents them in a way that highlights their implications for current LCM validation practices. Second, it translates the general results from Chapter 2 into the LCM context, proposing a validation approach that is better suited to the specific characteristics of land change. Finally, it offers an initial discussion of the empirical implications of the CSI framework, laying the foundation for the applied analysis developed in Chapter 4.

3.1 Introduction

Land use is a central component of territorial dynamics, shaping ecological processes, economic activity, and social interactions. Changes in land use alter landscape structure, environmental quality, and the spatial distribution of human activities, making them a key element in the study of coupled human–natural systems (Lambin and Meyfroidt 2011; Turner et al. 2007). To anticipate future pressures on land and to support planning and environmental assessment, a wide range of spatially explicit models has been developed. Among them, pattern-based land change models (LCMs) are now widely used to explore the dynamics of change, assess alternative futures, and inform policy decisions (Brown, Verburg, et al. 2013; Verburg et al. 2013).

As LCMs have gained influence in policy and planning contexts, concerns about the reliability of their outputs have become increasingly important (Müller et al. 2014). Validation plays a critical role in this regard, as it provides the basis for interpreting model results and for comparing simulated patterns to reference data patterns that can be trusted for analysis or decision-making. Despite this importance, validation is often treated as a procedural step rather than as a core component of model evaluation and interpretation (Magliocca et al. 2023). Empirical reviews show that validation practices in LCMs are frequently incomplete or poorly aligned with the characteristics of land change processes. For example, van Vliet, Bregt, Brown, et al. (2016) found that nearly one third of LCM studies did not report any form of validation. Tong and Feng (2020) further shows that many studies that reported validation relied on evaluation procedures poorly suited to the dynamics of land change. Similarly, Güneralp and Ahasan (2025) found that nearly half of the reviewed papers reported either outdated accuracy measures or no accuracy assessment at all.

A growing methodological literature has identified fundamental problems with the validation metrics most commonly used in LCM applications (Pontius, Castella, et al. 2018; Pontius and Si 2014; Vliet, Bregt, and Hagen-Zanker 2011; Vliet, Bregt, Brown, et al. 2016; Tong and Feng 2020; Pontius and Millones 2011; Lobo et al. 2008; Shao et al. 2019; Stehman and Foody 2019; Olofsson, Foody, Herold, et al. 2014). Measures such as overall agreement, Kappa coefficient, and the area under the ROC curve (AUC) are widely reported, yet they tend to provide overly optimistic

assessments of model performance because land change is a rare event (Hand and Anagnostopoulos 2013). As most locations do not change between observation time points, these metrics are dominated by the correct classification of land use persistence and are therefore misleading about a model's ability to locate actual change (Vliet, Bregt, and Hagen-Zanker 2011). Moreover, many commonly used metrics focus exclusively on observed agreement, without accounting for the level of agreement that can be attributed to the model performance. Under these conditions, commonly reported validation results often fail to distinguish models that meaningfully represent land change from models that do not (Pontius, Castella, et al. 2018). This disconnect between methodological understanding and routine practice can lead to misleading conclusions about model performance, overlook the size of model error, and ultimately weaken the evidence base used to support planning decisions (Shao et al. 2019).

The goal of this chapter is to support more consistent, transparent, and informative validation practices that better reflect the ability of models to represent land change patterns. To do so, the chapter first demonstrates the limitations of current validation practices in LCMs and then proposes the use of the CSI framework to improve how validation is conducted. For consistency with terminology in land use change science, CSI is now referred to as figure of merit (FOM), as well as the terms ACUSI as AUFOM and the CSI framework as FOM framework.

Through an application to a case study in Navarra, Spain, we illustrate how validation results are often driven more by validation practices, chosen metrics, and data characteristics than by model performance. The chapter shows why commonly used metrics such as overall agreement, Kappa, and AUC are poorly suited for LCM validation. The chapter proposes the adoption of the FOM framework, which introduces new metrics and structural validation in a way that is consistent with the characteristics of land change patterns. The framework allows for a clearer and more transparent assessment of model behavior by focusing explicitly on change, integrating hard and soft validation within a single approach, and interpreting agreement relative to a baseline level of performance.

3.2 Current LCM validation approaches and their limitations

As overviewed in Chapter 1, validation of LCM is commonly performed through direct comparison between simulated and observed maps. The most frequent blunder in validation is to compare the observed map at time t_1 with the simulated map at the same time point (Liu, Liu, et al. 2024; Tholo et al. 2026; Wu et al. 2021; Sankarrao et al. 2021). If the research goal is to simulate land change, this practice is conceptually inappropriate because it evaluates agreement in land presence rather than the model’s ability to reproduce change. The literature shows that this practice leads to unrealistically high validation results because land change is usually rare and land use persistence is easy to predict (Pontius, Castella, et al. 2018; Vliet, Bregt, and Hagen-Zanker 2011). As most locations persist over time, a model can achieve high agreement simply by reproducing persistence, even if the model performs poorly in simulating actual land changes. We will refer to this practice as land presence validation.

To rectify this flaw, a more appropriate approach is to validate changes during a time interval rather than land presence at a time point. The proper method involves comparing observed changes between t_0 and t_1 with the simulated changes between t_0 and t_1 . We will refer to this practice as land change validation.

Hard classified maps are validated using a contingency table, which distinguishes between correct and incorrect simulations. For land presence validation, a correct simulation consists of the agreement between the observed and simulated land presence categories. For land change validation, there are four possible outcomes, as we illustrated in Table 1.5.

A wide range of validation metrics used in LCMs are derived from these contingency tables. One of the most commonly reported measures is OA, which represents where simulated and observed outcomes match as a proportion of the study area. For land presence validation, OA is computed as the proportion of the extent with a correct land presence. For land change validation, OA is computed to express correctly simulated change and persistence as a proportion of all locations. Regardless of whether land presence or change is validated, OA primarily reflects the model’s

ability to reproduce persistence rather than its ability to simulate change (Shao et al. 2019). In land presence validation, most of the agreement arises from land that did not change, meaning that even a null model that predicts persistence everywhere would achieve high agreement. In land change validation, most of the agreement ($H + C$) is driven by correctly simulated persistence (C), which is the most frequent observation. Consequently, OA systematically reports high values in LCM applications, even when models provide little information about the spatial distribution of land change.

Many LCM applications incorporate the Kappa coefficient (so far referred to as HSS) to overcome the limitations of OA. Kappa introduces a reference level of OA expected under a random allocation of land presence and measures how much the model's OA exceeds this baseline. In this sense, the adoption of Kappa can be interpreted as an early attempt to move beyond pure agreement metrics toward skill-based evaluation.

In its practical application to LCMs, however, Kappa often fails to measure model performance effectively (Pontius and Millones 2011). Kappa is typically applied to land presence rather than land change, which mechanically leads to a weak and uninformative baseline (Vliet, Bregt, and Hagen-Zanker 2011). Although most locations are far more likely to maintain their previous land category than to change, the random allocation baseline of Kappa assumes that a proportion of locations at the final time point have their land category randomly reassigned. When this baseline map is overlaid with the initial map, it produces a large amount of simulated land change, something unlikely to occur in reality. As a result, this baseline tends to achieve lower agreement than even a naïve persistence model, making almost any model appear skillful by comparison. Consequently, Kappa can report high values while offering little insight into the model's ability to represent land change patterns.

Similar limitations arise in soft validation. Soft validation assesses whether higher values of a suitability map are associated with locations where land presence or land change is observed, without committing to a specific categorical allocation. The most widely used metric for this purpose is the Area Under the Receiver Operating Characteristic Curve (AUC). Beyond the problems associated with validating land

presence instead of changes, AUC introduces an additional limitation in LCM validation. Although AUC is often interpreted as a measure of predictive performance, it is strongly influenced by class imbalance and can report unrealistically high values in such cases (Hand and Anagnostopoulos 2013)). For this reason, its use has been questioned or avoided in several application domains, such as disease detection, meteorology, machine learning, or species distribution modeling (Lobo et al. 2008; Ma et al. 2013; Hand 2009). Because land change is inherently rare, AUC is dominated by the majority class of persistence and therefore mainly reflects the model’s ability to simulate persistence correctly. As a result, AUC tends to report high values in LCM applications even when models fail to accurately capture the spatial pattern of land change.

3.3 Methods

To illustrate the concepts developed in this chapter, we use a case study of land change in the region of Navarra, Spain. Figure 3.1 shows Forest presence in 2005 (a) and 2023 (b). In the remainder of this chapter, land presence refers to a land use at a time point, while land change refers to the change during a time interval. Forest presence in (b) is decomposed into Forest persistence and Forest gains between 2005 and 2023.

Table 3.1 summarizes the overlay of Forest and Non Forest presence between these two maps. For land presence validation, we consider the presence and absence of Forest as the reference binary variable. Here, the extent of analysis covers 10384 km², of which 3858 km² was Forest in 2023, resulting in prevalence of approximately 37%. Prevalence is defined as the proportion of observed units of the event of interest as a proportion of the total number of units considered in the analysis. For change validation, we focus on the presence or absence of Forest gains, defined as locations classified as Non-Forest in 2005 and Forest in 2023. In this case, the extent of analysis covers approximately 6800 km², of which about 350 km² experienced Forest gains during the observed interval, resulting in prevalence of approximately 5%. Forest presence in 2005 is excluded from the extent because these locations are not eligible for Forest gains.

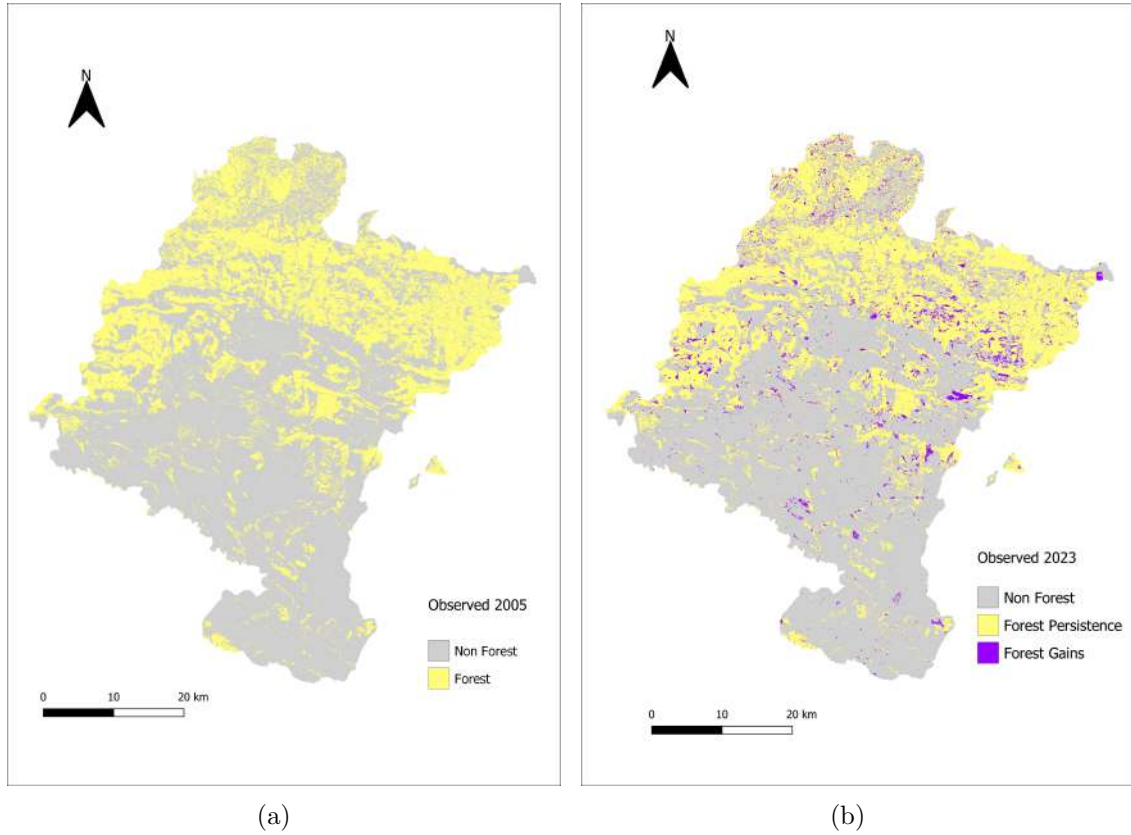


FIGURE 3.1: Observed Forest and Non-Forest presence at 2005 (a) and 2023 (b). Forest presence in (b) is decomposed into Forest persistence and Forest gains.

TABLE 3.1: Contingency table comparing the observed land presence in 2005 and 2023 in km^2 .

		2023		
		Non Forest	Forest	Sum
2005	Forest	3508	73	3581
	Non Forest	350	6453	6803
Sum		3858	6526	10384

Figure 3.2 shows the simulated soft and hard maps used in the illustration. Soft maps, or suitability maps, represent a continuous value of suitability of each location for Forest presence or Forest gain, while hard maps are categorical maps in which each location is assigned to a single class. The upper maps represent a simulation

focused on Forest presence in 2023. The suitability map 3.2(a) is derived from a Random Forest regression in which Forest presence is the dependent variable and altitude and annual precipitation are used as explanatory variables. The resulting map represents the relative suitability of each location for Forest according to the model. Panel 3.2(b) shows the corresponding hard-classified map used for validation, which is derived from the suitability map.

The lower maps represent a simulation focused on Forest gains during the period 2005–2023. The suitability map is identical to the former, except that locations with Forest presence in 2005 are excluded. Panel 3.2(d) shows the corresponding hard classified map, which is constrained to the same total quantity of simulated Forest presence as (b) to facilitate comparisons.

Using this setup as a common basis, we conduct a comparative analysis between conventional LUCM validation metrics and the CSI framework. The analysis is performed using the same observed maps and simulated outputs described above, ensuring that all comparisons are based on an identical set of inputs. This allows the evaluation results to be directly comparable across different validation approaches.

3.4 Results

3.4.1 Validation with conventional practices

Figure 3.3 represents the soft validation using the conventional approach with ROC curves. Panel (a) shows the ROC curve obtained when land presence in 2023 is validated, that is, by directly comparing maps 1(b) and 2(a). Panel (b) shows the ROC curve obtained when land change is validated through maps 1(a), 1(b), and 2(c). In both panels, the blue square represents the validation results obtained from the hard classification (Fig 3.2b and d).

Figure 3.4 and Table 3.2 show the hard validated results. The figure represents the agreement and disagreement in Forest presence (a) and Forest gains (b). The table represents the resulting confusion matrix for Forest presence, corresponding to the blue square marker on the curve in panel (a), and the one for Forest gain, corresponding to the marker in panel (b).

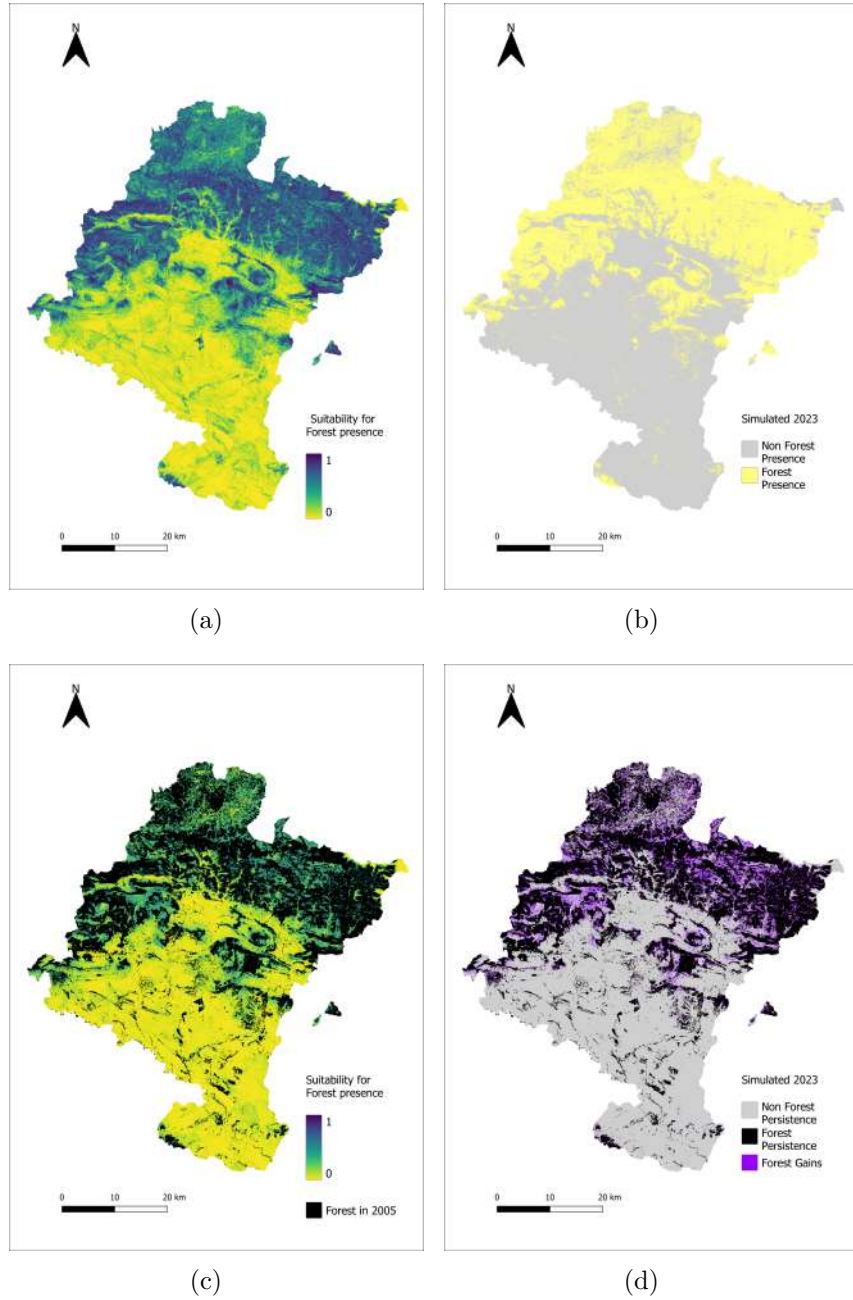


FIGURE 3.2: Simulated soft and hard maps for Forest presence (a)–(b) and Forest gains (c)–(d). Panels (a)–(b) show the suitability map and corresponding hard-classified map for Forest presence in 2023. Panels (c)–(d) show the suitability map and corresponding hard-classified map for Forest gains during 2005–2023. In the lower suitability map, locations with Forest presence in 2005 are excluded. The hard-classified maps are constrained to the same total quantity of simulated Forest presence to facilitate comparison.

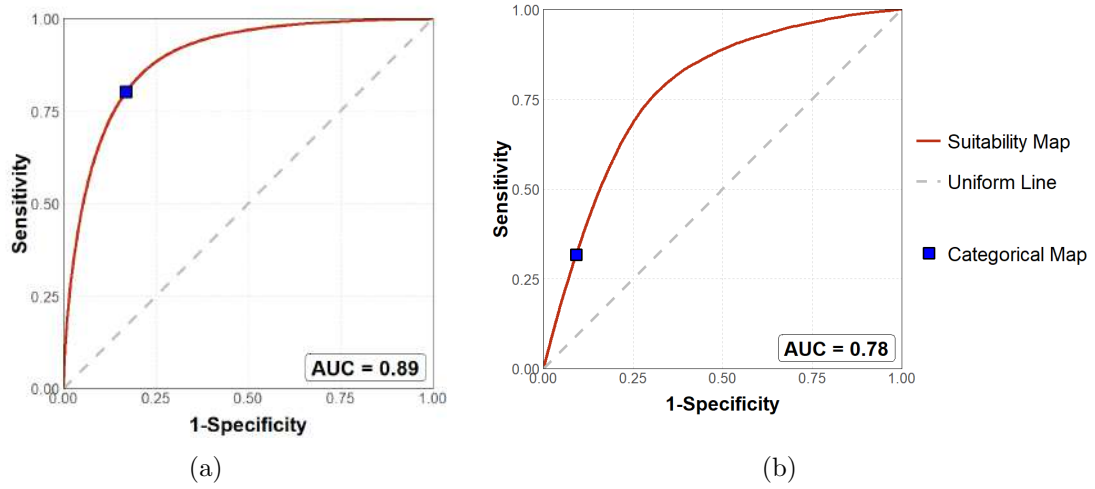


FIGURE 3.3: ROC curves corresponding to the validation of Forest presence (a) and Forest gain (b) of the suitability and categorical maps.

If we relied exclusively on these results, we would conclude that the model performs very well. When validating land presence, the suitability map achieves an AUC of 0.89, a value that is commonly interpreted as excellent. Under this interpretation, one could conclude that the suitability map reproduces the spatial distribution of Forest accurately, which it does, but the reason is persistence, not change. The hard classification reinforces this impression, with an OA of 82%, indicating that a large share of the map is classified correctly. As an additional robustness check, Kappa also reports a high value of 0.67, suggesting that the model performs substantially better than a random allocation.

TABLE 3.2: Contingency table of observed and simulated Forest presence and Forest gain in km².

		Observed						
		Forest	Non Forest	Sum		Gain	Non Gain	Sum
Simulated	Forest	3129	1151	4280	Gain	102	596	698
	Non Forest	729	5375	6104	No Gain	247	5858	6105
	Sum	3858	6526	10384		349	6454	6803
		OA = 82%, Kappa = 67%			OA = 88%, FOM = 11%, Kappa = 14%			

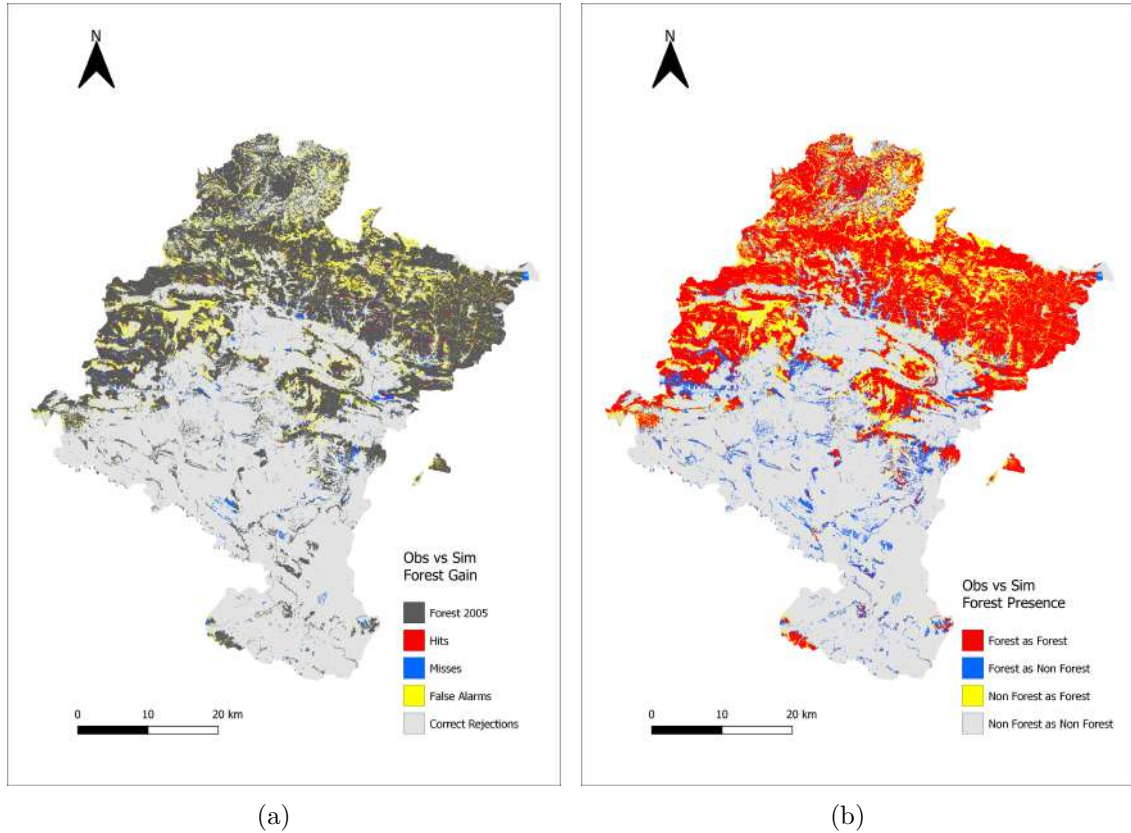


FIGURE 3.4: Agreement and disagreement between observed and simulated Forest presence (a) and Forest gain (b). In Panel (b) forest in 2005 is excluded from the extent as it is not eligible for Forest gain Source: Self elaboration.

Panel b appears to reinforce these conclusions. An AUC of 0.78 could be interpreted as evidence that the model captures not only the spatial distribution of Forest, but also the locations of Forest gains. The hard classification again reports a very high OA of 88%, even higher than when validating land presence alone. Based on conventional validation procedures, one would therefore conclude that the model performs very well in representing both Forest presence and Forest gains.

However, these favorable results reflect the limitations of the validation procedures rather than the actual performance of the model. A simple visual inspection of the maps in Figure 3.4 reveals the problem: even without using any metric, it is clear that hard simulation (a) does not represent observations accurately, because the Agreement in Forest presence at 2023 mainly corresponds to areas that were already Forest in 2005, while the model simulates a much larger amount of Forest gain

than actually occurred. Simulation (b) performs no better; it also over simulates the Quantity of Forest gains and allocates them to locations where Non Forest persistence was observed. This assessment is confirmed by the confusion matrices (Tab. 3.2), where the total disagreement in Forest presence ($729 + 1151 = 188 \text{ km}^2$) is several times larger than the entire observed extent of Forest gains (349 km^2).

When land presence at 2023 is validated (Fig. 3.4a, Tab. 3.2), both AUC and OA report high values that contradict the insights from the visual analysis of the maps. The reason for this is that, as this validation focuses on land presence rather than change, and land uses are highly persistent, a large share of agreement is achieved simply by reproducing Forest and Non-Forest where they already existed in 2005. This agreement does not imply good model performance: Land uses have strong temporal dependence, meaning that a location is far more likely to retain its previous use than to change, making land presence easy to predict. As a result, any simplistic model that considers this temporal dependence will tend to report high Agreement values when validating land presence. For example, even a naïve baseline, such as assuming the absence of change, would report an OA of 95%, resulting in a higher score than our model.

In the case of AUC, a second factor also contributes to the inflated values. AUC performs poorly under conditions of unbalanced datasets because it implicitly favors the majority class. In our case study, Forest covers 37% of the territory in 2023, meaning that AUC reflects the model’s ability to correctly identify Non-Forest in a larger extent. As a result, high AUC values can be obtained even when the model performs poorly in locating Forest, and therefore provide a misleading assessment of model performance. This limitation becomes even more pronounced in LCM applications involving multiple land use categories, because increasing the number of classes necessarily reduces their prevalence and further distorts the properties of AUC.

This problem is not resolved by applying conventional Skill metrics. Our model, like any LCM, implicitly incorporates temporal dependence, whereas the random-allocation baseline underlying Kappa does not. Kappa assumes that a pixel classified as Forest in 2005 is just as likely to be Forest in 2023 as a pixel classified as Non-Forest in 2005, which mechanically lowers the expected OA of the baseline. The result is an unrealistically weak benchmark that inflates Skill values.

In this case study, Forest accounts for 37% of the map. Under Kappa’s randomness assumption, this implies a random relocation of 63% of the Forest pixels to Non-Forest and 37% of the Non-Forest pixels to Forest¹. In aggregate, this is equivalent to a 2023 map in which $2 \cdot 0.37 \cdot (1 - 0.37) = 47\%$ of the territory has its classes rearranged, with 23% of Forest relocation. Given that the observed Forest gain is only 5%, overlaying the initial map with the map expected under randomness implies Forest gains between 18% and 28%, depending on how gains and losses spatially intersect. This is a land change rate clearly inconsistent with the observed land dynamics. Under these conditions, the baseline would achieve an OA of only 53%. This example illustrates the conceptual inconsistency of using Kappa to validate land presence in LCM.

There are two main ways to address the temporal dependence of land uses in validation: either by interpreting Agreement metrics accordingly and using alternative base-lines for Skill metrics that explicitly account for temporal dependence, or by validating land change instead of land presence. The confusion matrix of changes (Tab. 3.2) follows the latter approach. In this case, OA remains high (88%) because the model correctly classifies persistence, which dominates the landscape. High OA values therefore remain not because of temporal dependence, but because most cells do not change. Thus, when validation focuses on land change, Kappa’s baseline is no longer affected by temporal dependence and is therefore more appropriate for LCMs. The random-allocation baseline now yields an expected OA of 85%, effectively capturing the influence of a low prevalence on our results. This implies that a large share of the agreement achieved by the model can be obtained by chance alone. As a result, the model’s OA for Forest gains exceeds this baseline by only three percentage points, leading to a more realistic Kappa value of 0.14. This version of Kappa corresponds to the $\text{Kappa}_{\text{Simulation}}$ proposed by Vliet (2016), which can be useful in certain LCM applications if interpreted appropriately (Tong and Feng 2020).

The remaining question is why, despite the more realistic results obtained from hard validation when focusing on land change, the soft validation of change still reports a high AUC (Fig. 3.3b). This apparent contradiction is explained by the distortion of AUC under low-prevalence conditions discussed above. Forest gains are

¹This follows from the definition of random Agreement used by HSS in Equation 1.14.

a very rare event, covering only about 5% of the study area. Under such strong class imbalance, AUC becomes largely insensitive to how well Forest gains are identified and is driven mainly by the correct classification of persistence. This effect is visible in Figure 3.3b, where the blue square, as well as classifications to its right, contribute strongly to the AUC because they lie well above the uniform line. However, Table 3.2 shows that most locations of Forest gains are wrong and the model simulates roughly twice as many gains as observed. Moreover, classifications to the right of the blue square marker account for a large share of the AUC, while implying an even stronger over simulation of Forest gains. Taken together, these results show that AUC fails to reflect the model’s inability to accurately locate Forest gains and is therefore not an appropriate metric for validating LCMs.

3.4.2 Validation with the FOM framework

The limitations identified in conventional validation practices motivate the need for an alternative approach that is better suited to land change patterns. Applying the FOM framework to our case study shows how it addresses these limitations and provides a more appropriate validation procedure for land change models.

Figure 3.5 shows how the FOM framework overcomes the shortcomings of conventional validation metrics. In our case study, the low FOM value obtained for the hard classification (11%) at the blue marker clearly indicates that the model does not accurately simulate Forest gains. At the same time, FOM is an agreement-based metric and is therefore still influenced by prevalence. To account for this, the observed FOM value can be compared to the value of the uniform line, which is approximately 3% at the quantity that corresponds to the blue marker. This comparison yields a FOM Skill of 8%, indicating that Forest gains are allocated only slightly more efficiently than under a random allocation of change at that particular threshold that over simulates gain.

As for soft validation, the framework provides AUFOM as a metric that is directly comparable to AUC. For our suitability map, AUFOM reports a much lower value (0.48) than AUC. This result clearly reflects the inability of any hard classification derived from the suitability map to correctly locate Forest gains. In addition, the random allocation baseline yields an AUFOM value of 0.25, which implies an AUFOM

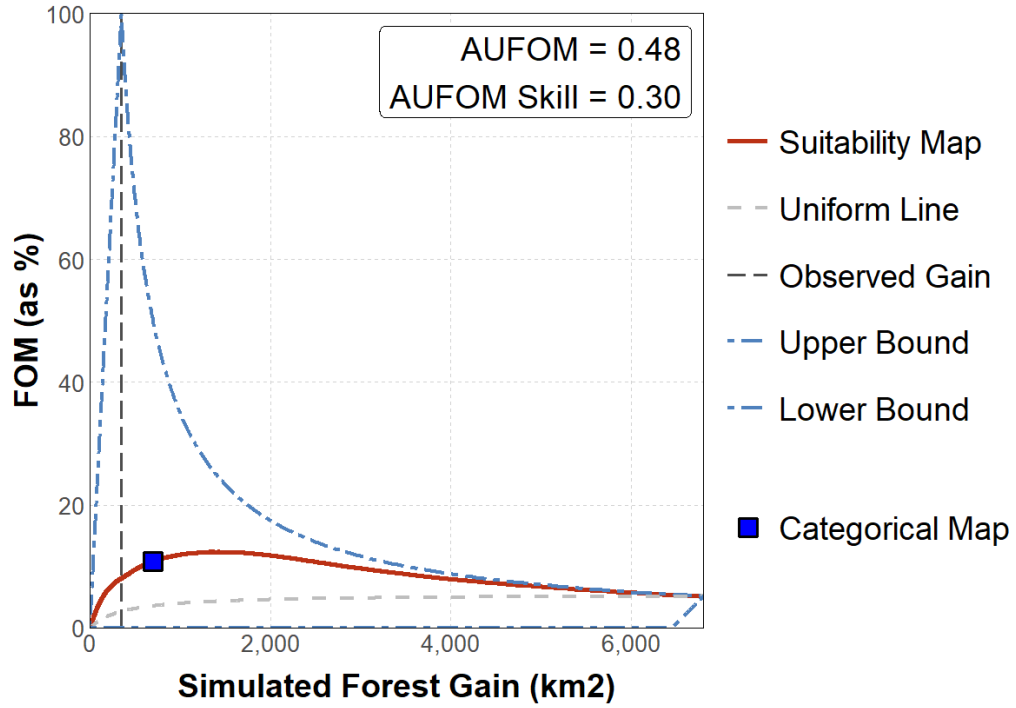


FIGURE 3.5: Graphical representation of the FOM framework applied to our suitability and categorical maps.

Skill of 0.30. This confirms that the suitability map improves only marginally over random allocation when evaluated in terms of land change.

Beyond the numerical results, the framework provides insights that conventional validation procedures do not offer. By integrating soft and hard validation within a single graphical representation, it allows suitability maps and hard classifications to be evaluated using comparable metrics. In our application, this reveals that any hard classification derived from the suitability map reaches at most a 12% FOM, which occurs when the simulated forest gain is about three times larger than the reference forest gain. This insight is crucial for interpreting model behavior, as it reveals the limitation of the suitability map to specify the spatial allocation. The framework also makes the prevalence of land change explicit, which we show is critical for interpreting agreement metrics such as OA, AUC, and FOM, and which is not conveyed by ROC curves or other conventional validation approaches. Finally, the framework helps to identify the sources of error in hard classifications by distinguishing between errors in

the quantity of gain and errors in the spatial allocation of gain. In our application, disagreement arises both from an over-simulation of Forest gain in the hard map and from the gain’s incorrect spatial allocation that derives from the suitability map.

The use of the FOM framework offers several benefits for LCM validation. Most importantly, it allows for a unified approach to both soft and hard validation through comparable metrics within the same graphical framework. By using AUFOM for soft validation and FOM for hard validation, we can compare the two results with metrics that exclusively focus on the Agreement of changes, instead of OA. This opens up the possibility for directly comparing soft and hard classified maps, which have fundamentally different outputs. Additionally, FOM graphs enable the visualization of both soft and hard results within a single plot, something that cannot be done when comparing AUC and FOM. Therefore, FOM graphs represent a substantial advancement toward the convergence of validation processes for LCMs.

Moreover, FOM graphs allow both a visual and numerical analysis of how results compare to the baseline. As demonstrated in Chapter 2, the FOM curve of the randomness baseline varies depending on prevalence. By plotting both the soft and hard classified maps on the same graph, we can assess how and to what extent the model diverges from the baseline. For soft maps, this is numerically measured by comparing the AUFOM of the suitability map to that of the baseline (Chapter 2), and visually by observing the area between the two curves. For hard classified maps, the Skill is numerically measured by comparing the FOM values of the map and the baseline at a specific threshold, and visually by comparing the curve heights. As we will see in the following Section, this analysis is crucial due to the influence that prevalence has in LCM applications.

This visual representation of the metrics enhances their comprehension and interpretation. In Chapter 2, we reflected on how FOM is a metric that might lead to Quantity overestimation. We concluded that the room for Quantity overestimation is remarkably large in cases with low prevalence and low FOM values, which are common characteristics observed in LCM literature. In our illustration, the rate at which the suitability map simulates Hits is so low that the FOM increases slowly but along a larger set of simulated Quantities (Fig.3.3). This makes the peak of the FOM curve to have a large Quantity disagreement. If the threshold selection criteria is the

maximization of FOM, trying to report the best model results possible, LCMs will probably incur in large Quantity disagreements. FOM curves offer an informative alternative for single threshold selection criteria.

In this regard, both TOC and FOM graphs display the Quantity disagreement at multiple thresholds. However, FOM curves also reveal the contribution of both Quantity and Allocation disagreement to FOM values. The distance between $y = 100\%$ and the upper bound represents Quantity disagreement; the distance between the upper bound and the FOM curve represents Allocation disagreement (Chapter 2, Fig. 2.1). This characteristic of FOM curves overcomes one of the main limitations of the FOM metric, which is that it combines Allocation and Quantity disagreement without distinguishing between them (Pontius, Boersma, et al. 2008). In our example, the same FOM value can be achieved with different combinations of Quantity and Allocation disagreement. If we simulate the observed amount of gains, our suitability map would yield a FOM of 9%, with an Allocation disagreement of 91% and no Quantity disagreement. The same FOM value could be obtained by simulating ten times the observed gains, but with 5% Allocation disagreement and 86% Quantity disagreement.

3.4.3 Insights on the current use of FOM in LCM

Since its introduction to LCM by Pontius, Boersma, et al. (2008), the FOM has been increasingly used as a validation metric. For the reasons discussed in the previous section, FOM is often presented as an appropriate metric for LCM validation. Across the literature, reported FOM values typically range between 5% and 20% (Pontius, Castella, et al. 2018), even for models that are usually claimed to perform well. Any FOM value less than 50% means that disagreement of change exceeds agreement of change.

To support the claims of acceptable model performance, many studies explicitly cite Pontius, Castella, et al. (2018) or Pontius, Boersma, et al. (2008), where the authors report a positive relationship between land change prevalence and FOM. In that work, the correlation is explained by the difficulty of predicting rare events, with the argument that when the signal to train and to validate a model is weak, then the model's predictive ability is low. Based on this reasoning, many studies argue

that low FOM values are "acceptable" in their applications because land changes are scarce (Liu, Liang, et al. 2017; Meentemeyer et al. 2013; Chen et al. 2020; Li et al. 2017; Peng et al. 2020). Conversely, studies that report FOM values that are still low but higher than in other case studies interpret this as evidence of better model performance (Zhang et al. 2022). This tendency to compare case studies to gain insight to model performance is faulty because the work of Pontius shows that the FOM value frequently reflects the landscape more than the model. The tendency is that, rather than encouraging a deeper interpretation of model behavior and a careful analysis of error types and their causes, these low FOM values are often justified as acceptable.

The FOM framework provides the tools needed to evaluate and critically assess these arguments. In their study, Pontius, Castella, et al. (2018) overlook an important point about FOM. FOM is an agreement metric, and its value is therefore directly affected by prevalence. This means that part of the FOM obtained by a model is simply due to the amount of land change in the study area, regardless of the model itself or the amount of information used.

Figure 3.6a shows the relationship between the prevalence of land change and the FOM for the 13 case studies reported by Pontius, Castella, et al. (2018). The figure also shows the FOM of the corresponding baseline for each case study. The dark dotted line represents the baseline FOM7, while the gray dashed line represents the 1:1 line. Panel 6b shows the relationship between prevalence and FOM skill, measured as the difference between the model FOM and the baseline FOM. As noted by Pontius, Castella, et al. (2018), panel a shows a strong positive correlation between prevalence and FOM, with a value of 0.75. However, the gray dots illustrate that the baseline explains a substantial share of the reported agreement captured by FOM. This implies that part of the observed correlation between prevalence and FOM is driven by the higher level of agreement that can be expected simply when there is more land change to agree with. In fact, with the exception of Cho Don, the baseline accounts for between 13 and 67 percentage points of the reported FOM values.

In panel (b), we remove the effect of prevalence from the FOM results, which allows us to isolate the true merit of the model. The correlation between skill and prevalence remains positive, with a value of 0.38, but it is much weaker than for

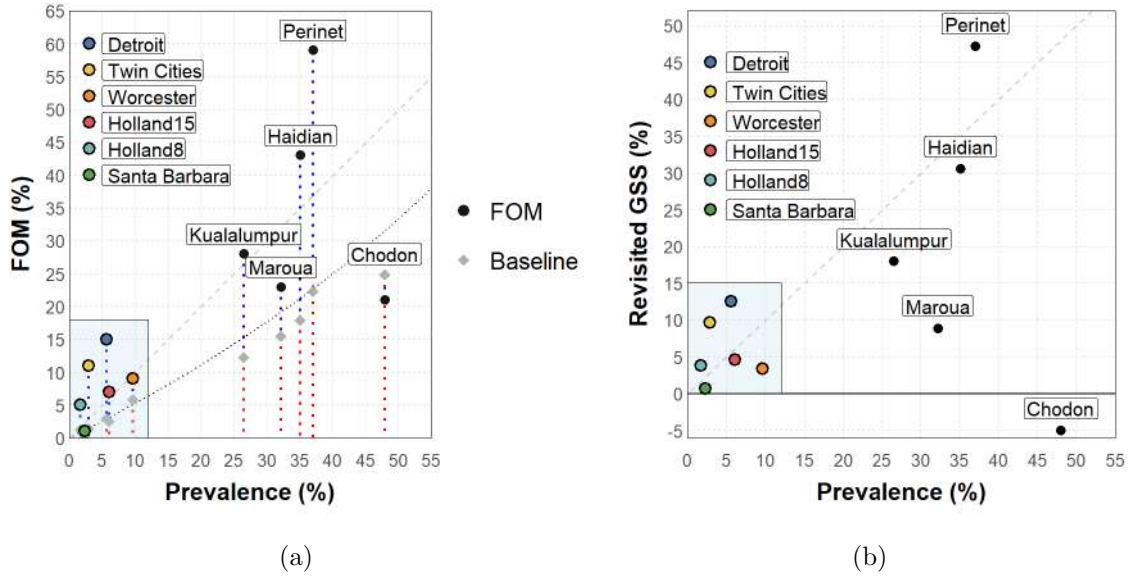


FIGURE 3.6: TOC (a) and FOM (b) plots showing model performance across different prevalence levels. *Source: Self elaboration.*

agreement. Some case studies with high prevalence, such as Maroua and Kuala Lumpur, show skill levels comparable to those with much lower prevalence. In the case of Cho Don, the reported FOM is even lower than the baseline FOM. This shows that higher FOM values should not be read as better model performance, but can mainly reflect the effect of prevalence on agreement.

As a result, we can deduce that direct comparisons of FOM values across applications can be misleading. In particular, claiming that one model performs better than another applied in a different region simply because it reports a higher FOM lacks interpretative basis. Without explicitly reporting and addressing differences in prevalence FOM may reflect characteristics of the study area rather than differences in model performance.

Removing the effect of prevalence clarifies that agreement and skill capture fundamentally different aspects of model behavior. However, prevalence is not the only factor that complicates the interpretation of FOM values. Even within the same application, comparing FOM across different hard classifications or models can be misleading, because FOM depends not only on prevalence but also on the amount of change that is simulated. As noted in previous studies, FOM is often maximized

when the quantity of a reference change class is overestimated (Baldwin and Kain 2006).

This issue is illustrated by our case study in Figure 3.5. A FOM value of 10% can be obtained by simulating either 560 km² or 2850 km² of Forest gains, even though these two simulations clearly differ in quantity. The first over simulates the observed change by 60%, whereas the second over simulates it by 700%. The baseline FOM differs between these cases, taking values of 5% and 7%, respectively. As a result, the two simulations outperform the baseline to different degrees. When 2850 km² of Forest gains are simulated, a larger share of the reported FOM is explained by the baseline, leaving less of it attributable to the actual performance of the suitability map.

These results also imply that comparing the FOM of two models or two simulations within the same study can be equally uninformative when the model runs have different quantities. A higher FOM does not necessarily indicate better spatial allocation, as it may simply reflect differences in the quantity of change simulated. When two models have a different quantity of simulated changes, their FOM values are not directly comparable unless Quantity disagreement is explicitly addressed.

These results qualify the conclusions in Pontius, Castella, et al. (2018), and therefore also qualify the way in which those conclusions have been used in subsequent applications. Liu, Liang, et al. (2017), for example, argue that their results are “acceptable” because their study region exhibits a relatively low prevalence of change (17%), yet they report a FOM of 19%, a value that would be approximated by the corresponding random baseline given that prevalence. They further compare their hard maps with those of other models and case studies, but do not report the prevalence of change or the simulated quantity of change in those applications. Without this information, differences in FOM cannot be interpreted as differences in model performance. Similarly, Chen et al. (2020) claim that their model is reliable because their low FOM values are similar to or greater than those reported in other studies, despite not reporting the prevalence of change in their application. They also assess the contribution of alternative model specifications to differences in FOM without reporting the associated Quantity disagreement, making it unclear whether these differences reflect improvements in spatial allocation or simply differences in simulated

quantity. In another example, Zhang et al. (2022) compare two FOM values obtained from different models to argue that one is superior, yet they do not report the number of changes simulated by each model. Without information on simulated quantity and baseline agreement, claims about relative model performance remain unsupported.

As an additional comment, according to Pontius, Castella, et al. (2018), "It does not make sense to use criteria that reward systematic underestimates or overestimates of the quantity of each category. This is a weakness of using percentage correct and the null model as benchmarks for predictive Agreement, and is a reason why Pontius, Boersma, et al. (2008) used the figure of merit as a criterion." However, the results in Chapter 2 demonstrate that FOM rewards systematic overestimation of the Quantity. It is striking that, contrary to these results, the majority of the 13 case studies simulate the same or fewer changes than the observed. Although in some of these applications the correct quantity was manually determined by the modelers, we suspect that the criterion for selecting the quantity may be based on overall agreement metrics. In LCM, due to the scarcity of changes, these metrics are optimized by undersimulating the quantities of changes. If the modelers were aiming to maximize the FOM, they would likely oversimulate the quantities of changes. (Pontius, Boersma, et al. 2008) points out that FOM values are higher the less Quantity disagreement the simulations have. However, this is not completely true. Since all the applications underestimate the quantity of changes, as the simulated Quantity increases, FOM values also increase. It is likely that these applications would achieve higher FOM values even with larger Quantity disagreements, as long as this disagreement was oversimulation.

3.4.4 Considering other alternative baselines

Section 3.4.3 highlights the central role of baselines in the interpretation of Agreement. The conventional baseline of randomness can even be a weak baseline even if land change is validated. The reason for this is that most land use processes exhibit clear spatial structure rather than randomness.

Hamill and Juras (2006) shows that CSI is highly sensitive to spatial heterogeneity, particularly under low prevalence. In LCM applications, land changes are often concentrated in specific areas, even when their overall prevalence is low. Under these

conditions, a baseline defined over the full spatial extent may be too weak, leading to inflated estimates of Skill. Thus, randomness is often not a sufficiently demanding benchmark for LCM evaluation.

To address this limitation, more informative baselines can be considered. Brown, Hall, et al. (2007) proposes constructing credible baselines tailored to the structure of the problem. In the presence of spatial heterogeneity, simple classifications that incorporate this structure may provide a more appropriate reference. Pontius (2022) introduces the concept of one-minute models, defined as naive assumptions that can be formulated with minimal effort. For example, Forest gain is unlikely to occur randomly across the landscape and is more likely to appear near existing Forest. While the randomness baseline assigns equal probability of gain across all locations (equal to prevalence), a simple distance-to-Forest model assigns higher probabilities to locations closer to existing Forest, thereby capturing spatial heterogeneity.

Figure 3.7 compares the randomness baseline with a distance-to-Forest baseline in our case study. The distance-based model achieves substantially higher Agreement than randomness and, in fact, yields results very similar to those of the suitability map. This comparison illustrates the importance of carefully defining Skill in LCM applications. Although AUFOM indicates that the suitability map improves upon randomness, it is difficult to interpret this as genuine Skill when a much simpler model achieves nearly identical Agreement. In this context, the apparent Skill relative to randomness largely reflects the spatial concentration of Forest gains rather than the added value of the model itself. Consequently, the randomness baseline provides a misleading benchmark in this application.

Despite their advantages, more demanding baselines introduce challenges for comparisons across case studies. Spatial heterogeneity varies between applications, so each baseline may capture a different share of the underlying structure. For instance, a distance-to-Forest model may provide a strong benchmark in one case but be nearly equivalent to randomness in another. In such situations, Skill values are not directly comparable. By contrast, the randomness baseline adapts mechanically to prevalence and therefore offers a more consistent, albeit sometimes weak, point of reference.

For LCM applications, we recommend using FOM curves that include both baselines. FOM curves provide a detailed representation of Agreement across thresholds

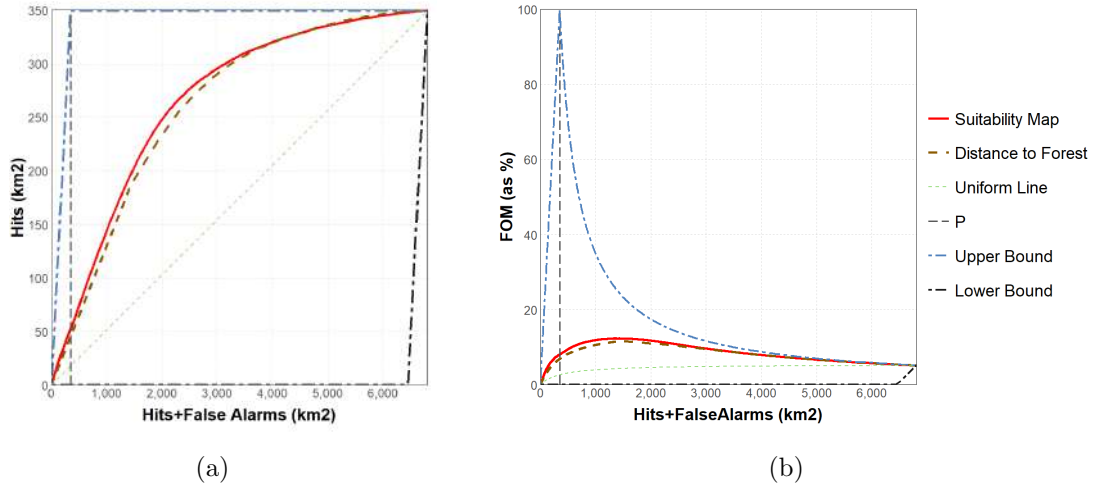


FIGURE 3.7: Representation of the TOC (a) and FOM (b) plots. *Source: Self elaboration.*

while making prevalence, as well as Quantity and Allocation disagreement, explicit. When comparing models within the same case study, one-minute models offer a more informative benchmark, as all models are evaluated against the same, more demanding baseline. When comparing results across case studies, the randomness baseline remains useful because it reflects the amount of Agreement that can be attributed to prevalence alone, allowing for more consistent interpretation.

The FOM framework enables the joint use of both baselines and supports a decomposition of Skill. Figure 3.7 illustrates this decomposition. The difference between the x-axis and the randomness baseline captures Agreement explained by prevalence. The gap between the randomness baseline and the one-minute model reflects the contribution of spatial heterogeneity. Finally, the difference between the one-minute baseline and the model represents the additional performance attributable to the model itself.

3.5 Discussion

This chapter contributes to a growing body of literature highlighting fundamental problems in the validation of LCMs (Olofsson, Foody, Herold, et al. 2014; Pontius, Castella, et al. 2018; Pontius and Millones 2011; Pontius and Parmentier 2014; Shao et al. 2019; Stehman and Foody 2019; Tong and Feng 2020; Vliet, Bregt, Brown, et al.

2016; Vliet, Bregt, and Hagen-Zanker 2011; Vliet 2016). Previous work has shown that commonly used validation procedures can yield misleadingly positive assessments of model performance because land change is rare and persistence dominates the landscape. Our results advance this discussion by systematically organizing the main limitations of LCM validation and by illustrating them in a way that makes their implications explicit and accessible to both researchers and practitioners.

This chapter shows how three issues interact to distort validation results in LCMs. First, the strong temporal dependence of land uses leads both Agreement and Skill metrics to report high values when validation focuses on land presence rather than on land change. Second, the low prevalence of land change causes many widely used metrics to be dominated by the majority class, making them largely insensitive to errors in the spatial location of change. Third, because Agreement metrics are strongly influenced by prevalence, differences in reported performance are often driven by the amount of change rather than by the actual Skill of the model. Together, these issues explain why LCMs are frequently reported to perform well even when they fail to accurately represent land changes.

Beyond identifying these problems, this chapter proposes a more coherent alternative for LCM validation based on the FOM framework. The contribution lies both in the introduction of suitable validation metrics and in the way these metrics are integrated into a structured validation framework. Approaching validation through a framework, rather than through isolated metrics, helps overcome the limitations associated with the partial and omitted information that individual validation metrics necessarily provide. By explicitly focusing on change, making prevalence visible, and separating Agreement from Skill, and distinguishing Quantity disagreement from allocation disagreement, the framework provides a unified conceptual structure that directly addresses the most common validation pitfalls in LCMs. This structure facilitates transparent communication of model performance and reduces the room for selectively relying on metrics that systematically report favorable results.

An important implication of this work is that using FOM alone, like any other Agreement metric, is not sufficient. As shown in both our case study and the literature review, FOM values are often low and highly sensitive to prevalence and to the simulated quantity of change. Interpretations of FOM are being used to justify

weak model performance, either by appealing to the scarcity of change or by making relative comparisons across case studies. The framework presented here addresses these limitations by offering tools for a clearer interpretation of FOM. We hope that this contribution helps move away from incorrect interpretations of FOM and from justifying low values by comparison rather than by their actual meaning.

The findings of this chapter, together with those of the broader literature (Olofsson, Foody, Stehman, et al. 2013; Viana et al. 2023), support a set of methodological recommendations for improving validation practices in LCMs. These recommendations are intended as general principles that follow directly from the limitations identified in current practices.

1. **Prioritize change-based validation over presence-based validation.** This recommendation has been repeatedly emphasized in the literature (Pontius, Castella, et al. 2018; Vliet, Bregt, Brown, et al. 2016; Vliet, Bregt, and Hagen-Zanker 2011; Tong and Feng 2020; Olofsson, Foody, Herold, et al. 2014; Olofsson, Foody, Stehman, et al. 2013), yet it has not been fully incorporated into common validation practice. If the research goal of LCMs is to simulate change, validation should focus explicitly on land change. Validation approaches that evaluate both change and persistence, and places that are not candidates for change, are poorly aligned with the objectives of LCMs because they are dominated by the correct prediction of persistence.
2. **Abandon the use of AUC for LUCM validation.** AUC is not informative for LCM applications because it tends to produce high values when the event of interest is rare. In LCMs, specific land uses usually occupy a small share of the territory, and land change itself is always rare. Under these conditions, AUC can suggest good performance even when the model does not simulate change accurately. Therefore, AUC does not provide meaningful information for evaluating LCMs and should not be used for validation in this context. The shape of the ROC curve, or more appropriately the TOC curve, can still provide useful information, such as identifying the thresholds at which the suitability map generates more hits. However, instead of relying on AUC as a summary metric, metrics designed for rare events, such as AUFOM, offer a more realistic assessment of how well changes are simulated.

3. **Report prevalence of change and simulated Quantity disagreement, and interpret metric results accordingly.** This recommendation is standard practice in fields with a long tradition of predicting rare events, such as meteorology (Hogan et al. 2010; Wang 2014; Hamill and Juras 2006). Prevalence should not be treated as a secondary descriptive statistic, but as a central component of validation in LCMs. Because land change is typically rare, agreement metrics are highly sensitive not only to the proportion of observed change, but also to the quantity of change simulated by the model. Quantity disagreement directly captures whether the model is over-simulating or under-simulating the amount of change, a dimension that agreement metrics alone cannot disentangle from spatial allocation performance. Reported differences in performance frequently reflect variation in simulated quantities rather than genuine differences in the model’s ability to allocate change accurately in space. Validation results should therefore always report both the prevalence of observed change and the simulated Quantity disagreement, and interpret agreement metrics explicitly in relation to them.
4. **Report agreement and skill jointly.** Agreement metrics describe how well simulated changes overlap with observed changes, but they do not indicate whether this agreement can be attributed to the model. Assessments of model performance should therefore rely on skill metrics defined relative to explicit and relevant baselines (Pontius, Castella, et al. 2018; Vliet, Bregt, and Hagen-Zanker 2011). Agreement and Skill should be reported together and interpreted consistently. As other authors (Pontius and Millones 2011; Vliet, Bregt, and Hagen-Zanker 2011; Olofsson, Foody, Herold, et al. 2014), we illustrated how Kappa should never be used for land use change validation.

Baselines should be treated as a central component of validation rather than as a technical detail (Pontius 2022). While random allocation provides an initial naive benchmark, the definition of skill proposed in this paper opens the door to more informative baselines. The FOM framework allows models to be evaluated against random allocation and other more realistic baselines, enabling more meaningful comparisons across applications.

5. **Use and report FOM curves in addition to or instead of single summary metrics.** Single FOM values provide limited insight into model behavior. The paper shows how the comparison of FOM values, both between case studies and within the same application, can be misleading. FOM curves place these values in the context of prevalence and show whether model error is mainly due to over-simulation or under-simulation of the amount of change, or to errors in its spatial allocation. The FOM framework offers the graphical and numerical tools for a more informed interpretation of FOM.
6. **Carry out soft and hard validations within a unified graphical and conceptual framework.** Unifying both soft and hard validation within a single framework allows suitability maps and categorical classifications to be evaluated using a consistent definition of agreement. This improves traceability between model components and clarifies how hard-classified outcomes relate to soft predictions.

As a conclusion, we hope that the clear communication of the limitations of current validation practices, together with the alternative framework proposed in this paper, contributes to improving how land change models are evaluated and interpreted. Improving validation practices is essential not only for methodological rigor but also for ensuring that LCM results are interpreted appropriately in applied and policy-relevant contexts.

References

- Baldwin, Michael E. and John S. Kain (Aug. 2006). “Sensitivity of Several Performance Measures to Displacement Error, Bias, and Event Frequency.” en. In: *Weather and Forecasting* 21.4. DOI: <https://doi.org/10.1175/WAF933.1>. URL: https://journals.ametsoc.org/view/journals/wefo/21/4/waf933_1.xml (visited on 08/01/2024).
- Brown, Daniel G., Peter H. Verburg, et al. (Oct. 2013). “Opportunities to improve impact, integration, and evaluation of land change models”. In: *Current opinion in environmental sustainability* 5.5, pp. 452–457. DOI: [10.1016/j.cosust.2013.07.012](https://doi.org/10.1016/j.cosust.2013.07.012). URL: <https://doi.org/10.1016/j.cosust.2013.07.012>.

- Brown, Sandra, Myrna Hall, et al. (2007). “Baselines for Land-Use Change in the Tropics: Application to Avoided Deforestation Projects”. In: *Mitigation and Adaptation Strategies for Global Change* 12, pp. 1001–1026. DOI: [10.1007/s11027-006-9073-3](https://doi.org/10.1007/s11027-006-9073-3).
- Chen, Guangzhao et al. (Jan. 2020). “Global projections of future urban land expansion under shared socioeconomic pathways”. en. In: *Nature Communications* 11.1, p. 537. ISSN: 2041-1723. DOI: [10.1038/s41467-020-14386-x](https://doi.org/10.1038/s41467-020-14386-x). URL: <https://www.nature.com/articles/s41467-020-14386-x> (visited on 02/18/2026).
- Güneralp, Burak and Rakibul Ahasan (Dec. 2025). “Urban land-change futures: current understanding, challenges, and implications”. en. In: *npj Urban Sustainability* 6.1, p. 7. ISSN: 2661-8001. DOI: [10.1038/s42949-025-00308-7](https://doi.org/10.1038/s42949-025-00308-7). URL: <https://www.nature.com/articles/s42949-025-00308-7> (visited on 03/11/2026).
- Hamill, Thomas M. and Josip Juras (2006). “Measuring forecast skill: is it real skill or is it the varying climatology?” en. In: *Quarterly Journal of the Royal Meteorological Society* 132.621C, pp. 2905–2923. ISSN: 1477-870X. DOI: [10.1256/qj.06.25](https://doi.org/10.1256/qj.06.25). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1256/qj.06.25> (visited on 08/16/2024).
- Hand, D. J. and C. Anagnostopoulos (Apr. 2013). “When is the area under the receiver operating characteristic curve an appropriate measure of classifier performance?” In: *Pattern Recognition Letters* 34.5, pp. 492–495. DOI: [10.1016/j.patrec.2012.12.004](https://doi.org/10.1016/j.patrec.2012.12.004). URL: <https://doi.org/10.1016/j.patrec.2012.12.004>.
- Hand, David J. (Oct. 2009). “Measuring classifier performance: a coherent alternative to the area under the ROC curve”. en. In: *Machine Learning* 77.1, pp. 103–123. ISSN: 1573-0565. DOI: [10.1007/s10994-009-5119-5](https://doi.org/10.1007/s10994-009-5119-5). URL: <https://doi.org/10.1007/s10994-009-5119-5> (visited on 02/18/2026).
- Hogan, Robin J. et al. (Apr. 2010). “Equitability Revisited: Why the “Equitable Threat Score” Is Not Equitable”. en. In: *Weather and Forecasting*. Section: Weather and Forecasting. DOI: [10.1175/2009WAF2222350.1](https://doi.org/10.1175/2009WAF2222350.1). URL: https://journals.ametsoc.org/view/journals/wefo/25/2/2009waf2222350_1.xml (visited on 08/14/2024).
- Lambin, Eric F. and Patrick Meyfroidt (Feb. 2011). “Global land use change, economic globalization, and the looming land scarcity”. In: *Proceedings of the National*

- Academy of Sciences* 108.9, pp. 3465–3472. DOI: [10.1073/pnas.1100480108](https://doi.org/10.1073/pnas.1100480108). URL: <https://doi.org/10.1073/pnas.1100480108>.
- Li, Xia et al. (Sept. 2017). “A New Global Land-Use and Land-Cover Change Product at a 1-km Resolution for 2010 to 2100 Based on Human–Environment Interactions”. In: *Annals of the American Association of Geographers* 107.5, pp. 1040–1059. ISSN: 2469-4452. DOI: [10.1080/24694452.2017.1303357](https://doi.org/10.1080/24694452.2017.1303357). URL: <https://doi.org/10.1080/24694452.2017.1303357> (visited on 02/18/2026).
- Liu, J., J. Liu, et al. (2024). “Enhancing credit risk prediction based on ensemble tree-based feature transformation and logistic regression”. In: *Journal of Forecasting* 43.2, pp. 429–455. DOI: [10.1002/for.3040](https://doi.org/10.1002/for.3040).
- Liu, Xiaoping, Xun Liang, et al. (2017). “A future land use simulation model (FLUS) for simulating multiple land use scenarios by coupling human and natural effects”. In: *Landscape and Urban Planning* 168, pp. 94–116. DOI: [10.1016/j.landurbplan.2017.08.002](https://doi.org/10.1016/j.landurbplan.2017.08.002). URL: <https://doi.org/10.1016/j.landurbplan.2017.08.002>.
- Lobo, Jorge M., Alberto Jiménez-Valverde, and Raimundo Real (Mar. 2008). “AUC: a misleading measure of the performance of predictive distribution models”. en. In: *Global Ecology and Biogeography* 17.2, pp. 145–151. ISSN: 1466-822X, 1466-8238. DOI: [10.1111/j.1466-8238.2007.00358.x](https://doi.org/10.1111/j.1466-8238.2007.00358.x). URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1466-8238.2007.00358.x> (visited on 04/02/2024).
- Ma, Hua et al. (2013). “On use of partial area under the ROC curve for evaluation of diagnostic performance”. In: *Statistics in Medicine* 32.6. Received 19 June 2012, Accepted 11 2 2013, Published online 18 March 2013, pp. 1102–1117. DOI: [10.1002/sim.5777](https://doi.org/10.1002/sim.5777). URL: <https://wileyonlinelibrary.com/doi/abs/10.1002/sim.5777>.
- Magliocca, Nicholas R., Pratik Dhungana, and Carter D. Sink (Dec. 2023). “Review of counterfactual land change modeling for causal inference in land system science”. In: *Journal of Land Use Science* 18.1, pp. 1–24. ISSN: 1747-423X. DOI: [10.1080/1747423X.2023.2173325](https://doi.org/10.1080/1747423X.2023.2173325). URL: <https://doi.org/10.1080/1747423X.2023.2173325> (visited on 06/09/2023).
- Meentemeyer, Ross K. et al. (2013). “FUTURES: Multilevel Simulations of Emerging Urban–Rural Landscape Structure Using a Stochastic Patch-Growing Algorithm”. In: *Annals of the Association of American Geographers* 103.4, pp. 785–807. DOI:

- 10.1080/00045608.2012.707591. URL: <https://doi.org/10.1080/00045608.2012.707591>.
- Müller, Daniel et al. (Sept. 2014). “Regime shifts limit the predictability of land-system change”. en. In: *Global Environmental Change* 28, pp. 75–83. ISSN: 0959-3780. DOI: 10.1016/j.gloenvcha.2014.06.003. URL: <https://www.sciencedirect.com/science/article/pii/S0959378014001083> (visited on 05/02/2023).
- Olofsson, Pontus, Giles M. Foody, Martin Herold, et al. (May 2014). “Good practices for estimating area and assessing accuracy of land change”. en. In: *Remote Sensing of Environment* 148, pp. 42–57. ISSN: 0034-4257. DOI: 10.1016/j.rse.2014.02.015. URL: <https://www.sciencedirect.com/science/article/pii/S0034425714000704> (visited on 04/19/2023).
- Olofsson, Pontus, Giles M. Foody, Stephen V. Stehman, et al. (Feb. 2013). “Making better use of accuracy data in land change studies: Estimating accuracy and area and quantifying uncertainty using stratified estimation”. en. In: *Remote Sensing of Environment* 129, pp. 122–131. ISSN: 0034-4257. DOI: 10.1016/j.rse.2012.10.031. URL: <https://www.sciencedirect.com/science/article/pii/S0034425712004191> (visited on 05/10/2023).
- Peng, Kaifeng et al. (Oct. 2020). “Simulating wetland changes under different scenarios based on integrating the random forest and CLUE-S models: A case study of Wuhan Urban Agglomeration”. In: *Ecological Indicators* 117, p. 106671. ISSN: 1470-160X. DOI: 10.1016/j.ecolind.2020.106671. URL: <https://www.sciencedirect.com/science/article/pii/S1470160X20306087> (visited on 02/18/2026).
- Pontius, Robert Gilmore (2022). *Metrics That Make a Difference: How to Analyze Change and Error*. en. Advances in Geographic Information Science. Cham: Springer International Publishing. ISBN: 978-3-030-70764-4. DOI: 10.1007/978-3-030-70765-1. URL: <https://link.springer.com/10.1007/978-3-030-70765-1> (visited on 01/08/2024).
- Pontius, Robert Gilmore, Wideke Boersma, et al. (Mar. 2008). “Comparing the input, output, and validation maps for several models of land change”. en. In: *The Annals of Regional Science* 42.1, pp. 11–37. ISSN: 0570-1864, 1432-0592. DOI: 10.1007/

- s00168-007-0138-2. URL: <http://link.springer.com/10.1007/s00168-007-0138-2> (visited on 06/04/2024).
- Pontius, Robert Gilmore, Jean-Christophe Castella, et al. (2018). “Lessons and Challenges in Land Change Modeling Derived from Synthesis of Cross-Case Comparisons”. en. In: *Trends in Spatial Analysis and modeling*. Ed. by Martin Behnisch and Gotthard Meinel. Vol. 19. Series Title: Geotechnologies and the Environment. Cham: Springer International Publishing, pp. 143–164. ISBN: 978-3-319-52520-4. DOI: [10.1007/978-3-319-52522-8_8](https://doi.org/10.1007/978-3-319-52522-8_8). URL: http://link.springer.com/10.1007/978-3-319-52522-8_8 (visited on 12/13/2022).
- Pontius, Robert Gilmore and Marco Millones (Aug. 2011). “Death to Kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment”. en. In: *International Journal of Remote Sensing* 32.15, pp. 4407–4429. ISSN: 0143-1161, 1366-5901. DOI: [10.1080/01431161.2011.552923](https://doi.org/10.1080/01431161.2011.552923). URL: <https://www.tandfonline.com/doi/full/10.1080/01431161.2011.552923> (visited on 01/08/2024).
- Pontius, Robert Gilmore and Benoit Parmentier (Mar. 2014). “Recommendations for using the relative operating characteristic (ROC)”. en. In: *Landscape Ecology* 29.3, pp. 367–382. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/s10980-013-9984-8](https://doi.org/10.1007/s10980-013-9984-8). URL: <http://link.springer.com/10.1007/s10980-013-9984-8> (visited on 01/08/2024).
- Pontius, Robert Gilmore and Kangping Si (Mar. 2014). “The total operating characteristic to measure diagnostic ability for multiple thresholds”. In: *International Journal of Geographical Information Science* 28.3, pp. 570–583. ISSN: 1365-8816. DOI: [10.1080/13658816.2013.862623](https://doi.org/10.1080/13658816.2013.862623). URL: <https://doi.org/10.1080/13658816.2013.862623> (visited on 10/09/2023).
- Sankarrao, Landa, Dillip Kumar Ghose, and Maheswaran Rathinsamy (Nov. 2021). “Predicting land-use change: Intercomparison of different hybrid machine learning models”. In: *Environmental modeling & Software* 145, p. 105207. ISSN: 1364-8152. DOI: [10.1016/j.envsoft.2021.105207](https://doi.org/10.1016/j.envsoft.2021.105207). URL: <https://www.sciencedirect.com/science/article/pii/S1364815221002498> (visited on 02/18/2026).
- Shao, Guofan, Lina Tang, and Jiangfu Liao (Nov. 2019). “Overselling overall map accuracy misinforms about research reliability”. en. In: *Landscape Ecology* 34.11,

- pp. 2487–2492. ISSN: 0921-2973, 1572-9761. DOI: [10.1007/s10980-019-00916-6](https://doi.org/10.1007/s10980-019-00916-6). URL: <http://link.springer.com/10.1007/s10980-019-00916-6> (visited on 11/09/2024).
- Stehman, Stephen V. and Giles M. Foody (Sept. 2019). “Key issues in rigorous accuracy assessment of land cover products”. In: *Remote Sensing of Environment* 231, p. 111199. ISSN: 0034-4257. DOI: [10.1016/j.rse.2019.05.018](https://doi.org/10.1016/j.rse.2019.05.018). URL: <https://www.sciencedirect.com/science/article/pii/S0034425719302111> (visited on 09/12/2024).
- Tholo, Harineck Mayamiko et al. (Jan. 2026). “modeling and prediction of land use and land cover change using machine learning in the Michiru Forest Reserve, Southern Malawi”. en. In: *Discover Environment* 4.1, p. 41. ISSN: 2731-9431. DOI: [10.1007/s44274-026-00542-5](https://doi.org/10.1007/s44274-026-00542-5). URL: <https://doi.org/10.1007/s44274-026-00542-5> (visited on 02/18/2026).
- Tong, Xiaohua and Yongjiu Feng (May 2020). “A review of assessment methods for cellular automata models of land-use change and urban growth”. In: *International Journal of Geographical Information Science* 34.5, pp. 866–898. ISSN: 1365-8816. DOI: [10.1080/13658816.2019.1684499](https://doi.org/10.1080/13658816.2019.1684499). URL: <https://doi.org/10.1080/13658816.2019.1684499> (visited on 07/29/2024).
- Turner, B. L., Eric F. Lambin, and Anette Reenberg (Dec. 2007). “The emergence of land change science for global environmental change and sustainability”. en. In: *Proceedings of the National Academy of Sciences* 104.52, pp. 20666–20671. ISSN: 0027-8424, 1091-6490. DOI: [10.1073/pnas.0704119104](https://doi.org/10.1073/pnas.0704119104). URL: <https://pnas.org/doi/full/10.1073/pnas.0704119104> (visited on 01/28/2023).
- Verburg, Peter H., Andrzej Tabeau, and Erez Hatna (2013). “Assessing spatial uncertainties of land allocation using a scenario approach and sensitivity analysis: a study for land use in Europe”. In: *Journal of Environmental Management* 127. Epub 2012 Sep 29. PMID: 23026356, S132–S144. DOI: [10.1016/j.jenvman.2012.08.038](https://doi.org/10.1016/j.jenvman.2012.08.038). URL: <https://doi.org/10.1016/j.jenvman.2012.08.038>.
- Viana, Cláudia M., Robert Gilmore Pontius, and Jorge Rocha (Aug. 2023). “Four Fundamental Questions to Evaluate Land Change Models with an Illustration of a Cellular Automata–Markov Model”. en. In: *Annals of the American Association of Geographers*, pp. 1–15. ISSN: 2469-4452, 2469-4460. DOI: [10.1080/24694452.2023](https://doi.org/10.1080/24694452.2023).

2232435. URL: <https://www.tandfonline.com/doi/full/10.1080/24694452.2023.2232435> (visited on 09/07/2023).
- Vliet, Jasper van (2016). “Calibration and validation of land use models”. PhD thesis. Wageningen University.
- Vliet, Jasper van, Arnold K. Bregt, Daniel G. Brown, et al. (Aug. 2016). “A review of current calibration and validation practices in land-change modeling”. In: *Environmental modeling & Software* 82, pp. 174–182. ISSN: 1364-8152. DOI: [10.1016/j.envsoft.2016.04.017](https://doi.org/10.1016/j.envsoft.2016.04.017). URL: <https://www.sciencedirect.com/science/article/pii/S1364815216301128> (visited on 07/23/2024).
- Vliet, Jasper van, Arnold K. Bregt, and Alex Hagen-Zanker (Apr. 2011). “Revisiting Kappa to account for change in the accuracy assessment of land-use change models”. en. In: *Ecological modeling* 222.8, pp. 1367–1375. ISSN: 03043800. DOI: [10.1016/j.ecolmodel.2011.01.017](https://doi.org/10.1016/j.ecolmodel.2011.01.017). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0304380011000494> (visited on 01/31/2023).
- Wang, Chung-Chieh (Aug. 2014). “On the Calculation and Correction of Equitable Threat Score for Model Quantitative Precipitation Forecasts for Small Verification Areas: The Example of Taiwan”. en. In: *Weather and Forecasting* 29.4. Section: Weather and Forecasting. DOI: [10.1175/WAF-D-13-00087.1](https://doi.org/10.1175/WAF-D-13-00087.1). URL: https://journals.ametsoc.org/view/journals/wefo/29/4/waf-d-13-00087_1.xml (visited on 08/14/2024).
- Wu, Xia et al. (Dec. 2021). “Simulating spatiotemporal land use change in middle and high latitude regions using multiscale fusion and cellular automata: The case of Northeast China”. In: *Ecological Indicators* 133, p. 108449. ISSN: 1470-160X. DOI: [10.1016/j.ecolind.2021.108449](https://doi.org/10.1016/j.ecolind.2021.108449). URL: <https://www.sciencedirect.com/science/article/pii/S1470160X21011146> (visited on 02/18/2026).
- Zhang, Shihe et al. (2022). “Landscape ecological risk projection based on the PLUS model under the localized shared socioeconomic pathways in the Fujian Delta region”. In: *Ecological Indicators* 136, p. 108642. DOI: [10.1016/j.ecolind.2022.108642](https://doi.org/10.1016/j.ecolind.2022.108642). URL: <https://doi.org/10.1016/j.ecolind.2022.108642>.

Chapter 4

The economic value of erosion: Integrating land changes and nutrient prices

This chapter is based on a paper recently accepted for publication in *Remote Sensing Letters*¹:

Perales Barriendo, J., Campión Arrastia, M. J., Domínguez Irastorza, E., and Osés Eraso, N. (2026) *The economic value of erosion: Integrating land changes and nutrient prices*. Forthcoming in *Remote Sensing Letters*.

¹Please note that there have been modifications to the original paper to ensure consistency with the rest of the dissertation.

Preface

This chapter provides the empirical and applied counterpart to the methodological developments presented in the previous chapters. The thesis has so far focused on the evaluation of binary classification models, identifying limitations in widely used metrics and proposing a more coherent validation framework. While these contributions are developed at a general level, their relevance ultimately depends on their ability to inform applied analyses in contexts where model outputs are used to support economic and environmental decision-making.

In this sense, the present chapter complements the contributions of the thesis in several ways. First, it brings together the insights and knowledge developed around LCMs to provide an application with a clearer economic orientation, thereby bringing these models closer to the domain of public policy design. Second, it translates the theoretical and methodological contributions into an empirical setting, allowing them to be assessed in terms of their practical usefulness. This shift from abstraction to application makes it possible to contextualize the previous findings and to explore how they can support the design of empirically grounded approaches for economic valuation and policy analysis.

To do so, the chapter develops and applies a methodological framework to economically evaluate soil erosion and evaluate it. It combines three well-established components: LCMs, the Revised Universal Soil Loss Equation (RUSLE), and economic valuation through the replacement cost approach. Rather than focusing on each component in isolation, the chapter integrates them into a single empirical setting, allowing the analysis to move from abstract modeling outputs to economically interpretable results. This chapter, therefore, places into perspective the importance of evaluation metrics, the different types of errors they entail, and their implications for practical applications, beyond the specific results obtained from any particular metric.

Abstract

This chapter presents a methodological framework for estimating the economic value of soil erosion with a practical application to the region of Navarre in Spain. This framework is based on the coordinated use of land change models (LCM), the revised universal soil loss equation (RUSLE) and replacement cost (RC) models. Previous erosion literature has considered either LCM and RUSLE type models, overlooking the relevance of economic valuation, or RUSLE and RC type models, ignoring the influence of land changes. In addition, those that cover economic valuation tend to provide a static valuation. This methodology overcomes these limitations.

Based on observed data (2005–2023) and projections (2024–2030), the applied analysis shows the quantification and economic valuation of the impact of land changes on erosion in Navarre, showing a slight decrease in erosion with a slight increase in erosion valuation. It also produces spatially explicit, dynamic, and aggregated results that capture the spatial and temporal heterogeneity of erosion changes in Navarre, varying from -19% to -4.6% , depending on the areas. Moreover, the methodology identifies critical geographical areas where erosion is intense and economically significant. This is particularly valuable as it provides a basis for precise, cost-efficient decision-making.

Keywords: Economic valuation, soil erosion, environmental economics, land change.

JEL Classification: Q51, Q24, O13

4.1 Introduction

Soil erosion is a critical environmental challenge with significant economic and social consequences; as a result, it has received increasing attention in scientific literature in recent decades, as can be seen in the reviews of García-Ruiz et al. (2013) and Borrelli, Alewell, et al. (2021). Erosion leads to the loss of fertile soil essential for agriculture and natural ecosystems, affecting crop productivity, water quality, and terrestrial habitats (Pimentel et al. 1995; Sartori et al. 2019). The economic impacts of erosion are substantial, encompassing both direct costs to agriculture and broader economic implications (Martínez-Casasnovas and Ramos 2006). In Europe, erosion annually costs the agricultural sector approximately €295 million and results in a gross domestic product loss of about €155 million (Panagos, Standardi, et al. 2018). These economic evaluations provide insights for stakeholders and policymakers to understand erosion’s financial implications, guiding decisions on soil conservation investments (Senanayake, Pradhan, Wedathanthirige, et al. 2024).

Anthropogenic activities have been identified as the main drivers of accelerated soil erosion processes (Borrelli, Robinson, et al. 2017). The most direct pathway through which humans influence erosion is land use and land change, and thus, it represents the primary means for its mitigation (Pacheco et al. 2014; Panagos, Borrelli, Poesen, et al. 2015). Numerous studies have used the Revised Universal Soil Loss Equation (RUSLE) and land use/land cover change (LCM) models to predict future soil erosion at various scales, local (Paiboonvorachet and Oyana 2011; Pacheco et al. 2014; Berteni and Grossi 2020; Senanayake, Pradhan, Huete, et al. 2022; Richi 2025), and global (Podmanicky et al. 2011; Panagos, Borrelli, Poesen, et al. 2015; Borrelli, Robinson, et al. 2017; Vanwalleghe et al. 2017; Xiong and Leng 2024). While these studies provide valuable insights into the spatial distribution of erosion, they rarely incorporate an explicit valuation of its economic costs, limiting their ability to inform stakeholders about where erosion is most critical for an effective decision-making.

The economic impact of erosion has been examined from various angles in recent decades. Essentially, there are three approaches to assessing the economic effects of erosion. The first considers lost agricultural productivity (Asfaw et al. 2020; Crosson 1997; Panagos, Standardi, et al. 2018; Pimentel et al. 1995; Sartori et al. 2019). The

second approach attempts to evaluate the economic value of environmental services (Almansa et al. 2012; Bianchi et al. 2018; Lucchesi et al. 2024; Pandey et al. 2024). The third approach uses replacement costs (RC), focusing mainly on nutrients but also considering the replacement of plant mass (Getaneh et al. 2015; Hansen and Hellerstein 2007; Pandey et al. 2024; Pimentel et al. 1995; San and Rapera 2010; Selassie and Belay 2013). All of these approaches can be incorporated into cost-benefit analyses to assess erosion mitigation projects (Lucchesi et al. 2024; Marta-Pedroso et al. 2007; Martínez-Casasnovas and Ramos 2006). However, most of these assessments are conducted at aggregated scales, overlooking the spatial heterogeneity of erosion processes and the influence of land use dynamics on erosion-related costs.

In summary, existing studies assessing the causes and effects of soil erosion are, to some extent, incomplete. Many fail to address the economic relevance of their results, while others operate at scales or levels of aggregation that prevent the effective identification of areas where human activity most strongly influences erosion patterns. Therefore, the main contribution of this work is methodological. It is based on the coordinated use of LCM, RUSLE, and RC models, which are well-established in the field, to bridge existing gaps in the assessment and evaluation of soil erosion. First, this approach facilitates the quantification and economic valuation of the effects of land changes on erosion, ultimately providing insights into the social and economic relevance of erosion. Second, it produces spatially explicit, dynamic, and aggregated results that capture the spatial and temporal heterogeneity of erosion, providing an improved analytical framework. Third, it enables the identification of critical geographical areas where erosion is not only intense but also economically significant. This is particularly valuable, as it offers a basis for precision and cost-efficient decision-making. As an example, the methodology is applied to Navarre, a European region whose erosion patterns exhibit spatial heterogeneity influenced by diverse land uses and topography.

4.2 Methods

The methodology is designed to predict the evolution of the economic valuation of erosion, combining future land uses and nutrient prices in a three-step process. Each

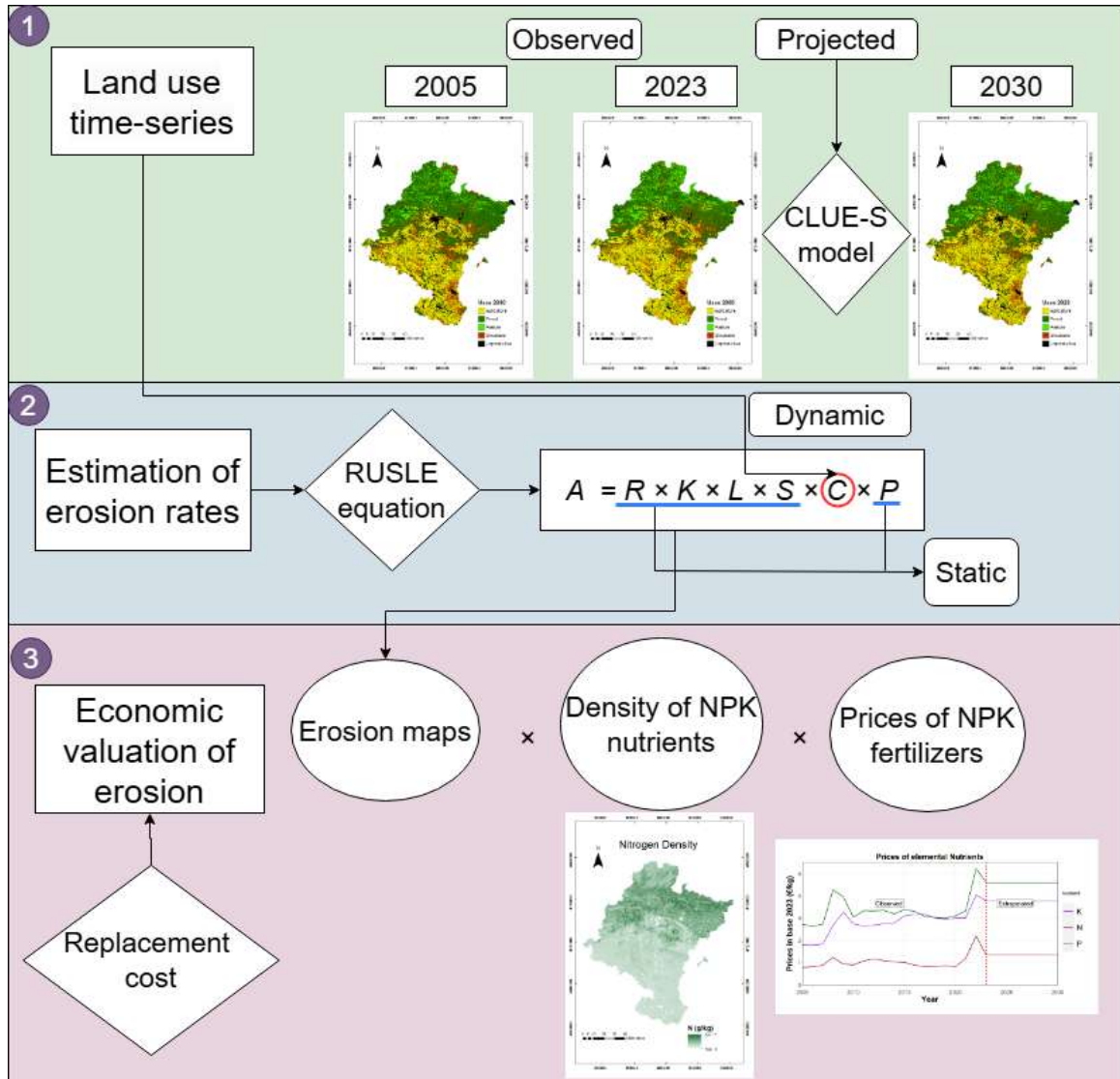


FIGURE 4.1: Graphical summary of the steps involved in the methodology. 1 application of CLUE-S to estimate a land use time-series. 2 application of the RUSLE equation to estimate erosion rates. 3 application of the replacement cost method to estimate the economic valuation of erosion. Source: Self elaboration.

of the outputs of the first two steps is used as an input for the subsequent step (Figure 4.1).

The first step focuses on land use and land change. The integration of land use maps into the Conversion of Land Use and its Effects at Small regional extent (CLUE-S) model (Verburg et al. 2002), a widely utilized tool for simulating land use dynamics, generates an observed and predicted spatiotemporal series of land uses.

The second step involves the application of the RUSLE (Renard et al. 1997) to generate spatio-temporal series of erosion. This methodology establishes the erosion amount (in tonnes), A , as the product of a set of key factors: the rainfall erosivity, R , the soil erodibility, K , the slope length, L , the slope steepness, S , the erosion coefficient associated with the land use, C , and the erosion control practice, P .

$$A = R \cdot K \cdot L \cdot S \cdot C \cdot P \quad (4.1)$$

The third step addresses the environmental valuation of erosion using the RC methodology, which evaluates the costs that society would incur to replace environmental resources that are no longer available (Holl and Howarth 2000). The basic premise of the RC method is that costs incurred in replacing productive assets can be interpreted as benefits when damage is prevented (Dixon et al. 1994). However, this method has conceptual shortcomings since soil nutrients can be lost through means other than soil erosion, as well as advantages since the method only requires estimates of soil erosion rates, nutrient analysis of eroded soils and nutrient prices (Gunatilake and Vieth 2000). This methodology allows the estimation of the valuation of erosion in terms of the costs required to replace the nutrients lost due to erosion events (Equation 4.2).

$$RC_{i,t} = A_{i,t} \sum_{j=1}^K N_{i,j} p_{j,t} \quad (4.2)$$

where $RC_{i,t}$ represents the nutrient replacement cost associated with soil erosion in pixel i in year t , expressed in euros; $A_{i,t}$ refers to the erosion in pixel i , in tonnes, in year t ; $N_{i,j}$ corresponds to the density of nutrient j in pixel i (kg/t); and $p_{j,t}$ refers to the price of nutrient j in year t (€/kg).

In summary, integrating the CLUE-S model enables the projection of future land use scenarios, which, when incorporated into the RUSLE model, generate soil erosion maps. These erosion estimates are then valued using the RC approach, producing a spatiotemporal series of erosion valuations.

4.3 Case study

As a case study, we propose the analysis of Navarre, a European region located in the north of Spain, with an area of 10,391 km² and a population of 678,333 inhabitants in 2024. Navarre is characterized by a great variety of landscapes, due to the great differences in orography and climate. The north is mountainous and rugged, with a humid continental climate, followed by a transitional zone with a smoother topography and a southern zone with large plains and a Mediterranean climate.

4.3.1 Data

The dataset used in the CLUE-S model consists of the available temporal series of land use maps for Navarre covering 2005–2023. These maps were obtained from Navarre’s Sistema de Información Geográfica de Parcelas Agrícolas (SIGPAC), the official agricultural geographic information system. The original maps, provided in vector format, were rasterized at a spatial resolution of 100 m. Land use categories were then reclassified into five classes: agriculture, forest, pasture, scrubland, and unproductive. These classes represent the main components of the regional land use mosaic and are associated with differentiated erosion rates.

The spatial information required by the RUSLE model is obtained from Panagos, Borrelli, Meusburger, et al. (2015) and Panagos, Borrelli, Poesen, et al. (2015). Specifically, the erosion coefficients for Navarre are 0.2 for agriculture, 0.09 for pasture, 0.04 for shrubland, 0.0011 for forest, and 0 for unproductive land.

The RC method requires information on nutrient density. We consider only essential soil nutrients: nitrogen, phosphorus, and potassium (NPK). We obtain the nutrient density data from the European Soil Database (ESDAC). These nutrients are unevenly distributed across the region, both in terms of density and geographical distribution (Figure 4.2). The prices of these nutrients, as well as their projection, are extracted from Campión et al., (2024).

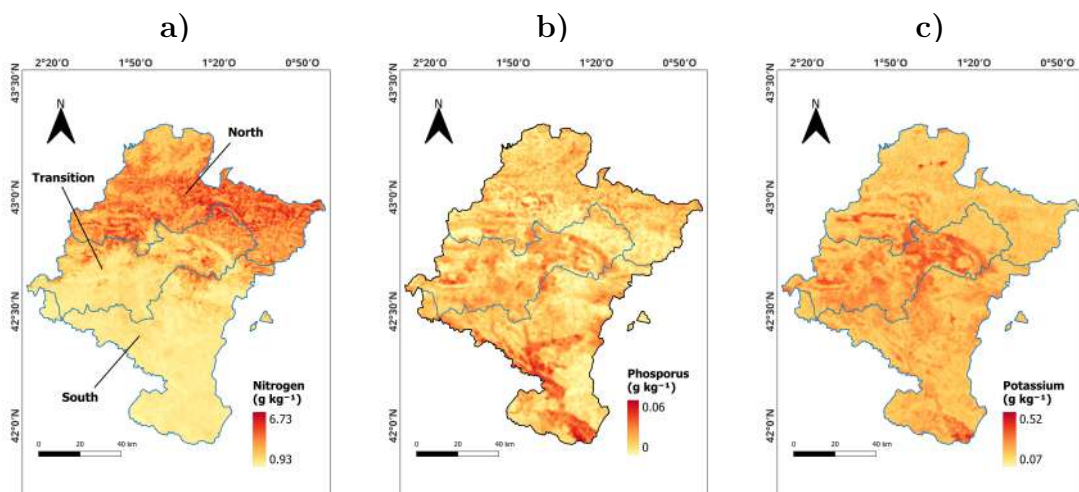


FIGURE 4.2: Spatial distribution of (a) Nitrogen, (b) Phosphorus and (c) Potassium densities (normalized colour scales). North, Transition and South areas are marked by a blue line. Source: Self elaboration based on ESDAC.

4.3.2 Results

Based on these data, projections were generated for the period 2024–2030. The year 2030 was chosen as the projection horizon as this is the target date for achieving the Sustainable Development Goals, for which soil quality has been identified as a key factor (Lal et al. 2021; Senanayake, Pradhan, Wedathanthirige, et al. 2024). The results are summarized in Figure 4.3, where each column corresponds to a step in the methodology, while the rows represent different time points. Table 4.1 complements Figure 4.3 by showing aggregate results for the region and the three aforementioned areas (North, Transition and South), which are also marked in the maps. The observed and projected land changes are small, as is the case in most developed regions (Figures 4.3a, 4.3d, and 4.3g). Across the region as a whole, forest cover has increased, and agricultural land has decreased between 2005 and 2023, and this trend is expected to continue until 2030. This change is the result of a transition from agriculture and pasture to shrubland in some locations, and from scrubland to forest in others. The maps that show these intermediate transitions are not displayed but are available upon request. However, this change is not uniform across all areas due to differences in land use, soil, and climate characteristics (see Table 4.1).

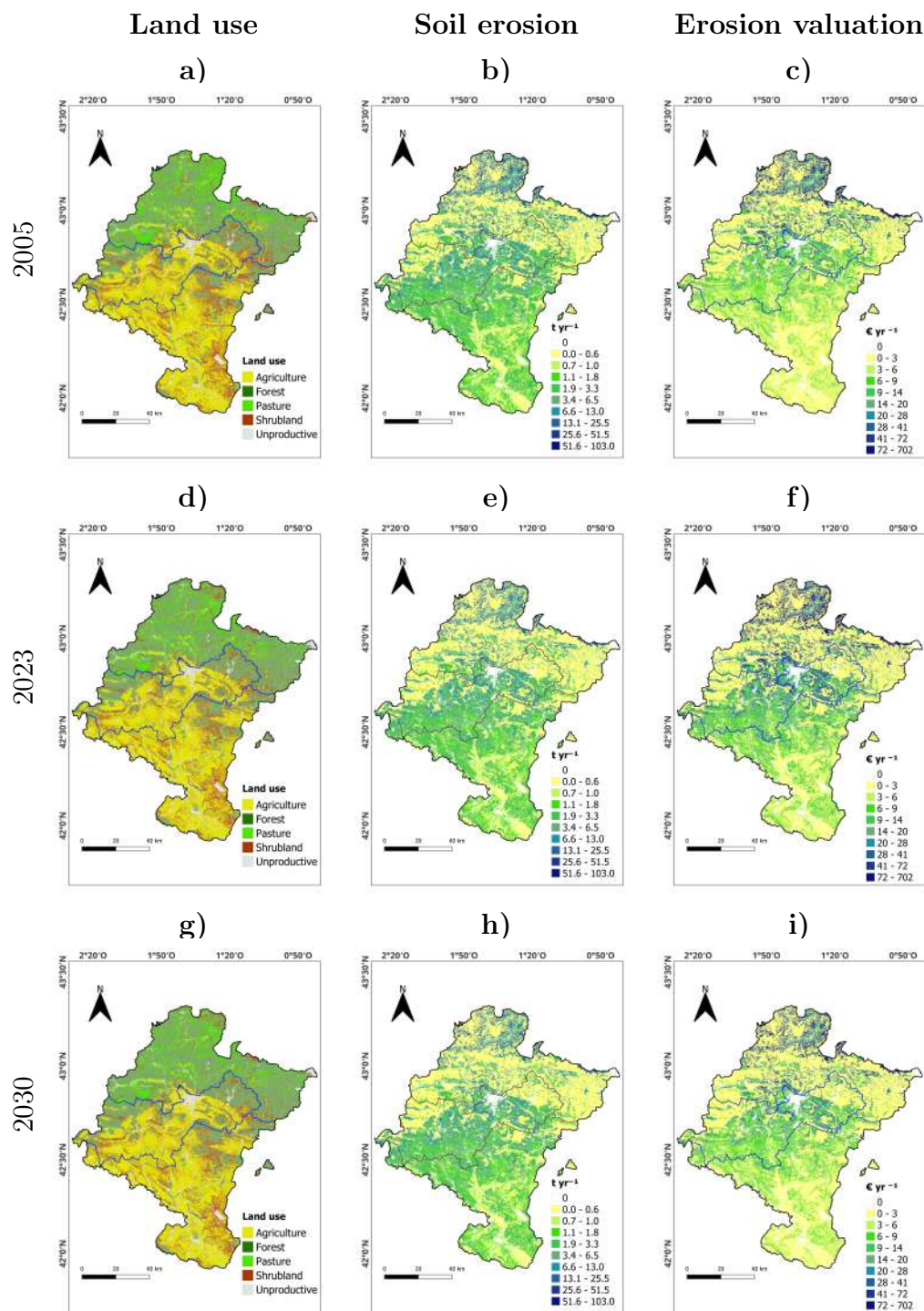


FIGURE 4.3: Evolution of land uses (column 1), soil erosion (column 2), and valuation of erosion (column 3), in years 2005 (row 1), 2023 (row 2) and 2030 (row 3). North, Transition and South areas are marked by a coloured line. Source: Self elaboration.

TABLE 4.1: Temporal evolution (2005, 2023 and 2030) of land use (percentage of the total area for each land use type within a specific region); erosion (in tonnes per hectare in a specific area); and valuation (in millions € in a specific area).

		Total	North	Transition	South
Extentsion (ha)		1,038,201	372,319	277,543	388,339
2005	Agriculture	34.0%	5.5%	41.7%	55.8%
	Forest	34.5%	59.2%	29.6%	14.2%
	Pasture	8.5%	22.2%	1.7%	0.2%
	Shrubland	16.4%	8.2%	19.3%	22.1%
	Unproductive	6.6%	4.8%	7.6%	7.7%
	Erosion (t ha ⁻¹)	2.94	3.93	3.04	1.92
	Valuation (million €)	7.58	4.44	1.86	1.28
2023	Agriculture	32.6%	4.0%	40.8%	54.2%
	Forest	37.1%	62.9%	32.7%	15.6%
	Pasture	7.5%	19.3%	1.6%	0.5%
	Shrubland	14.8%	8.4%	16.3%	19.9%
	Unproductive	7.9%	5.4%	8.6%	9.7%
	Erosion (t ha ⁻¹)	2.59	3.18	2.87	1.83
	Valuation (million €)	11.87	6.47	3.17	2.22
2030	Agriculture	32.1%	3.5%	40.1%	53.7%
	Forest	38.2%	64.9%	33.6%	15.8%
	Pasture	7.1%	18.5%	1.4%	0.3%
	Shrubland	14.2%	7.3%	15.7%	19.8%
	Unproductive	8.4%	5.7%	9.1%	10.4%
	Erosion (t ha ⁻¹)	2.48	2.97	2.78	1.80
	Valuation (million €)	7.81	4.17	2.12	1.52

The spatially aggregated results for the entire region and time span under consideration are presented in Figure 4.4. Figure 4.4a shows the aggregate evolution and

projection of land uses. In general terms, changes in land uses have occurred gradually over time, forest and unproductive uses increase at the expense of agricultural, pastures, and shrubland uses. Consequently, land uses with lower erosion coefficients occupy a larger area by the end of the study period. Additionally, agricultural uses, which have the highest erosion coefficients, show the greatest reduction in total area.

Land changes explain the changes observed in erosion maps (Figure 4.3b, 4.3e and 4.3h). In general, erosion decreases due to the transition to land uses with lower erosion coefficients. However, in the North area, this transition process is more intense than in the South where the stability of agricultural land prevents further reductions in erosion. According to the figures in Table 1, erosion decreased by 19% in the North and by just 4.6% in the South between 2005 and 2023. In addition, Figure 4.4b shows that the decrease in erosion in the whole region occurs for all essential nutrients, especially nitrogen. To facilitate comparability, Figure 4.4b expresses erosion as an index $2005 = 100$.

The distribution of nutrients (Figure 4.2) and the evolution of nutrient prices (Figure 4.4c) are two fundamental factors involved in the transition from erosion maps to economic valuation maps. The North area, where the main land use is forest, presents the highest nitrogen density and the highest reduction in erosion. The highest phosphorus density is in the South area, and the highest potassium density is in the Transition area. Agriculture is the main land use in these areas, and therefore, the decrease in erosion is similar. This heterogeneity in the distribution of nutrients has an additional effect on the economic valuation when the relative prices of nutrients change.

The evolution of prices is shown in Figure 4.4c. Comparing 2005 with 2023, a general increase in prices is observed and from 2023 to 2030, prices are expected to decrease to levels similar to 2005, except for potassium, leading to a change in relative prices. The notable 56.6% increase in erosion valuation in the region between 2005 and 2023 (Table 4.1) is due to the exceptional increase in fertilizer prices in 2022 (Campión et al. 2024). This anomalous price behaviour justifies comparing the results for 2030 with those for 2005: the erosion valuation in the region increases by 3%.

It is important to note that changes in the valuation of erosion are not uniform

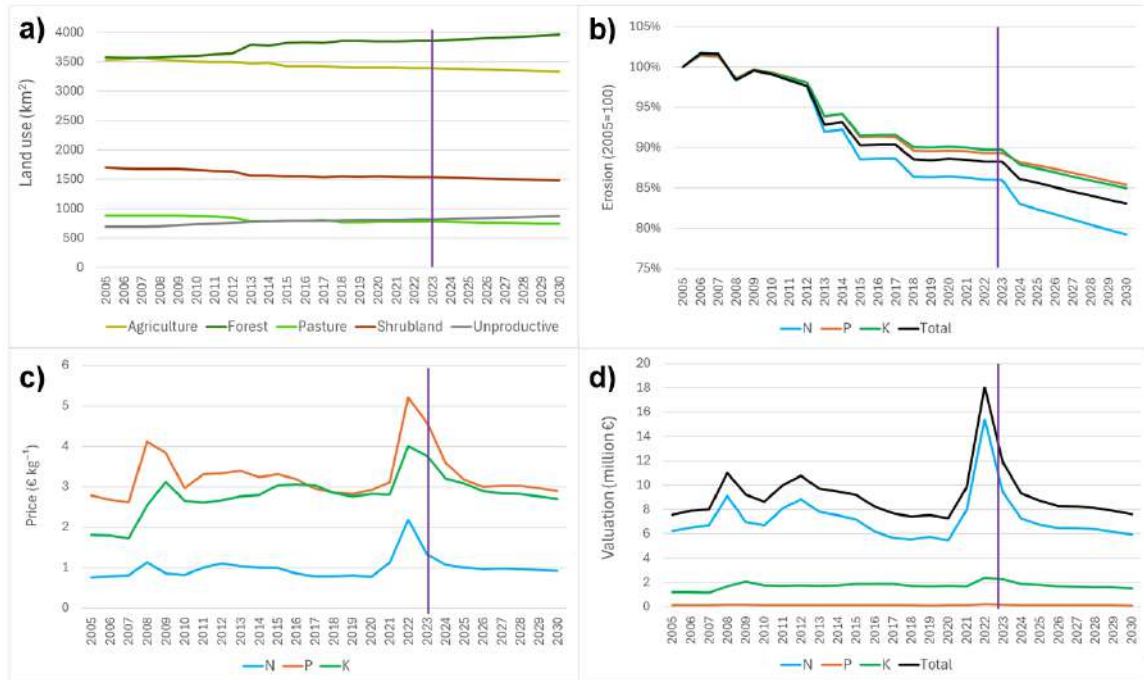


FIGURE 4.4: Evolution of (a) land uses, (b) soil erosion, (c) prices and (d) erosion valuation for the whole region. Panel (b) expresses erosion as an index 2005 = 100. The vertical line delimits the observation (2005-2023) and projection periods (2024-2030). Source: (a), (b) and (d), Self elaboration based on the results; (c) adapted from Campión et al (2024).

across the region (Figures 4.3c and 4.3i). In the North area, the erosion valuation decreases, whereas in the Transition and South areas, it increases (see Table 4.1). This discrepancy is due not only to the different evolution of erosion, but also to the concentration and price of nutrients. The North area is characterised by the greatest reduction in erosion, the highest nitrogen concentration and the lowest potassium concentration (Figure 4.2). The combination of these three factors means that the change in the relative price of these two nutrients (Figure 4.4c) does not compensate for the reduction in erosion, resulting in a decrease in the erosion valuation. The opposite is true in the other defined areas. These areas show smaller decreases in erosion, and potassium is relatively more important. This means that the change in relative prices leads to an increase in the valuation of erosion.

A more detailed analysis is facilitated by the spatial results, which permit the combination of Figures 4.3c and 4.3i in order to highlight changes in erosion valuation during this period, as shown in Figure 4.5. This figure shows two different but highly illustrative cases. The first refers to the eastern half of the Transition area. Most of

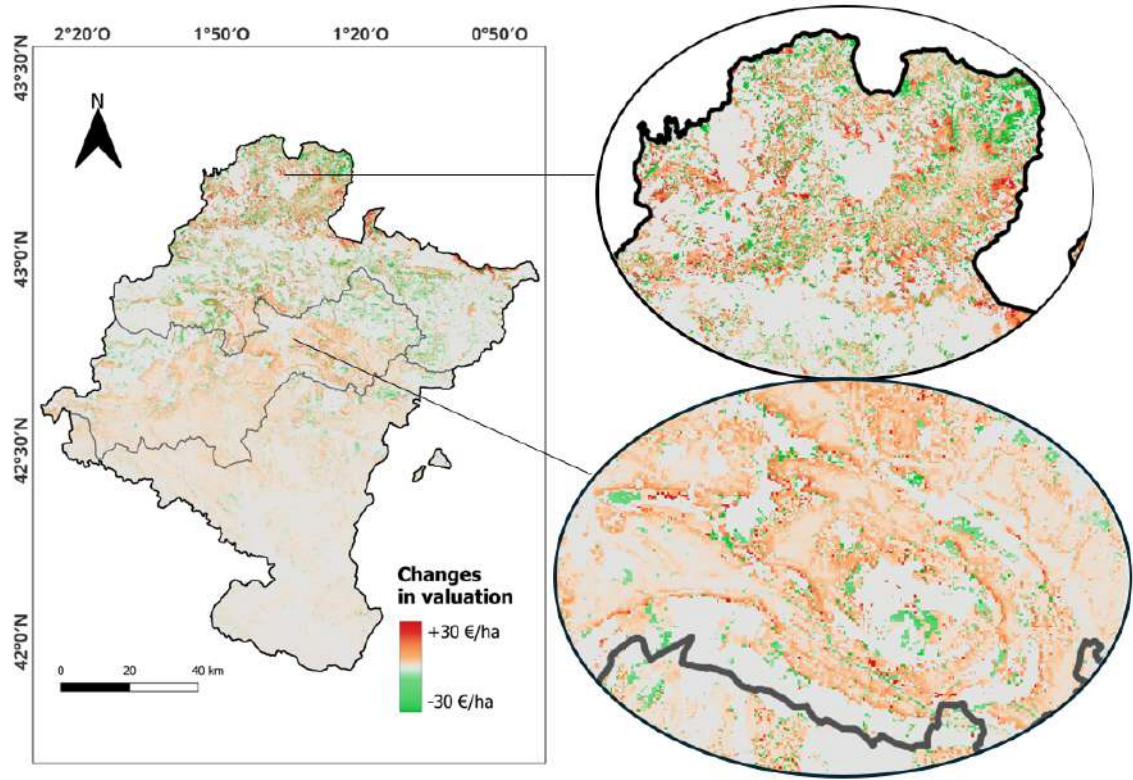


FIGURE 4.5: Changes in valuation of erosion between 2005 and 2030. Map of the region and two zoomed areas. North, Transition and South areas are marked by a coloured line.

this area shows a moderate to high increase in the economic valuation of erosion. The second case is observed in the northernmost part of the North area, where specific locations experience very high increases in the economic valuation of erosion, while others experience high decreases. Both areas could be subjected to different analyses, aggregated in the first case and highly disaggregated in the second.

In summary, the changes in erosion and erosion valuation from 2005 to 2030 are small when considered across the whole region. However, when the analysis focuses on the three natural areas (North, Transition and South), the changes are notable and allow the behaviour of the North area to be distinguished from that of the other two. Finally, spatial analysis has been used to identify specific areas that could be crucial in mitigation erosion policies.

4.4 Discussion

Erosion in Navarre, has been considered in previous studies (De Santisteban et al. 2006; Casalí, Gastesi, et al. 2008; Casalí, Giménez, et al. 2009). Unlike these studies, which analyze specific areas of the region or specific crops, this chapter allows for analysis at different spatial scales. Furthermore, the analysis is complemented by an economic assessment, a feature absent not only from the aforementioned studies, but also from many other erosion studies. However, accurately estimating the costs of soil erosion is an effective way of alerting agricultural producers, society and the government to the need for erosion control measures (Senanayake, Pradhan, Wedathanthirige, et al. 2024; Telles et al. 2013).

A common feature of the studies that have considered economic valuation is that they either focus on very specific locations or crops (e.g. Almansa et al. (2012), Bianchi et al. (2018), Lucchesi et al. (2024), and Pandey et al. (2024), among many others), or analyse geopolitically defined areas, regions, countries, continents or the world as a whole (e.g. Pimentel et al. (1995), Panagos, Standardi, et al. (2018), Sartori et al. (2019), and Asfaw et al. (2020), among many others). The methodology proposed in this study provides sufficient information to conduct a study of the region as a whole or of different pre-selected geographical areas. It can even be used to identify areas of special interest that would not initially have been considered.

In addition to incorporating economic valuation and enabling diverse spatial scales, the present research extends beyond the scope of conventional erosion studies by not only considering land use as a determining factor of erosion (usually by the RUSLE model or similar) but also exploring the impact of land changes on erosion dynamics.

Although all models used in this methodology have strengths and weaknesses, the RC model is of particular concern. The RC method has conceptual shortcomings. On the one hand, the RC method assumes the perfect substitution of resources (Dixon et al. 1994; Blockeel et al. 2013). However, merely replenishing lost nutrients does not fully restore the ecosystem services provided by the soil. Soil biology, structure and organic matter are also critical factors in maintaining soil functionality (Larney

et al. 2000). On the other hand, the RC method used in this article only considers the 'on-site cost' of soil nutrient loss. However, it ignores the 'off-site cost' of soil loss that spread beyond the area under consideration. This mainly occurs through water pollution, sedimentation processes and effects on biodiversity (Telles et al. 2013). Nevertheless, we use the RC model because data is available, accessible and reliable for most territories (Gunatilake and Vieth 2000). This broadens the reach and usage of proposed methodology.

Given the public budget constraints, the application of cost-effective policies is becoming mandatory for public authorities. In this context, the economic valuation of erosion can inform the design of effective erosion mitigation policies. These policies go beyond the replacement of nutrients, and our methodology can be used to evaluate the effectiveness of other public interventions, such as land use policies. This can be key to achieve the land degradation-neutral world as stated in Sustainable Development Goal 15.

The case study has been developed using historical data in a 'business as usual' scenario. However, society is currently facing important challenges that may affect erosion patterns, such as climate change and demographic shifts. The methodology proposed in this chapter could be used to develop future erosion analyses that incorporate climate change and demographic scenarios.

References

- Almansa, C., J. Calatrava, and J. M. Martínez-Paz (2012). "Extending the framework of the economic evaluation of erosion control actions in Mediterranean basins". In: *Land Use Policy* 29.2, pp. 294–308. DOI: [10.1016/j.landusepol.2011.06.013](https://doi.org/10.1016/j.landusepol.2011.06.013).
- Asfaw, S., G. Pallante, and A. Palma (2020). "Distributional impacts of soil erosion on agricultural productivity and welfare in Malawi". In: *Ecological Economics* 177, p. 106764. DOI: [10.1016/j.ecolecon.2020.106764](https://doi.org/10.1016/j.ecolecon.2020.106764).
- Berteni, F. and G. Grossi (2020). "Water Soil Erosion Evaluation in a Small Alpine Catchment Located in Northern Italy". In: *Geosciences* 10.10, p. 386. DOI: [10.3390/geosciences10100386](https://doi.org/10.3390/geosciences10100386).

- Bianchi, E. et al. (2018). “The Economic Evaluation of Forest Protection Service Against Rockfall”. In: *Ecological Economics* 154, pp. 409–418. DOI: [10.1016/j.ecolecon.2018.08.021](https://doi.org/10.1016/j.ecolecon.2018.08.021).
- Blockeel, Hendrik et al., eds. (2013). *Machine Learning and Knowledge Discovery in Databases*. en. 1st ed. Vol. 8190. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer Berlin Heidelberg. ISBN: 978-3-642-40993-6. DOI: [10.1007/978-3-642-40994-3](https://doi.org/10.1007/978-3-642-40994-3). URL: <http://link.springer.com/10.1007/978-3-642-40994-3> (visited on 08/14/2024).
- Borrelli, P., C. Alewell, P. Alvarez, et al. (2021). “Soil erosion modeling: A global review and statistical analysis”. In: *Science of The Total Environment* 780, p. 146494. DOI: [10.1016/j.scitotenv.2021.146494](https://doi.org/10.1016/j.scitotenv.2021.146494).
- Borrelli, P., D. A. Robinson, L. R. Fleischer, et al. (2017). “An Assessment of the Global Impact of 21st Century Land Use Change on Soil Erosion”. In: *Nature Communications* 8.1, p. 2013. DOI: [10.1038/s41467-017-02142-7](https://doi.org/10.1038/s41467-017-02142-7).
- Casalí, J., R. Gastesi, et al. (2008). “Runoff erosion and water quality of agricultural watersheds in central Navarre (Spain)”. In: *Agricultural Water Management* 95.10, pp. 1111–1128. DOI: [10.1016/j.agwat.2008.06.013](https://doi.org/10.1016/j.agwat.2008.06.013).
- Casalí, J., R. Giménez, et al. (2009). “Determination of long-term erosion rates in vineyards of Navarre (Spain) using botanical benchmarks”. In: *Catena* 78.1, pp. 12–19. DOI: [10.1016/j.catena.2009.02.015](https://doi.org/10.1016/j.catena.2009.02.015).
- Crosson, P. (1997). “The On-Farm Economic Costs of Soil Erosion”. In: *Methods for Assessment of Soil Degradation*. CRC Press.
- De Santisteban, L. M., J. Casalí, and J. J. López (2006). “Assessing soil erosion rates in cultivated areas of Navarre (Spain)”. In: *Earth Surface Processes and Landforms* 31, pp. 487–506. DOI: [10.1002/esp.1281](https://doi.org/10.1002/esp.1281).
- Dixon, R. K. et al. (1994). “Integrated Land-Use Systems”. In: *Climatic Change* 27.1, pp. 71–92. DOI: [10.1007/BF01098474](https://doi.org/10.1007/BF01098474).
- García-Ruiz, J. M. et al. (2013). “Erosion in Mediterranean Landscapes: Changes and Future Challenges”. In: *Geomorphology* 198, pp. 20–36. DOI: [10.1016/j.geomorph.2013.05.023](https://doi.org/10.1016/j.geomorph.2013.05.023).

- Getaneh, K., A. Gessesse, M. Addisie, et al. (2015). “The economic cost of upland and gully erosion”. In: *African Journal of Agricultural and Resource Economics* 10, pp. 265–278.
- Gunatilake, H. M. and G. R. Vieth (2000). “Estimation of on-site cost of soil erosion”. In: *Journal of Soil and Water Conservation* 55.2, pp. 197–204. DOI: [10.1080/00224561.2000.12436394](https://doi.org/10.1080/00224561.2000.12436394).
- Hansen, L. and D. Hellerstein (2007). “The Value of the Reservoir Services Gained with Soil Conservation”. In: *Land Economics* 83.3, pp. 285–301. DOI: [10.3368/le.83.3.285](https://doi.org/10.3368/le.83.3.285).
- Holl, K. D. and R. B. Howarth (2000). “Paying for Restoration”. In: *Restoration Ecology* 8.3, pp. 260–267. DOI: [10.1046/j.1526-100x.2000.80037.x](https://doi.org/10.1046/j.1526-100x.2000.80037.x).
- Lal, R., J. Bouma, E. Brevik, et al. (2021). “Soils and sustainable development goals of the United Nations”. In: *Geoderma Regional* 25, e00398. DOI: [10.1016/j.geodrs.2021.e00398](https://doi.org/10.1016/j.geodrs.2021.e00398).
- Larney, Francis J. et al. (2000). “Early Impact of Topsoil Removal and Soil Amendments on Crop Productivity”. en. In: *Agronomy Journal* 92.5, pp. 948–956. ISSN: 1435-0645. DOI: [10.2134/agronj2000.925948x](https://doi.org/10.2134/agronj2000.925948x). (Visited on 06/24/2024).
- Lucchesi, A., M. Khanna, P. C. Pereda, et al. (2024). “Araguaia biodiversity corridor cost benefit analysis”. In: *Land Use Policy* 141, p. 107122. DOI: [10.1016/j.landusepol.2024.107122](https://doi.org/10.1016/j.landusepol.2024.107122).
- Marta-Pedroso, C. et al. (2007). “Cost–benefit analysis of the Zonal Program of Castro Verde”. In: *Soil and Tillage Research* 97.1, pp. 79–90. DOI: [10.1016/j.still.2007.08.010](https://doi.org/10.1016/j.still.2007.08.010).
- Martínez-Casasnovas, J. A. and M. C. Ramos (2006). “The cost of soil erosion in vineyard fields”. In: *Catena* 68.2, pp. 194–199. DOI: [10.1016/j.catena.2006.04.007](https://doi.org/10.1016/j.catena.2006.04.007).
- Pacheco, F. A. L. et al. (2014). “Soil losses in rural watersheds with environmental land use conflicts”. In: *Science of The Total Environment* 485–486, pp. 110–120. DOI: [10.1016/j.scitotenv.2014.03.069](https://doi.org/10.1016/j.scitotenv.2014.03.069).
- Paiboonvorachart, C. and T. J. Oyana (2011). “Land-cover changes and potential impacts on soil erosion in Thailand”. In: *International Journal of Remote Sensing* 32.21, pp. 6587–6609. DOI: [10.1080/01431161.2010.512935](https://doi.org/10.1080/01431161.2010.512935).

- Panagos, P., P. Borrelli, K. Meusburger, et al. (2015). “Estimating the soil erosion cover-management factor at the European scale”. In: *Land Use Policy* 48, pp. 38–50. DOI: [10.1016/j.landusepol.2015.05.021](https://doi.org/10.1016/j.landusepol.2015.05.021).
- Panagos, P., P. Borrelli, J. Poesen, et al. (2015). “The new assessment of soil loss by water erosion in Europe”. In: *Environmental Science & Policy* 54, pp. 438–447. DOI: [10.1016/j.envsci.2015.08.012](https://doi.org/10.1016/j.envsci.2015.08.012).
- Panagos, P., G. Standardi, et al. (2018). “Cost of Agricultural Productivity Loss Due to Soil Erosion in the European Union”. In: *Land Degradation & Development* 29.3, pp. 471–484. DOI: [10.1002/ldr.2879](https://doi.org/10.1002/ldr.2879).
- Pandey, R. et al. (2024). “Quantifying soil erosion and soil organic carbon conservation services”. In: *Ecological Indicators* 163, p. 112086. DOI: [10.1016/j.ecolind.2024.112086](https://doi.org/10.1016/j.ecolind.2024.112086).
- Pimentel, D., C. Harvey, P. Resosudarmo, et al. (1995). “Environmental and Economic Costs of Soil Erosion and Conservation Benefits”. In: *Science* 267.5201, pp. 1117–1123. DOI: [10.1126/science.267.5201.1117](https://doi.org/10.1126/science.267.5201.1117).
- Podmanicky, L. et al. (2011). “modeling soil quality changes in Europe”. In: *Ecological Indicators* 11.1, pp. 4–15. DOI: [10.1016/j.ecolind.2009.08.002](https://doi.org/10.1016/j.ecolind.2009.08.002).
- Renard, K. G. et al. (1997). *Predicting Soil Erosion by Water: A Guide to Conservation Planning with the Revised Universal Soil Loss Equation (RUSLE)*. Agricultural Research Service.
- Richi, S. (2025). “Integrated RUSLE-GIS Modeling for Enhancing Soil Erosion Management”. In: *DYSONA–Applied Science* 6.1, pp. 104–112.
- San, C. C. and C. Rapera (2010). “The On-site Cost of Soil Erosion by the Replacement Cost Methods”. In: *Journal of Environmental Science and Management* 13, pp. 67–81.
- Sartori, M., G. Philippidis, E. Ferrari, et al. (2019). “Assessing the global market impacts of soil erosion”. In: *Land Use Policy* 86, pp. 299–312. DOI: [10.1016/j.landusepol.2019.05.014](https://doi.org/10.1016/j.landusepol.2019.05.014).
- Selassie, Y. and Y. Belay (2013). “Costs of Nutrient Losses in Priceless Soils”. In: *Journal of Agricultural Science* 5.7, p. 227. DOI: [10.5539/jas.v5n7p227](https://doi.org/10.5539/jas.v5n7p227).

- Senanayake, S., B. Pradhan, A. Huete, et al. (2022). “Spatial modeling of soil erosion hazards”. In: *Science of The Total Environment* 806, p. 150405. DOI: [10.1016/j.scitotenv.2021.150405](https://doi.org/10.1016/j.scitotenv.2021.150405).
- Senanayake, S., B. Pradhan, H. Wedathanthirige, et al. (2024). “Monitoring soil erosion in support of achieving SDGs”. In: *Catena* 234, p. 107537. DOI: [10.1016/j.catena.2023.107537](https://doi.org/10.1016/j.catena.2023.107537).
- Telles, Tiago Santos et al. (June 2013). “Valuation and assessment of soil erosion costs”. en. In: *Scientia Agricola* 70, pp. 209–216. ISSN: 1678-992X. DOI: [10.1590/S0103-90162013000300010](https://doi.org/10.1590/S0103-90162013000300010). (Visited on 07/16/2024).
- Vanwalleghe, T., J. A. Gómez, J. Infante Amate, et al. (2017). “Impact of historical land use and soil management change”. In: *Anthropocene* 17, pp. 13–29. DOI: [10.1016/j.ancene.2017.01.002](https://doi.org/10.1016/j.ancene.2017.01.002).
- Verburg, Peter H. et al. (2002). “Modeling the Spatial Dynamics of Regional Land Use: The CLUE-S Model”. In: *Environmental Management* 30.3, pp. 391–405. ISSN: 0364-152X. DOI: [10.1007/s00267-002-2630-x](https://doi.org/10.1007/s00267-002-2630-x).
- Xiong, M. and G. Leng (2024). “Global soil water erosion responses to climate and land use changes”. In: *Catena* 241, p. 108043. DOI: [10.1016/j.catena.2024.108043](https://doi.org/10.1016/j.catena.2024.108043).

General Conclusions and Further Research Lines

4.5 General conclusions

This dissertation reviews, revisits, and develops evaluation metrics for binary classification, with a special focus on land change models. It addresses two main research objectives. The first is to make the concepts and methods of binary classification evaluation more accessible and useful for end users. To this end, the dissertation reviews and organizes the core ideas in the field and introduces a methodology designed to improve clarity and interpretability.

The second objective is to examine how these concepts and methods can improve the validation of land change models. By incorporating insights from other disciplines and applying the proposed methodology, this dissertation aims to refine and update validation practices in this field.

Regarding the **first objective**, **Chapter 1** provides a structured review of the concepts, methodologies, and metrics used in binary classification evaluation. This contribution addresses the fragmented nature of the field, where similar concepts and approaches have often been developed independently across disciplines under different names. By bringing this literature together under a more homogeneous language, the chapter facilitates a deeper understanding of binary classification evaluation and encourages the transfer of methods across fields of application. This synthesis is particularly valuable for introducing underused but suitable methodologies into areas of research where they have received little attention.

The chapter also serves as a guide for both researchers who are new to the topic and those who already apply these techniques but may not be familiar with the wider range of available validation approaches. In this sense, it functions as a bridge between disciplines and creates the basis for more informed and methodologically consistent applications of classification evaluation tools.

In line with this objective, **Chapter 2** introduces the CSI framework as an intuitive and adaptable tool for applications with low prevalence. The framework complements existing methodologies in several ways. First, it provides graphical and numerical tools based on the same family of metrics for the validation of both single-threshold and multiple-threshold classifications. This improves the comparability of

soft and hard outputs and helps clarify whether the errors observed in a final classification are mainly driven by the design of the classifier or by the choice of threshold. The joint use of AUCSI, AUCSIS, CSI, and RGSS supports the analysis of both Agreement and Skill, making model behavior easier to interpret and validation results easier to communicate.

Second, the framework makes it possible to distinguish between Quantity and Allocation errors. This is a substantial advantage over the ROC/AUC paradigm, especially in applications where this distinction is essential for interpretation. In practice, it is often important to know whether a classifier fails because it predicts too many or too few occurrences of the event, or because it allocates them poorly across observations. The CSI framework makes this distinction explicit for both single and multiple classifications.

A broader contribution of the framework is that it offers an accessible alternative to the deeply rooted ROC/AUC paradigm in low-prevalence applications. AUC tends to report high values under low-prevalence conditions, which helps explain its persistence in many fields despite its limitations in such settings. This creates an important barrier to the adoption of more suitable metrics, which often provide less optimistic but more informative assessments. While alternatives such as Precision–Recall curves are available, they have not achieved the same level of diffusion among end users. In this context, the CSI framework provides a more interpretable and more coherent alternative for the evaluation of rare events.

At the intersection of methodological development and scientific practice, this is an important contribution. Clear, informative, and interpretable methodologies can help researchers move away from metrics that are popular but not well-suited to their data. In this regard, the CSI framework integrates information on single-threshold and multiple-threshold metrics, while also distinguishing between Quantity and Allocation errors within a unified graphical representation. This makes the framework not only methodologically useful but also easier to adopt in practice.

All in all, the CSI framework has the potential to challenge the dominance of ROC/AUC in applications where it is not appropriate, encouraging a shift toward evaluation practices that better reflect the demands of low-prevalence settings. Such a shift can lead to more faithful assessments of model behavior and, ultimately, to

more robust research across different fields.

As for the **second objective**, **Chapter 3** applies these concepts to the validation of land change models and critically revises many current practices in the field. This chapter contributes to the literature by identifying important limitations in existing validation methods and by introducing the CSI framework as a basis for unifying soft and hard validation. Since land-use change is inherently a low-prevalence phenomenon, AUC is not well-suited to assess these changes, even though it remains the dominant validation tool for suitability maps. In this context, AUCSI offers a more appropriate alternative for evaluating those outputs, which strongly influence the final results of land change models.

The chapter also provides a critical review of the Figure of Merit (FOM), one of the most widely used metrics in land change model validation. In addition, it revisits two influential studies that have often been used to justify the low FOM values typically reported in the literature (Pontius, Boersma, et al. 2008; Pontius, Castella, et al. 2018). While FOM is one of the most appropriate Agreement metrics for land change models because it focuses on changes, its interpretation and comparison are often problematic (Liu et al. 2017; Meentemeyer et al. 2013; Zhang et al. 2022). **Chapter 3** shows that FOM depends on prevalence and on the quantity of simulated change, so comparisons across models and applications can be misleading. Even when a correlation exists between prevalence and FOM, an important share of that relationship arises simply because more changes create more opportunities for Agreement.

In this sense, the incorporation of the concept of Skill into land change model validation represents a substantial advance. In low-prevalence contexts, Skill makes it possible to distinguish between Agreement due to actual model performance and Agreement due merely to the abundance of changes available to match. This distinction is essential for qualifying the conclusions drawn from validation results. **Chapter 3** shows that modelers cannot infer the goodness of fit or superiority of a model from FOM values alone, because FOM is a metric of Agreement. Such claims require metrics of Skill, such as the Revised Gilbert Skill Score.

The closing section of **Chapter 3** addresses both objectives of the dissertation simultaneously by proposing a series of recommendations for better validation practices in land change models, and more generally in binary classification problems.

These recommendations include replacing AUC with AUFOM (or AUCSI) for model validation, adopting FOM (or CSI) curves as complements to TOC curves, and reporting both Agreement and Skill metrics. They also include the unification of soft and hard validation within FOM (or CSI) graphs. These recommendations aim to promote more rigorous validation procedures and more informative interpretations in low-prevalence contexts. In the specific case of land change models, their adoption could improve model understanding and encourage a more critical assessment of actual model Skill.

Chapter 4 complements these contributions by placing the theoretical and methodological developments of the dissertation in an empirical setting. While the previous chapters focus on the evaluation and interpretation of binary classification metrics, this chapter shows how these insights can be embedded in a concrete application linking land change model outputs with economic valuation and policy-relevant analysis. By developing a framework to estimate the economic value of soil erosion through the integration of land change models, RUSLE, and replacement cost methods, the chapter illustrates how model outputs acquire practical meaning when connected to real-world processes.

In doing so, the chapter highlights the importance of evaluation metrics, the errors they reflect, and the implications of those errors beyond the numerical values of particular indicators. When model outputs are used to estimate economic costs and inform policy decisions, differences in model performance and error structure translate directly into differences in valuation and prioritization. Validation is therefore not merely a technical exercise, but a necessary condition for producing reliable and actionable information.

Moreover, the chapter shows how the integration of modeling, spatial analysis, and economic valuation can support the design of more targeted and efficient policy interventions. By identifying areas where erosion is both physically significant and economically relevant, the proposed approach helps bridge the gap between model evaluation and decision-making. In this way, **Chapter 4** reinforces the central argument of the dissertation: improving the evaluation of binary classification models is essential not only for methodological rigor, but also for enhancing their usefulness in applied economic and environmental contexts.

Beyond its initial objectives, this dissertation makes several additional contributions. The revision and extension of the Gilbert Skill Score addresses a fundamental inconsistency in its formulation. The Gilbert Skill Score has been a widely used metric since its inception in 1884. This revision improves its applicability across disciplines such as meteorology, hydrology, climate science, and economics, particularly in the study of rare events. Although the correction does not alter the direction of the Skill measurement, it modifies its magnitude and may therefore lead to different substantive conclusions. The design of RGSS also opens the door to the development of new Skill metrics focused on a single class, which may be useful for a broader range of research questions.

In addition, the representation of single-threshold metrics across multiple thresholds proves to be a powerful tool for interpreting conventional binary classification metrics. This approach makes it possible to build new frameworks such as the proposed CSI framework, which evaluates both single and multiple thresholds under a unified definition of Agreement². The analyses developed in this dissertation show the analytical value of this perspective, which remains largely unexplored in the literature.

Finally, the theoretical simulations in **Chapter 2** provide useful insights into the relationships between different metrics. For example, we show that a classifier with a given AUC value can achieve a maximum CSI value of $2 \cdot \text{AUC} - 1$, regardless of prevalence. Understanding these relationships is important for promoting the appropriate use of metrics and for improving their interpretation among end users. Future research could extend this line of inquiry by studying the relationships among other multiple-threshold metrics, among single-threshold metrics, and between single-threshold and multiple-threshold metrics.

All the results of this dissertation point to a common conclusion: the selection of a classifier or a classification threshold must be guided by the underlying research question. Thresholds are not arbitrary choices, but decisions that should reflect the objectives and trade-offs of the specific problem under study. Likewise, the choice of evaluation metrics should not be determined by popularity or disciplinary convention, but by their suitability for the application at hand. More generally, metrics should

²Figure 2.9 provides an initial illustration of this idea using OA as a reference metric.

not be interpreted as isolated summary values. They should be understood together with their graphical representation, which reveals how performance changes across thresholds and clarifies the balance between errors and successes under different decision criteria. This broader perspective makes evaluation more rigorous, transparent, and useful for end users, leading to more reliable and more meaningful interpretations of classification results.

4.6 Conclusiones generales

Esta tesis revisa, reexamina y desarrolla métricas de evaluación para la clasificación binaria, con un enfoque especial en los modelos de cambio de uso del suelo. Aborda dos objetivos principales de investigación. El primero es hacer que los conceptos y métodos de evaluación de la clasificación binaria sean más accesibles y útiles para los usuarios finales. Para ello, la tesis revisa y organiza las ideas centrales del campo e introduce una metodología diseñada para mejorar la claridad y la interpretabilidad.

El segundo objetivo es analizar cómo estos conceptos y métodos pueden mejorar la validación de los modelos de cambio de uso del suelo. Mediante la incorporación de ideas procedentes de otras disciplinas y la aplicación de la metodología propuesta, esta tesis pretende refinar y actualizar las prácticas de validación en este ámbito.

En relación con el **primer objetivo**, el **Capítulo 1** presenta una revisión estructurada de los conceptos, metodologías y métricas empleadas en la evaluación de clasificaciones binarias. Esta contribución aborda el carácter fragmentado del campo, donde conceptos y enfoques similares se han desarrollado de forma independiente en distintas disciplinas bajo diferentes denominaciones. Al reunir esta literatura bajo un lenguaje más homogéneo, el capítulo facilita una comprensión más profunda de la evaluación de la clasificación binaria y fomenta la transferencia de metodologías entre áreas de aplicación. Esta síntesis resulta especialmente valiosa para introducir metodologías infrautilizadas pero adecuadas en áreas de investigación donde han recibido poca atención.

El capítulo también actúa como guía tanto para investigadores que se inician en el tema como para aquellos que ya aplican estas técnicas, pero no están familiarizados con la amplia variedad de enfoques de validación disponibles. En este sentido,

funciona como un puente entre disciplinas y sienta las bases para aplicaciones más informadas y metodológicamente consistentes de las herramientas de evaluación.

En línea con este objetivo, el **Capítulo 2** introduce el marco CSI como una herramienta intuitiva y adaptable para aplicaciones con baja prevalencia. El marco complementa las metodologías existentes de varias maneras. En primer lugar, proporciona herramientas gráficas y numéricas basadas en la misma familia de métricas para la validación de clasificaciones con umbral único y con múltiples umbrales. Esto mejora la comparabilidad entre salidas soft y hard, y ayuda a clarificar si los errores observados en una clasificación final se deben principalmente al diseño del clasificador o a la elección del umbral. El uso conjunto de AUCSI, AUCSIS, CSI y RGSS permite analizar tanto Agreement como Skill, facilitando la interpretación del comportamiento del modelo y la comunicación de los resultados de validación.

En segundo lugar, el marco permite distinguir entre errores de Cantidad (Quantity) y de Asignación (Allocation). Esta es una ventaja sustancial frente al paradigma ROC/AUC, especialmente en aplicaciones donde esta distinción es clave para la interpretación. En la práctica, a menudo es fundamental saber si un clasificador falla porque predice demasiadas o muy pocas ocurrencias del evento, o porque las asigna incorrectamente a las observaciones. El marco CSI hace explícita esta distinción tanto para clasificaciones de umbral único como para clasificaciones de múltiples umbrales.

Una contribución más amplia del marco es que ofrece una alternativa accesible al arraigado paradigma ROC/AUC en aplicaciones con baja prevalencia. El AUC tiende a reportar valores elevados en este tipo de contextos, lo que ayuda a explicar su persistencia en muchos campos a pesar de sus limitaciones. Esto genera una barrera importante para la adopción de métricas más adecuadas, que a menudo proporcionan evaluaciones menos optimistas pero más informativas. Aunque existen alternativas como las curvas Precision–Recall, estas no han alcanzado el mismo grado de difusión entre los usuarios finales. En este contexto, el marco CSI proporciona una alternativa más interpretable y coherente para la evaluación de eventos raros.

En la intersección entre el desarrollo metodológico y la práctica científica, esta constituye una contribución relevante. Metodologías claras, informativas e interpretables pueden ayudar a los investigadores a abandonar métricas que son populares pero no adecuadas para sus datos. En este sentido, el marco CSI integra la información de

métricas de umbral único y múltiple, al tiempo que distingue entre errores de Cantidad y Asignación dentro de una representación gráfica unificada. Esto hace que el marco no solo sea útil desde un punto de vista metodológico, sino también más fácil de adoptar en la práctica.

En conjunto, el marco CSI tiene el potencial de cuestionar la dominancia de ROC/AUC en aplicaciones donde no resulta apropiado, fomentando una transición hacia prácticas de evaluación más acordes con las exigencias de contextos de baja prevalencia. Este cambio puede conducir a evaluaciones más fieles del comportamiento de los modelos y, en última instancia, a una investigación más sólida en distintos campos.

En cuanto al **segundo objetivo**, el **Capítulo 3** aplica estos conceptos a la validación de modelos de cambio de uso del suelo y revisa críticamente muchas de las prácticas actuales en este ámbito. Este capítulo contribuye a la literatura al identificar limitaciones relevantes en los métodos de validación existentes e introducir el marco CSI como base para unificar la validación soft y hard. Dado que el cambio de uso del suelo es inherentemente un fenómeno de baja prevalencia, el AUC no es adecuado para evaluarlo, aunque sigue siendo la herramienta de validación predominante para los mapas de idoneidad. En este contexto, AUCSI ofrece una alternativa más apropiada para evaluar estas salidas, que influyen de manera decisiva en los resultados finales de los modelos.

El capítulo también presenta una revisión crítica del Figure of Merit (FOM), una de las métricas más utilizadas en la validación de modelos de cambio de uso del suelo. Asimismo, revisa dos estudios influyentes que han sido frecuentemente utilizados para justificar los bajos valores de FOM observados en la literatura (Pontius, Boersma, et al. 2008; Pontius, Castella, et al. 2018). Aunque el FOM es una de las métricas de Agreement más adecuadas en este contexto, ya que se centra en los cambios, su interpretación y comparación presentan problemas (Liu et al. 2017; Meentemeyer et al. 2013; Zhang et al. 2022). El **Capítulo 3** muestra que el FOM depende de la prevalencia y de la cantidad de cambio simulado, por lo que las comparaciones entre modelos y aplicaciones pueden resultar engañosas. Incluso cuando existe una correlación entre prevalencia y FOM, una parte importante de esta relación se debe simplemente a que un mayor número de cambios genera más oportunidades de Agreement.

En este sentido, la incorporación del concepto de Skill en la validación de modelos de cambio de uso del suelo supone un avance sustancial. En contextos de baja prevalencia, el Skill permite distinguir entre el Agreement derivado del desempeño real del modelo y aquel que resulta únicamente de la existencia de más cambios con los que coincidir. Esta distinción es fundamental para matizar las conclusiones extraídas de los resultados de validación. El **Capítulo 3** muestra que no es posible inferir la bondad de ajuste o la superioridad de un modelo a partir de valores de FOM, ya que este es un indicador de Agreement. Este tipo de conclusiones requiere el uso de métricas de Skill, como el Revised Gilbert Skill Score.

La sección final del **Capítulo 3** aborda simultáneamente ambos objetivos de la tesis proponiendo una serie de recomendaciones para mejorar las prácticas de validación en modelos de cambio de uso del suelo y, en general, en problemas de clasificación binaria. Entre estas recomendaciones se incluyen la sustitución del AUC por AUFOM (o AUCSI) para la validación de modelos, la adopción de curvas FOM (o CSI) como complemento de las curvas TOC, y la presentación conjunta de métricas de Agreement y Skill. Asimismo, se propone utilizar referencias más exigentes que el azar, interpretar las métricas junto con sus representaciones gráficas y unificar la validación soft y hard dentro de gráficos FOM (o CSI). Estas propuestas buscan promover procedimientos de validación más rigurosos y interpretaciones más informativas en contextos de baja prevalencia. En el caso específico de los modelos de cambio de uso del suelo, su adopción podría mejorar la comprensión de los modelos y fomentar una evaluación más crítica de su Skill real.

El **Capítulo 4** complementa estas contribuciones situando los desarrollos teóricos y metodológicos de la tesis en un contexto empírico. Mientras que los capítulos anteriores se centran en la evaluación e interpretación de métricas de clasificación binaria, este capítulo muestra cómo estos avances pueden integrarse en una aplicación concreta que vincula los resultados de modelos de cambio de uso del suelo con la valoración económica y el análisis orientado a políticas públicas. Mediante el desarrollo de un marco para estimar el valor económico de la erosión del suelo a través de la integración de modelos de cambio de uso del suelo, RUSLE y métodos de coste de reposición, el capítulo ilustra cómo los resultados de los modelos adquieren significado práctico cuando se conectan con procesos reales.

Al hacerlo, el capítulo pone de relieve la importancia de las métricas de evaluación, los errores que reflejan y sus implicaciones más allá de los valores numéricos de indicadores concretos. Cuando los resultados de los modelos se utilizan para estimar costes económicos y orientar decisiones de política pública, las diferencias en el rendimiento de los modelos y en la estructura de sus errores se traducen directamente en diferencias en la valoración y la priorización. Por tanto, la validación no es únicamente un ejercicio técnico, sino una condición necesaria para generar información fiable y útil.

Asimismo, el capítulo muestra cómo la integración de modelización, análisis espacial y valoración económica puede contribuir al diseño de intervenciones de política más eficientes y focalizadas. Al identificar áreas donde la erosión es tanto físicamente relevante como económicamente significativa, el enfoque propuesto ayuda a cerrar la brecha entre la evaluación de modelos y la toma de decisiones. De este modo, el **Capítulo 4** refuerza el argumento central de la tesis: mejorar la evaluación de modelos de clasificación binaria es fundamental no solo para el rigor metodológico, sino también para aumentar su utilidad en contextos económicos y ambientales aplicados.

Más allá de los objetivos iniciales, esta tesis realiza varias contribuciones adicionales. La revisión y extensión del Gilbert Skill Score corrige una inconsistencia fundamental en su formulación. Esta revisión mejora su aplicabilidad en disciplinas como la meteorología, la hidrología, la ciencia del clima y la economía, especialmente en el análisis de eventos raros. Aunque la corrección no altera la dirección de la medición del Skill, sí modifica su magnitud, lo que puede dar lugar a conclusiones distintas en su aplicación. El desarrollo de RGSS también abre la puerta al diseño de nuevas métricas de Skill centradas en una sola clase, lo que puede resultar útil para una mayor variedad de preguntas de investigación.

Además, la representación de métricas de umbral único a lo largo de múltiples umbrales se muestra como una herramienta poderosa para interpretar las métricas convencionales de clasificación binaria. Este enfoque permite construir nuevos marcos como el CSI propuesto, que evalúa conjuntamente umbrales únicos y múltiples bajo una definición unificada de Agreement³. Los análisis desarrollados en esta tesis

³La Figura 2.9 ofrece una ilustración inicial de esta idea utilizando OA como métrica de referencia.

muestran el valor analítico de esta perspectiva, que sigue estando poco explorada en la literatura.

Por último, las simulaciones teóricas del **Capítulo 2** aportan información relevante sobre las relaciones entre diferentes métricas. Por ejemplo, se demuestra que un clasificador con un determinado valor de AUC puede alcanzar un valor máximo de CSI de $2 \cdot \text{AUC} - 1$, independientemente de la prevalencia. Comprender estas relaciones es clave para fomentar el uso adecuado de las métricas y mejorar su interpretación por parte de los usuarios finales. Futuras investigaciones podrían ampliar esta línea de trabajo analizando las relaciones entre otras métricas de múltiples umbrales, entre métricas de umbral único y entre ambos tipos de métricas.

Todos los resultados de esta tesis convergen en una conclusión común: la selección de un clasificador o de un umbral de clasificación debe estar guiada por la pregunta de investigación subyacente. Los umbrales no son decisiones arbitrarias, sino elecciones que deben reflejar los objetivos y compromisos propios del problema estudiado. Del mismo modo, la elección de métricas de evaluación no debe basarse en su popularidad o en convenciones disciplinares, sino en su adecuación al contexto de aplicación. Más aún, las métricas no deben interpretarse como valores aislados, sino en conjunto con su representación gráfica, que permite entender cómo varía el rendimiento a lo largo de distintos umbrales y clarifica el equilibrio entre errores y aciertos bajo diferentes criterios de decisión. Esta perspectiva más amplia hace que el proceso de evaluación sea más riguroso, transparente y útil para el usuario final, dando lugar a interpretaciones más fiables y significativas de los resultados de clasificación.

4.7 Further research lines

The literature review and analyses conducted throughout this dissertation have revealed several promising directions for further research. In addition to the specific contributions derived from this work, a number of gaps in the existing literature have been identified that can be effectively addressed using the methodologies and knowledge developed here.

Research derived from Chapter 1

From **Chapter 1**, the literature review highlights one of the main limitations in the application of binary classifiers: the significant fragmentation of theoretical and practical knowledge across different disciplines. This fragmentation leads to a lack of connectivity between fields, limiting the exchange of knowledge that could otherwise be mutually beneficial. A potential extension of this chapter involves the systematic compilation and standardization of metrics and their nomenclatures across various disciplines. Such an effort could facilitate greater knowledge transfer among fields, resulting in theoretical advancements inspired by diverse practical needs.

Additionally, this chapter reveals that the selection of metrics within each field of application often responds to disciplinary practices rather than to the actual requirements of specific research questions. This highlights the need to establish clearer relationships between metrics and develop more pedagogical systems for understanding their areas of application. By addressing these limitations, researchers could bridge the gap between experts in statistical methodologies and end-users of validation procedures. Comparative studies of metrics could further illuminate their interpretation and usability, providing more accessible knowledge to practitioners lacking advanced statistical expertise. Moreover, the methodological advancements in **Chapter 2** offer a great toolset to address these types of comparisons.

Research derived from Chapter 2

Chapter 2, which represents the primary methodological contribution of this dissertation, opens the door to a variety of highly productive research avenues. On one hand, the analysis in this chapter focuses exclusively on the Critical Success Index (CSI) due to its relevance in evaluating rare events. However, the approach detailed in **Section 2.2** is generalizable and can be applied to any binary classification metric, such as Overall Accuracy (OA), Specificity, Sensitivity, or F_1 score. By adopting any other metric as a reference, multiple classifications can be represented in terms of Quantity and Allocation differences, with areas under the curve calculated to obtain single and multi-threshold metrics unified by a consistent definition of Agreement. Moreover, by incorporating baseline representations, this framework can be extended

to include Skill metrics, providing a flexible and generalizable approach to model evaluation across various contexts.

This opens the door to designing different binary evaluation frameworks suited to the specific needs of various fields and applications. For example, if OA is a standard metric in a particular field, developing such frameworks using our approach would provide a tool to analyze Quantity and Allocation differences, prevalence, single- and multiple-threshold Agreement, and Skill, all through comparable metrics and a unified graphical representation.

In relation to this, the analysis presented in Section 2.3.3, where we relate AUC, AUCSI, and CSI, paves the way for theoretical results regarding the mathematical relationships between single- and multiple-threshold metrics. These results could be valuable for a deeper understanding of the metrics, their differences, and their interpretation and contextualization. Through this approach, further research can be pursued to converge single- and multiple-threshold metrics, which we consider crucial for improving classification model assessment.

One limitation of this chapter, and thus a potential future research direction, is the lack of an application of the CSI framework to real-world data. While the application to simulations provides an intuition of how the theoretical results translate into practice, it would be valuable to compare these findings with existing datasets and applications. This type of analysis will be included in the publication derived from this chapter.

This chapter also touches, even if somewhat superficially, on the insufficiency of randomness as a suitable baseline for many applications and how alternative baselines can be more informative about the true performance of classifications. All the metrics and analytical tools presented in this chapter assume randomness as a baseline but can be easily adapted to other types of baselines. Consequently, a highly relevant line of research involves the more systematic implementation of such baselines and the adaptation of traditional Skill metrics to them.

In this regard, a key short-term task is the development of software packages to implement the concepts, metrics, and graphical elements of the CSI framework for proper application. The Binary Evaluation Metrics R package (<https://github.com/JulenPerales/Binary-Evaluation-Metrics>) will provide a summary toolset

for calculating all the metrics and visualizations described in Chapters 1 and 2. The further development of this package constitutes one of the main research priorities for disseminating the results of this chapter.

Finally, one of the major limitations of the CSI framework, and therefore a potential path for future research, is that it still evaluates unrealistic classifications, even if it assigns them a small weight. Partial measures of AUC address this limitation by focusing on specific areas under the curve. However, these measures rely on performance-based cutoff criteria, such as Sensitivity or Specificity (Peterson et al. 2008; Dodd and Pepe 2003), which makes the evaluated areas inconsistent across classifiers and case studies and hinders the comparison of results.

CSI graphs, which change shape based on prevalence, present an opportunity to develop new partial AUCSI metrics that are influenced by prevalence rather than classifier performance. The shape or slope of the baseline, as well as the upper and lower bounds of the CSI curves, all of which depend exclusively on prevalence, could serve as criteria for defining these partial measures. For example, a partial AUCSI measure could be derived by considering the curves only up to the point where the slope of the upper bound equals -1 , which would exclude the rightmost region of the graph in proportion to its decreasing relevance. By focusing on prevalence-dependent partial measures, it becomes possible to compare classifiers both within the same application and across different case studies. Future research could explore the derivation of these prevalence-based partial AUCSI measures directly from the CSI graph, offering a more comparable and meaningful evaluation framework for binary classification tasks.

Research derived from Chapter 3

Chapter 3 provides an alternative approach to validating LCMs. First, the incorporation of the CSI framework has proven to provide a highly useful set of tools for the validation of LCM. Future research could focus on the development of standardized validation procedures for LCM based on this framework. Such efforts would aim to facilitate the flow of theoretical knowledge into practical applications, particularly for LCM modelers and practitioners. To this end, creating accessible and educational contributions, such as leveraging tools like the R package developed in

this dissertation, could significantly enhance the understanding and adoption of the methodologies proposed in this work.

Second, this chapter identifies the relationship between the prevalence of change used for model calibration and its outcome as one of the main arguments for justifying poor model performance, particularly through the work of Pontius, Boersma, et al. (2008) and Pontius, Castella, et al. (2018). As argued, the incorporation of Skill into the evaluation of models significantly qualifies this relationship. However, the population used to analyze this relationship remains limited, as only the original 13 case studies analyzed by Pontius, Boersma, et al. (2008) were evaluated. Therefore, a promising line of research would be a more detailed study of the relationship between prevalence and model performance, expanding the analysis to a larger population of real-world applications. In this regard, it would be interesting to reevaluate the results reported by different studies by applying our methodology to see if it influences the conclusions drawn from the models.

Finally, it would be valuable to build on the research directions introduced in Chapter 1 and apply them to LCM contexts. For instance, integrating the concept of Skill alongside baselines not based on randomness could be particularly relevant for phenomena as spatially heterogeneous as land changes. Similarly, exploring Skill metrics developed and applied in other disciplines could offer insights into whether any of these metrics might address the specific research challenges faced in LCMs. Such investigations could lead to a broader set of tools and knowledge from which researchers could benefit, contributing to the reduction of the use of inadequate validation procedures in LCMs.

Research derived from Chapter 4

Chapter 4 develops a methodological framework for the economic valuation of soil erosion through the integration of LCM models, RUSLE, and replacement cost approaches. While the results provide a spatially explicit and dynamic assessment of erosion and its economic implications, several avenues for future research emerge from this work.

First, an important extension of this chapter would be the application of the proposed methodology under alternative climate change scenarios. Since both land

use dynamics and erosion processes are highly sensitive to climatic conditions, incorporating different climate projections would allow for the assessment of how erosion patterns and their economic valuation may evolve under future environmental conditions. This would provide a more robust basis for policy design, enabling the evaluation of mitigation and adaptation strategies under uncertainty and supporting long-term planning efforts.

Second, future research could focus on extending the spatially explicit approach to economic valuation developed in this chapter to other environmental processes. The combination of spatial modeling and economic assessment offers a promising framework for identifying areas where environmental degradation is not only physically significant but also economically relevant. Applying similar methodologies to other environmental challenges—such as biodiversity loss, water quality degradation, or carbon dynamics—could contribute to more targeted and efficient policy interventions, improving the allocation of limited resources and enhancing the effectiveness of environmental management strategies.

Lastly, further work could explore the integration of this framework with decision-support systems for policymakers. By linking spatially detailed economic valuations with policy instruments, such as subsidies, taxes, or land management regulations, future research could contribute to the development of operational tools that facilitate the translation of modeling results into concrete policy actions. This would reinforce the role of integrated modeling approaches as a bridge between scientific analysis and practical decision-making.

References

- Dodd, Lori E. and Margaret S. Pepe (Sept. 2003). “Partial AUC Estimation and Regression”. en. In: *Biometrics* 59.3, pp. 614–623. ISSN: 0006-341X, 1541-0420. DOI: [10.1111/1541-0420.00071](https://doi.org/10.1111/1541-0420.00071). URL: <https://academic.oup.com/biometrics/article/59/3/614-623/7286731> (visited on 02/01/2024).
- Liu, Xiaoping et al. (2017). “A future land use simulation model (FLUS) for simulating multiple land use scenarios by coupling human and natural effects”. In: *Landscape and Urban Planning* 168, pp. 94–116. DOI: [10.1016/j.landurbplan.2017.08.002](https://doi.org/10.1016/j.landurbplan.2017.08.002). URL: <https://doi.org/10.1016/j.landurbplan.2017.08.002>.
- Meentemeyer, Ross K. et al. (2013). “FUTURES: Multilevel Simulations of Emerging Urban–Rural Landscape Structure Using a Stochastic Patch-Growing Algorithm”. In: *Annals of the Association of American Geographers* 103.4, pp. 785–807. DOI: [10.1080/00045608.2012.707591](https://doi.org/10.1080/00045608.2012.707591). URL: <https://doi.org/10.1080/00045608.2012.707591>.
- Peterson, A. Townsend, Monica Papeş, and Jorge Soberón (Apr. 2008). “Rethinking receiver operating characteristic analysis applications in ecological niche modeling”. In: *Ecological modeling* 213.1, pp. 63–72. ISSN: 0304-3800. DOI: [10.1016/j.ecolmodel.2007.11.008](https://doi.org/10.1016/j.ecolmodel.2007.11.008). URL: <https://www.sciencedirect.com/science/article/pii/S0304380007006163> (visited on 08/15/2024).
- Pontius, Robert Gilmore, Wideke Boersma, et al. (Mar. 2008). “Comparing the input, output, and validation maps for several models of land change”. en. In: *The Annals of Regional Science* 42.1, pp. 11–37. ISSN: 0570-1864, 1432-0592. DOI: [10.1007/s00168-007-0138-2](https://doi.org/10.1007/s00168-007-0138-2). URL: <http://link.springer.com/10.1007/s00168-007-0138-2> (visited on 06/04/2024).
- Pontius, Robert Gilmore, Jean-Christophe Castella, et al. (2018). “Lessons and Challenges in Land Change Modeling Derived from Synthesis of Cross-Case Comparisons”. en. In: *Trends in Spatial Analysis and modeling*. Ed. by Martin Behnisch and Gotthard Meinel. Vol. 19. Series Title: Geotechnologies and the Environment. Cham: Springer International Publishing, pp. 143–164. ISBN: 978-3-319-52520-4. DOI: [10.1007/978-3-319-52522-8_8](https://doi.org/10.1007/978-3-319-52522-8_8). URL: http://link.springer.com/10.1007/978-3-319-52522-8_8 (visited on 12/13/2022).

Zhang, Shihe et al. (2022). “Landscape ecological risk projection based on the PLUS model under the localized shared socioeconomic pathways in the Fujian Delta region”. In: *Ecological Indicators* 136, p. 108642. DOI: [10.1016/j.ecolind.2022.108642](https://doi.org/10.1016/j.ecolind.2022.108642). URL: <https://doi.org/10.1016/j.ecolind.2022.108642>.

Appendix A

Appendix Chapter 1

MCC

$$\text{MCC} = \frac{\text{Cov}(o, c)}{\sigma_o \sigma_c},$$

where o represents the observed binary value and c is the classified binary outcome.

The covariance between o and c is expressed as:

$$\text{Cov}(o, c) = E[(o - E[o])(c - E[c])].$$

For binary classifications, where $o, c \in \{0, 1\}$, the terms simplify as follows:

- $E[o]$: The expectation of o , equivalent to $P(o = 1) = \frac{H + M}{E}$.
- $E[c]$: The expectation of c , equivalent to $P(c = 1) = \frac{H + F}{E}$.
- $E[rc]$: The expectation of the product oc , which equals $P(o = 1, c = 1) = \frac{H}{E}$.

Expanding the covariance formula for binary variables:

$$\begin{aligned}\text{Cov}(o, c) &= P(o = 1, c = 1) - P(o = 1)P(c = 1) \\ &= \frac{H}{E} - \frac{(H + M)(H + F)}{E^2} \\ &= \frac{HC - MF}{E^2}.\end{aligned}$$

The variance of a binary variable $o \in \{0, 1\}$ is defined as:

$$\text{Var}(o) = E[(o - E[o])^2] = E[o^2] - (E[o])^2.$$

For binary variables, $o^2 = o$, so:

$$\text{Var}(o) = E[o] - (E[o])^2 = \frac{H + M}{E} - \left(\frac{H + M}{E}\right)^2.$$

$$\text{Var}(o) = \frac{E(H + M) - (H + M)^2}{E^2} = \frac{(H + M)(C + F)}{E^2}.$$

Similarly, for $c \in \{0, 1\}$:

$$\text{Var}(c) = \frac{(H + F)(C + M)}{E^2}.$$

Therefore, we can express the MCC in terms of classification outcomes:

$$\text{MCC} = \frac{\text{Cov}(o, c)}{\sigma_o \sigma_c} = \frac{HC - MF}{\sqrt{(H + F)(H + M)(C + F)(C + M)}}$$

Appendix B

Appendix Chapter 2

B.1 The math behind CSI curves

B.1.1 Reconstruction of Contingency Table from CSI curves

The design of the CSI plot provides two bits of information:

- P can be obtained from the point where the upper bound reaches $y = 1$.
- Pr can be obtained from the point where all the curves converge at $x = E$.

Similarly, CSI curves offer two additional bits of information: the CSI value of one classification and its Hits+False alarms (x). From those four values, we can recompose the contingency table of any classification:

$$\begin{aligned} CSI &= \frac{H_x}{H_x + F_x + M_x} = \frac{H_x}{F_x + P} \\ CSI &= \frac{H_x}{x - H_x + P} \\ H_x &= CSI(x - H_x + P) \\ \mathbf{H_x} &= \frac{CSI}{1 + CSI}(x + P) \\ \mathbf{F_x} &= x - H_x \\ \mathbf{M_x} &= P - H_x \end{aligned}$$

$$C_x = E - H_x - F_x - M_x$$

B.1.2 Slopes of CSI curves

The slope of a CSI curve is given by:

$$\begin{aligned}\Delta \text{CSI} &= \Delta \left(\frac{H_t}{F_t + P} \right) = \frac{H_t}{F_t + P} - \frac{H_{t-1}}{F_{t-1} + P} \\ &= \frac{H_t(F_{t-1} + P) - H_{t-1}(F_t + P)}{(F_t + P)(F_{t-1} + P)} \\ &= \frac{H_t(F_t - \Delta F_t + P) - (H_t - \Delta H_t)(F_t + P)}{(F_t + P)(F_t - \Delta F_t + P)} \\ &= \frac{H_t F_t - H_t \Delta F_t + H_t P - H_t F_t - H_t P + \Delta H_t F_t + \Delta H_t P}{(F_t + P)^2 - \Delta F_t(F_t + P)} \\ &= \frac{-H_t \Delta F_t + \Delta H_t F_t + \Delta H_t P}{(F_t + P)(F_t - \Delta F_t + P)}\end{aligned}$$

The maximum of a CSI curve ($\Delta \text{CSI} = 0$) is given by:

$$0 = -H_t \Delta F_t + \Delta H_t F_t + \Delta H_t P;$$

$$H_t \Delta F_t = \Delta H_t (F_t + P);$$

$$\frac{H_t}{F_t + P} = \frac{\Delta H_t}{\Delta F_t};$$

$$\text{CSI}_t = \frac{\Delta H_t}{\Delta(H_t + F_t) - \Delta H_t};$$

$$\frac{\Delta H_t}{\Delta(H_t + F_t)} = \frac{\text{CSI}_t}{\text{CSI}_t + 1} = \Delta \text{TOC}_t$$

Since $\frac{\Delta H_t}{\Delta(H_t + F_t)}$ is the slope of TOC curves, the CSI curve reaches its maximum when:

$$\Delta \text{TOC}_t = \frac{\text{CSI}_t}{\text{CSI}_t + 1}$$

B.1.3 Equation of the baseline CSI curve

The baseline generates Hits at the rate of prevalence. Thus, out of the x simulated presences, the baseline will result in:

$$\begin{aligned}H_x &= Prx \\F_x &= (1 - Pr)x\end{aligned}$$

Consequently, the equation of the CSI curve of the baseline is:

$$CSI(x, Pr) = \frac{Prx}{(1 - Pr)x + P}$$

As a result, at $x = P$, the baseline gets the value:

$$CSI(P, Pr) = \frac{Pr}{2 - Pr}$$

B.2 Construction of AUC and AUCS

Area between the Upper Bound and Lower Bound TOC

$$\text{Agreement Area}(x) = \min\left(\frac{x^2}{2}, \frac{P^2}{2}\right) + \max(0, P(x - P)) - \max\left(0, \frac{(x - Q)^2}{2}\right)$$

$$\text{Agreement Area}(E) = PQ$$

Area of the Baseline TOC

$$\text{Baseline Agreement Area}(x) = \frac{Prx^2}{2} - \max\left(0, \frac{(x - Q)^2}{2}\right)$$

$$\text{Baseline Agreement Area}(E) = \frac{PQ}{2}$$

AUC as a Proportion of the Area Between the Upper and Lower Bound

AUC is calculated as the proportion of the Agreement area that is covered at the point x . This proves that the proportion of AUC explained only depends on x/E and Pr . As a result, these results apply to any size of population. From this equation, derive figures 2.4 and 2.3.

$$\begin{aligned} \text{AUC}(x, Pr) &= \frac{\text{Agreement Area}(x, Pr)}{\text{Agreement Area}(E, Pr)} \\ &= \frac{\min\left(\frac{x^2}{2}, \frac{P^2}{2}\right) + \max(0, P(x - P)) - \frac{\max(0, x - Q)^2}{2}}{PQ} \\ &= \frac{\min\left(\frac{1}{2} \left(\frac{x}{E}\right)^2, \frac{Pr^2}{2}\right) + Pr \max\left(0, \frac{x}{E} - Pr\right) - \frac{\max\left(0, \frac{x}{E} - (1 - Pr)\right)^2}{2}}{Pr(1 - Pr)} \end{aligned}$$

***AUCS* as a Proportion of the Area Between the Upper Bound and the Baseline**

AUCS is calculated as the proportion of the Skill area that is covered at the point x . This proves that the proportion of AUCS explained only depends on x/E and Pr . As a result, these results apply to any size of population. This equation refers to Equation 2.4, from which figures 2.6 and 2.5 are derived.

$$\begin{aligned}
 AUCS(x, Pr) &= \frac{AUC(x, Pr) - BAUC(x, Pr)}{AUC(E) - AUC(E)} \\
 &= \frac{\min\left(\frac{x^2}{2}, \frac{P^2}{2}\right) + \max(0, P(x - P)) - \frac{Prx^2}{2}}{\frac{PQ}{2}} \\
 &= \frac{\min\left(\left(\frac{x}{E}\right)^2, Pr^2\right) + 2\max\left(0, Pr\left(\frac{x}{E} - Pr\right)\right) - Pr\left(\frac{x}{E}\right)^2}{Pr(1 - Pr)}
 \end{aligned}$$

B.3 Construction of AUCSI and AUCSIS

Area between the Upper Bound and Lower Bound CSI

The Upper Bound of CSI follows the equation:

$$\left\{ \begin{array}{l} \text{if } x < P, \text{ then } F_x = 0; H_x = x \\ \text{if } x \geq P, \text{ then } H_x = P; F_x = x - P \end{array} \right\}$$

The area is equal to the sum of the area minus the lower right triangle.

$$\begin{aligned} \text{Agreement Area}(x) &= \min \left(\int_0^x \frac{x}{P} dx, \frac{P}{2} \right) + \max \left(0, \int_P^x \frac{P}{x} dx \right) - \frac{\min(0, x - Q)^2}{2E} \\ &= \min \left(\frac{x^2}{2P}, \frac{P}{2} \right) + \max \left(0, P \ln \left(\frac{x}{P} \right) \right) - \frac{\min(0, x - Q)^2}{2E} \\ \text{Agreement Area}(E) &= \frac{P}{2} - P \ln(P) - \frac{P^2}{2E} \\ &= P \left[\frac{1}{2} - \ln(P) - \frac{P}{2} \right] \end{aligned}$$

Area of the Baseline CSI

$$\text{Baseline Agreement Area}(x) = \int_0^x \frac{Prx}{(1 - Pr)x + P} dx - \frac{\min(0, x - Q)^2}{2E}$$

$$* \text{Let } u = (1 - Pr)x + P \quad \text{and} \quad du = (1 - Pr) dx.$$

$$* \int \frac{Prx}{(1 - Pr)x + P} dx = \frac{Pr((1 - Pr)x - P \ln((1 - Pr)x + P))}{(1 - Pr)^2}$$

$$\begin{aligned} \text{Baseline Agreement Area}(x) &= \frac{Pr((1 - Pr)x - P \ln((1 - Pr)x + P))}{(1 - Pr)^2} \\ &\quad + \frac{Pr(P \ln(P))}{(1 - Pr)^2} - \frac{\min(0, x - Q)^2}{2E} \\ &= \frac{Pr}{(1 - Pr)^2} \left((1 - Pr)x + P \ln \left(\frac{P}{(1 - Pr)x + P} \right) \right) \end{aligned}$$

$$\text{Baseline Agreement Area}(E) = \frac{Pr}{(1 - Pr)^2} (P \ln(Pr) + Q) - \frac{PrP}{2} - \frac{\min(0, x - Q)^2}{2E}$$

***AUCSI* as a Proportion of the Area Between the Upper and Lower Bound**

AUCSI is calculated as the proportion of the Agreement area that is covered at the point x . This proves that the proportion of AUCSI explained only depends on x/E and Pr . As a result, these results apply to any size of population. This equation refers to Equation 2.3, from which figures 2.4 and 2.3 are derived.

$$\begin{aligned} \text{ASCI}(x, Pr) &= \frac{\text{Agreement Area}(x, Pr)}{\text{Agreement Area}(E, Pr)} \\ &= \frac{\min\left(\frac{x^2}{2P}, \frac{P}{2}\right) + \max\left(0, P \ln\left(\frac{x}{P}\right)\right) - \frac{\min(0, x - Q)^2}{2E}}{P \left[\frac{1}{2} - \ln(Pr) - Pr\right]} \\ \text{ASCI}\left(\frac{x}{E}, Pr\right) &= \frac{\min\left(\frac{1}{2Pr^2} \left(\frac{x}{E}\right)^2, \frac{1}{2}\right) + \max\left(0, \ln\left(\frac{1}{Pr} \frac{x}{E}\right)\right) - \frac{\max\left(0, \frac{x}{E} - (1 - Pr)\right)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} \end{aligned}$$

***AUCSIS* as a Proportion of the Area Between the Upper Bound and the Baseline**

AUCSIS is calculated as the proportion of the Skill area that is covered at the point x . This proves that the proportion of AUCSIS explained only depends on x/E and Pr . As a result, these results apply to any size of population. This equation refers to Equation 2.5, from which figures 2.6 and 2.5 are derived.

$$\text{ASCIS}(x, Pr) = \frac{\text{Agreement Area}(x, Pr) - \text{Baseline Agreement Area}(x, Pr)}{\text{Agreement Area}(E, Pr) - \text{Baseline Agreement Area}(E, Pr)} =$$

$$\frac{\min\left(\frac{x^2}{2P}, \frac{P}{2}\right) + \max\left(0, P \ln\left(\frac{x}{P}\right)\right) - \frac{Pr}{(1-Pr)^2} \left(P \ln\left(\frac{P}{(1-Pr)x + P}\right) + (1-Pr)x\right)}{\frac{P}{2} - P \ln(Pr) - \frac{Pr}{(1-Pr)^2} (P \ln(Pr) + Q)}$$

$$\text{ASCIS}\left(\frac{x}{E}, Pr\right) =$$

$$\frac{\min\left(\frac{1}{2Pr^2} \left(\frac{x}{E}\right)^2, \frac{1}{2}\right) + \max\left(0, \ln\left(\frac{1}{Pr} \frac{x}{E}\right)\right) + \frac{Pr \ln\left(\frac{1-Pr}{Pr} \frac{x}{E} + 1\right) - (1-Pr) \frac{x}{E}}{(1-Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}}$$

B.4 Question 1

Do CSI-derived metrics weigh the far right region of the plot less than TOC-derived metrics?

We calculate the contribution that AUC and ACIS get from the classifications up to P (left region), from which we can deduce the contribution from P to E (right region). We analyze Agreement and Skill metrics.

B.4.1 AUC

Proportion at $x = P$:

$$\text{AUC}(P, Pr) = \frac{\frac{Pr^2}{2} - \frac{\max(0, 2Pr - 1)^2}{2}}{Pr(1 - Pr)}$$

$$\text{AUC}(P, Pr) = \begin{cases} \frac{3Pr - 1}{2Pr} & \text{if } Pr > \frac{1}{2}, \\ \frac{Pr}{2(1 - Pr)} & \text{if } Pr \leq \frac{1}{2}. \end{cases}$$

B.4.2 AUCSI

Proportion at $x = P$:

$$\text{AUCSI}(P, Pr) = \frac{\frac{1}{2} - \frac{\max(0, 2Pr - 1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}}$$

$$\text{AUCSI}(P, Pr) = \begin{cases} \frac{1 - \frac{(2Pr - 1)^2}{Pr}}{1 - 2\ln(Pr) - Pr} & \text{if } Pr > \frac{1}{2}, \\ \frac{1}{1 - 2\ln(Pr) - Pr} & \text{if } Pr \leq \frac{1}{2}. \end{cases}$$

B.4.3 AUCS

Proportion at $x = P$:

$$\begin{aligned}\text{AUCS}(P, Pr) &= \frac{Pr^2 - Pr^3}{Pr(1 - Pr)} \\ &= Pr\end{aligned}$$

B.4.4 AUCSIS

Proportion at $x = P$:

$$\text{AUCSI}(P, Pr) = \frac{\frac{1}{2} + \frac{Pr \ln(2 - Pr) - (1 - Pr)Pr}{(1 - Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1 - Pr)}{(1 - Pr)^2}}$$

B.5 Question 2

Do metrics concentrate more area around P than metrics?

To answer this, we analyze the derivative of each metric at $x = P$. As all the areas depend only on x/E and Pr , we can calculate the derivative with respect to x/E , and thus, compare elasticities for any population size. We analyze Agreement and Skill metrics.

B.5.1 AUC

$$\frac{\partial \text{AUC}(P, Pr)}{\partial \left(\frac{x}{E}\right)} = \begin{cases} \frac{1}{Pr} & \text{if } Pr > \frac{1}{2}, \\ \frac{1}{1-Pr} & \text{if } Pr \leq \frac{1}{2}. \end{cases}$$

B.5.2 AUCSI

$$\frac{\partial \text{AUCSI}(P, Pr)}{\partial \left(\frac{x}{E}\right)} = \begin{cases} \frac{\frac{1}{Pr} - \frac{(2Pr-1)^2}{Pr}}{\frac{1}{2} - \ln Pr - \frac{Pr}{2}} = -\frac{4(2Pr-2)}{1-Pr-2\ln(Pr)} & \text{if } Pr > \frac{1}{2}, \\ \frac{\frac{1}{Pr}}{\frac{1}{2} - \ln Pr - \frac{Pr}{2}} = \frac{2}{Pr(1-Pr-2\ln(Pr))} & \text{if } Pr \leq \frac{1}{2}. \end{cases}$$

B.5.3 AUCS

$$\frac{\partial \text{AUCS}(x, Pr)}{\partial \left(\frac{x}{E}\right)} = \frac{2\frac{x}{E} - 2Pr\frac{x}{E}}{Pr(1-Pr)}$$

$$\frac{\partial \text{AUCS}(P, Pr)}{\partial \left(\frac{x}{E}\right)} = \frac{2Pr - 2Pr^2}{Pr(1-Pr)} = 2$$

B.5.4 AUCSIS

$$\frac{\partial \text{AUCSIS}(x, Pr)}{\partial \left(\frac{x}{E}\right)} = \frac{\frac{x}{EPr^2} - \frac{\frac{x}{E}}{(1-Pr)\frac{x}{E} + Pr}}{\frac{1}{2} - \ln Pr - \frac{Pr \ln Pr + (1-Pr)}{(1-Pr)^2}}$$

$$\frac{\partial \text{AUCSIS}(P, Pr)}{\partial \left(\frac{x}{E}\right)} = \frac{\frac{1}{Pr} - \frac{1}{2-Pr}}{\frac{1}{2} - \ln Pr - \frac{Pr \ln Pr + (1-Pr)}{(1-Pr)^2}}$$

B.6 Question 3

Does any of the metrics favor over or underestimation of the quantity of presence?

To analyze the difference in areas, we calculate the areas to the right and left of P for both metrics and determine their ratio.

Let $h = \min(P, Q)$, which is the maximum size of the window we can consider. The window selection varies the magnitude of the difference but not its direction. The asymmetries are calculated as:

$$\text{Ratio between areas} = \begin{cases} \frac{\text{Right Area}}{\text{Left Area}} - 1, & \text{if Right Area} > \text{Left Area}, \\ -\frac{\text{Left Area}}{\text{Right Area}} + 1, & \text{if Left Area} \leq \text{Right Area}. \end{cases}$$

From the window size we consider, we deduce that:

$$h = \min(P, Q) = \begin{cases} P, & \text{if } P \leq Q, \\ Q, & \text{if } P \geq Q. \end{cases}$$

B.6.1 AUC

Right Area

$$\begin{aligned} \text{Right Area} &= \text{AUC}(P + h, Pr) - \text{AUC}(P, Pr) \\ &= \frac{\left(\frac{P + h}{E}\right)^2 - Pr^2 - \max\left(0, \frac{P + h - (1 - Pr)}{E}\right) + \max(0, 2Pr - 1)}{2Pr(1 - Pr)} \\ &= \frac{2Pr \frac{h}{E} + \left(\frac{h}{E}\right)^2 - \max(0, 2Pr + \frac{h}{E} - 1) + \max(0, 2Pr - 1)}{2Pr(1 - Pr)} \end{aligned}$$

$$\text{Right Area} = \begin{cases} \frac{Pr}{(1-Pr)}, & \text{if } 2P \leq Q; h = P, \\ \frac{3Pr^2 - 3Pr + 1}{2Pr(1-Pr)}, & \text{if } 2P > Q > P; h = P, \\ \frac{1}{2}, & \text{if } P \geq Q; h = Q. \end{cases}$$

Left Area

$$\begin{aligned} \text{Left Area} &= \text{AUC}(P, Pr) - \text{AUC}(P - h, Pr) \\ &= \frac{Pr^2 - \left(\frac{P-h}{E}\right)^2 - \max(0, 2Pr - 1) + \max(0, \frac{P-h}{E} - (1-Pr))}{2Pr(1-Pr)} \\ &= \frac{2Pr\frac{h}{E} - \left(\frac{h}{E}\right)^2 - \max(0, 2Pr - 1) + \max(0, 2Pr - \frac{h}{E} - 1)}{2Pr(1-Pr)} \end{aligned}$$

$$\text{Left Area} = \begin{cases} \frac{Pr}{2(1-Pr)}, & \text{if } P \leq Q; h = P, \\ -\frac{3Pr - 2}{2(1-Pr)}, & \text{if } \frac{2P}{3} < Q < P; h = Q, \\ \frac{3Pr - 2}{2Pr}, & \text{if } \frac{2P}{3} \geq Q; h = Q. \end{cases}$$

B.6.2 AUCSI

Right Area

$$\begin{aligned} \text{Right Area} &= \text{AUCSI}(P + h, Pr) - \text{AUCSI}(P, Pr) \\ &= \frac{\frac{1}{2} + \ln\left(\frac{P+h}{PrE}\right) - \frac{\left(2Pr - 1 + \frac{h}{E}\right)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} - \frac{\frac{1}{2} - \frac{\max(0, 2Pr - 1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} \end{aligned}$$

$$\begin{aligned}
&= \frac{\ln\left(\frac{P+h}{PrE}\right) - \frac{\left(2Pr-1+\frac{h}{E}\right)^2}{2Pr} + \frac{\max(0, 2Pr-1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} \\
\text{Right Area} &= \begin{cases} \frac{\ln(2) - \frac{(3Pr-1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}}, & \text{if } 2P \leq Q; h = P, \\ \frac{2\ln(2) - 5Pr + 2}{1 - 2\ln(Pr) - Pr}, & \text{if } 2P > Q > P; h = P, \\ \frac{\ln\left(\frac{1}{Pr}\right) - \frac{Pr}{2} + \frac{(2Pr-1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}}, & \text{if } P \geq Q; h = Q. \end{cases}
\end{aligned}$$

Left Area

$$\text{Left Area} = \text{AUCSI}(P, Pr) - \text{AUCSI}(P - h, Pr)$$

$$\begin{aligned} &= \frac{\frac{1}{2} - \frac{\max(0, 2Pr - 1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} - \frac{\frac{1}{2Pr^2} \left(Pr - \frac{h}{E}\right)^2 - \frac{\max\left(0, 2Pr - 1 - \frac{h}{E}\right)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} \\ &= \frac{\frac{1}{2} - \frac{1}{2Pr^2} \left(Pr - \frac{h}{E}\right)^2 + \frac{\max\left(0, 2Pr - 1 - \frac{h}{E}\right)^2 - \max(0, 2Pr - 1)^2}{2Pr}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}} \\ \text{Left Area} &= \begin{cases} \frac{1}{1 - 2\ln(Pr) - Pr}, & \text{if } P \leq Q; h = P, \\ \frac{\frac{1}{2} - \frac{(2Pr - 1)^2(1 + Pr)}{2Pr^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}}, & \text{if } \frac{2P}{3} < Q < P; h = Q, \\ \frac{\frac{1}{2} - \frac{(2x - 1)^2}{2x^2} + \frac{5x^2 - 8x + 3}{2x}}{\frac{1}{2} - \ln(Pr) - \frac{Pr}{2}}, & \text{if } \frac{2P}{3} \geq Q; h = Q. \end{cases} \end{aligned}$$

B.6.3 AUCS

Right Area

$$\text{Right Area} = \text{AUCS}(P + h, Pr) - \text{AUCS}(P, Pr)$$

$$\begin{aligned} &= \frac{Pr^2 + 2Prh - Pr \left(Pr + \frac{h}{E}\right)^2}{Pr(1 - Pr)} - Pr \\ &= \frac{Pr + 2h - \left(Pr + \frac{h}{E}\right)^2}{(1 - Pr)} - Pr \end{aligned}$$

$$\text{Right Area} = \begin{cases} \frac{Pr + 2PrE - 4Pr^2}{(1 - Pr)} - Pr, & \text{if } Q > P, \\ \frac{Pr + 2(1 - Pr)E - 1}{(1 - Pr)} - Pr, & \text{if } Q \leq P. \end{cases}$$

Left Area

$$\begin{aligned} \text{Left Area} &= \text{AUCS}(P, Pr) - \text{AUCS}(P - h, Pr) \\ &= Pr - \frac{\left(\frac{P - h}{E}\right)^2 - Pr \left(\frac{P - h}{E}\right)^2}{Pr(1 - Pr)} \\ &= Pr - \frac{\left(Pr - \frac{h}{E}\right)^2}{Pr} \\ &= \frac{\left(2Pr - \frac{h}{E}\right)h}{PrE} \end{aligned}$$

$$\text{Left Area} = \begin{cases} \frac{Pr}{E}, & \text{if } Q > P, \\ \frac{3Pr - 1}{E}, & \text{if } Q \leq P. \end{cases}$$

B.6.4 AUCSIS

Right Area

$$\begin{aligned} \text{Right Area} &= \text{AUCSIS}(P + h, Pr) - \text{AUCSIS}(P, Pr) \\ &= \frac{\ln\left(1 + \frac{h}{PrE}\right) + \frac{Pr \ln\left(\frac{(1 - Pr)(P + h) + PrE}{PrE}\right) - (1 - Pr)\frac{h}{E} - Pr \ln(2 - Pr)}{(1 - Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1 - Pr)}{(1 - Pr)^2}} \end{aligned}$$

$$= \frac{\ln\left(1 + \frac{h}{PrE}\right) + \frac{Pr \ln\left(\frac{(1-Pr)(P+h) + PrE}{PrE(2-Pr)}\right) - (1-Pr)\frac{h}{E}}{(1-Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}}$$

$$\text{Right Area} = \begin{cases} \frac{\ln(2) + \frac{Pr \left(\ln\left(\frac{3-2Pr}{2-Pr}\right) - (1-Pr) \right)}{(1-Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}}, & \text{if } Q > P, \\ \frac{\ln\left(\frac{1}{Pr}\right) + \frac{Pr}{(1-Pr)^2} \ln\left(\frac{1}{Pr(2-Pr)}\right) - 1}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}}, & \text{if } Q \leq P. \end{cases}$$

Left Area

$$\text{Left Area} = \text{AUCSIS}(P, Pr) - \text{AUCSIS}(P-h, Pr)$$

$$\begin{aligned} &= \frac{\frac{1}{2} + \frac{Pr \ln(2-Pr) - (1-Pr)Pr}{(1-Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}} \\ &- \frac{\frac{1}{2Pr^2} \left(\frac{P-h}{E}\right)^2 + \frac{Pr \ln\left(\frac{1-Pr}{Pr} \frac{P-h}{E} + 1\right) - (1-Pr)\frac{P-h}{E}}{(1-Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}} \\ &= \frac{\frac{1}{2} - \frac{1}{2Pr^2} \left(\frac{P-h}{E}\right)^2 + \frac{Pr \ln\left(\frac{(2-Pr)PrE}{(1-Pr)(P-h) + PrE}\right) - (1-Pr)\frac{h}{E}}{(1-Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1-Pr)}{(1-Pr)^2}} \end{aligned}$$

$$\text{Left Area} = \begin{cases} \frac{1}{2} + \frac{Pr(\ln(2 - Pr) - 1 + Pr)}{(1 - Pr)^2}, & \text{if } Q > P, \\ \frac{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1 - Pr)}{(1 - Pr)^2}}{\frac{1}{2} - \ln(Pr) - \frac{Pr \ln(Pr) + (1 - Pr)}{(1 - Pr)^2}} \cdot \left(\frac{(2 - Pr)Pr}{(1 - Pr) \left(\frac{2Pr - 1}{E^2} \right) + Pr} \right) - 1, & \text{if } Q \leq P. \end{cases}$$

B.7 Design of the simulated classifiers

A ROC curve is defined as concave down in this dissertation if, for any two points A and B on the curve and the straight line ss that connects them, no point on the curve between A and B lies below s . concave down curves ensure that the AUC accurately reflects the strength of the monotonic relationship between the index variable and the binary variable's presence, as these curves do not intersect the uniform line (Pontius 2022; Lloyd 2002).

The concave hull of an ROC curve refers to the smallest concave boundary that encloses all the points on the curve. By definition, this boundary dominates all other curves that could be constructed using linear interpolation between the given points (Davis and Goadrich 2006). This concept is essential for identifying the subset of classifiers that are potentially optimal within ROC space (Provost and Fawcett 2001). The relationship between concavity in ROC curves and the monotonicity of the likelihood ratio has been extensively studied (Lloyd 2002).

The concave hull can also be used to analyze classification performance with respect to the Critical Success Index (CSI). The curve with the smallest AUC that achieves a given maximum CSI is represented by curve A in Figure B.1. This curve initially classifies all presences correctly until reaching the maximum CSI and subsequently maintains a slope defined by $\Delta TOC_t = \text{CSI}_t / (\text{CSI}_t + 1)$ until it intersects the upper bound. Curve A is equivalent to the concave hull formed exclusively by points A and B in Figure B.1. Conversely, the curve with the largest AUC that achieves the same maximum CSI is represented by curve B in Figure B.1. This curve reaches its

maximum CSI at point B and remains along the upper bound thereafter. Curve B corresponds to the concave hull constructed solely from point B .

B.7.1 Calculation of the upper and lower boundaries of Figure 2.12

In Figure 2.12, the bold upper boundary represents the envelope derived from all possible curves of type A , or the concave hulls formed by points A and B . Similarly, the bold lower boundary represents the envelope derived from all possible curves of type B , or the concave hulls constructed from point B alone.

The same analysis is repeated for Skill metrics (AUCS, AUCSIS, and RGSS), considering the TOC curves that result in the same AUC but different RGSS. In this case, A and B are built under the same logic, but considering equal distances to baselines in B.1 (b).

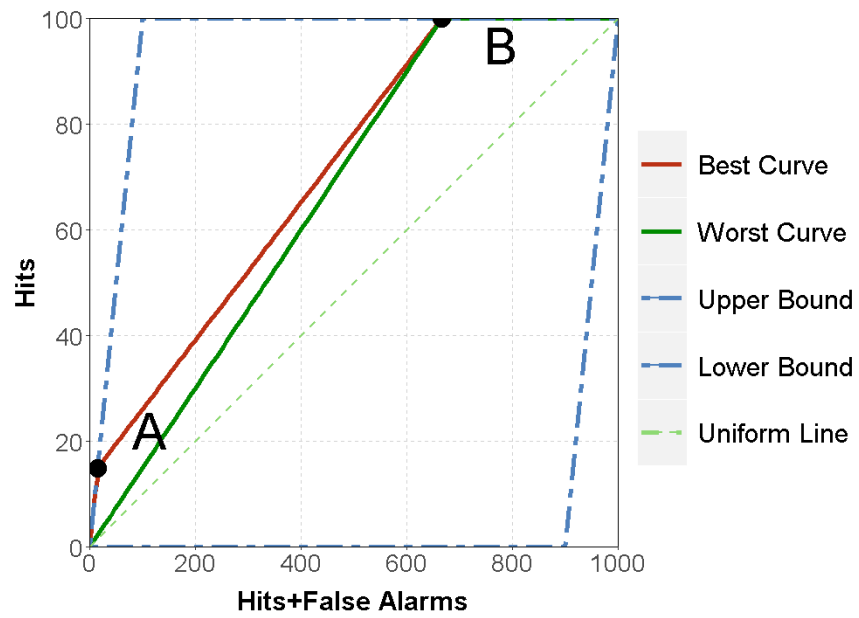
B.7.2 Simulated TOC Curves

The simulated TOC curves are designed as strictly concave downward curves, each composed of two piecewise-defined exponential functions. To ensure continuity, differentiability and concavity at the breakpoint, each curve is constructed with a distinct breakpoint $B = (B_x, B_y)$, where $0 \leq B_x \leq E$ and $Prx \leq B_y \leq P$. At this breakpoint, the following conditions hold:

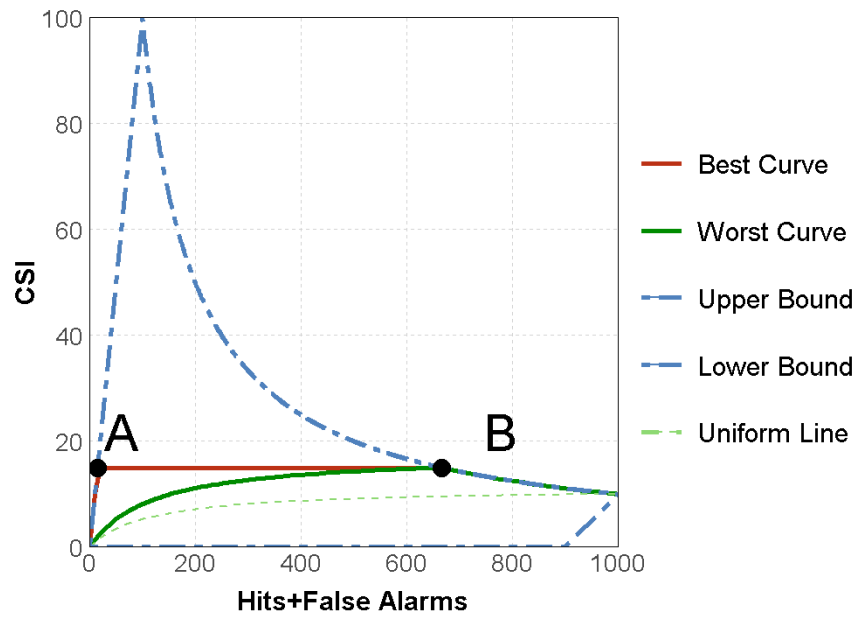
$$\lim_{x \rightarrow B_x^-} f(x) = \lim_{x \rightarrow B_x^+} f(x) = f(B_x).$$

$$\lim_{x \rightarrow B_x^-} f'(x) = \lim_{x \rightarrow B_x^+} f'(x) = Pr.$$

The derivative of the function is set to Pr at the breakpoint to ensure that the TOC curve begins to converge toward the uniform line after B_x . Through the consideration of multiple combinations of B_x and B_y , the design allows for the generation of TOC curves with different shapes while maintaining the same AUC values.



(a)



(b)

FIGURE B.1: TOC (a) and CSI (b) plots of the most and less efficient possible concave TOC curve. *Source: Own elaboration.*

Thus, the simulated TOC curves, which pass through the origin $(0, 0)$ and the upper-right endpoint (E, P) while satisfying the continuity and differentiability conditions, are defined as:

$$f(x) = \begin{cases} \min(x, a_1 + k_1 e^{b_1 x}), & x > B_x; \\ \min(P, a_2 + k_2 e^{b_2 x}), & x \leq B_x. \end{cases}$$

The parameters a_1, k_1, b_1, a_2, k_2 , and b_2 are determined by the constraints:

- $f(0) = 0$,
- $f(E) = P$,
- $\lim_{x \rightarrow B_x^-} f(x) = \lim_{x \rightarrow B_x^+} f(x) = f(B_x)$,
- $\lim_{x \rightarrow B_x^-} f'(x) = \lim_{x \rightarrow B_x^+} f'(x) = Pr$.

Appendix C

Appendix Chapter 3

C.1 Data sources and data processing for the case study

The data used for the forest gain map between 2005 and 2023 (Fig.3.1 (a)) come from the SIGPAC Navarra platform (Gobierno de Navarra 2025c). To generate this map, the original vector format maps were first rasterized and reclassified in ArcGIS for processing. The rasterization is done at a resolution of 100m, and the categories “FO-Forestal” and “PA-Pasto con Arbolado” are considered as forested areas. Two-category maps are generated, one for forest and one for non-forest areas. The binary forest gain map is calculated by overlapping the maps from 2005 and 2023, with all the forested areas from 2005 excluded from the sample as they are not eligible for forest gain.

The data used for calculating the suitability map (Fig.3.1 (b)) include annual average precipitation and altitude. The precipitation map is calculated using data provided by the Climate and Meteorology Department of the Government of Navarra (Gobierno de Navarra 2025b), with historical records disaggregated by municipality. The altitude map is the original one downloadable from Gobierno de Navarra (2025a). All maps were resampled to a resolution of 100m.

The suitability map is calculated through a Random Forest regression in R, using the Ranger package (Wright et al. 2024). Forest presence or absence is used as the

dependent variable, while precipitation and altitude serve as independent variables to estimate the probability of forest presence.

Definitions Index

Agreement, 39	Prevalence, 34
Allocation disagreement, 30	Quantity disagreement, 30
AUC, 44	Random classifier, 44
AUCSI, 98	Reference Area, 99
AUCSIS, 99	ROC, 236
Binary classification evaluation, 27	Skill, 39
Contingency table, 29	Soft classification, 29
Critical Success Index, 34	Stationarity, 25
Hard classification, 29	TOC, 43
Imbalanced datasets, 27	Total Area, 98
Land change model, 6	Total disagreement, 30

Equations Index

AUCSI, 99

CSI, 37

Gilbert Skill Score, 50, 121

Heidke Skill Score, 52

HSS, 51

Kappa Index, 52

Overall Agreement, 36

Pierce Skill Score, 52, 53

Revised Gilbert Skill Score, 123

Skill, 50